

First-Transition Prediction for Spatially Propagated Gaussian Disturbances

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Abstract

Predicting boundary-layer transition from stochastic disturbances requires accounting for the streamwise history of each disturbance realization. A stationwise exceedance probability,

$$\mathbb{P}(e(x) \geq e_{\text{thres}}(x)),$$

does not distinguish between realizations that first exceed the transition threshold at the current station and realizations that exceeded the threshold upstream. This limitation is important when disturbances undergo transient amplification, decay, and possible recrossing of the threshold.

This work develops a first-transition formulation for disturbances propagated by a linear spatial evolution model, with particular emphasis on the strongly non-normal One-Way Navier–Stokes (OWNS) equations. The initial disturbance is modeled as Gaussian and written as

$$\mathbf{q}(0) = \mathbf{B}_0 \mathbf{w}, \quad \mathbf{w} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}),$$

while the propagated state is represented in the common initial stochastic coordinates as

$$\mathbf{q}(x) = \mathbf{B}(x) \mathbf{w}.$$

For a quadratic disturbance energy, the normalized energy at each streamwise station is therefore a quadratic form in the same Gaussian vector,

$$\frac{e(x)}{e_{\text{thres}}(x)} = \mathbf{w}^* \mathbf{G}(x) \mathbf{w}.$$

The first-transition location is defined by the first streamwise threshold crossing,

$$X_{\text{tr}} = \inf \{x : \mathbf{w}^* \mathbf{G}(x) \mathbf{w} \geq 1\}.$$

A radial–directional decomposition of the Gaussian vector analytically integrates its radial component and reduces the transition probability to an expectation over directions in the initial stochastic space. For a real Gaussian vector of dimension r , the resulting first-transition cumulative distribution is

$$F_{X_{\text{tr}}}(x) = \mathbb{E}_{\mathbf{u}} \left[\overline{F}_{\chi_r^2} \left(\frac{1}{\max_{0 \leq \xi \leq x} \mathbf{u}^* \mathbf{G}(\xi) \mathbf{u}} \right) \right].$$

The running maximum supplies the required streamwise memory, while the common stochastic basis retains the transient growth and rotation induced by the non-normal propagator. The resulting algorithm computes the survival function, transition-location distribution, and right-censoring probability without reconstructing a large ensemble of Gaussian disturbance trajectories.

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Notation and Conventions

Unless otherwise stated, bold lowercase symbols denote vectors, bold uppercase symbols denote matrices, and nonbold symbols denote scalars. The superscripts $\top$ and $*$ denote the transpose and conjugate transpose, respectively. The main text uses a real standard Gaussian coordinate even when the Fourier-transformed physical state is complex.

The propagated factor is written in synthesis orientation:

$$\boldsymbol{q}(x) = \boldsymbol{B}(x)\boldsymbol{w}, \quad \boldsymbol{B}(x) \in \mathbb{C}^{n_q \times r}, \quad (1)$$

so that

$$\boldsymbol{C}(x, x) = \boldsymbol{B}(x)\boldsymbol{B}(x)^*. \quad (2)$$

If a numerical implementation instead stores a row-oriented factor $\boldsymbol{B}_{\text{row}}$ satisfying $\boldsymbol{C} = \boldsymbol{B}_{\text{row}}^* \boldsymbol{B}_{\text{row}}$, then the synthesis factor used in this document is

$$\boldsymbol{B} = \boldsymbol{B}_{\text{row}}^*. \quad (3)$$

Table 1: Principal notation used in the first-transition formulation.

Symbol	Type or dimension	Definition or interpretation
Spatial and spectral coordinates		
x	Scalar	Streamwise coordinate. It plays the mathematical role usually assigned to time in a first-passage problem.
x_0	Scalar	Inlet or reference station for the stochastic propagation problem. The main derivation sets $x_0 = 0$.
$x_{\max}$	Scalar	Terminal station of the computational domain.
y	Scalar	Wall-normal coordinate. In equations where $y(x)$ denotes normalized energy, the meaning is distinguished by context.
z	Scalar	Spanwise physical coordinate. The same letter is also occasionally used as a generic chi-square argument; the meaning is clear from context.
t	Scalar	Physical time.
ω	Scalar	Temporal angular frequency after Fourier transformation in time.
β	Scalar	Spanwise wavenumber after Fourier transformation in z .
OWNS state and propagation		

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Table 1 continued from the previous page

Symbol	Type or dimension	Definition or interpretation
$\mathbf{q}(x)$	$\mathbb{C}^{n_q}$	Discrete Fourier-transformed disturbance state at streamwise station x .
n_q	Integer	Dimension of the wall-normal-discretized OWNS state.
$\mathbf{L}(x)$	$\mathbb{C}^{n_q \times n_q}$	Effective downstream spatial generator after applying the OWNS projection and eliminating or consistently treating constraints.
$\mathbf{E}(x)$	Matrix	Mass or descriptor matrix in the generalized spatial system $\mathbf{E}\mathbf{q}_x = \mathbf{K}\mathbf{q}$.
$\mathbf{K}(x)$	Matrix	Spatial operator in the generalized OWNS system. This symbol is distinct from the aggregate reduction matrix $\mathbf{K}_\omega$.
$\mathbf{M}(x_a, x_b)$	$\mathbb{C}^{n_q \times n_q}$	Spatial propagator mapping a state at x_a to a state at x_b : $\mathbf{q}(x_b) = \mathbf{M}(x_a, x_b)\mathbf{q}(x_a)$.
$\mathbf{H}(x)$	$\mathbb{C}^{n_q \times n_q}$	Hermitian positive-semidefinite physical disturbance-energy matrix, including variable weights and wall-normal quadrature weights.
$\mathbf{W}(x)$	Matrix	Energy weighting factor satisfying $\mathbf{H}(x) = \mathbf{W}(x)^*\mathbf{W}(x)$.
Gaussian initial-condition model		
$\mathbf{C}(x_a, x_b)$	$\mathbb{C}^{n_q \times n_q}$	Two-point streamwise correlation $\mathbb{E}[\mathbf{q}(x_a)\mathbf{q}(x_b)^*]$.
$\mathbf{C}_0$	$\mathbb{C}^{n_q \times n_q}$	Initial correlation matrix $\mathbf{C}(0, 0)$.
$\mathbf{B}_0$	$\mathbb{C}^{n_q \times r}$	Initial synthesis factor satisfying $\mathbf{C}_0 = \mathbf{B}_0\mathbf{B}_0^*$.
$\mathbf{B}(x)$	$\mathbb{C}^{n_q \times r}$	Propagated synthesis factor $\mathbf{B}(x) = \mathbf{M}(0, x)\mathbf{B}_0$. It preserves the identity of the initial stochastic coordinates.
$\mathbf{B}_{\text{row}}(x)$	$\mathbb{C}^{r \times n_q}$	Optional row-oriented stored factor satisfying $\mathbf{C} = \mathbf{B}_{\text{row}}^*\mathbf{B}_{\text{row}}$. It is related to the document's synthesis factor by $\mathbf{B} = \mathbf{B}_{\text{row}}^*$.
$\mathbf{w}$	$\mathbb{R}^r$	Real standard Gaussian coordinate used in the main text: $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_r)$. The same realization is used at every streamwise station.
r	Integer	Retained real stochastic dimension before any transition-oriented reduction.
$\mathbf{z}$	$\mathbb{C}^r$	Proper complex Gaussian coordinate used only under the alternative convention $\mathbf{z} \sim \mathcal{CN}(\mathbf{0}, \mathbf{I}_r)$.
$\mathbf{P}$	Matrix	Pseudo-covariance of a complex random vector, $\mathbb{E}[(\mathbf{q} - \boldsymbol{\mu})(\mathbf{q} - \boldsymbol{\mu})^\top]$. It vanishes for a proper complex Gaussian vector.

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Symbol	Type or dimension	Definition or interpretation
Energy and normalized quadratic forms		
$e(x)$	Scalar	Disturbance energy $\mathbf{q}(x)^* \mathbf{H}(x) \mathbf{q}(x)$.
$e_{\text{thres}}(x)$	Positive scalar	Prescribed energy threshold used to define transition.
$\mathbf{A}(x)$	$\mathbb{R}^{r \times r}$	Energy matrix in the real initial stochastic coordinates: $\mathbf{A}(x) = \Re[\mathbf{B}(x)^* \mathbf{H}(x) \mathbf{B}(x)].$
$\mathbf{G}(x)$	$\mathbb{R}^{r \times r}$	The real-part operation is unnecessary in a fully real representation.
		Threshold-normalized quadratic-form matrix: $\mathbf{G}(x) = \mathbf{A}(x)/e_{\text{thres}}(x).$
$y(x; \mathbf{w})$	Scalar	Normalized disturbance energy: $y(x; \mathbf{w}) = \frac{e(x)}{e_{\text{thres}}(x)} = \mathbf{w}^\top \mathbf{G}(x) \mathbf{w}.$
$\mathbf{D}(x)$	Matrix	Normalized energy-output map $\mathbf{D}(x) = \mathbf{W}(x) \mathbf{B}(x) / \sqrt{e_{\text{thres}}(x)},$
for which $\mathbf{G}$ is the corresponding Gram matrix.		
First-transition statistics		
X_{tr}	Random variable	First-transition location: $X_{\text{tr}} = \inf\{x : e(x) \geq e_{\text{thres}}(x)\}.$
$Y_{\text{max}}(x; \mathbf{w})$	Scalar	A realization that does not transition by x_{max} is treated as right-censored or assigned $X_{\text{tr}} = \infty$.
		Running maximum of the normalized energy realization: $Y_{\text{max}}(x; \mathbf{w}) = \max_{\xi \leq x} \mathbf{w}^\top \mathbf{G}(\xi) \mathbf{w}.$
$S(x)$	Probability	Survival probability $\mathbb{P}(X_{\text{tr}} > x)$.
$F_{X_{\text{tr}}}(x)$	Probability	First-transition cumulative distribution $\mathbb{P}(X_{\text{tr}} \leq x) = 1 - S(x).$

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Symbol	Type or dimension	Definition or interpretation
$f_{X_{\text{tr}}}(x)$	Probability per unit x	First-transition-location density, where it exists: $f_{X_{\text{tr}}}(x) = dF_{X_{\text{tr}}}(x)/dx.$
$\lambda(x)$	Inverse streamwise length	Conditional transition hazard: $\lambda(x) = f_{X_{\text{tr}}}(x)/S(x).$
$p_{\text{local}}(x)$	Probability	Stationwise threshold-exceedance probability $\mathbb{P}[e(x) \geq e_{\text{thres}}(x)]$. It does not retain previous crossings.
p_n	Probability	Probability of first transition in $(x_{n-1}, x_n]$.
p_{censored}	Probability	Probability of no transition by x_{max} : $p_{\text{censored}} = S(x_{\text{max}}).$
h_n	Probability	Discrete conditional transition probability over $(x_{n-1}, x_n]$, given survival through x_{n-1} .
$\mathcal{S}(x)$	Subset of $\mathbb{R}^r$	Survivor set in the initial stochastic coordinates: $\mathcal{S}(x) = \bigcap_{\xi \leq x} \{\mathbf{w} : \mathbf{w}^\top \mathbf{G}(\xi) \mathbf{w} < 1\}.$
$\mathcal{D}(x)$	Subset of state space	Instantaneous physical safe domain $\{\mathbf{q} : \mathbf{q}^* \mathbf{H}(x) \mathbf{q} < e_{\text{thres}}(x)\}.$
Radial-directional representation		
R	Nonnegative scalar	Gaussian radius in $\mathbf{w} = R\mathbf{u}$. Under the real convention, $R^2 \sim \chi_r^2$.
$\mathbf{u}$	$\mathbb{S}^{r-1}$	Uniform initial stochastic direction, independent of R .
$\mathbb{S}^{r-1}$	Set	Unit sphere $\{\mathbf{u} \in \mathbb{R}^r : \ \mathbf{u}\ _2 = 1\}.$
$a_{\mathbf{u}}(x)$	Nonnegative scalar	Directional normalized gain: $a_{\mathbf{u}}(x) = \mathbf{u}^\top \mathbf{G}(x) \mathbf{u}.$
$m_{\mathbf{u}}(x)$	Nonnegative scalar	Directional running maximum: $m_{\mathbf{u}}(x) = \max_{\xi \leq x} a_{\mathbf{u}}(\xi).$

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Symbol	Type or dimension	Definition or interpretation
$\rho_{\mathbf{u}}(x)$	Nonnegative scalar	Critical Gaussian radius for survival through x : $\rho_{\mathbf{u}}(x) = 1/\sqrt{m_{\mathbf{u}}(x)}.$
$F_{\chi_r^2}$	Function	CDF of a chi-square random variable with r degrees of freedom.
$\overline{F}_{\chi_r^2}$	Function	Chi-square survival function: $1 - F_{\chi_r^2}$.
$\Phi_x(\mathbf{u})$	Probability	Directional survival integrand: $\Phi_x(\mathbf{u}) = F_{\chi_r^2}[1/m_{\mathbf{u}}(x)].$
$\Psi_x(\mathbf{u})$	Probability	Directional transition integrand: $\Psi_x(\mathbf{u}) = \overline{F}_{\chi_r^2}[1/m_{\mathbf{u}}(x)].$
Angular integration		
$\mathbf{u}_k$	$\mathbb{S}^{r-1}$	The k -th spherical quadrature, design, Monte Carlo, or RQMC direction.
ω_k	Nonnegative scalar	Normalized angular weight, with $\sum_{k=1}^K \omega_k = 1$. This symbol is distinct from the temporal frequency ω .
K	Integer	Number of angular directions in one quadrature or RQMC replicate.
J	Integer	Number of independent RQMC randomizations or scramblings.
$m_k(x)$	Nonnegative scalar	Resolved running maximum for angular direction $\mathbf{u}_k$.
Transition-oriented reduction		
$\mathbf{V}_s$	$\mathbb{R}^{r \times s}$	Fixed orthonormal transition-oriented stochastic basis.
s	Integer	Reduced stochastic dimension, with $s \leq r$.
$\mathbf{P}_s$	$\mathbb{R}^{r \times r}$	Orthogonal projector onto the reduced stochastic subspace: $\mathbf{P}_s = \mathbf{V}_s \mathbf{V}_s^\top$.
$\mathbf{B}_s(x)$	$\mathbb{C}^{n_q \times s}$	Reduced propagated factor $\mathbf{B}_s(x) = \mathbf{B}(x) \mathbf{V}_s$.
$\mathbf{G}_s(x)$	$\mathbb{R}^{s \times s}$	Reduced normalized energy matrix $\mathbf{G}_s(x) = \mathbf{V}_s^\top \mathbf{G}(x) \mathbf{V}_s.$

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Table 1 continued from the previous page

Symbol	Type or dimension	Definition or interpretation
$\mathbf{K}_\omega$	$\mathbb{R}^{r \times r}$	Weighted streamwise aggregate used for reduction: $\mathbf{K}_\omega = \int \omega(x) \mathbf{G}(x) dx.$ The subscript ω here denotes a weighting function and should not be confused with temporal frequency.
$\varepsilon_{G,s}$	Nonnegative scalar	Uniform reduced-matrix residual: $\max_x \ \mathbf{G}(x) - \mathbf{P}_s \mathbf{G}(x) \mathbf{P}_s\ _2.$
Stabilized factor propagation		
$\mathbf{Q}(x)$	$\mathbb{C}^{n_q \times r}$	Orthonormal propagated physical basis in $\mathbf{B}(x) = \mathbf{Q}(x) \mathbf{T}(x)$.
$\mathbf{T}(x)$	$\mathbb{C}^{r \times r}$	Accumulated coordinate map from the original Gaussian coordinate to the current orthonormal physical basis.
$\mathbf{R}_n$	$\mathbb{C}^{r \times r}$	Local upper-triangular factor generated by a periodic QR reorthogonalization.
$\ell(x)$	Scalar	Optional logarithmic scale in $\mathbf{B}(x) = \exp[\ell(x)] \mathbf{Q}(x) \overline{\mathbf{T}}(x)$.
$\overline{\mathbf{T}}(x)$	$\mathbb{C}^{r \times r}$	Scaled coordinate factor maintained at a moderate numerical norm.
Absorbing-density and level-crossing notation		
$h(\mathbf{q}, x)$	Scalar	Physical limit-state function: $h(\mathbf{q}, x) = \mathbf{q}^* \mathbf{H}(x) \mathbf{q} - e_{\text{thres}}(x).$
$g(\mathbf{w}, x)$	Scalar	Normalized limit-state function in initial coordinates: $g(\mathbf{w}, x) = \mathbf{w}^\top \mathbf{G}(x) \mathbf{w} - 1.$
$p(\mathbf{q}, x)$	Density	Unconditional physical-state probability density.
$\rho(\mathbf{q}, x)$	Unnormalized density	Survivor density in physical state space. Its total mass is $S(x)$.
$\rho_w(\mathbf{w}, x)$	Unnormalized density	Initial-coordinate survivor density $\phi_r(\mathbf{w}) \mathbf{1}_{S(x)}(\mathbf{w})$.

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Table 1 continued from the previous page

Symbol	Type or dimension	Definition or interpretation
$\nu_+(x)$	Crossings per unit x	Unconditional positive level-crossing intensity from Rice theory. It counts all positive crossings rather than only first crossings.
$N_+(0, x)$	Integer-valued random variable	Number of positive threshold crossings over $(0, x]$.
$\delta(\cdot)$	Distribution	Dirac delta, used in boundary-flux and Rice formulas.
$[a]_+$	Scalar	Positive part: $[a]_+ = \max(a, 0)$.
Numerical discretization and error notation		
x_n	Scalar	The n -th stored or adaptively inserted streamwise station.
$N + 1$	Integer	Number of stored streamwise stations.
N_y	Integer	Number of wall-normal degrees of freedom or grid points.
K_b	Integer	Number of angular directions processed in one computational block.
ε_{abs}	Positive scalar	Absolute tolerance for streamwise maximum refinement.
ε_{rel}	Positive scalar	Relative tolerance for streamwise maximum refinement.
$\varepsilon_F^{\text{ang}}$	Positive scalar	Target error associated with angular integration.
$\varepsilon_F^{\text{stream}}$	Positive scalar	Target CDF error associated with streamwise maximum detection.
$\varepsilon_F^{\text{red}}$	Positive scalar	Target CDF error associated with stochastic dimension reduction.
$\varepsilon_F^{\text{OWNS}}$	Positive scalar	Target CDF error associated with OWNS and physical-space discretization.
$\mathbb{E}[\cdot]$	Operator	Expectation.
$\mathbb{P}(\cdot)$	Operator	Probability.
$\text{tr}(\cdot)$	Operator	Matrix trace.
$\ \cdot\ _2$	Norm	Euclidean vector norm or induced matrix spectral norm, according to the argument.
$\ \cdot\ _F$	Norm	Matrix Frobenius norm.
$\lambda_{\text{max}}(\cdot)$	Operator	Largest eigenvalue of a real symmetric or Hermitian matrix.
$\sigma_{\text{max}}(\cdot)$	Operator	Largest singular value.
$\mathbf{1}_{\mathcal{A}}$	Function	Indicator of event or set $\mathcal{A}$.

1 Introduction

Boundary-layer transition is sensitive to the amplitudes, structures, and spectral content of disturbances entering the flow. When these disturbances are uncertain, transition is naturally described probabilistically. Consider a disturbance state $\mathbf{q}(x)$ at streamwise station x , with associated energy $e(x)$, and let $e_{\text{thres}}(x)$ denote an energy threshold used as an operational transition criterion. A common stationwise prediction is the local exceedance probability

$$p_{\text{local}}(x) = \mathbb{P}(e(x) \geq e_{\text{thres}}(x)). \quad (4)$$

This quantity describes the fraction of the ensemble above the threshold at a particular station. It does not, however, describe the fraction that has transitioned at or before that station.

The distinction arises because the energy of an individual realization need not increase monotonically downstream. In a non-normal boundary-layer system, a disturbance can experience transient amplification, exceed the prescribed threshold, and subsequently decay below it. Such a realization contributes to p_{local} while it is above the threshold but ceases to contribute after it falls below the threshold. If the first threshold crossing is interpreted as transition, the realization should instead remain classified as transitioned at all downstream stations. Consequently, a transition prediction based solely on eq. (4) does not possess the required streamwise memory.

To introduce this memory, define the first-transition location

$$X_{\text{tr}} = \inf \{x \in [0, x_{\text{max}}] : e(x) \geq e_{\text{thres}}(x)\}. \quad (5)$$

If the threshold is not reached within the computational interval, the realization is treated as right-censored at x_{max} , or, equivalently, assigned $X_{\text{tr}} = \infty$. The cumulative first-transition probability is

$$\begin{aligned} F_{X_{\text{tr}}}(x) &= \mathbb{P}(X_{\text{tr}} \leq x) \\ &= \mathbb{P}\left(\sup_{0 \leq \xi \leq x} \frac{e(\xi)}{e_{\text{thres}}(\xi)} \geq 1\right). \end{aligned} \quad (6)$$

Unlike the local exceedance probability, $F_{X_{\text{tr}}}(x)$ is necessarily nondecreasing in x . Its complement,

$$S(x) = 1 - F_{X_{\text{tr}}}(x) = \mathbb{P}(e(\xi) < e_{\text{thres}}(\xi) \text{ for every } \xi \in [0, x]), \quad (7)$$

is the probability that a realization remains below the transition threshold over its entire upstream history. The function $S(x)$ is referred to as the survival probability.

The present problem has additional structure that can be exploited. The initial disturbance is assumed Gaussian and represented as

$$\mathbf{q}(0) = \mathbf{B}_0 \mathbf{w}, \quad \mathbf{w} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad (8)$$

where $\mathbf{B}_0$ is a factor of the initial correlation matrix. Under a linear spatial propagator $\mathbf{M}(0, x)$, the state becomes

$$\mathbf{q}(x) = \mathbf{M}(0, x) \mathbf{B}_0 \mathbf{w} = \mathbf{B}(x) \mathbf{w}. \quad (9)$$

The same latent vector $\mathbf{w}$ therefore identifies a realization at every streamwise station. This common stochastic coordinate system contains information that is not available from the marginal correlation matrix at each station alone. In particular, independently diagonalizing the propagated correlation matrix at every x retains the one-point energy distribution but generally discards the streamwise correspondence between realizations.

For a quadratic energy norm,

$$e(x) = \mathbf{q}(x)^* \mathbf{H}(x) \mathbf{q}(x), \quad (10)$$

substitution of eq. (9) yields

$$e(x) = \mathbf{w}^* \mathbf{A}(x) \mathbf{w}, \quad \mathbf{A}(x) = \mathbf{B}(x)^* \mathbf{H}(x) \mathbf{B}(x). \quad (11)$$

Thus, the energy history is a family of correlated Gaussian quadratic forms generated by a single random vector. Defining

$$\mathbf{G}(x) = \frac{\mathbf{A}(x)}{e_{\text{thres}}(x)}, \quad (12)$$

the first-transition event becomes

$$\{X_{\text{tr}} \leq x\} = \left\{ \sup_{0 \leq \xi \leq x} \mathbf{w}^* \mathbf{G}(\xi) \mathbf{w} \geq 1 \right\}. \quad (13)$$

Direct trajectory Monte Carlo could estimate this event by repeatedly sampling $\mathbf{w}$, propagating the corresponding realization, and recording its first threshold crossing. The objective here is instead to exploit the Gaussian quadratic-form structure to avoid reconstructing a large ensemble of random energy trajectories. Writing a real r -dimensional standard Gaussian vector as

$$\mathbf{w} = R\mathbf{u}, \quad R^2 \sim \chi_r^2, \quad \mathbf{u} \sim \text{Unif}(\mathbb{S}^{r-1}), \quad (14)$$

separates the random magnitude from the initial disturbance direction. For a fixed direction $\mathbf{u}$, define

$$m_{\mathbf{u}}(x) = \max_{0 \leq \xi \leq x} \mathbf{u}^* \mathbf{G}(\xi) \mathbf{u}. \quad (15)$$

The radial variable can then be integrated analytically, leaving an angular expectation for the transition probability:

$$F_{X_{\text{tr}}}(x) = \mathbb{E}_{\mathbf{u}} \left[\overline{F}_{\chi_r^2} \left(\frac{1}{m_{\mathbf{u}}(x)} \right) \right]. \quad (16)$$

The running maximum in eq. (15) retains the largest normalized energy previously attained by each initial direction. A downstream energy peak contributes to the transition probability only when it exceeds all previous peaks for that direction. Threshold recrossings are therefore handled without an additional Markov or independence assumption.

This formulation is particularly suitable for the One-Way Navier–Stokes equations. Boundary-layer spatial propagators can be strongly non-normal, producing large transient growth and substantial rotation of the most amplified disturbance directions. Such behavior complicates a direct absorbing-density calculation in the full propagated state space because the density may become highly anisotropic and the relevant state dimension is typically large. In the common initial coordinates, by contrast, the Gaussian measure is fixed. The effects of non-normal propagation are retained in the streamwise evolution of $\mathbf{G}(x)$ and in the directional gain histories appearing in eq. (15).

The objectives of this document are therefore to:

- (i) formulate boundary-layer transition as a first-threshold crossing of a Gaussian quadratic-form process;
- (ii) show how the propagated factor $\mathbf{B}(x)$ preserves the streamwise dependence required for a history-aware prediction;

- (iii) derive a radial–directional representation of the survival and first-transition probabilities;
- (iv) provide a numerical algorithm for computing the distribution of X_{tr} from $\mathbf{B}(x)$, the energy metric, and the transition threshold;
- (v) address streamwise maximum detection, angular integration, dimension reduction, and numerical conditioning in strongly non-normal OWNS calculations; and
- (vi) describe verification and validation procedures, including comparison with trajectory Monte Carlo as an independent reference.

The remainder of the document is organized as follows. Section 2 introduces the spatial OWNS propagation model and the disturbance-energy norm. ?? defines the Gaussian initial-condition model and its representation in common stochastic coordinates. Sections 3 and 4 develop the quadratic-form and first-transition formulations. Section 5 derives the radial–directional reduction, and section 7 gives the corresponding computational algorithm. Subsequent sections address continuous streamwise maximum detection, angular integration, transition-oriented dimension reduction, numerical conditioning, and the relation to absorbing Liouville and first-passage methods. The final sections present verification procedures, the boundary-layer application, and the principal conclusions.

2 Spatial Evolution Model

This section defines the deterministic spatial evolution model used to propagate the uncertain inlet disturbance. The development is stated for a linear One-Way Navier–Stokes (OWNS) formulation at a prescribed temporal frequency and spanwise wavenumber. The probabilistic method developed in later sections does not require a particular OWNS projection or numerical discretization; it requires only a linear mapping from the initial disturbance to each downstream station.

2.1 One-Way Navier–Stokes formulation

Consider a boundary-layer base flow that is steady and homogeneous in the spanwise coordinate z . A perturbation may be decomposed into temporal-frequency and spanwise-wavenumber components according to

$$\mathbf{q}'(x, y, z, t) = \hat{\mathbf{q}}(x, y; \omega, \beta) \exp(i\beta z - i\omega t) + \text{c.c.}, \quad (17)$$

where ω is the angular frequency, β is the spanwise wavenumber, and c.c. denotes the complex-conjugate contribution required to produce a real physical field. In what follows, ω and β are fixed unless otherwise indicated, and their dependence is suppressed for notational clarity.

After discretization in the wall-normal coordinate and application of the appropriate OWNS projection, the downstream-marching system may be written in generalized form as

$$\mathbf{E}(x) \frac{d\mathbf{q}}{dx} = \mathbf{K}(x) \mathbf{q}(x), \quad (18)$$

where $\mathbf{q}(x) \in \mathbb{C}^{n_q}$ is the discrete perturbation state. Depending on the particular OWNS formulation, $\mathbf{q}$ may contain primitive variables, characteristic variables, auxiliary projection variables, or a reduced set of downstream-propagating degrees of freedom.

Equation (18) is assumed to have been restricted to a consistent downstream-marching state space. After eliminating algebraic constraints and applying the one-way projection, its evolution is

represented as

$$\frac{d\mathbf{q}}{dx} = \mathbf{L}(x)\mathbf{q}(x). \quad (19)$$

Here, $\mathbf{L}(x)$ denotes the effective spatial generator. If $\mathbf{E}(x)$ is nonsingular on the admissible state space, then formally

$$\mathbf{L}(x) = \mathbf{E}(x)^{-1}\mathbf{K}(x). \quad (20)$$

In practice, $\mathbf{L}(x)$ need not be explicitly formed. The OWNS marching algorithm may instead apply the action of the spatial evolution operator while enforcing its projection and constraint equations.

The propagator from a station x_0 to a downstream station x is denoted by

$$\mathbf{M}(x_0, x), \quad (21)$$

and is defined by

$$\frac{\partial}{\partial x}\mathbf{M}(x_0, x) = \mathbf{L}(x)\mathbf{M}(x_0, x), \quad (22)$$

$$\mathbf{M}(x_0, x_0) = \mathbf{I}. \quad (23)$$

The disturbance state consequently satisfies

$$\mathbf{q}(x) = \mathbf{M}(x_0, x)\mathbf{q}(x_0). \quad (24)$$

For three ordered stations $x_0 \leq x_1 \leq x_2$, the propagator has the composition property

$$\mathbf{M}(x_0, x_2) = \mathbf{M}(x_1, x_2)\mathbf{M}(x_0, x_1). \quad (25)$$

The reference station is taken to be $x_0 = 0$, so that

$$\mathbf{q}(x) = \mathbf{M}(0, x)\mathbf{q}(0). \quad (26)$$

The coordinate $x = 0$ denotes the inlet of the probabilistic propagation problem and need not coincide with the physical leading edge of the boundary layer.

The formulation in eq. (19) is linear. Thus, all uncertainty considered here enters through the initial state $\mathbf{q}(0)$, while the base flow, OWNS operator, and energy metric are treated as deterministic. Distributed stochastic forcing can be included in a more general model,

$$\frac{d\mathbf{q}}{dx} = \mathbf{L}(x)\mathbf{q}(x) + \mathbf{f}(x), \quad (27)$$

but is not considered in the present derivation. The initial-condition formulation is important because it allows every downstream state to be expressed using the same finite-dimensional Gaussian coordinate introduced in ??.

2.2 Non-normality of the spatial propagator

Boundary-layer disturbance dynamics are generally strongly non-normal. In a fixed Euclidean coordinate system, a matrix $\mathbf{L}$ is non-normal when

$$\mathbf{L}^*\mathbf{L} \neq \mathbf{L}\mathbf{L}^*. \quad (28)$$

For a spatially varying system, the amplification behavior depends on the complete ordered evolution generated by $\mathbf{L}(x)$, rather than on the eigensystem at any single streamwise station.

Non-normality permits substantial finite-distance amplification even when the local modal behavior is asymptotically or spatially decaying. In particular, the eigenvectors of the generator need not be orthogonal in the disturbance-energy inner product. Destructive interference between nonorthogonal components at the initial station can become constructive downstream, producing transient growth through mechanisms such as the lift-up effect.

For a constant Euclidean energy metric, the maximum amplification over the interval from x_0 to x is related to the largest singular value of the propagator:

$$\mathcal{G}_{\max}(x_0, x) = \max_{\mathbf{q}_0 \neq \mathbf{0}} \frac{\|\mathbf{M}(x_0, x)\mathbf{q}_0\|_2^2}{\|\mathbf{q}_0\|_2^2} = \sigma_{\max}^2(\mathbf{M}(x_0, x)). \quad (29)$$

The maximizing right singular vector identifies the initial disturbance direction producing the largest amplification at the specified terminal station. Because the propagator is spatially varying and non-normal, this optimal initial direction may change substantially with the terminal location x .

A corresponding expression in the physical energy norm is developed in section 2.3. The essential consequence for the probabilistic problem is that the relative orientation of propagated initial directions must be retained. Local eigenvalues of $\mathbf{L}(x)$, and even the eigenvalues of the one-point correlation matrix at each station, do not by themselves determine the streamwise history of an individual disturbance realization.

The non-normality also implies that the disturbance energy need not be monotone:

$$e(x_1) < e(x_2) > e(x_3), \quad x_1 < x_2 < x_3. \quad (30)$$

A realization may therefore cross an energy threshold during a transient-growth interval and subsequently return below the threshold. This behavior is the principal reason that the stationwise probability $\mathbb{P}(e(x) \geq e_{\text{thres}}(x))$ cannot be interpreted as the cumulative probability of transition.

Strong non-normality does not invalidate a first-transition formulation. Instead, it makes it necessary to use the full propagator, or an equivalent propagation of the initial stochastic basis. In the present method, non-normal growth and rotation are contained in the propagated factor

$$\mathbf{B}(x) = \mathbf{M}(0, x)\mathbf{B}_0. \quad (31)$$

The factor $\mathbf{B}(x)$ maps the same initial stochastic coordinate to the physical state at every station. It therefore preserves the pathwise information needed to determine first threshold crossings.

2.3 Energy inner product

Let the disturbance energy at station x be defined by the Hermitian quadratic form

$$e(x) = \mathbf{q}(x)^* \mathbf{H}(x) \mathbf{q}(x), \quad (32)$$

where

$$\mathbf{H}(x) = \mathbf{H}(x)^*, \quad \mathbf{H}(x) \succeq \mathbf{0}. \quad (33)$$

The matrix $\mathbf{H}(x)$ contains the physical energy weights, wall-normal quadrature weights, variable scalings, and any streamwise-dependent factors required by the selected norm. If nonphysical or constrained variables are included in the numerical state, $\mathbf{H}$ may be positive semidefinite rather than positive definite. It is assumed to be positive definite on the physically admissible disturbance subspace relevant to the transition criterion.

For two states $\mathbf{q}_1$ and $\mathbf{q}_2$, the associated streamwise-dependent inner product is

$$\langle \mathbf{q}_1, \mathbf{q}_2 \rangle_{\mathbf{H}(x)} = \mathbf{q}_1^* \mathbf{H}(x) \mathbf{q}_2. \quad (34)$$

The corresponding norm is

$$\|\mathbf{q}\|_{\mathbf{H}(x)} = \sqrt{\mathbf{q}^* \mathbf{H}(x) \mathbf{q}}. \quad (35)$$

For a nonzero initial state $\mathbf{q}_0 = \mathbf{q}(x_0)$, the finite-distance energy amplification is

$$\mathcal{G}(x_0, x; \mathbf{q}_0) = \frac{\mathbf{q}_0^* \mathbf{M}(x_0, x)^* \mathbf{H}(x) \mathbf{M}(x_0, x) \mathbf{q}_0}{\mathbf{q}_0^* \mathbf{H}(x_0) \mathbf{q}_0}. \quad (36)$$

The maximum possible amplification is

$$\mathcal{G}_{\max}(x_0, x) = \max_{\mathbf{q}_0 \neq \mathbf{0}} \mathcal{G}(x_0, x; \mathbf{q}_0). \quad (37)$$

If $\mathbf{H}(x_0)$ is positive definite, this maximum is the largest generalized eigenvalue of

$$\mathbf{M}(x_0, x)^* \mathbf{H}(x) \mathbf{M}(x_0, x) \mathbf{v} = \gamma \mathbf{H}(x_0) \mathbf{v}. \quad (38)$$

Equivalently,

$$\mathcal{G}_{\max}(x_0, x) = \sigma_{\max}^2 \left[\mathbf{H}(x)^{1/2} \mathbf{M}(x_0, x) \mathbf{H}(x_0)^{-1/2} \right]. \quad (39)$$

This expression shows explicitly that the relevant measure of non-normal amplification depends on both the propagator and the physical energy norm.

Differentiating eq. (32) along a solution of eq. (19) gives

$$\begin{aligned} \frac{de}{dx} &= \left(\frac{d\mathbf{q}}{dx} \right)^* \mathbf{H} \mathbf{q} + \mathbf{q}^* \mathbf{H} \frac{d\mathbf{q}}{dx} + \mathbf{q}^* \mathbf{H}' \mathbf{q} \\ &= \mathbf{q}^* [\mathbf{L}^* \mathbf{H} + \mathbf{H} \mathbf{L} + \mathbf{H}'] \mathbf{q}. \end{aligned} \quad (40)$$

Define the Hermitian instantaneous energy-growth operator

$$\mathbf{Q}(x) = \mathbf{L}(x)^* \mathbf{H}(x) + \mathbf{H}(x) \mathbf{L}(x) + \mathbf{H}'(x). \quad (41)$$

Then

$$\frac{de}{dx} = \mathbf{q}^* \mathbf{Q}(x) \mathbf{q}. \quad (42)$$

Although the local spatial eigenvalues of $\mathbf{L}(x)$ may correspond to modal decay, $\mathbf{Q}(x)$ can possess positive directions. Such directions exhibit instantaneous energy growth and contribute to the finite-distance transient amplification represented by $\mathbf{M}(x_0, x)$.

For a spatially varying transition threshold $e_{\text{thres}}(x) > 0$, define the normalized energy

$$y(x) = \frac{e(x)}{e_{\text{thres}}(x)}. \quad (43)$$

Its derivative along a trajectory is

$$\frac{dy}{dx} = \frac{\mathbf{q}^* \mathbf{Q}(x) \mathbf{q}}{e_{\text{thres}}(x)} - \frac{e'_{\text{thres}}(x)}{e_{\text{thres}}(x)} y(x). \quad (44)$$

At a threshold crossing, $y(x) = 1$. An outward crossing from the safe region $y < 1$ to the transitioned region $y \geq 1$ therefore satisfies

$$\left. \frac{dy}{dx} \right|_{y=1} > 0. \quad (45)$$

The derivative in eq. (44) will later be useful for detecting streamwise energy maxima and resolving crossings that occur between stored OWNS stations.

The energy definition in eq. (32) is assumed to be consistent across all streamwise stations. In particular, changes in the base flow, wall-normal grid, quadrature, or state normalization must be incorporated into $\mathbf{H}(x)$. This consistency is necessary for comparing the energy of a single propagated realization at different values of x and for interpreting the first threshold crossing as a streamwise transition location.

3 Quadratic-Form Representation of Energy

The common stochastic-coordinate representation developed in ?? converts the disturbance energy into a quadratic form of a single Gaussian vector. This representation separates the deterministic OWNS propagation from the probabilistic calculation and retains the dependence between all streamwise stations.

3.1 Energy in the initial stochastic coordinates

Recall that

$$\mathbf{q}(x) = \mathbf{B}(x)\mathbf{w}, \quad \mathbf{w} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_r), \quad (46)$$

where the same realization of $\mathbf{w}$ is used at every streamwise station. Substitution into the energy definition gives

$$\begin{aligned} e(x) &= \mathbf{q}(x)^* \mathbf{H}(x) \mathbf{q}(x) \\ &= \mathbf{w}^\top \mathbf{B}(x)^* \mathbf{H}(x) \mathbf{B}(x) \mathbf{w}. \end{aligned} \quad (47)$$

The transpose on $\mathbf{w}$ is used because the stochastic coordinates are real under the convention adopted in ??.

If $\mathbf{B}$ and $\mathbf{H}$ are real, define

$$\mathbf{A}(x) = \mathbf{B}(x)^\top \mathbf{H}(x) \mathbf{B}(x). \quad (48)$$

If the physical Fourier state is complex while $\mathbf{w}$ remains real, then

$$\mathbf{K}(x) = \mathbf{B}(x)^* \mathbf{H}(x) \mathbf{B}(x) \quad (49)$$

is Hermitian. Write

$$\mathbf{K}(x) = \Re \mathbf{K}(x) + i \Im \mathbf{K}(x). \quad (50)$$

For a Hermitian matrix, $\Re \mathbf{K}$ is real symmetric and $\Im \mathbf{K}$ is real skew-symmetric. Consequently, for every real vector $\mathbf{w}$,

$$\mathbf{w}^\top \Im \mathbf{K}(x) \mathbf{w} = 0. \quad (51)$$

The effective real quadratic-form matrix is therefore

$$\boxed{\mathbf{A}(x) = \Re [\mathbf{B}(x)^* \mathbf{H}(x) \mathbf{B}(x)]}. \quad (52)$$

With either a real physical state or a complex Fourier state, $\mathbf{A}(x) \in \mathbb{R}^{r \times r}$ is symmetric positive semidefinite, and the energy can be written as

$$\boxed{e(x) = \mathbf{w}^\top \mathbf{A}(x) \mathbf{w}. \quad (53)$$

Equivalently, if the state has first been realified as described in ??, then

$$\mathbf{A}(x) = \mathbf{B}_R(x)^\top \mathbf{H}_R(x) \mathbf{B}_R(x). \quad (54)$$

Equations (52) and (54) produce the same quadratic form when their Gaussian-coordinate conventions are consistent.

The matrix $\mathbf{A}(x)$ acts on the initial stochastic-coordinate space rather than on the full OWNS state space. Its dimension is $r \times r$, where r is the retained stochastic rank, even when the physical state dimension n_q is much larger. It contains the combined effects of:

- (i) the initial disturbance covariance through $\mathbf{B}_0$;
- (ii) the non-normal spatial propagation through $\mathbf{M}(0, x)$;
- (iii) the physical energy definition through $\mathbf{H}(x)$; and
- (iv) the relative orientation of all propagated initial disturbance directions.

Using $\mathbf{B}(x) = \mathbf{M}(0, x) \mathbf{B}_0$, the energy matrix may also be written as

$$\mathbf{A}(x) = \Re [\mathbf{B}_0^* \mathbf{M}(0, x)^* \mathbf{H}(x) \mathbf{M}(0, x) \mathbf{B}_0]. \quad (55)$$

This expression makes the role of finite-distance non-normal amplification explicit.

3.2 Normalization by the transition threshold

Let

$$e_{\text{thres}}(x) > 0 \quad (56)$$

be the prescribed transition-energy threshold. Define the normalized energy matrix

$$\boxed{\mathbf{G}(x) = \frac{\mathbf{A}(x)}{e_{\text{thres}}(x)}}. \quad (57)$$

The normalized disturbance energy is then

$$y(x) = \frac{e(x)}{e_{\text{thres}}(x)} = \mathbf{w}^\top \mathbf{G}(x) \mathbf{w}. \quad (58)$$

The safe and threshold-exceeded conditions are, respectively,

$$y(x) < 1 \iff \mathbf{w}^\top \mathbf{G}(x) \mathbf{w} < 1, \quad (59)$$

$$y(x) \geq 1 \iff \mathbf{w}^\top \mathbf{G}(x) \mathbf{w} \geq 1. \quad (60)$$

For a fixed station x , the safe set in stochastic-coordinate space is

$$\mathcal{E}(x) = \left\{ \mathbf{w} \in \mathbb{R}^r : \mathbf{w}^\top \mathbf{G}(x) \mathbf{w} < 1 \right\}. \quad (61)$$

If $\mathbf{G}(x)$ is positive definite, this set is the interior of an ellipsoid. If $\mathbf{G}(x)$ is positive semidefinite and rank deficient, it is an unbounded elliptic cylinder along the null space of $\mathbf{G}(x)$. These null-space directions carry no energy under the selected metric at that station, although they may become energetic downstream as $\mathbf{G}(x)$ evolves.

The normalization in eq. (57) permits a spatially varying threshold without modifying the subsequent first-transition derivation. Variation in the threshold is incorporated into the same matrix family that represents the propagated disturbance energy.

3.3 One-point energy distribution

At any fixed station, diagonalize the real symmetric matrix $\mathbf{A}(x)$ as

$$\mathbf{A}(x) = \mathbf{U}(x)\mathbf{\Lambda}_A(x)\mathbf{U}(x)^\top, \quad (62)$$

where

$$\mathbf{\Lambda}_A(x) = \text{diag}(\alpha_1(x), \dots, \alpha_r(x)), \quad \alpha_j(x) \geq 0. \quad (63)$$

Because the standard Gaussian distribution is rotationally invariant,

$$\mathbf{z}(x) = \mathbf{U}(x)^\top \mathbf{w} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_r) \quad (64)$$

at the individual station x . Therefore,

$$e(x) \stackrel{d}{=} \sum_{j=1}^r \alpha_j(x) Z_j^2, \quad Z_j \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1). \quad (65)$$

Thus, the one-point energy follows a weighted chi-square, or generalized chi-square, distribution.

Similarly, if

$$\mathbf{G}(x) = \mathbf{U}(x)\mathbf{\Lambda}_G(x)\mathbf{U}(x)^\top, \quad \mathbf{\Lambda}_G(x) = \text{diag}(\gamma_1(x), \dots, \gamma_r(x)), \quad (66)$$

then

$$y(x) \stackrel{d}{=} \sum_{j=1}^r \gamma_j(x) Z_j^2. \quad (67)$$

The local exceedance probability is consequently

$$p_{\text{local}}(x) = \mathbb{P} \left[\sum_{j=1}^r \gamma_j(x) Z_j^2 \geq 1 \right]. \quad (68)$$

This probability can be evaluated using standard numerical methods for weighted sums of chi-square variables, such as characteristic-function inversion.

The first two moments of the energy follow directly from standard Gaussian quadratic-form identities:

$$\mathbb{E}[e(x)] = \text{tr} \mathbf{A}(x), \quad (69)$$

$$\text{Var}[e(x)] = 2 \text{tr} [\mathbf{A}(x)^2]. \quad (70)$$

Likewise,

$$\mathbb{E}[y(x)] = \text{tr} \mathbf{G}(x), \quad (71)$$

$$\text{Var}[y(x)] = 2 \text{tr} [\mathbf{G}(x)^2]. \quad (72)$$

These formulas apply to the real standard Gaussian convention adopted here. The corresponding factors differ if a proper complex Gaussian representation is used without realification.

3.4 Streamwise dependence of the energy process

Although the distribution at a fixed station depends only on the eigenvalues of $\mathbf{A}(x)$, the joint energy history depends on the relative orientations of the matrices at different stations. For two stations x_1 and x_2 ,

$$e(x_1) = \mathbf{w}^\top \mathbf{A}(x_1) \mathbf{w}, \quad e(x_2) = \mathbf{w}^\top \mathbf{A}(x_2) \mathbf{w}. \quad (73)$$

Their covariance is

$$\boxed{\text{Cov}[e(x_1), e(x_2)] = 2 \text{tr}[\mathbf{A}(x_1) \mathbf{A}(x_2)]}. \quad (74)$$

For the normalized energies,

$$\text{Cov}[y(x_1), y(x_2)] = 2 \text{tr}[\mathbf{G}(x_1) \mathbf{G}(x_2)]. \quad (75)$$

To expose the dependence on relative orientation, write

$$\mathbf{A}(x_1) = \mathbf{U}_1 \mathbf{\Lambda}_1 \mathbf{U}_1^\top, \quad (76)$$

$$\mathbf{A}(x_2) = \mathbf{U}_2 \mathbf{\Lambda}_2 \mathbf{U}_2^\top. \quad (77)$$

Then

$$\text{tr}[\mathbf{A}(x_1) \mathbf{A}(x_2)] = \text{tr}[\mathbf{\Lambda}_1 \mathbf{R}_{12} \mathbf{\Lambda}_2 \mathbf{R}_{12}^\top], \quad \mathbf{R}_{12} = \mathbf{U}_1^\top \mathbf{U}_2. \quad (78)$$

The eigenvalue matrices determine the marginal distributions, while $\mathbf{R}_{12}$ describes the rotation of energetic directions between the two stations.

Independent diagonalization at each station replaces the fixed Gaussian vector $\mathbf{w}$ by station-dependent rotated coordinates,

$$\mathbf{z}(x) = \mathbf{U}(x)^\top \mathbf{w}. \quad (79)$$

Although every $\mathbf{z}(x)$ is marginally standard Gaussian, the vectors at different stations are not independent and are not identical:

$$\mathbf{z}(x_2) = \mathbf{U}(x_2)^\top \mathbf{U}(x_1) \mathbf{z}(x_1). \quad (80)$$

Discarding this relationship preserves eq. (68) but changes the joint process and, therefore, its first-transition distribution.

A simple example demonstrates the distinction. Consider two two-dimensional normalized energy matrices with equal eigenvalues,

$$\mathbf{G}(x_1) = \begin{bmatrix} \gamma_1 & 0 \\ 0 & \gamma_2 \end{bmatrix}, \quad (81)$$

$$\mathbf{G}(x_2) = \mathbf{R}(\theta) \begin{bmatrix} \gamma_1 & 0 \\ 0 & \gamma_2 \end{bmatrix} \mathbf{R}(\theta)^\top, \quad (82)$$

where $\mathbf{R}(\theta)$ is a planar rotation. The local normalized energy distributions at x_1 and x_2 are identical for every θ , because the eigenvalues are unchanged. Their covariance, however, depends on θ :

$$\text{Cov}[y(x_1), y(x_2)] = 2 \text{tr}[\mathbf{G}(x_1) \mathbf{G}(x_2)]. \quad (83)$$

Consequently, the probability of exceeding the threshold at either station also depends on θ . This orientation information is retained by constructing every $\mathbf{G}(x)$ from the commonly propagated factor $\mathbf{B}(x)$.

3.5 Streamwise derivative of the quadratic-form matrix

The derivative of $\mathbf{A}(x)$ is useful for locating transient energy maxima and detecting threshold crossings between stored streamwise stations. Suppose that

$$\mathbf{B}'(x) = \mathbf{L}(x)\mathbf{B}(x). \quad (84)$$

For a real physical representation,

$$\begin{aligned} \mathbf{A}'(x) &= \mathbf{B}'(x)^\top \mathbf{H}(x) \mathbf{B}(x) + \mathbf{B}(x)^\top \mathbf{H}'(x) \mathbf{B}(x) \\ &\quad + \mathbf{B}(x)^\top \mathbf{H}(x) \mathbf{B}'(x) \\ &= \mathbf{B}(x)^\top \left[\mathbf{L}(x)^\top \mathbf{H}(x) + \mathbf{H}(x) \mathbf{L}(x) + \mathbf{H}'(x) \right] \mathbf{B}(x). \end{aligned} \quad (85)$$

For a complex Fourier-state representation with real stochastic coordinates,

$$\boxed{\mathbf{A}'(x) = \Re \left\{ \mathbf{B}(x)^* \left[\mathbf{L}(x)^* \mathbf{H}(x) + \mathbf{H}(x) \mathbf{L}(x) + \mathbf{H}'(x) \right] \mathbf{B}(x) \right\}.} \quad (86)$$

The normalized matrix derivative is

$$\boxed{\mathbf{G}'(x) = \frac{\mathbf{A}'(x)}{e_{\text{thres}}(x)} - \frac{e'_{\text{thres}}(x)}{e_{\text{thres}}(x)} \mathbf{G}(x).} \quad (87)$$

For a fixed realization $\mathbf{w}$, the normalized energy derivative is therefore

$$y'(x) = \mathbf{w}^\top \mathbf{G}'(x) \mathbf{w}. \quad (88)$$

These expressions avoid finite differencing the energy of each realization and will be used in ??.

3.6 Basis invariance

The quadratic-form representation is invariant under a fixed orthogonal change of the real stochastic coordinates. Let

$$\tilde{\mathbf{w}} = \mathbf{U}^\top \mathbf{w}, \quad \tilde{\mathbf{B}}(x) = \mathbf{B}(x) \mathbf{U}, \quad \mathbf{U}^\top \mathbf{U} = \mathbf{I}. \quad (89)$$

Then

$$\tilde{\mathbf{w}} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_r), \quad (90)$$

and

$$\tilde{\mathbf{G}}(x) = \mathbf{U}^\top \mathbf{G}(x) \mathbf{U}. \quad (91)$$

The normalized energy is unchanged:

$$\tilde{\mathbf{w}}^\top \tilde{\mathbf{G}}(x) \tilde{\mathbf{w}} = \mathbf{w}^\top \mathbf{G}(x) \mathbf{w}. \quad (92)$$

Consequently, the first-transition distribution is independent of the particular fixed square-root factor used for the initial correlation.

This invariance permits QR or singular-value factorizations to be used for numerical stabilization, provided all associated transformations are accumulated consistently. It does not permit an arbitrary streamwise-dependent reset of the stochastic coordinates. The latter would replace the common realization $\mathbf{w}$ by a different coordinate vector at each station and would alter the history of the quadratic-form process.

4 First-Transition Formulation

The quadratic-form representation defines the normalized energy of each realization at every streamwise station using the same Gaussian coordinate vector. This section uses that representation to define the first-transition location, survival probability, transition distribution, and associated safe set. The resulting quantities retain the complete upstream threshold history.

4.1 Definition of the first-transition location

For a realization $\mathbf{w}$, define the normalized energy path

$$y(x; \mathbf{w}) = \mathbf{w}^\top \mathbf{G}(x) \mathbf{w}, \quad x \in [0, x_{\max}]. \quad (93)$$

The transition criterion is the first attainment or exceedance of the normalized threshold $y = 1$. The first-transition location is therefore

$$X_{\text{tr}}(\mathbf{w}) = \inf \{x \in [0, x_{\max}] : y(x; \mathbf{w}) \geq 1\}. \quad (94)$$

Equivalently,

$$X_{\text{tr}}(\mathbf{w}) = \inf \left\{ x \in [0, x_{\max}] : \mathbf{w}^\top \mathbf{G}(x) \mathbf{w} \geq 1 \right\}. \quad (95)$$

The convention

$$\inf \emptyset = \infty \quad (96)$$

is adopted. Thus, a realization that never reaches the transition threshold on the computational interval is assigned $X_{\text{tr}} = \infty$. In a finite-domain statistical analysis, this outcome is equivalently interpreted as right censoring at $x_{\max}$.

If the realization is already at or above the threshold at the inlet, then

$$y(0; \mathbf{w}) \geq 1 \implies X_{\text{tr}}(\mathbf{w}) = 0. \quad (97)$$

The probability distribution can therefore contain a point mass at $x = 0$. It can also contain a point mass at infinity representing realizations that do not transition over the modeled domain.

Throughout this document, threshold attainment counts as transition, as indicated by the inequality $y \geq 1$. If $\mathbf{G}(x)$ varies continuously with x , then any realization that begins below the threshold and later exceeds it satisfies

$$y(X_{\text{tr}}; \mathbf{w}) = 1. \quad (98)$$

The first contact need not be a transverse upcrossing. For example, a trajectory may touch the threshold at a local maximum without crossing into $y > 1$. Under the operational criterion adopted here, such tangential contact is classified as transition. For smooth nondegenerate random models, exact tangencies commonly have probability zero, but the distinction should be retained in numerical implementations with finite tolerances.

The definition in eq. (94) treats transition as irreversible: once a realization reaches the threshold, its transition location is fixed even if the energy later decreases below the threshold. No post-transition model is required because the calculation terminates at the first threshold encounter.

4.2 Running-maximum representation

Define the running maximum of the normalized energy for a realization $\mathbf{w}$ by

$$Y_{\max}(x; \mathbf{w}) = \sup_{0 \leq \xi \leq x} y(\xi; \mathbf{w}). \quad (99)$$

If $y(\cdot; \mathbf{w})$ is continuous on the compact interval $[0, x]$, the supremum is attained and can be written as a maximum:

$$Y_{\max}(x; \mathbf{w}) = \max_{0 \leq \xi \leq x} \mathbf{w}^\top \mathbf{G}(\xi) \mathbf{w}. \quad (100)$$

The first-transition event can then be expressed as

$$\boxed{\{X_{\text{tr}} \leq x\} = \{Y_{\max}(x; \mathbf{w}) \geq 1\}}. \quad (101)$$

Similarly,

$$\boxed{\{X_{\text{tr}} > x\} = \{Y_{\max}(x; \mathbf{w}) < 1\}}, \quad (102)$$

subject to the convention that threshold attainment itself counts as transition.

The function $Y_{\max}(x; \mathbf{w})$ is nondecreasing in x , even when the instantaneous normalized energy $y(x; \mathbf{w})$ is strongly nonmonotone. It is therefore the path-dependent state variable that supplies the desired streamwise memory. If a realization reaches a large transient-growth peak and then decays, the running maximum retains the value of the earlier peak.

4.3 Survival function and transition distribution

The survival function is defined as

$$S(x) = \mathbb{P}(X_{\text{tr}} > x). \quad (103)$$

Using eq. (102),

$$S(x) = \mathbb{P}[Y_{\max}(x; \mathbf{w}) < 1] \quad (104)$$

$$= \mathbb{P}\left[\mathbf{w}^\top \mathbf{G}(\xi) \mathbf{w} < 1 \quad \text{for every } \xi \in [0, x]\right]. \quad (105)$$

Because the event of surviving to a later station is contained in the event of surviving to any earlier station,

$$x_1 \leq x_2 \implies S(x_2) \leq S(x_1). \quad (106)$$

The cumulative distribution function of the first-transition location is

$$F_{X_{\text{tr}}}(x) = \mathbb{P}(X_{\text{tr}} \leq x). \quad (107)$$

It is related to the survival function by

$$\boxed{F_{X_{\text{tr}}}(x) = 1 - S(x)}. \quad (108)$$

Equivalently,

$$F_{X_{\text{tr}}}(x) = \mathbb{P}\left[\sup_{0 \leq \xi \leq x} \mathbf{w}^\top \mathbf{G}(\xi) \mathbf{w} \geq 1\right]. \quad (109)$$

The transition CDF is necessarily nondecreasing:

$$x_1 \leq x_2 \implies F_{X_{\text{tr}}}(x_1) \leq F_{X_{\text{tr}}}(x_2). \quad (110)$$

At a fixed station, the local threshold-exceedance event is a subset of the event that the threshold has been exceeded at some upstream location:

$$\{y(x) \geq 1\} \subseteq \{X_{\text{tr}} \leq x\}. \quad (111)$$

Consequently,

$$\boxed{p_{\text{local}}(x) = \mathbb{P}[y(x) \geq 1] \leq F_{X_{\text{tr}}}(x)}. \quad (112)$$

Equality holds when all realizations that crossed previously remain above the threshold at x , as occurs when every normalized energy path is nondecreasing. For transiently amplified disturbances that can decay after crossing, the inequality can be strict.

The probability of no transition over the complete computational domain is

$$p_{\text{censored}} = \mathbb{P}(X_{\text{tr}} > x_{\text{max}}) = S(x_{\text{max}}). \quad (113)$$

Equivalently, under the convention $X_{\text{tr}} = \infty$ for nontransitioning realizations,

$$\mathbb{P}(X_{\text{tr}} = \infty) = S(x_{\text{max}}). \quad (114)$$

4.4 Transition density, interval probabilities, and hazard

If $F_{X_{\text{tr}}}$ is absolutely continuous over an interval in the interior of the streamwise domain, the first-transition-location density is

$$f_{X_{\text{tr}}}(x) = \frac{d}{dx} F_{X_{\text{tr}}}(x) = -\frac{d}{dx} S(x). \quad (115)$$

This density measures the probability of first transition per unit streamwise distance. It should not be confused with the derivative of the local exceedance probability, which may be negative during downstream decay.

The probability of first transition within a streamwise interval $(x_a, x_b]$ is

$$\mathbb{P}(x_a < X_{\text{tr}} \leq x_b) = F_{X_{\text{tr}}}(x_b) - F_{X_{\text{tr}}}(x_a) \quad (116)$$

$$= S(x_a) - S(x_b). \quad (117)$$

For a computational grid

$$0 = x_0 < x_1 < \cdots < x_N = x_{\text{max}}, \quad (118)$$

define

$$p_n = \mathbb{P}(x_{n-1} < X_{\text{tr}} \leq x_n) = F_{X_{\text{tr}}}(x_n) - F_{X_{\text{tr}}}(x_{n-1}), \quad n = 1, \dots, N. \quad (119)$$

The inlet probability mass is

$$p_0 = \mathbb{P}(X_{\text{tr}} = 0) = F_{X_{\text{tr}}}(0), \quad (120)$$

and probability conservation requires

$$p_0 + \sum_{n=1}^N p_n + p_{\text{censored}} = 1. \quad (121)$$

Where $S(x) > 0$ and a transition density exists, define the streamwise hazard rate by

$$\boxed{\lambda(x) = \frac{f_{X_{\text{tr}}}(x)}{S(x)}}. \quad (122)$$

The hazard has the conditional interpretation

$$\lambda(x) = \lim_{\Delta x \rightarrow 0^+} \frac{\mathbb{P}(x < X_{\text{tr}} \leq x + \Delta x \mid X_{\text{tr}} > x)}{\Delta x}. \quad (123)$$

It is therefore a transition probability per unit downstream distance conditioned on the realization having remained below the threshold up to x .

For an absolutely continuous survival function,

$$\lambda(x) = -\frac{d}{dx} \log S(x), \quad (124)$$

and

$$S(x) = S(0^+) \exp \left[-\int_{0^+}^x \lambda(\xi) d\xi \right]. \quad (125)$$

The factor $S(0^+) = 1 - p_0$ accounts for any immediate-transition probability at the inlet.

A discrete conditional transition probability over $(x_{n-1}, x_n]$ is

$$h_n = \mathbb{P}(x_{n-1} < X_{\text{tr}} \leq x_n \mid X_{\text{tr}} > x_{n-1}) = \frac{S(x_{n-1}) - S(x_n)}{S(x_{n-1})}, \quad (126)$$

provided $S(x_{n-1}) > 0$. Unlike a model that assumes independent stationwise exceedances, this conditional probability is derived from the common Gaussian realization and its complete running-maximum history.

4.5 Safe set in the initial stochastic coordinates

At a single station ξ , define the instantaneous safe set

$$\mathcal{E}(\xi) = \left\{ \mathbf{w} \in \mathbb{R}^r : \mathbf{w}^\top \mathbf{G}(\xi) \mathbf{w} < 1 \right\}. \quad (127)$$

To remain untransitioned through station x , a realization must belong to the instantaneous safe set at every upstream station. Accordingly, define the survivor set

$$\boxed{\mathcal{S}(x) = \bigcap_{0 \leq \xi \leq x} \mathcal{E}(\xi) = \bigcap_{0 \leq \xi \leq x} \left\{ \mathbf{w} : \mathbf{w}^\top \mathbf{G}(\xi) \mathbf{w} < 1 \right\}.} \quad (128)$$

The survivor sets are nested:

$$x_1 \leq x_2 \implies \mathcal{S}(x_2) \subseteq \mathcal{S}(x_1). \quad (129)$$

This geometric nesting is the stochastic-coordinate-space expression of irreversible transition.

Let

$$\phi_r(\mathbf{w}) = \frac{1}{(2\pi)^{r/2}} \exp \left(-\frac{1}{2} \mathbf{w}^\top \mathbf{w} \right) \quad (130)$$

denote the standard Gaussian density. The survival probability is the Gaussian probability content of the survivor set:

$$\boxed{S(x) = \int_{\mathcal{S}(x)} \phi_r(\mathbf{w}) d\mathbf{w}.} \quad (131)$$

The transition CDF can equivalently be expressed as

$$F_{X_{\text{tr}}}(x) = \int_{\mathbb{R}^r \setminus \mathcal{S}(x)} \phi_r(\mathbf{w}) d\mathbf{w}. \quad (132)$$

Each instantaneous safe set $\mathcal{E}(\xi)$ is an ellipsoid or elliptic cylinder. Their continuously indexed intersection $\mathcal{S}(x)$ is centrally symmetric and convex because each $\mathbf{G}(\xi)$ is positive semidefinite. In particular,

$$\mathbf{w} \in \mathcal{S}(x) \iff -\mathbf{w} \in \mathcal{S}(x). \quad (133)$$

The boundary of $\mathcal{S}(x)$, however, need not be a single ellipsoid. Different portions of the boundary can be determined by the threshold ellipsoids associated with different upstream stations. Strong non-normality can cause these active ellipsoids to rotate and change aspect ratio substantially with x .

The safe-set representation is equivalent to an absorbing probability-density formulation. In the physical OWNS state space, probability is transported by the linear dynamics and removed at the energy boundary. In the initial stochastic coordinates, the Gaussian density ϕ_r is stationary while the family of safe ellipsoids evolves. The latter representation avoids solving a high-dimensional Liouville equation and forms the basis of the radial-directional reduction in section 5.

4.6 Discrete and continuous first-transition events

If $\mathbf{G}(x)$ is available only at stored stations $\{x_n\}_{n=0}^N$, a stationwise approximation to the first-transition location is

$$X_{\text{tr}}^{(N)}(\mathbf{w}) = \min \left\{ x_n : \mathbf{w}^\top \mathbf{G}(x_n) \mathbf{w} \geq 1 \right\}, \quad (134)$$

with $X_{\text{tr}}^{(N)} = \infty$ if the set is empty. The corresponding discrete survivor set is

$$\mathcal{S}_N(x_n) = \bigcap_{j=0}^n \left\{ \mathbf{w} : \mathbf{w}^\top \mathbf{G}(x_j) \mathbf{w} < 1 \right\}. \quad (135)$$

Because the discrete grid examines only a subset of the continuous streamwise interval,

$$\mathcal{S}(x_n) \subseteq \mathcal{S}_N(x_n). \quad (136)$$

It follows that

$$S(x_n) \leq S_N(x_n), \quad (137)$$

$$F_{X_{\text{tr}}}(x_n) \geq F_{X_{\text{tr}}^{(N)}}(x_n). \quad (138)$$

Thus, using only stored stations generally underestimates the continuous first-transition probability. A disturbance may exceed the threshold between two stations and decay below it before the next stored station, particularly when non-normal amplification creates a narrow streamwise peak.

If the streamwise grids are nested and become dense in $[0, x]$, and if $\mathbf{G}(\xi)$ is continuous, the discrete running maximum converges monotonically to the continuous running maximum:

$$\max_{x_j \leq x} \mathbf{w}^\top \mathbf{G}(x_j) \mathbf{w} \nearrow \max_{0 \leq \xi \leq x} \mathbf{w}^\top \mathbf{G}(\xi) \mathbf{w}. \quad (139)$$

Accordingly, the discrete transition probability converges from below, subject to standard measure-zero qualifications at exact threshold contact. Continuous maximum detection and adaptive streamwise refinement are developed in ??.

5 Radial–Directional Reduction

The first-transition probability in eq. (109) is an r -dimensional Gaussian integral over the complement of an evolving intersection of ellipsoids. The isotropy of the standard Gaussian distribution permits the radial coordinate to be integrated analytically. The remaining calculation is an expectation over directions in the initial stochastic-coordinate space.

This reduction is exact for the finite-dimensional, zero-mean, real-Gaussian model developed in the preceding sections. It does not assume stationwise independence, monotonic disturbance energy, or normality of the OWNS propagator.

5.1 Gaussian radial decomposition

Let

$$\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_r) \quad (140)$$

and write

$$\boxed{\mathbf{w} = R\mathbf{u}}, \quad (141)$$

where

$$R = \|\mathbf{w}\|_2 \quad (142)$$

is the radial coordinate and

$$\mathbf{u} = \frac{\mathbf{w}}{\|\mathbf{w}\|_2} \in \mathbb{S}^{r-1} \quad (143)$$

is the direction. The event $\mathbf{w} = \mathbf{0}$, for which $\mathbf{u}$ is undefined, has probability zero and does not affect the following results.

For a standard Gaussian vector, the radius and direction are independent:

$$R \perp\!\!\!\perp \mathbf{u}. \quad (144)$$

Their distributions are

$$R^2 \sim \chi_r^2, \quad (145)$$

$$\mathbf{u} \sim \text{Unif}(\mathbb{S}^{r-1}), \quad (146)$$

where

$$\mathbb{S}^{r-1} = \{\mathbf{u} \in \mathbb{R}^r : \|\mathbf{u}\|_2 = 1\}. \quad (147)$$

The probability density of $Z = R^2$ is

$$f_{\chi_r^2}(z) = \frac{z^{r/2-1} \exp(-z/2)}{2^{r/2} \Gamma(r/2)}, \quad z > 0. \quad (148)$$

Its cumulative distribution and survival functions are denoted by

$$F_{\chi_r^2}(z) = \mathbb{P}(R^2 \leq z), \quad (149)$$

$$\overline{F}_{\chi_r^2}(z) = 1 - F_{\chi_r^2}(z) = \mathbb{P}(R^2 > z). \quad (150)$$

They may be expressed in terms of the regularized incomplete gamma functions:

$$F_{\chi_r^2}(z) = P\left(\frac{r}{2}, \frac{z}{2}\right), \quad (151)$$

$$\overline{F}_{\chi_r^2}(z) = Q\left(\frac{r}{2}, \frac{z}{2}\right). \quad (152)$$

The radial decomposition may also be obtained directly from the Gaussian measure. In spherical coordinates,

$$\phi_r(\mathbf{w}) d\mathbf{w} = \frac{1}{(2\pi)^{r/2}} \exp\left(-\frac{R^2}{2}\right) R^{r-1} dR d\sigma(\mathbf{u}), \quad (153)$$

where $d\sigma$ is the surface measure on $\mathbb{S}^{r-1}$. The factorization into a function of R and a constant angular density establishes eq. (144).

5.2 Directional energy gain

Substituting $\mathbf{w} = R\mathbf{u}$ into the normalized energy gives

$$\begin{aligned} y(x; R, \mathbf{u}) &= (R\mathbf{u})^\top \mathbf{G}(x) (R\mathbf{u}) \\ &= R^2 \mathbf{u}^\top \mathbf{G}(x) \mathbf{u}. \end{aligned} \quad (154)$$

For a fixed direction $\mathbf{u}$, define the directional normalized energy gain

$$\boxed{a_{\mathbf{u}}(x) = \mathbf{u}^\top \mathbf{G}(x) \mathbf{u}.} \quad (155)$$

Because $\mathbf{G}(x) \succeq \mathbf{0}$,

$$a_{\mathbf{u}}(x) \geq 0. \quad (156)$$

The normalized energy along the ray $\{R\mathbf{u} : R \geq 0\}$ is therefore

$$y(x; R, \mathbf{u}) = R^2 a_{\mathbf{u}}(x). \quad (157)$$

The function $a_{\mathbf{u}}(x)$ has a direct physical interpretation. Using the propagated factor,

$$a_{\mathbf{u}}(x) = \frac{[\mathbf{B}(x)\mathbf{u}]^* \mathbf{H}(x) [\mathbf{B}(x)\mathbf{u}]}{e_{\text{thres}}(x)}, \quad (158)$$

with the real part understood as in eq. (52) when required. The vector $\mathbf{B}_0\mathbf{u}$ specifies the structure of a unit-radius initial disturbance in stochastic-coordinate space, and $\mathbf{B}(x)\mathbf{u}$ is its OWNS-propagated state. Thus, $a_{\mathbf{u}}(x)$ is the normalized energy history produced by that initial direction at unit stochastic radius.

For a fixed x , the largest possible directional gain is

$$\max_{\|\mathbf{u}\|_2=1} a_{\mathbf{u}}(x) = \lambda_{\max}[\mathbf{G}(x)]. \quad (159)$$

The corresponding eigenvector is the most dangerous initial stochastic direction for threshold exceedance at that particular station. The direction that is most important for first transition over an interval need not coincide with the dominant eigenvector at any preselected terminal station, because the first-transition calculation depends on the entire gain history.

5.3 Directional running maximum

Define the running maximum of the directional gain by

$$\boxed{m_{\mathbf{u}}(x) = \sup_{0 \leq \xi \leq x} a_{\mathbf{u}}(\xi) = \sup_{0 \leq \xi \leq x} \mathbf{u}^\top \mathbf{G}(\xi) \mathbf{u}.} \quad (160)$$

If $\mathbf{G}(\xi)$ is continuous, the supremum is attained:

$$m_{\mathbf{u}}(x) = \max_{0 \leq \xi \leq x} \mathbf{u}^\top \mathbf{G}(\xi) \mathbf{u}. \quad (161)$$

For every direction,

$$x_1 \leq x_2 \implies m_{\mathbf{u}}(x_1) \leq m_{\mathbf{u}}(x_2). \quad (162)$$

The maximum normalized energy attained by a realization $\mathbf{w} = R\mathbf{u}$ through station x is

$$\begin{aligned} Y_{\max}(x; R, \mathbf{u}) &= \sup_{0 \leq \xi \leq x} R^2 a_{\mathbf{u}}(\xi) \\ &= R^2 m_{\mathbf{u}}(x). \end{aligned} \quad (163)$$

It follows that transition by x occurs if and only if

$$\boxed{R^2 m_{\mathbf{u}}(x) \geq 1}. \quad (164)$$

Likewise, survival through x requires

$$R^2 m_{\mathbf{u}}(x) < 1. \quad (165)$$

When $m_{\mathbf{u}}(x) > 0$, define the critical squared radius

$$\rho_{\mathbf{u}}^2(x) = \frac{1}{m_{\mathbf{u}}(x)}. \quad (166)$$

The corresponding critical radius is

$$\rho_{\mathbf{u}}(x) = \frac{1}{\sqrt{m_{\mathbf{u}}(x)}}. \quad (167)$$

For the initial direction $\mathbf{u}$,

$$R < \rho_{\mathbf{u}}(x) \implies X_{\text{tr}} > x, \quad (168)$$

$$R \geq \rho_{\mathbf{u}}(x) \implies X_{\text{tr}} \leq x. \quad (169)$$

If

$$m_{\mathbf{u}}(x) = 0, \quad (170)$$

then the selected direction produces zero energy over the entire interval $[0, x]$, and every finite radius survives through x . For notational convenience, define

$$\rho_{\mathbf{u}}(x) = \infty \quad \text{when } m_{\mathbf{u}}(x) = 0. \quad (171)$$

The critical radius is nonincreasing downstream:

$$x_1 \leq x_2 \implies \rho_{\mathbf{u}}(x_2) \leq \rho_{\mathbf{u}}(x_1). \quad (172)$$

Geometrically, new streamwise stations can only remove radial portions of the survivor set; they cannot restore realizations that crossed the threshold upstream.

5.4 Exact survival and transition probabilities

Conditioning on the initial direction $\mathbf{u}$, the survival probability is

$$\begin{aligned} S(x | \mathbf{u}) &= \mathbb{P} [R^2 m_{\mathbf{u}}(x) < 1 | \mathbf{u}] \\ &= \mathbb{P} \left[R^2 < \frac{1}{m_{\mathbf{u}}(x)} \right]. \end{aligned} \quad (173)$$

Since $R^2 \sim \chi_r^2$ and is independent of $\mathbf{u}$,

$$S(x | \mathbf{u}) = F_{\chi_r^2} \left(\frac{1}{m_{\mathbf{u}}(x)} \right). \quad (174)$$

The convention

$$F_{\chi_r^2}(\infty) = 1 \quad (175)$$

handles directions for which $m_{\mathbf{u}}(x) = 0$.

Averaging over the uniformly distributed direction gives the exact survival probability:

$$S(x) = \mathbb{E}_{\mathbf{u}} \left[F_{\chi_r^2} \left(\frac{1}{m_{\mathbf{u}}(x)} \right) \right]. \quad (176)$$

Similarly, the conditional probability of transition by x is

$$F_{X_{\text{tr}}}(x | \mathbf{u}) = \bar{F}_{\chi_r^2} \left(\frac{1}{m_{\mathbf{u}}(x)} \right), \quad (177)$$

and the unconditional first-transition CDF is

$$F_{X_{\text{tr}}}(x) = \mathbb{E}_{\mathbf{u}} \left[\bar{F}_{\chi_r^2} \left(\frac{1}{m_{\mathbf{u}}(x)} \right) \right]. \quad (178)$$

Equations (176) and (178) are the central probability formulas used by the numerical algorithm.

Let

$$|\mathbb{S}^{r-1}| = \frac{2\pi^{r/2}}{\Gamma(r/2)} \quad (179)$$

denote the surface area of the unit sphere. The expectation in eq. (178) can be written explicitly as

$$F_{X_{\text{tr}}}(x) = \frac{1}{|\mathbb{S}^{r-1}|} \int_{\mathbb{S}^{r-1}} \bar{F}_{\chi_r^2} \left[\frac{1}{\sup_{0 \leq \xi \leq x} \mathbf{u}^\top \mathbf{G}(\xi) \mathbf{u}} \right] d\sigma(\mathbf{u}). \quad (180)$$

Thus, the original Gaussian integral over $\mathbb{R}^r$ is reduced to a bounded integral over the $(r-1)$ -dimensional unit sphere, with the radial probability evaluated analytically.

The distinction between strict and non-strict inequalities does not alter these formulas because the chi-square distribution is continuous. For a fixed direction with $m_{\mathbf{u}}(x) > 0$,

$$\mathbb{P} \left[R^2 = \frac{1}{m_{\mathbf{u}}(x)} \right] = 0. \quad (181)$$

5.5 Geometric interpretation

The survivor set from eq. (128) can be represented in radial coordinates as

$$\boxed{\mathcal{S}(x) = \{R\mathbf{u} : \mathbf{u} \in \mathbb{S}^{r-1}, 0 \leq R < \rho_{\mathbf{u}}(x)\}}. \quad (182)$$

The function $\rho_{\mathbf{u}}(x)$ is therefore the radial function of the survivor set.

For an individual instantaneous ellipsoid,

$$\mathcal{E}(\xi) = \{R\mathbf{u} : R^2 a_{\mathbf{u}}(\xi) < 1\}, \quad (183)$$

the radial boundary is

$$R = \frac{1}{\sqrt{a_{\mathbf{u}}(\xi)}}. \quad (184)$$

Taking the intersection over all upstream stations selects the most restrictive radial boundary:

$$\begin{aligned} \rho_{\mathbf{u}}(x) &= \inf_{0 \leq \xi \leq x} \frac{1}{\sqrt{a_{\mathbf{u}}(\xi)}} \\ &= \frac{1}{\sqrt{\sup_{0 \leq \xi \leq x} a_{\mathbf{u}}(\xi)}}. \end{aligned} \quad (185)$$

This identity explains why the running maximum appears naturally in the first-transition probability.

Although $\mathcal{S}(x)$ may have a boundary assembled from many different streamwise threshold ellipsoids, every ray from the origin intersects that boundary at at most one finite positive radius. Consequently, the radial variable can be integrated without explicitly constructing the full intersection geometry.

Because the quadratic form is even in the direction,

$$a_{-\mathbf{u}}(x) = a_{\mathbf{u}}(x), \quad (186)$$

which implies

$$m_{-\mathbf{u}}(x) = m_{\mathbf{u}}(x) \quad (187)$$

and

$$\rho_{-\mathbf{u}}(x) = \rho_{\mathbf{u}}(x). \quad (188)$$

The angular integrand is therefore antipodally symmetric. Numerical integration may exploit this property by using only one representative from each pair $\{\mathbf{u}, -\mathbf{u}\}$, provided the angular weights are adjusted consistently. Equivalently, the angular domain may be viewed as the real projective space $\mathbb{RP}^{r-1}$.

5.6 Directional first-transition location

For a fixed direction $\mathbf{u}$ and squared radius $z = R^2$, define

$$X_{\text{tr}}(z, \mathbf{u}) = \inf \{x \in [0, x_{\text{max}}] : z a_{\mathbf{u}}(x) \geq 1\}. \quad (189)$$

Equivalently,

$$X_{\text{tr}}(z, \mathbf{u}) = \inf \left\{ x \in [0, x_{\text{max}}] : m_{\mathbf{u}}(x) \geq \frac{1}{z} \right\}. \quad (190)$$

This expression makes clear that, for a given direction, the first-transition location is the first station at which the running maximum reaches the radius-dependent level $1/z$.

The conditional CDF in eq. (177) can alternatively be derived from

$$\{X_{\text{tr}}(R^2, \mathbf{u}) \leq x\} = \left\{R^2 \geq \frac{1}{m_{\mathbf{u}}(x)}\right\}. \quad (191)$$

No random energy trajectory must be reconstructed: for each direction, the complete probability over all Gaussian amplitudes is determined by the single scalar running-maximum function $m_{\mathbf{u}}(x)$.

If a scalar realization of X_{tr} is desired after its distribution has been computed, it can be generated by inverse-CDF sampling as described in section 7. Such a draw is distinct from trajectory Monte Carlo: it samples from the already evaluated first-transition distribution rather than propagating a new random OWNS realization.

5.7 First-transition density from the running maximum

Where $m_{\mathbf{u}}(x) > 0$ is differentiable, the conditional transition CDF is

$$F_{X_{\text{tr}}}(x | \mathbf{u}) = \overline{F}_{\chi_r^2}(m_{\mathbf{u}}(x)^{-1}). \quad (192)$$

Differentiating gives

$$\frac{\partial}{\partial x} F_{X_{\text{tr}}}(x | \mathbf{u}) = f_{\chi_r^2}\left(\frac{1}{m_{\mathbf{u}}(x)}\right) \frac{m'_{\mathbf{u}}(x)}{m_{\mathbf{u}}(x)^2}. \quad (193)$$

Formally, when differentiation and angular expectation may be interchanged,

$$f_{X_{\text{tr}}}(x) = \mathbb{E}_{\mathbf{u}} \left[f_{\chi_r^2}\left(\frac{1}{m_{\mathbf{u}}(x)}\right) \frac{m'_{\mathbf{u}}(x)}{m_{\mathbf{u}}(x)^2} \right]. \quad (194)$$

The derivative of the running maximum is nonzero only when the current directional gain establishes a new upstream record. At regular points,

$$m'_{\mathbf{u}}(x) = \begin{cases} a'_{\mathbf{u}}(x), & a_{\mathbf{u}}(x) = m_{\mathbf{u}}(x) \text{ and } a'_{\mathbf{u}}(x) > 0, \\ 0, & a_{\mathbf{u}}(x) < m_{\mathbf{u}}(x). \end{cases} \quad (195)$$

At points where a new record begins, where a record maximum becomes stationary, or where competing maxima exchange activity, $m_{\mathbf{u}}$ may fail to be classically differentiable.

Equation (194) is useful for interpretation, but direct numerical differentiation can be sensitive to streamwise resolution and nonsmooth changes in the active maximum. In computation, it is generally more robust to evaluate the monotone CDF using eq. (178) and then obtain interval transition probabilities from conservative CDF differences.

5.8 Discrete streamwise approximation

Suppose $\mathbf{G}(x)$ is initially available at the grid

$$0 = x_0 < x_1 < \cdots < x_N = x_{\text{max}}. \quad (196)$$

For an angular direction $\mathbf{u}$, define

$$a_{\mathbf{u},n} = \mathbf{u}^{\text{T}} \mathbf{G}(x_n) \mathbf{u} \quad (197)$$

and update the discrete running maximum recursively:

$$m_{\mathbf{u},0} = a_{\mathbf{u},0}, \quad (198)$$

$$m_{\mathbf{u},n} = \max(m_{\mathbf{u},n-1}, a_{\mathbf{u},n}), \quad n = 1, \dots, N. \quad (199)$$

The corresponding stationwise approximation to the conditional survival probability is

$$S_N(x_n | \mathbf{u}) = F_{\chi_r^2} \left(\frac{1}{m_{\mathbf{u},n}} \right). \quad (200)$$

Because

$$m_{\mathbf{u},n} \leq m_{\mathbf{u}}(x_n), \quad (201)$$

the discrete-grid approximation satisfies

$$S_N(x_n | \mathbf{u}) \geq S(x_n | \mathbf{u}), \quad (202)$$

$$F_{X_{\text{tr}}^{(N)}}(x_n | \mathbf{u}) \leq F_{X_{\text{tr}}}(x_n | \mathbf{u}). \quad (203)$$

Averaging over directions preserves these bounds. Thus, a stationwise maximum gives a lower estimate of the continuous first-transition probability unless all relevant directional maxima are resolved by the grid.

This issue is particularly important for a strongly non-normal propagator, which may produce narrow amplification peaks. The continuous maximum-detection procedures in ?? refine the calculation of $m_{\mathbf{u}}(x)$ between stored OWNS stations.

5.9 Numerical angular approximation

Let

$$\{(\mathbf{u}_k, \omega_k)\}_{k=1}^K \quad (204)$$

be an angular integration rule on $\mathbb{S}^{r-1}$, with normalized weights satisfying

$$\omega_k \geq 0, \quad \sum_{k=1}^K \omega_k = 1. \quad (205)$$

Then eqs. (176) and (178) are approximated by

$$S_K(x) = \sum_{k=1}^K \omega_k F_{\chi_r^2} \left(\frac{1}{m_k(x)} \right), \quad (206)$$

$$F_{X_{\text{tr}},K}(x) = \sum_{k=1}^K \omega_k \bar{F}_{\chi_r^2} \left(\frac{1}{m_k(x)} \right), \quad (207)$$

where

$$m_k(x) = \max_{0 \leq \xi \leq x} \mathbf{u}_k^\top \mathbf{G}(\xi) \mathbf{u}_k. \quad (208)$$

If the same normalized angular weights are used in both expressions, then

$$S_K(x) + F_{X_{\text{tr}},K}(x) = 1 \quad (209)$$

up to floating-point error.

The approximation in eq. (207) evaluates one deterministic directional gain history for each quadrature point and analytically accounts for every Gaussian radius along that direction. It is therefore distinct from direct trajectory Monte Carlo, in which both the direction and radius are randomly sampled and each sampled trajectory contributes only a binary transition outcome.

Deterministic spherical quadrature, spherical designs, sparse grids, and randomized quasi-Monte Carlo may all be used to evaluate the angular expectation. Their suitability depends strongly on the stochastic dimension and on the angular localization of the dangerous transient-growth directions. Detailed choices are discussed in section 8.

5.10 Proper complex Gaussian extension

If the stochastic coordinates are modeled directly as a proper complex Gaussian vector

$$\mathbf{z} \sim \mathcal{CN}(\mathbf{0}, \mathbf{I}_r), \quad (210)$$

write

$$\mathbf{z} = R_{\mathbb{C}} \mathbf{u}_{\mathbb{C}}, \quad \|\mathbf{u}_{\mathbb{C}}\|_2 = 1. \quad (211)$$

Under the convention in ??,

$$R_{\mathbb{C}}^2 = \|\mathbf{z}\|_2^2 \sim \text{Gamma}(r, 1) = \frac{1}{2} \chi_{2r}^2. \quad (212)$$

The complex direction is uniformly distributed on the unit sphere in $\mathbb{C}^r$, and is independent of $R_{\mathbb{C}}$.

For a Hermitian normalized energy matrix $\mathbf{G}_{\mathbb{C}}(x)$, define

$$m_{\mathbf{u}_{\mathbb{C}}}(x) = \max_{0 \leq \xi \leq x} \mathbf{u}_{\mathbb{C}}^* \mathbf{G}_{\mathbb{C}}(\xi) \mathbf{u}_{\mathbb{C}}. \quad (213)$$

The transition CDF is then

$$F_{X_{\text{tr}}}(x) = \mathbb{E}_{\mathbf{u}_{\mathbb{C}}} \left[\overline{F}_{\text{Gamma}(r,1)} \left(\frac{1}{m_{\mathbf{u}_{\mathbb{C}}}(x)} \right) \right]. \quad (214)$$

Equivalently,

$$\overline{F}_{\text{Gamma}(r,1)} \left(\frac{1}{m} \right) = \overline{F}_{\chi_{2r}^2} \left(\frac{2}{m} \right). \quad (215)$$

The remainder of this document uses the real standard Gaussian formulation in eq. (178). A proper complex model may instead be realified as described in ??, after which the real formula applies with dimension $2r$ and the appropriately scaled realified energy matrix.

6 Computation of the First-Transition Location

This section converts the radial–directional formulation into a numerical procedure for computing the distribution of X_{tr} from the propagated correlation factor $\mathbf{B}(x)$. Two related computational tasks are distinguished:

- (i) evaluating the distribution of X_{tr} , including its survival function, CDF, interval probabilities, and right-censoring probability; and
- (ii) drawing a scalar realization of X_{tr} from the computed distribution.

The first task is the principal deterministic or quasi-deterministic calculation. The second can subsequently be performed by inverse-CDF sampling without propagating an additional random OWNS trajectory.

6.1 Inputs and outputs

Let

$$0 = x_0 < x_1 < \cdots < x_N = x_{\max} \quad (216)$$

be the streamwise stations at which the propagated factors are available. The basic algorithm requires the following inputs:

- (i) the streamwise stations $\{x_n\}_{n=0}^N$;
- (ii) the propagated synthesis factors

$$\mathbf{B}_n = \mathbf{B}(x_n) \in \mathbb{C}^{n_q \times r}; \quad (217)$$

- (iii) the energy matrices

$$\mathbf{H}_n = \mathbf{H}(x_n); \quad (218)$$

- (iv) the positive transition thresholds

$$e_{\text{thres},n} = e_{\text{thres}}(x_n) > 0; \quad (219)$$

- (v) angular integration directions

$$\mathbf{u}_k \in \mathbb{S}^{r-1}, \quad k = 1, \dots, K; \quad (220)$$

and

- (vi) normalized nonnegative angular weights

$$\omega_k \geq 0, \quad \sum_{k=1}^K \omega_k = 1. \quad (221)$$

The principal outputs are:

- (i) the survival probabilities

$$S_n \approx S(x_n); \quad (222)$$

- (ii) the transition CDF

$$F_n \approx F_{X_{\text{tr}}}(x_n); \quad (223)$$

- (iii) the probability of first transition in each streamwise interval;

- (iv) the probability of no transition by $x_{\max}$; and

- (v) optionally, the hazard, transition-location density, transition quantiles, or scalar draws of X_{tr} .

The algorithms below are first stated using the normalized quadratic-form matrices

$$\mathbf{G}_n = \frac{\Re(\mathbf{B}_n^* \mathbf{H}_n \mathbf{B}_n)}{e_{\text{thres},n}}. \quad (224)$$

For a fully real state-space representation, the real-part operator is unnecessary and the conjugate transpose is replaced by a transpose. A matrix-free variant that avoids explicitly forming $\mathbf{G}_n$ is given in section 7.4.

6.2 Preprocessing of the normalized energy matrices

Finite-precision arithmetic can introduce a small asymmetric component into the computed matrix in eq. (288). The normalized energy matrix should therefore be symmetrized:

$$\mathbf{G}_n \leftarrow \frac{1}{2} (\mathbf{G}_n + \mathbf{G}_n^\top). \quad (225)$$

In exact arithmetic, $\mathbf{G}_n \succeq \mathbf{0}$. Small negative eigenvalues may nevertheless appear because of roundoff, inaccurate quadrature weights in $\mathbf{H}_n$, or errors in the propagated factor. If

$$\lambda_{\min}(\mathbf{G}_n) \geq -\varepsilon_{\text{psd}} \|\mathbf{G}_n\|_2 \quad (226)$$

for a prescribed tolerance ε_{psd} , the negative eigenvalues may be projected to zero. Larger negative eigenvalues indicate an inconsistency that should be investigated rather than silently corrected.

For example, if

$$\mathbf{G}_n = \mathbf{V}_n \text{diag}(\gamma_{1,n}, \dots, \gamma_{r,n}) \mathbf{V}_n^\top, \quad (227)$$

a positive-semidefinite projection is

$$\mathbf{G}_n^+ = \mathbf{V}_n \text{diag}(\max(\gamma_{1,n}, 0), \dots, \max(\gamma_{r,n}, 0)) \mathbf{V}_n^\top. \quad (228)$$

This operation should be used only to remove numerical violations that are demonstrably small relative to the physical matrix scale.

The angular directions should also be normalized before use:

$$\mathbf{u}_k \leftarrow \frac{\mathbf{u}_k}{\|\mathbf{u}_k\|_2}. \quad (229)$$

If the supplied quadrature weights sum to a value other than one, they may be normalized according to

$$\omega_k \leftarrow \frac{\omega_k}{\sum_{\ell=1}^K \omega_\ell}. \quad (230)$$

6.3 Discrete-streamwise transition-distribution algorithm

For each angular direction, the algorithm evaluates the directional gain and maintains its running maximum:

$$a_{k,n} = \mathbf{u}_k^\top \mathbf{G}_n \mathbf{u}_k, \quad (231)$$

$$m_{k,n} = \max_{0 \leq j \leq n} a_{k,j}. \quad (232)$$

The conditional survival probability represented by direction $\mathbf{u}_k$ is

$$s_{k,n} = F_{\chi_r^2} \left(\frac{1}{m_{k,n}} \right). \quad (233)$$

When $m_{k,n} = 0$, the convention $s_{k,n} = 1$ is used.

The angularly integrated survival probability is

$$S_n = \sum_{k=1}^K \omega_k s_{k,n}, \quad (234)$$

and the transition CDF is evaluated by complementarity:

$$F_n = 1 - S_n. \quad (235)$$

Computing only one of S_n and F_n and obtaining the other by complementarity avoids small inconsistencies between separately evaluated lower- and upper-tail functions.

The update of m_k is irreversible: once a directional gain has reached a given value, a later decrease in $a_{k,n}$ does not increase the directional survival probability. This single scalar update per direction provides the streamwise memory.

Because the running maximum is nondecreasing,

$$m_{k,n+1} \geq m_{k,n}, \quad (236)$$

the directional survival probabilities are nonincreasing:

$$s_{k,n+1} \leq s_{k,n}. \quad (237)$$

For nonnegative angular weights, the computed survival curve and CDF therefore satisfy

$$S_{n+1} \leq S_n, \quad (238)$$

$$F_{n+1} \geq F_n \quad (239)$$

up to floating-point error. No post hoc monotonic regression should be needed if the calculation is implemented consistently.

Algorithm 5 uses the maxima over stored stations. It therefore evaluates the discrete-grid first-transition distribution from eq. (134). As established in section 4.6, this distribution generally underestimates the continuous-streamwise transition probability. The algorithm can be upgraded by replacing each stationwise value $a_{k,n}$ with a maximum resolved over the intervening streamwise interval, as described in section 7.5 and ??.

6.4 Matrix-free directional evaluation

If r is large or storing every $\mathbf{G}_n \in \mathbb{R}^{r \times r}$ is undesirable, the directional gain may be evaluated without forming $\mathbf{G}_n$. For each angular direction, define the propagated directional state

$$\mathbf{v}_{k,n} = \mathbf{B}_n \mathbf{u}_k. \quad (240)$$

Then

$$a_{k,n} = \frac{\mathbf{v}_{k,n}^* \mathbf{H}_n \mathbf{v}_{k,n}}{e_{\text{thres},n}}. \quad (241)$$

The result is real and nonnegative in exact arithmetic.

The matrix-free calculation may be performed in blocks. If

$$\mathbf{U} = [\mathbf{u}_1 \quad \cdots \quad \mathbf{u}_K] \in \mathbb{R}^{r \times K}, \quad (242)$$

then all directional states at station x_n are

$$\mathbf{V}_n = \mathbf{B}_n \mathbf{U}. \quad (243)$$

The gains are the diagonal entries of

$$\frac{\mathbf{V}_n^* \mathbf{H}_n \mathbf{V}_n}{e_{\text{thres},n}}, \quad (244)$$

but the full $K \times K$ matrix need not be formed. Each column can instead be weighted by $\mathbf{H}_n$, and the corresponding diagonal inner products can be evaluated directly.

The explicit- $\mathbf{G}$ and matrix-free approaches have complementary cost profiles:

- Forming $\mathbf{G}_n$ is advantageous when r is moderate, K is large, and many angular rules or postprocessing operations will reuse the same reduced matrices.
- Matrix-free evaluation is advantageous when r is large, K is moderate, or $\mathbf{H}_n$ can be applied efficiently without forming a dense matrix.
- If the OWNS solver can propagate selected vectors directly, $\mathbf{B}(x)\mathbf{u}_k$ may be marched without storing the complete factor. This reduces storage but may require additional OWNS marches when the angular rule is refined.

6.5 Continuous-streamwise maximum upgrade

To approximate the continuous first-transition distribution, define the maximum directional gain over each marching interval:

$$\hat{a}_{k,n} = \max_{x \in [x_{n-1}, x_n]} a_{\mathbf{u}_k}(x), \quad n = 1, \dots, N. \quad (245)$$

For the inlet, set

$$\hat{a}_{k,0} = a_{\mathbf{u}_k}(x_0). \quad (246)$$

The running maximum is then updated as

$$m_{k,n} = \max(m_{k,n-1}, \hat{a}_{k,n}). \quad (247)$$

The interval maximum may be estimated using an interpolant based on the endpoint gains and derivatives. The directional derivative is

$$a'_{\mathbf{u}_k}(x) = \mathbf{u}_k^\top \mathbf{G}'(x) \mathbf{u}_k. \quad (248)$$

A cubic Hermite interpolant on $[x_{n-1}, x_n]$ uses

$$a_{k,n-1}, \quad a_{k,n}, \quad a'_{k,n-1}, \quad a'_{k,n}. \quad (249)$$

The interval maximum is obtained by evaluating the interpolant at both endpoints and at all real stationary points in the interval.

The interval maximum should be considered converged only after the interpolation or adaptive-refinement error is small compared with the required probability accuracy. A more complete treatment of streamwise maximum detection is provided in ??.

6.6 Interval probabilities and right censoring

Once the CDF has been computed, the inlet transition probability is

$$p_0 = F_0. \quad (250)$$

For $n = 1, \dots, N$, the probability of first transition in $(x_{n-1}, x_n]$ is

$$\boxed{p_n = F_n - F_{n-1} = S_{n-1} - S_n.} \quad (251)$$

The right-censoring probability is

$$\boxed{p_{\text{censored}} = S_N = 1 - F_N.} \quad (252)$$

These quantities satisfy

$$p_0 + \sum_{n=1}^N p_n + p_{\text{censored}} = 1 \quad (253)$$

up to numerical error.

The interval probabilities can be computed more stably from survival differences when F_n is close to one, and from CDF differences when S_n is close to one. For example,

$$p_n = \begin{cases} F_n - F_{n-1}, & F_n \leq \frac{1}{2}, \\ S_{n-1} - S_n, & F_n > \frac{1}{2}. \end{cases} \quad (254)$$

Small negative values caused solely by floating-point cancellation may be set to zero, followed by a probability-conservation check. Material negative values indicate a computational error because the running maximum construction guarantees monotonicity analytically.

A discrete conditional transition probability, or interval hazard, is

$$h_n = \frac{p_n}{S_{n-1}} = 1 - \frac{S_n}{S_{n-1}}, \quad S_{n-1} > 0. \quad (255)$$

An average hazard rate over the interval may be defined as

$$\bar{\lambda}_n = -\frac{\log S_n - \log S_{n-1}}{x_n - x_{n-1}}, \quad S_n > 0. \quad (256)$$

This logarithmic estimate is consistent with a piecewise-constant hazard model.

6.7 Transition quantiles

For $p \in (0, 1)$, define the unconditional generalized quantile

$$x_p = \inf \{x \in [0, x_{\max}] : F_{X_{\text{tr}}}(x) \geq p\}. \quad (257)$$

A finite quantile exists within the modeled domain only if

$$p \leq F_{X_{\text{tr}}}(x_{\max}). \quad (258)$$

If $p > F_{X_{\text{tr}}}(x_{\max})$, then the corresponding unconditional quantile is beyond the computational domain or belongs to the point mass at infinity.

A quantile conditioned on transition occurring by $x_{\max}$ uses the conditional CDF

$$F_{\text{tr}|\text{tr by } x_{\max}}(x) = \frac{F_{X_{\text{tr}}}(x)}{F_{X_{\text{tr}}}(x_{\max})}, \quad 0 \leq x \leq x_{\max}, \quad (259)$$

provided $F_{X_{\text{tr}}}(x_{\max}) > 0$. Conditional and unconditional quantiles should be labeled explicitly because they answer different questions.

On a discrete grid, the quantile is first bracketed by finding the smallest index n such that

$$F_n \geq p. \quad (260)$$

A within-interval estimate may then be obtained from a monotone interpolant of the CDF. Piecewise-linear interpolation gives

$$x_p \approx x_{n-1} + (x_n - x_{n-1}) \frac{p - F_{n-1}}{F_n - F_{n-1}}, \quad (261)$$

when $F_n > F_{n-1}$. If $F_n = F_{n-1}$, the interval carries no transition probability and cannot contain an interior quantile.

6.8 Generating a scalar realization of X_{tr}

After the transition CDF has been evaluated, a scalar transition location can be drawn using inverse-transform sampling. Let

$$U \sim \text{Unif}(0, 1). \quad (262)$$

If

$$U > F_N, \quad (263)$$

the draw corresponds to a realization that does not transition by x_{max} . It is returned as either $X_{\text{tr}} = \infty$ or as a right-censored observation at x_{max} . Otherwise, the transition location is the generalized inverse

$$X_{\text{tr}} = F_{X_{\text{tr}}}^{-1}(U) = \inf \{x : F_{X_{\text{tr}}}(x) \geq U\}. \quad (264)$$

Algorithm 8 samples from a piecewise-linear approximation to the computed CDF. If only interval-resolved transition locations are desired, one may instead return the interval index n with probability p_n , without assuming any within-interval distribution.

Drawing X_{tr} in this manner does not generate a corresponding disturbance realization $\mathbf{w}$ or physical trajectory $\mathbf{q}(x)$. If the transition-causing state or initial disturbance direction is also required, one must sample from the conditional distribution of $(R, \mathbf{u})$ given a transition-location interval. That conditional sampling problem is distinct from evaluating the marginal distribution of X_{tr} .

6.9 Numerical evaluation of chi-square tails

The directional transition probability can be extremely small when the critical radius is large. In that regime, it is preferable to evaluate

$$\bar{F}_{\chi_r^2}(z) \quad (265)$$

using a dedicated upper-tail or regularized upper-incomplete-gamma routine rather than computing $1 - F_{\chi_r^2}(z)$, which can suffer catastrophic cancellation.

Conversely, if the survival probability is very small, the lower-tail CDF should be evaluated directly. A stable implementation may compute both aggregate quantities according to

$$F_n = \sum_{k=1}^K \omega_k \bar{F}_{\chi_r^2} \left(\frac{1}{m_{k,n}} \right), \quad (266)$$

$$S_n = \sum_{k=1}^K \omega_k F_{\chi_r^2} \left(\frac{1}{m_{k,n}} \right), \quad (267)$$

using stable library routines for each tail. Since separately summed values may not add to exactly one, the smaller and more accurately resolved aggregate probability may be retained and the larger obtained by complementarity.

For probabilities below ordinary floating-point range, logarithmic tail functions and a weighted log-sum-exp calculation may be required. If

$$\ell_{k,n} = \log \bar{F}_{\chi_r^2} \left(\frac{1}{m_{k,n}} \right), \quad (268)$$

then

$$\log F_n = \text{LSE}_k [\log \omega_k + \ell_{k,n}], \quad (269)$$

where

$$\text{LSE}_k(z_k) = z_{\max} + \log \left[\sum_k \exp(z_k - z_{\max}) \right], \quad z_{\max} = \max_k z_k. \quad (270)$$

6.10 Basic algorithmic checks

Before interpreting the computed transition distribution, the following checks should be applied:

(i) **Unit directions:**

$$||\mathbf{u}_k||_2 - 1 \leq \varepsilon_u. \quad (271)$$

(ii) **Normalized weights:**

$$\left| \sum_{k=1}^K \omega_k - 1 \right| \leq \varepsilon_\omega, \quad \omega_k \geq 0. \quad (272)$$

(iii) **Nonnegative gains:**

$$a_{k,n} \geq -\varepsilon_a. \quad (273)$$

(iv) **Running-maximum monotonicity:**

$$m_{k,n+1} \geq m_{k,n}. \quad (274)$$

(v) **Probability bounds:**

$$0 \leq S_n \leq 1, \quad 0 \leq F_n \leq 1. \quad (275)$$

(vi) **CDF and survival monotonicity:**

$$F_{n+1} \geq F_n, \quad S_{n+1} \leq S_n. \quad (276)$$

(vii) **Complementarity:**

$$|S_n + F_n - 1| \leq \varepsilon_p. \quad (277)$$

(viii) **Total probability:**

$$\left| p_0 + \sum_{n=1}^N p_n + p_{\text{censored}} - 1 \right| \leq \varepsilon_p. \quad (278)$$

(ix) **Local-probability lower bound:** when the local exceedance probability is independently evaluated,

$$p_{\text{local}}(x_n) \leq F_n \quad (279)$$

should hold to within the combined numerical tolerances.

These checks detect implementation errors but do not establish convergence with respect to streamwise resolution, angular integration, or stochastic rank. Those convergence questions are addressed in ?? and sections 8 and 9.

7 Computation of the First-Transition Location

This section converts the radial–directional formulation into a numerical procedure for computing the distribution of X_{tr} from the propagated correlation factor $\mathbf{B}(x)$. Two related computational tasks are distinguished:

- (i) evaluating the distribution of X_{tr} , including its survival function, CDF, interval probabilities, and right-censoring probability; and
- (ii) drawing a scalar realization of X_{tr} from the computed distribution.

The first task is the principal deterministic or quasi-deterministic calculation. The second can subsequently be performed by inverse-CDF sampling without propagating an additional random OWNS trajectory.

7.1 Inputs and outputs

Let

$$0 = x_0 < x_1 < \cdots < x_N = x_{\max} \quad (280)$$

be the streamwise stations at which the propagated factors are available. The basic algorithm requires the following inputs:

- (i) the streamwise stations $\{x_n\}_{n=0}^N$;
- (ii) the propagated synthesis factors

$$\mathbf{B}_n = \mathbf{B}(x_n) \in \mathbb{C}^{n_q \times r}; \quad (281)$$

- (iii) the energy matrices

$$\mathbf{H}_n = \mathbf{H}(x_n); \quad (282)$$

- (iv) the positive transition thresholds

$$e_{\text{thres},n} = e_{\text{thres}}(x_n) > 0; \quad (283)$$

- (v) angular integration directions

$$\mathbf{u}_k \in \mathbb{S}^{r-1}, \quad k = 1, \dots, K; \quad (284)$$

and

- (vi) normalized nonnegative angular weights

$$\omega_k \geq 0, \quad \sum_{k=1}^K \omega_k = 1. \quad (285)$$

The principal outputs are:

- (i) the survival probabilities

$$S_n \approx S(x_n); \quad (286)$$

(ii) the transition CDF

$$F_n \approx F_{X_{\text{tr}}}(x_n); \quad (287)$$

(iii) the probability of first transition in each streamwise interval;

(iv) the probability of no transition by x_{max} ; and

(v) optionally, the hazard, transition-location density, transition quantiles, or scalar draws of X_{tr} .

The algorithms below are first stated using the normalized quadratic-form matrices

$$\mathbf{G}_n = \frac{\Re(\mathbf{B}_n^* \mathbf{H}_n \mathbf{B}_n)}{e_{\text{thres},n}}. \quad (288)$$

For a fully real state-space representation, the real-part operator is unnecessary and the conjugate transpose is replaced by a transpose. A matrix-free variant that avoids explicitly forming $\mathbf{G}_n$ is given in section 7.4.

7.2 Preprocessing of the normalized energy matrices

Finite-precision arithmetic can introduce a small asymmetric component into the computed matrix in eq. (288). The normalized energy matrix should therefore be symmetrized:

$$\mathbf{G}_n \leftarrow \frac{1}{2} (\mathbf{G}_n + \mathbf{G}_n^\top). \quad (289)$$

In exact arithmetic, $\mathbf{G}_n \succeq \mathbf{0}$. Small negative eigenvalues may nevertheless appear because of roundoff, inaccurate quadrature weights in $\mathbf{H}_n$, or errors in the propagated factor. If

$$\lambda_{\min}(\mathbf{G}_n) \geq -\varepsilon_{\text{psd}} \|\mathbf{G}_n\|_2 \quad (290)$$

for a prescribed tolerance ε_{psd} , the negative eigenvalues may be projected to zero. Larger negative eigenvalues indicate an inconsistency that should be investigated rather than silently corrected.

For example, if

$$\mathbf{G}_n = \mathbf{V}_n \text{diag}(\gamma_{1,n}, \dots, \gamma_{r,n}) \mathbf{V}_n^\top, \quad (291)$$

a positive-semidefinite projection is

$$\mathbf{G}_n^+ = \mathbf{V}_n \text{diag}(\max(\gamma_{1,n}, 0), \dots, \max(\gamma_{r,n}, 0)) \mathbf{V}_n^\top. \quad (292)$$

This operation should be used only to remove numerical violations that are demonstrably small relative to the physical matrix scale.

The angular directions should also be normalized before use:

$$\mathbf{u}_k \leftarrow \frac{\mathbf{u}_k}{\|\mathbf{u}_k\|_2}. \quad (293)$$

If the supplied quadrature weights sum to a value other than one, they may be normalized according to

$$\omega_k \leftarrow \frac{\omega_k}{\sum_{\ell=1}^K \omega_\ell}. \quad (294)$$

7.3 Discrete-streamwise transition-distribution algorithm

For each angular direction, the algorithm evaluates the directional gain and maintains its running maximum:

$$a_{k,n} = \mathbf{u}_k^\top \mathbf{G}_n \mathbf{u}_k, \quad (295)$$

$$m_{k,n} = \max_{0 \leq j \leq n} a_{k,j}. \quad (296)$$

The conditional survival probability represented by direction $\mathbf{u}_k$ is

$$s_{k,n} = F_{\chi_r^2} \left(\frac{1}{m_{k,n}} \right). \quad (297)$$

When $m_{k,n} = 0$, the convention $s_{k,n} = 1$ is used.

The angularly integrated survival probability is

$$S_n = \sum_{k=1}^K \omega_k s_{k,n}, \quad (298)$$

and the transition CDF is evaluated by complementarity:

$$F_n = 1 - S_n. \quad (299)$$

Computing only one of S_n and F_n and obtaining the other by complementarity avoids small inconsistencies between separately evaluated lower- and upper-tail functions.

The update of m_k is irreversible: once a directional gain has reached a given value, a later decrease in $a_{k,n}$ does not increase the directional survival probability. This single scalar update per direction provides the streamwise memory.

Because the running maximum is nondecreasing,

$$m_{k,n+1} \geq m_{k,n}, \quad (300)$$

the directional survival probabilities are nonincreasing:

$$s_{k,n+1} \leq s_{k,n}. \quad (301)$$

For nonnegative angular weights, the computed survival curve and CDF therefore satisfy

$$S_{n+1} \leq S_n, \quad (302)$$

$$F_{n+1} \geq F_n \quad (303)$$

up to floating-point error. No post hoc monotonic regression should be needed if the calculation is implemented consistently.

Algorithm 5 uses the maxima over stored stations. It therefore evaluates the discrete-grid first-transition distribution from eq. (134). As established in section 4.6, this distribution generally underestimates the continuous-streamwise transition probability. The algorithm can be upgraded by replacing each stationwise value $a_{k,n}$ with a maximum resolved over the intervening streamwise interval, as described in section 7.5 and ??.

7.4 Matrix-free directional evaluation

If r is large or storing every $\mathbf{G}_n \in \mathbb{R}^{r \times r}$ is undesirable, the directional gain may be evaluated without forming $\mathbf{G}_n$. For each angular direction, define the propagated directional state

$$\mathbf{v}_{k,n} = \mathbf{B}_n \mathbf{u}_k. \quad (304)$$

Then

$$a_{k,n} = \frac{\mathbf{v}_{k,n}^* \mathbf{H}_n \mathbf{v}_{k,n}}{e_{\text{thres},n}}. \quad (305)$$

The result is real and nonnegative in exact arithmetic.

The matrix-free calculation may be performed in blocks. If

$$\mathbf{U} = [\mathbf{u}_1 \quad \cdots \quad \mathbf{u}_K] \in \mathbb{R}^{r \times K}, \quad (306)$$

then all directional states at station x_n are

$$\mathbf{V}_n = \mathbf{B}_n \mathbf{U}. \quad (307)$$

The gains are the diagonal entries of

$$\frac{\mathbf{V}_n^* \mathbf{H}_n \mathbf{V}_n}{e_{\text{thres},n}}, \quad (308)$$

but the full $K \times K$ matrix need not be formed. Each column can instead be weighted by $\mathbf{H}_n$, and the corresponding diagonal inner products can be evaluated directly.

The explicit- $\mathbf{G}$ and matrix-free approaches have complementary cost profiles:

- Forming $\mathbf{G}_n$ is advantageous when r is moderate, K is large, and many angular rules or postprocessing operations will reuse the same reduced matrices.
- Matrix-free evaluation is advantageous when r is large, K is moderate, or $\mathbf{H}_n$ can be applied efficiently without forming a dense matrix.
- If the OWNS solver can propagate selected vectors directly, $\mathbf{B}(x)\mathbf{u}_k$ may be marched without storing the complete factor. This reduces storage but may require additional OWNS marches when the angular rule is refined.

7.5 Continuous-streamwise maximum upgrade

To approximate the continuous first-transition distribution, define the maximum directional gain over each marching interval:

$$\hat{a}_{k,n} = \max_{x \in [x_{n-1}, x_n]} a_{\mathbf{u}_k}(x), \quad n = 1, \dots, N. \quad (309)$$

For the inlet, set

$$\hat{a}_{k,0} = a_{\mathbf{u}_k}(x_0). \quad (310)$$

The running maximum is then updated as

$$m_{k,n} = \max(m_{k,n-1}, \hat{a}_{k,n}). \quad (311)$$

The interval maximum may be estimated using an interpolant based on the endpoint gains and derivatives. The directional derivative is

$$a'_{\mathbf{u}_k}(x) = \mathbf{u}_k^\top \mathbf{G}'(x) \mathbf{u}_k. \quad (312)$$

A cubic Hermite interpolant on $[x_{n-1}, x_n]$ uses

$$a_{k,n-1}, \quad a_{k,n}, \quad a'_{k,n-1}, \quad a'_{k,n}. \quad (313)$$

The interval maximum is obtained by evaluating the interpolant at both endpoints and at all real stationary points in the interval.

The interval maximum should be considered converged only after the interpolation or adaptive-refinement error is small compared with the required probability accuracy. A more complete treatment of streamwise maximum detection is provided in ??.

7.6 Interval probabilities and right censoring

Once the CDF has been computed, the inlet transition probability is

$$p_0 = F_0. \quad (314)$$

For $n = 1, \dots, N$, the probability of first transition in $(x_{n-1}, x_n]$ is

$$\boxed{p_n = F_n - F_{n-1} = S_{n-1} - S_n.} \quad (315)$$

The right-censoring probability is

$$\boxed{p_{\text{censored}} = S_N = 1 - F_N.} \quad (316)$$

These quantities satisfy

$$p_0 + \sum_{n=1}^N p_n + p_{\text{censored}} = 1 \quad (317)$$

up to numerical error.

The interval probabilities can be computed more stably from survival differences when F_n is close to one, and from CDF differences when S_n is close to one. For example,

$$p_n = \begin{cases} F_n - F_{n-1}, & F_n \leq \frac{1}{2}, \\ S_{n-1} - S_n, & F_n > \frac{1}{2}. \end{cases} \quad (318)$$

Small negative values caused solely by floating-point cancellation may be set to zero, followed by a probability-conservation check. Material negative values indicate a computational error because the running maximum construction guarantees monotonicity analytically.

A discrete conditional transition probability, or interval hazard, is

$$h_n = \frac{p_n}{S_{n-1}} = 1 - \frac{S_n}{S_{n-1}}, \quad S_{n-1} > 0. \quad (319)$$

An average hazard rate over the interval may be defined as

$$\bar{\lambda}_n = -\frac{\log S_n - \log S_{n-1}}{x_n - x_{n-1}}, \quad S_n > 0. \quad (320)$$

This logarithmic estimate is consistent with a piecewise-constant hazard model.

7.7 Transition quantiles

For $p \in (0, 1)$, define the unconditional generalized quantile

$$x_p = \inf \{x \in [0, x_{\max}] : F_{X_{\text{tr}}}(x) \geq p\}. \quad (321)$$

A finite quantile exists within the modeled domain only if

$$p \leq F_{X_{\text{tr}}}(x_{\max}). \quad (322)$$

If $p > F_{X_{\text{tr}}}(x_{\max})$, then the corresponding unconditional quantile is beyond the computational domain or belongs to the point mass at infinity.

A quantile conditioned on transition occurring by $x_{\max}$ uses the conditional CDF

$$F_{\text{tr}|\text{tr by } x_{\max}}(x) = \frac{F_{X_{\text{tr}}}(x)}{F_{X_{\text{tr}}}(x_{\max})}, \quad 0 \leq x \leq x_{\max}, \quad (323)$$

provided $F_{X_{\text{tr}}}(x_{\max}) > 0$. Conditional and unconditional quantiles should be labeled explicitly because they answer different questions.

On a discrete grid, the quantile is first bracketed by finding the smallest index n such that

$$F_n \geq p. \quad (324)$$

A within-interval estimate may then be obtained from a monotone interpolant of the CDF. Piecewise-linear interpolation gives

$$x_p \approx x_{n-1} + (x_n - x_{n-1}) \frac{p - F_{n-1}}{F_n - F_{n-1}}, \quad (325)$$

when $F_n > F_{n-1}$. If $F_n = F_{n-1}$, the interval carries no transition probability and cannot contain an interior quantile.

7.8 Generating a scalar realization of X_{tr}

After the transition CDF has been evaluated, a scalar transition location can be drawn using inverse-transform sampling. Let

$$U \sim \text{Unif}(0, 1). \quad (326)$$

If

$$U > F_N, \quad (327)$$

the draw corresponds to a realization that does not transition by $x_{\max}$. It is returned as either $X_{\text{tr}} = \infty$ or as a right-censored observation at $x_{\max}$. Otherwise, the transition location is the generalized inverse

$$X_{\text{tr}} = F_{X_{\text{tr}}}^{-1}(U) = \inf \{x : F_{X_{\text{tr}}}(x) \geq U\}. \quad (328)$$

Algorithm 8 samples from a piecewise-linear approximation to the computed CDF. If only interval-resolved transition locations are desired, one may instead return the interval index n with probability p_n , without assuming any within-interval distribution.

Drawing X_{tr} in this manner does not generate a corresponding disturbance realization $\mathbf{w}$ or physical trajectory $\mathbf{q}(x)$. If the transition-causing state or initial disturbance direction is also required, one must sample from the conditional distribution of $(R, \mathbf{u})$ given a transition-location interval. That conditional sampling problem is distinct from evaluating the marginal distribution of X_{tr} .

7.9 Numerical evaluation of chi-square tails

The directional transition probability can be extremely small when the critical radius is large. In that regime, it is preferable to evaluate

$$\bar{F}_{\chi_r^2}(z) \quad (329)$$

using a dedicated upper-tail or regularized upper-incomplete-gamma routine rather than computing $1 - F_{\chi_r^2}(z)$, which can suffer catastrophic cancellation.

Conversely, if the survival probability is very small, the lower-tail CDF should be evaluated directly. A stable implementation may compute both aggregate quantities according to

$$F_n = \sum_{k=1}^K \omega_k \bar{F}_{\chi_r^2} \left(\frac{1}{m_{k,n}} \right), \quad (330)$$

$$S_n = \sum_{k=1}^K \omega_k F_{\chi_r^2} \left(\frac{1}{m_{k,n}} \right), \quad (331)$$

using stable library routines for each tail. Since separately summed values may not add to exactly one, the smaller and more accurately resolved aggregate probability may be retained and the larger obtained by complementarity.

For probabilities below ordinary floating-point range, logarithmic tail functions and a weighted log-sum-exp calculation may be required. If

$$\ell_{k,n} = \log \bar{F}_{\chi_r^2} \left(\frac{1}{m_{k,n}} \right), \quad (332)$$

then

$$\log F_n = \text{LSE}_k [\log \omega_k + \ell_{k,n}], \quad (333)$$

where

$$\text{LSE}_k(z_k) = z_{\max} + \log \left[\sum_k \exp(z_k - z_{\max}) \right], \quad z_{\max} = \max_k z_k. \quad (334)$$

7.10 Basic algorithmic checks

Before interpreting the computed transition distribution, the following checks should be applied:

(i) **Unit directions:**

$$\| \mathbf{u}_k \|_2 - 1 \leq \varepsilon_u. \quad (335)$$

(ii) **Normalized weights:**

$$\left| \sum_{k=1}^K \omega_k - 1 \right| \leq \varepsilon_\omega, \quad \omega_k \geq 0. \quad (336)$$

(iii) **Nonnegative gains:**

$$a_{k,n} \geq -\varepsilon_a. \quad (337)$$

(iv) **Running-maximum monotonicity:**

$$m_{k,n+1} \geq m_{k,n}. \quad (338)$$

(v) **Probability bounds:**

$$0 \leq S_n \leq 1, \quad 0 \leq F_n \leq 1. \quad (339)$$

(vi) **CDF and survival monotonicity:**

$$F_{n+1} \geq F_n, \quad S_{n+1} \leq S_n. \quad (340)$$

(vii) **Complementarity:**

$$|S_n + F_n - 1| \leq \varepsilon_p. \quad (341)$$

(viii) **Total probability:**

$$\left| p_0 + \sum_{n=1}^N p_n + p_{\text{censored}} - 1 \right| \leq \varepsilon_p. \quad (342)$$

(ix) **Local-probability lower bound:** when the local exceedance probability is independently evaluated,

$$p_{\text{local}}(x_n) \leq F_n \quad (343)$$

should hold to within the combined numerical tolerances.

These checks detect implementation errors but do not establish convergence with respect to streamwise resolution, angular integration, or stochastic rank. Those convergence questions are addressed in ?? and sections 8 and 9.

8 Angular Integration

After the Gaussian radius has been integrated analytically, the first-transition CDF is an expectation over a uniformly distributed direction on the unit sphere:

$$F_{X_{\text{tr}}}(x) = \mathbb{E}_{\mathbf{u}} [\Psi_x(\mathbf{u})], \quad \mathbf{u} \sim \text{Unif}(\mathbb{S}^{r-1}), \quad (344)$$

where

$$\Psi_x(\mathbf{u}) = \overline{F}_{\chi_r^2} \left[\frac{1}{m_{\mathbf{u}}(x)} \right] \quad (345)$$

and

$$m_{\mathbf{u}}(x) = \max_{0 \leq \xi \leq x} \mathbf{u}^\top \mathbf{G}(\xi) \mathbf{u}. \quad (346)$$

The convention

$$\Psi_x(\mathbf{u}) = 0 \quad \text{if } m_{\mathbf{u}}(x) = 0 \quad (347)$$

is used.

The corresponding survival integrand is

$$\Phi_x(\mathbf{u}) = F_{\chi_r^2} \left[\frac{1}{m_{\mathbf{u}}(x)} \right] = 1 - \Psi_x(\mathbf{u}). \quad (348)$$

Thus,

$$F_{X_{\text{tr}}}(x) = \frac{1}{|\mathbb{S}^{r-1}|} \int_{\mathbb{S}^{r-1}} \Psi_x(\mathbf{u}) \, d\sigma(\mathbf{u}), \quad (349)$$

$$S(x) = \frac{1}{|\mathbb{S}^{r-1}|} \int_{\mathbb{S}^{r-1}} \Phi_x(\mathbf{u}) \, d\sigma(\mathbf{u}). \quad (350)$$

The quality of the angular approximation depends on both the stochastic dimension r and the localization of the integrand. Strong non-normal amplification can produce narrow regions of directions with large running maximum and disproportionately large transition probability. An angular rule that integrates smooth low-order polynomials accurately may still require substantial refinement to resolve these localized transition-producing regions.

8.1 Properties of the angular integrand

For each fixed x , the directional gain is a quadratic function of $\mathbf{u}$:

$$a_{\mathbf{u}}(\xi) = \mathbf{u}^\top \mathbf{G}(\xi) \mathbf{u}. \quad (351)$$

If $\mathbf{G}(\xi)$ is continuous in ξ , then $m_{\mathbf{u}}(x)$ is continuous in $\mathbf{u}$. In particular, for unit directions $\mathbf{u}$ and $\mathbf{v}$,

$$\begin{aligned} |m_{\mathbf{u}}(x) - m_{\mathbf{v}}(x)| &\leq \max_{0 \leq \xi \leq x} \left| \mathbf{u}^\top \mathbf{G}(\xi) \mathbf{u} - \mathbf{v}^\top \mathbf{G}(\xi) \mathbf{v} \right| \\ &\leq 2 \left[\max_{0 \leq \xi \leq x} \|\mathbf{G}(\xi)\|_2 \right] \|\mathbf{u} - \mathbf{v}\|_2. \end{aligned} \quad (352)$$

Thus, the running maximum is Lipschitz continuous on the sphere when the matrix family is uniformly bounded.

The transition-tail mapping

$$\Psi_r(m) = \overline{F}_{\chi_r^2}(1/m), \quad m > 0, \quad (353)$$

is nondecreasing, with derivative

$$\Psi'_r(m) = \frac{f_{\chi_r^2}(1/m)}{m^2}. \quad (354)$$

The angular transition integrand can therefore vary rapidly in regions where both:

- (i) the running maximum changes rapidly with direction; and
- (ii) the chi-square tail is sensitive to the corresponding critical radius.

Although $m_{\mathbf{u}}(x)$ is continuous, it need not be differentiable everywhere. Nondifferentiability can occur when two different streamwise locations attain equal directional maxima and exchange which one is active. Consequently, the angular integrand is generally continuous but only piecewise smooth. This feature should be considered when choosing high-order deterministic quadrature.

The integrand has antipodal symmetry:

$$\Psi_x(-\mathbf{u}) = \Psi_x(\mathbf{u}), \quad \Phi_x(-\mathbf{u}) = \Phi_x(\mathbf{u}). \quad (355)$$

A quadrature rule may therefore pair antipodal points or integrate over one representative of each projective direction.

Finally, the transition integrand is monotone in x :

$$x_1 \leq x_2 \implies \Psi_{x_1}(\mathbf{u}) \leq \Psi_{x_2}(\mathbf{u}). \quad (356)$$

The same angular directions should be used at every streamwise station so that this directional monotonicity is preserved exactly in the numerical CDF.

8.2 General angular quadrature approximation

Let

$$\mathcal{Q}_K = \{(\mathbf{u}_k, \omega_k)\}_{k=1}^K \quad (357)$$

be an integration rule on $\mathbb{S}^{r-1}$, where

$$\|\mathbf{u}_k\|_2 = 1, \quad \omega_k \geq 0, \quad \sum_{k=1}^K \omega_k = 1. \quad (358)$$

The first-transition CDF is approximated by

$$F_K(x) = \sum_{k=1}^K \omega_k \Psi_x(\mathbf{u}_k). \quad (359)$$

The survival probability is

$$S_K(x) = \sum_{k=1}^K \omega_k \Phi_x(\mathbf{u}_k) = 1 - F_K(x). \quad (360)$$

Positive weights are preferable because they preserve probability bounds and streamwise monotonicity:

$$0 \leq F_K(x) \leq 1 \quad (361)$$

and

$$x_1 \leq x_2 \implies F_K(x_1) \leq F_K(x_2). \quad (362)$$

Rules containing negative weights can achieve high polynomial exactness but may produce values outside the probability interval or amplify errors near localized transition regions.

If the rule explicitly contains both members of every antipodal pair, then

$$\mathbf{u}_{k-} = -\mathbf{u}_{k+} \quad (363)$$

and the two function evaluations are identical. They may be replaced by one representative with combined weight

$$\tilde{\omega}_k = \omega_{k+} + \omega_{k-}. \quad (364)$$

This can reduce the required OWNS directional evaluations by nearly a factor of two.

8.3 Deterministic spherical quadrature

Deterministic quadrature is attractive when the effective stochastic dimension is low and a sequence of increasingly accurate spherical rules is available.

8.3.1 Low-dimensional angular parameterizations

For $r = 1$, the sphere $\mathbb{S}^0 = \{-1, +1\}$ consists of two points. Antipodal symmetry makes the angular integration trivial:

$$F_{X_{\text{tr}}}(x) = \Psi_x(1) = \Psi_x(-1). \quad (365)$$

For $r = 2$, parameterize the circle by

$$\mathbf{u}(\theta) = \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix}, \quad 0 \leq \theta < 2\pi. \quad (366)$$

The angular expectation is

$$F_{X_{\text{tr}}}(x) = \frac{1}{2\pi} \int_0^{2\pi} \Psi_x[\mathbf{u}(\theta)] d\theta. \quad (367)$$

Because of antipodal symmetry, this can be reduced to

$$F_{X_{\text{tr}}}(x) = \frac{1}{\pi} \int_0^\pi \Psi_x[\mathbf{u}(\theta)] d\theta. \quad (368)$$

For smooth periodic integrands, the equally spaced trapezoidal rule is particularly effective:

$$\theta_k = \frac{\pi(k-1)}{K}, \quad \omega_k = \frac{1}{K}, \quad k = 1, \dots, K. \quad (369)$$

For $r = 3$, the direction may be parameterized as

$$\mathbf{u}(\vartheta, \varphi) = \begin{bmatrix} \sin \vartheta \cos \varphi \\ \sin \vartheta \sin \varphi \\ \cos \vartheta \end{bmatrix}, \quad (370)$$

where

$$0 \leq \vartheta \leq \pi, \quad 0 \leq \varphi < 2\pi. \quad (371)$$

Then

$$F_{X_{\text{tr}}}(x) = \frac{1}{4\pi} \int_0^{2\pi} \int_0^\pi \Psi_x[\mathbf{u}(\vartheta, \varphi)] \sin \vartheta d\vartheta d\varphi. \quad (372)$$

A change of variable

$$\mu = \cos \vartheta \quad (373)$$

gives

$$F_{X_{\text{tr}}}(x) = \frac{1}{4\pi} \int_0^{2\pi} \int_{-1}^1 \Psi_x[\mathbf{u}(\mu, \varphi)] d\mu d\varphi. \quad (374)$$

A Gauss–Legendre rule in μ combined with a periodic trapezoidal rule in φ provides a simple product quadrature. Lebedev rules are another effective option specifically for $\mathbb{S}^2$.

Direct hyperspherical-coordinate product rules become less attractive as r increases. Their Jacobians contain powers of sine functions, coordinate singularities occur at the poles, and the number of tensor points grows rapidly with dimension.

8.3.2 Spherical designs

A spherical t -design is a set of points

$$\{\mathbf{u}_k\}_{k=1}^K \subset \mathbb{S}^{r-1} \quad (375)$$

such that equal-weight averaging integrates every spherical polynomial of total degree at most t exactly:

$$\frac{1}{K} \sum_{k=1}^K p(\mathbf{u}_k) = \frac{1}{|\mathbb{S}^{r-1}|} \int_{\mathbb{S}^{r-1}} p(\mathbf{u}) d\sigma(\mathbf{u}) \quad (376)$$

for all such p . The corresponding angular weights are

$$\omega_k = \frac{1}{K}. \quad (377)$$

The first-transition integrand is not a polynomial because it contains both a streamwise maximum and a chi-square tail. Polynomial exactness therefore does not directly imply exact transition probability. Nevertheless, spherical designs provide well-distributed, coordinate-free directions and are useful for low- to moderate- dimensional deterministic integration.

Antipodally symmetric designs are especially suitable because the integrand is even. If both $\mathbf{u}$ and $-\mathbf{u}$ are present, one representative may be retained with twice its original weight.

8.3.3 Nested and non-nested rules

Convergence is easiest to assess with a nested sequence

$$\mathcal{Q}_{K_1} \subset \mathcal{Q}_{K_2} \subset \cdots, \quad (378)$$

because previously computed directional gain histories can be reused. Many high-quality spherical designs and symmetric cubature rules are not nested. In that case, convergence should be evaluated by comparing independent rules of increasing resolution:

$$\eta_{K_j}^F = \max_{0 \leq n \leq N} |F_{K_j}(x_n) - F_{K_{j-1}}(x_n)|. \quad (379)$$

A relative indicator may also be used away from zero:

$$\eta_{K_j}^{F,\text{rel}} = \max_{n: F_{K_j}(x_n) \geq F_{\min}} \frac{|F_{K_j}(x_n) - F_{K_{j-1}}(x_n)|}{F_{K_j}(x_n)}. \quad (380)$$

Convergence of only a terminal probability is insufficient if the complete transition-location distribution is required.

8.4 Sparse-grid angular integration

A direction can be generated from hyperspherical angles or from a mapping of a point in a unit hypercube. This permits the use of sparse grids in place of a full tensor-product rule.

Sparse grids can substantially reduce the number of points needed for smooth functions with low effective dimension. Their application here requires care for three reasons:

- (i) coordinate parameterizations of the sphere have singular or nonuniform Jacobians;
- (ii) the running maximum can make the angular integrand nonsmooth across active-station boundaries; and
- (iii) some sparse-grid combination rules contain negative weights, which can violate finite-level probability bounds and monotonicity.

If sparse grids are used, a measure-preserving map to the uniform sphere and a positive-weight rule are preferable. The computed CDF should be explicitly checked for

$$0 \leq F_K(x_n) \leq 1 \quad (381)$$

and

$$F_K(x_{n+1}) \geq F_K(x_n). \quad (382)$$

Failure of these conditions can result from negative quadrature weights even if every directional integrand is a valid monotone probability.

8.5 Randomized quasi-Monte Carlo

For moderate or large stochastic dimension, deterministic spherical cubature becomes difficult to construct and refine. Randomized quasi-Monte Carlo (RQMC) provides a dimension-independent alternative that retains unbiased error estimation under suitable randomization.

Let

$$\mathbf{v}_k \in (0, 1)^r, \quad k = 1, \dots, K, \quad (383)$$

be a randomized low-discrepancy point set, such as a scrambled Sobol' sequence. Transform each component to a standard Gaussian:

$$z_{k,j} = \Phi^{-1}(v_{k,j}), \quad j = 1, \dots, r, \quad (384)$$

where Φ^{-1} is the inverse standard-normal CDF. Normalize the resulting vector:

$$\mathbf{u}_k = \frac{\mathbf{z}_k}{\|\mathbf{z}_k\|_2}. \quad (385)$$

If $\mathbf{z}_k$ were independent standard Gaussian vectors, this normalization would generate independent uniform directions on the sphere. Applied to a randomized low-discrepancy sequence, it provides a practical low-discrepancy spherical point set.

The RQMC estimator with equal weights is

$$\hat{F}_K^{(j)}(x) = \frac{1}{K} \sum_{k=1}^K \Psi_x(\mathbf{u}_k^{(j)}), \quad (386)$$

where $j = 1, \dots, J$ indexes independent randomizations of the underlying digital sequence. The final estimator is

$$\hat{F}_{K,J}(x) = \frac{1}{J} \sum_{j=1}^J \hat{F}_K^{(j)}(x). \quad (387)$$

An empirical standard error is

$$\widehat{\text{SE}}[\hat{F}_{K,J}(x)] = \sqrt{\frac{1}{J(J-1)} \sum_{j=1}^J [\hat{F}_K^{(j)}(x) - \hat{F}_{K,J}(x)]^2}. \quad (388)$$

The complete CDF should be compared across replicates. A conservative summary is

$$\widehat{\text{SE}}_{\max} = \max_{0 \leq n \leq N} \widehat{\text{SE}}[\hat{F}_{K,J}(x_n)]. \quad (389)$$

Because the same angular points and randomizations are used at every station, each replicate CDF remains nondecreasing in x .

The Gaussian inverse transform should not be evaluated at exactly zero or one. In practice, the randomized points are clipped to

$$v_{k,j} \in [\varepsilon_\Phi, 1 - \varepsilon_\Phi] \quad (390)$$

for a small machine-dependent value ε_Φ .

RQMC remains an angular sampling method, but it differs substantially from direct trajectory Monte Carlo. Every angular point analytically integrates the complete Gaussian radial distribution. It therefore does not spend samples resolving variation in the Gaussian amplitude and does not reduce each OWNS evaluation to a binary transition indicator.

8.6 Comparison with ordinary directional Monte Carlo

Independent uniform directions

$$\mathbf{u}_k \stackrel{\text{iid}}{\sim} \text{Unif}(\mathbb{S}^{r-1}) \quad (391)$$

give the unbiased estimator

$$\hat{F}_K^{\text{dir}}(x) = \frac{1}{K} \sum_{k=1}^K \Psi_x(\mathbf{u}_k). \quad (392)$$

Its variance is

$$\text{Var} \left[\hat{F}_K^{\text{dir}}(x) \right] = \frac{\text{Var}_{\mathbf{u}}[\Psi_x(\mathbf{u})]}{K}. \quad (393)$$

Since

$$0 \leq \Psi_x(\mathbf{u}) \leq 1, \quad (394)$$

the variance is bounded by

$$\text{Var} \left[\hat{F}_K^{\text{dir}}(x) \right] \leq \frac{1}{4K}. \quad (395)$$

Even ordinary directional Monte Carlo generally has lower variance than direct trajectory Monte Carlo. Let

$$I_x(R, \mathbf{u}) = \mathbf{1} \{ R^2 m_{\mathbf{u}}(x) \geq 1 \} \quad (396)$$

be the binary transition indicator for a direct Gaussian trajectory. The angular integrand is its conditional expectation:

$$\Psi_x(\mathbf{u}) = \mathbb{E} [I_x(R, \mathbf{u}) \mid \mathbf{u}]. \quad (397)$$

By the law of total variance,

$$\begin{aligned} \text{Var}[I_x] &= \mathbb{E}_{\mathbf{u}} [\text{Var}(I_x \mid \mathbf{u})] + \text{Var}_{\mathbf{u}} [\Psi_x(\mathbf{u})] \\ &\geq \text{Var}_{\mathbf{u}} [\Psi_x(\mathbf{u})]. \end{aligned} \quad (398)$$

Thus, analytically integrating the radius is a Rao–Blackwellization of direct trajectory sampling and cannot increase estimator variance for a fixed number of independent directions.

RQMC can provide an additional improvement by distributing the angular points more uniformly than independent sampling. Its realized convergence rate depends on the smoothness and effective dimension of the mapped angular integrand and should be assessed empirically.

8.7 Dangerous directions and angular localization

For each terminal station x , define the maximum interval gain over all unit directions:

$$\begin{aligned} \Gamma(x) &= \max_{\|\mathbf{u}\|_2=1} m_{\mathbf{u}}(x) \\ &= \max_{\|\mathbf{u}\|_2=1} \max_{0 \leq \xi \leq x} \mathbf{u}^\top \mathbf{G}(\xi) \mathbf{u} \\ &= \max_{0 \leq \xi \leq x} \lambda_{\max} [\mathbf{G}(\xi)]. \end{aligned} \quad (399)$$

The interchange of maxima is valid because both operations are maxima over a product set. Let

$$\xi_* \in \arg \max_{0 \leq \xi \leq x} \lambda_{\max} [\mathbf{G}(\xi)] \quad (400)$$

and let

$$\mathbf{u}_* \in \arg \max_{\|\mathbf{u}\|_2=1} \mathbf{u}^\top \mathbf{G}(\xi_*) \mathbf{u}. \quad (401)$$

Then $\mathbf{u}_*$ is a direction with the largest possible normalized energy gain over the interval.

The most dangerous direction does not alone determine the transition probability. The probability depends on the angular measure of all directions with appreciable running gain. A very large but extremely narrow peak in direction space can contribute less probability than a moderate gain spread over a broad set of directions.

Near a station where the dominant eigenvalue is simple, write

$$\mathbf{G} = \mathbf{V} \text{diag}(\gamma_1, \gamma_2, \dots, \gamma_r) \mathbf{V}^\top, \quad \gamma_1 > \gamma_2 \geq \dots \geq 0. \quad (402)$$

A direction near the dominant eigenvector can be represented as

$$\mathbf{u} = \cos \theta \mathbf{v}_1 + \sin \theta \mathbf{s}, \quad \mathbf{s}^\top \mathbf{v}_1 = 0, \quad \|\mathbf{s}\|_2 = 1. \quad (403)$$

Its gain is

$$\mathbf{u}^\top \mathbf{G} \mathbf{u} = \gamma_1 \cos^2 \theta + \sin^2 \theta \mathbf{s}^\top \mathbf{G} \mathbf{s}. \quad (404)$$

A large gap between γ_1 and the remaining eigenvalues can therefore produce a sharply localized high-gain region around $\pm \mathbf{v}_1$.

Because the dominant eigenvectors can rotate with x , several separated angular regions may contribute to transition. A useful diagnostic set is obtained by collecting the leading eigenvectors of $\mathbf{G}(x_n)$ at stations where $\lambda_{\max}[\mathbf{G}(x_n)]$ establishes a new streamwise record. These directions identify candidate transition mechanisms but do not replace angular integration.

8.8 Adaptive angular refinement

Adaptive angular refinement should target the contribution to the integral rather than only the maximum value of $m_{\mathbf{u}}(x)$. The appropriate strategy depends on whether a geometrically local tessellation of the sphere is available.

8.8.1 Cell-based refinement in low dimensions

For $r = 2$, the projective angular domain is an interval in θ , and ordinary adaptive one-dimensional quadrature can be used. For $r = 3$, the sphere can be partitioned into spherical triangles. A cell $\mathcal{T}$ with normalized spherical measure $\mu_{\mathcal{T}}$ contributes

$$I_{\mathcal{T}}(x) = \frac{1}{|\mathbb{S}^{r-1}|} \int_{\mathcal{T}} \Psi_x(\mathbf{u}) \, d\sigma(\mathbf{u}). \quad (405)$$

A coarse cell estimate can be compared with the sum of estimates after subdivision:

$$\eta_{\mathcal{T}}(x) = \left| I_{\mathcal{T}}^{\text{fine}}(x) - I_{\mathcal{T}}^{\text{coarse}}(x) \right|. \quad (406)$$

To control error in the complete CDF, define

$$\eta_{\mathcal{T}} = \max_{0 \leq n \leq N} \eta_{\mathcal{T}}(x_n). \quad (407)$$

Cells with the largest indicators are subdivided until

$$\sum_{\mathcal{T}} \eta_{\mathcal{T}} \leq \varepsilon_{\text{ang}}. \quad (408)$$

A refinement indicator based only on variation in the running maximum,

$$\max_{\mathbf{u}, \mathbf{v} \in \mathcal{T}} |m_{\mathbf{u}}(x) - m_{\mathbf{v}}(x)|, \quad (409)$$

may over-refine regions where the chi-square transition tail is insensitive. Evaluating the probability integrand Ψ_x directly gives a more goal-oriented criterion.

8.8.2 Adaptive enrichment in general dimension

A local tessellation becomes expensive in higher dimensions. A practical enrichment strategy is to combine a space-filling base rule with targeted directions:

1. Generate an initial spherical design or RQMC point set.
2. Compute $m_{\mathbf{u}_k}(x_n)$ and $\Psi_{x_n}(\mathbf{u}_k)$ over the full streamwise grid.
3. Identify stations where $\lambda_{\max}[\mathbf{G}(x)]$ establishes a new record or where the transition CDF increases rapidly.
4. Compute several leading eigenvectors of $\mathbf{G}(x)$ at those stations.
5. Add directions in neighborhoods of these eigenvectors and their competing transient-growth subspaces.
6. Recompute the angular integral using a valid weighted integration or importance-sampling correction.
7. Continue until the CDF changes by less than the prescribed angular tolerance.

Appending targeted directions to an equal-weight rule without changing the weights generally biases the integral. Targeted enrichment must therefore be accompanied by one of the following:

- a recomputed spherical cubature rule;
- a geometric cell partition with known cell measures;
- an importance-sampling density with likelihood-ratio weights; or
- a control-variate or stratified estimator.

8.9 Importance sampling on the sphere

If transition is dominated by a small angular region, uniform directional sampling may be inefficient. Let

$$\pi(\mathbf{u}) = \frac{1}{|\mathbb{S}^{r-1}|} \quad (410)$$

be the uniform density with respect to surface measure, and let $g(\mathbf{u}) > 0$ be an alternative sampling density satisfying

$$\int_{\mathbb{S}^{r-1}} g(\mathbf{u}) \, d\sigma(\mathbf{u}) = 1. \quad (411)$$

Then

$$F_{X_{\text{tr}}}(x) = \mathbb{E}_g \left[\Psi_x(\mathbf{u}) \frac{\pi(\mathbf{u})}{g(\mathbf{u})} \right]. \quad (412)$$

An importance-sampling estimator is

$$\hat{F}_K^{\text{IS}}(x) = \frac{1}{K} \sum_{k=1}^K \Psi_x(\mathbf{u}_k) \frac{\pi(\mathbf{u}_k)}{g(\mathbf{u}_k)}, \quad \mathbf{u}_k \sim g. \quad (413)$$

Because the integrand is antipodally symmetric, the proposal should normally satisfy

$$g(-\mathbf{u}) = g(\mathbf{u}). \quad (414)$$

A mixture distribution centered around several pairs of dangerous directions can represent competing transition mechanisms. A uniform component should be retained:

$$g(\mathbf{u}) = \alpha \pi(\mathbf{u}) + (1 - \alpha) g_{\text{target}}(\mathbf{u}), \quad 0 < \alpha < 1, \quad (415)$$

to ensure that the complete sphere remains sampled and importance weights remain finite.

Possible spherical proposal families include the von Mises–Fisher distribution and the more general Bingham distribution. Since the integrand is axial rather than directional,

$$\Psi_x(-\mathbf{u}) = \Psi_x(\mathbf{u}), \quad (416)$$

an antipodally symmetric Bingham-type proposal is especially natural.

Importance sampling is most useful for rare transition probabilities whose angular contributions are highly localized. It introduces additional modeling choices and should be validated against a space-filling rule at reduced dimension or moderate thresholds.

8.10 Control variates based on local quadratic forms

A simpler approximation can be used as a control variate for the first-transition integrand. For example, choose a representative station x_* and define

$$\Psi_x^{\text{cv}}(\mathbf{u}) = \overline{F}_{\chi_r^2} \left[\frac{1}{\mathbf{u}^\top \mathbf{G}(x_*) \mathbf{u}} \right]. \quad (417)$$

If its spherical expectation

$$\mu_{\text{cv}} = \mathbb{E}_{\mathbf{u}} [\Psi_x^{\text{cv}}(\mathbf{u})] \quad (418)$$

is known or can be evaluated accurately, then

$$F_{X_{\text{tr}}}(x) = \mu_{\text{cv}} + \mathbb{E}_{\mathbf{u}} [\Psi_x(\mathbf{u}) - \Psi_x^{\text{cv}}(\mathbf{u})]. \quad (419)$$

The difference may have substantially smaller angular variation if one station dominates the running maximum over most directions.

More generally, several local exceedance integrands can be combined using coefficients fitted from pilot angular evaluations. Control variates preserve unbiasedness only when their expectations are known or evaluated independently to greater accuracy than the target calculation.

8.11 Angular convergence procedure

Angular convergence should be assessed after the streamwise maxima are sufficiently resolved. Otherwise, changes attributed to angular refinement may be contaminated by unresolved streamwise peaks.

For a sequence of angular resolutions $K_1 < K_2 < \dots$, define

$$\Delta_j^F = \max_{0 \leq n \leq N} |F_{K_j}(x_n) - F_{K_{j-1}}(x_n)|. \quad (420)$$

The interval-probability convergence indicator is

$$\Delta_j^p = \sum_{n=0}^N |p_n^{(K_j)} - p_n^{(K_{j-1})}|, \quad (421)$$

where the inlet mass is included as p_0 , and the censoring probability may be included as an additional term. The quantile indicator is

$$\Delta_j^{x_p} = |x_p^{(K_j)} - x_p^{(K_{j-1})}| \quad (422)$$

for each transition quantile of interest.

A calculation is considered angularly converged when the relevant indicators satisfy prescribed tolerances:

$$\Delta_j^F \leq \varepsilon_F^{\text{ang}}, \quad (423)$$

$$\Delta_j^p \leq \varepsilon_p^{\text{ang}}, \quad (424)$$

$$\Delta_j^{x_p} \leq \varepsilon_x^{\text{ang}}. \quad (425)$$

Only the criteria associated with reported outputs need be imposed, although convergence of the complete CDF is generally preferable.

For RQMC, both point-set refinement and replicate uncertainty should be checked:

$$\max_n |\widehat{F}_{2K,J}(x_n) - \widehat{F}_{K,J}(x_n)| \leq \varepsilon_F^{\text{ang}} \quad (426)$$

and

$$\max_n \widehat{\text{SE}} [\widehat{F}_{2K,J}(x_n)] \leq \varepsilon_{\text{SE}}^{\text{ang}}. \quad (427)$$

8.12 Recommended angular-integration strategy

The following hierarchy is recommended:

1. **Very low stochastic dimension.** For $r = 1$, evaluate the single projective direction exactly. For $r = 2$, use a nested periodic trapezoidal rule. For $r = 3$, use a positive-weight spherical rule, such as a suitable Lebedev rule or a product rule in $(\cos \vartheta, \varphi)$.
2. **Low to moderate stochastic dimension.** Use a sequence of well-distributed spherical designs or other positive-weight cubature rules. Exploit antipodal symmetry and reuse directions whenever the rule sequence permits.
3. **Moderate to high stochastic dimension.** Use independently scrambled Sobol' point sets mapped to the sphere by Gaussian normalization. Evaluate several randomizations to obtain an empirical standard error.
4. **Localized rare-event contributions.** Diagnose angular localization using the leading eigenvectors of $\mathbf{G}(x)$ at record-gain stations. If a uniform angular rule converges too slowly, use stratification or an antipodally symmetric importance density with a defensive uniform component.

5. **Convergence assessment.** Refine the angular rule until the complete transition CDF, interval probabilities, and any reported quantiles are stable. Do not infer angular convergence solely from agreement at $x_{\max}$.
6. **Independent verification.** At selected cases, compare deterministic quadrature, RQMC, and independent directional Monte Carlo. Agreement among methods with different point constructions provides a stronger check than refinement of a single rule alone.

The effective difficulty of the angular integral is controlled less by the nominal rank r than by the number and width of the transition-producing directional regions. The dimension-reduction methods in section 9 seek to reduce r while preserving those regions.

9 Transition-Oriented Dimension Reduction

The radial-directional method is exact for the retained Gaussian stochastic dimension r , but angular integration becomes more difficult as r increases. In addition, storing and manipulating the reduced energy matrices

$$\mathbf{G}(x) \in \mathbb{R}^{r \times r} \quad (428)$$

can become expensive when the initial correlation factor has high rank. Dimension reduction is therefore desirable.

For a strongly non-normal propagator, reduction based solely on the initial covariance eigenvalues can be unreliable. An initially low-variance direction can undergo substantial downstream amplification and make an important contribution to transition. The reduced basis should instead be selected using the complete family of normalized propagated energy matrices,

$$\mathbf{G}(x) = \frac{\Re[\mathbf{B}(x)^* \mathbf{H}(x) \mathbf{B}(x)]}{e_{\text{thres}}(x)}. \quad (429)$$

This section describes fixed stochastic-coordinate projections that preserve common streamwise coordinates. Several transition-oriented basis constructions are presented, followed by error indicators and a greedy basis algorithm.

9.1 Fixed projection in the stochastic-coordinate space

Let

$$\mathbf{V}_s \in \mathbb{R}^{r \times s}, \quad s < r, \quad (430)$$

have orthonormal columns:

$$\mathbf{V}_s^\top \mathbf{V}_s = \mathbf{I}_s. \quad (431)$$

The associated orthogonal projector is

$$\mathbf{P}_s = \mathbf{V}_s \mathbf{V}_s^\top. \quad (432)$$

The reduced initial Gaussian coordinate is

$$\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_s), \quad (433)$$

and the reduced disturbance model is

$$\mathbf{q}_s(x) = \mathbf{B}_s(x) \mathbf{z}, \quad (434)$$

where

$$\boxed{\mathbf{B}_s(x) = \mathbf{B}(x)\mathbf{V}_s.} \quad (435)$$

Because $\mathbf{V}_s$ is independent of x ,

$$\mathbf{B}_s(x) = \mathbf{M}(0, x)\mathbf{B}_0\mathbf{V}_s. \quad (436)$$

Thus, the reduced columns are propagated images of fixed combinations of the original initial stochastic directions.

The reduced initial covariance is

$$\mathbf{C}_{0,s} = \mathbf{B}_0\mathbf{P}_s\mathbf{B}_0^*, \quad (437)$$

and the propagated reduced covariance is

$$\mathbf{C}_s(x, x) = \mathbf{B}(x)\mathbf{P}_s\mathbf{B}(x)^*. \quad (438)$$

The reduced normalized energy matrix is

$$\begin{aligned} \mathbf{G}_s(x) &= \frac{\Re[\mathbf{B}_s(x)^*\mathbf{H}(x)\mathbf{B}_s(x)]}{e_{\text{thres}}(x)} \\ &= \boxed{\mathbf{V}_s^\top \mathbf{G}(x) \mathbf{V}_s.} \end{aligned} \quad (439)$$

The reduced normalized energy is therefore

$$\mathbf{y}_s(x) = \mathbf{z}^\top \mathbf{G}_s(x) \mathbf{z}. \quad (440)$$

The corresponding reduced first-transition location is

$$X_{\text{tr},s} = \inf \left\{ x \in [0, x_{\text{max}}] : \mathbf{z}^\top \mathbf{G}_s(x) \mathbf{z} \geq 1 \right\}. \quad (441)$$

Its CDF is

$$\boxed{F_{X_{\text{tr},s}}(x) = \mathbb{E}_{\mathbf{u}_s} \left[\overline{F}_{\chi_s^2} \left(\frac{1}{\max_{0 \leq \xi \leq x} \mathbf{u}_s^\top \mathbf{G}_s(\xi) \mathbf{u}_s} \right) \right]}, \quad (442)$$

where

$$\mathbf{u}_s \sim \text{Unif}(\mathbb{S}^{s-1}). \quad (443)$$

The chi-square degrees of freedom in eq. (442) are s , not r . The reduced model contains s independent standard Gaussian coordinates and therefore has a different radial distribution from the full r -dimensional model. Retaining χ_r^2 after truncating to s stochastic coordinates would be inconsistent.

9.2 Requirement of a streamwise-independent basis

The reduced basis must be fixed over the complete streamwise interval. If a different basis

$$\mathbf{V}_s(x) \quad (444)$$

is chosen independently at each station, then writing

$$\mathbf{q}_s(x) = \mathbf{B}(x)\mathbf{V}_s(x)\mathbf{z} \quad (445)$$

assigns a different physical initial disturbance to the same coefficient vector $\mathbf{z}$ at different values of x . This destroys the pathwise correspondence required for first-transition prediction.

A streamwise-dependent basis can be used only if the coordinate transformations are propagated consistently. If

$$\mathbf{w} = \mathbf{V}_s(x)\mathbf{z}(x), \quad (446)$$

then $\mathbf{z}(x)$ generally evolves and cannot be treated as a fixed standard Gaussian vector. The resulting moving-coordinate dynamics must include the derivative of the basis. The fixed-basis construction in eq. (435) avoids this complication.

9.3 Why initial covariance truncation may fail

Suppose the initial covariance is diagonalized as

$$\mathbf{C}(0, 0) = \mathbf{V}_C \mathbf{\Lambda}_C \mathbf{V}_C^*, \quad (447)$$

and the initial factor is

$$\mathbf{B}_0 = \mathbf{V}_C \mathbf{\Lambda}_C^{1/2}. \quad (448)$$

Truncating the smallest eigenvalues minimizes the initial covariance error in standard matrix norms. It does not necessarily minimize the error in the transition probability.

For an initial stochastic direction $\mathbf{u}$, the normalized downstream energy is

$$a_{\mathbf{u}}(x) = \frac{[\mathbf{M}(0, x)\mathbf{B}_0\mathbf{u}]^* \mathbf{H}(x) [\mathbf{M}(0, x)\mathbf{B}_0\mathbf{u}]}{e_{\text{thres}}(x)}. \quad (449)$$

The factor $\mathbf{B}_0$ accounts for initial variance, while the propagator and energy norm account for downstream amplification. A small initial covariance eigenvalue can therefore be offset by a large non-normal gain.

A covariance-energy criterion such as

$$\frac{\sum_{j=1}^s \lambda_j}{\sum_{j=1}^r \lambda_j} \geq 1 - \varepsilon_C \quad (450)$$

is useful as an initial diagnostic, but it should not be the sole rank selection criterion. Transition-oriented reduction must examine the propagated and threshold-normalized matrices $\mathbf{G}(x)$.

9.4 Output-map factorization

The normalized energy matrix can be expressed as a Gram matrix. Let

$$\mathbf{H}(x) = \mathbf{W}(x)^* \mathbf{W}(x) \quad (451)$$

be an energy-metric factorization, and define the normalized propagated output map

$$\boxed{\mathbf{D}(x) = \frac{\mathbf{W}(x)\mathbf{B}(x)}{\sqrt{e_{\text{thres}}(x)}}} \quad (452)$$

Then

$$\mathbf{G}(x) = \Re[\mathbf{D}(x)^* \mathbf{D}(x)] \quad (453)$$

for real stochastic coordinates. In a fully real representation,

$$\mathbf{G}(x) = \mathbf{D}(x)^\top \mathbf{D}(x). \quad (454)$$

The directional gain becomes

$$a_{\mathbf{u}}(x) = \|\mathbf{D}(x)\mathbf{u}\|_2^2. \quad (455)$$

Dimension reduction can therefore be viewed as finding a low-dimensional right subspace that preserves the action of the family $\{\mathbf{D}(x)\}$.

This factorization is useful because the right singular vectors of $\mathbf{D}(x)$ are the eigenvectors of $\mathbf{G}(x)$. They identify the initial stochastic directions producing the largest normalized energy at station x .

9.5 Union of dominant transient-growth subspaces

A direct transition-oriented construction collects dominant eigenvectors of $\mathbf{G}(x)$ over a set of representative stations. At station x_n , compute

$$\mathbf{G}(x_n)\mathbf{v}_{j,n} = \gamma_{j,n}\mathbf{v}_{j,n}, \quad \gamma_{1,n} \geq \gamma_{2,n} \geq \dots \geq 0. \quad (456)$$

Retain the first p_n eigenvectors according to a stationwise criterion, such as

$$\gamma_{j,n} \geq \varepsilon_\gamma \gamma_{1,n} \quad (457)$$

or

$$\frac{\sum_{j=1}^{p_n} \gamma_{j,n}}{\sum_{j=1}^r \gamma_{j,n}} \geq 1 - \varepsilon_{\text{loc}}. \quad (458)$$

Form the snapshot matrix

$$\mathbf{Z} = [\mathbf{v}_{1,0} \quad \dots \quad \mathbf{v}_{p_0,0} \quad \mathbf{v}_{1,1} \quad \dots \quad \mathbf{v}_{p_1,1} \quad \dots \quad \mathbf{v}_{1,N} \quad \dots \quad \mathbf{v}_{p_N,N}]. \quad (459)$$

A thin singular-value decomposition gives

$$\mathbf{Z} = \mathbf{U}_Z \mathbf{\Sigma}_Z \mathbf{V}_Z^\top. \quad (460)$$

The first s columns of $\mathbf{U}_Z$ define the fixed reduced basis:

$$\mathbf{V}_s = \mathbf{U}_Z(:, 1:s). \quad (461)$$

This method preserves directions that are dominant at different streamwise stations, including rotating transient-growth mechanisms. It is important that the station set resolve narrow gain peaks. Otherwise, a transition-relevant eigenspace may be absent from the snapshot matrix.

Stations at which

$$\lambda_{\max}[\mathbf{G}(x)] \quad (462)$$

establish new streamwise records are especially important because they can create new first-transition probability. However, retaining only the leading eigenvector at each record station may be insufficient when several comparable directions contribute appreciable angular measure.

9.6 Integrated finite-horizon output Gramian

A compact basis can be obtained from a weighted streamwise aggregate:

$$\mathbf{K}_\omega = \int_0^{x_{\max}} \omega(x) \mathbf{G}(x) \, dx, \quad (463)$$

where

$$\omega(x) \geq 0. \quad (464)$$

For a discrete streamwise rule,

$$\mathbf{K}_\omega \approx \sum_{n=0}^N \nu_n \omega_n \mathbf{G}(x_n), \quad (465)$$

where ν_n are streamwise quadrature weights.

The directional quadratic form of $\mathbf{K}_\omega$ is

$$\begin{aligned} \mathbf{u}^\top \mathbf{K}_\omega \mathbf{u} &= \int_0^{x_{\max}} \omega(x) \mathbf{u}^\top \mathbf{G}(x) \mathbf{u} \, dx \\ &= \int_0^{x_{\max}} \omega(x) a_{\mathbf{u}}(x) \, dx. \end{aligned} \quad (466)$$

The leading eigenvectors of $\mathbf{K}_\omega$ therefore preserve directions with large integrated normalized energy.

Equivalently, define a weighted stacked output matrix

$$\mathbf{D}_{\text{stack}} = \begin{bmatrix} \sqrt{\nu_0 \omega_0} \mathbf{D}(x_0) \\ \sqrt{\nu_1 \omega_1} \mathbf{D}(x_1) \\ \vdots \\ \sqrt{\nu_N \omega_N} \mathbf{D}(x_N) \end{bmatrix}. \quad (467)$$

Then

$$\Re[\mathbf{D}_{\text{stack}}^* \mathbf{D}_{\text{stack}}] = \mathbf{K}_\omega. \quad (468)$$

The leading right singular vectors of $\mathbf{D}_{\text{stack}}$ are the leading eigenvectors of $\mathbf{K}_\omega$.

The integrated basis is computationally simple and is closely related to a finite-horizon observability Gramian. Its principal limitation is that integration can suppress narrow but high-amplitude transient peaks. A direction that is dangerous over a short interval can have a small integrated contribution despite dominating first transition. Possible remedies include:

- choosing $\omega(x)$ to emphasize transition-sensitive regions;
- supplementing the integrated basis with eigenvectors at record-gain stations; or
- using the greedy envelope method developed below.

9.7 Maximum-envelope approximation

The first-transition event depends on the maximum directional gain

$$\mathcal{M}(\mathbf{u}) = \max_{0 \leq x \leq x_{\max}} \mathbf{u}^\top \mathbf{G}(x) \mathbf{u}. \quad (469)$$

A reduction aligned with this objective should preserve large values of $\mathcal{M}(\mathbf{u})$, including peaks localized in x .

For a current reduced basis $\mathbf{V}_s$, let

$$\mathbf{P}_s = \mathbf{V}_s \mathbf{V}_s^\top, \quad \mathbf{P}_s^\perp = \mathbf{I} - \mathbf{P}_s. \quad (470)$$

A neglected-direction indicator at station x is

$$\epsilon_\perp(x; \mathbf{V}_s) = \lambda_{\max} \left[\mathbf{P}_s^\perp \mathbf{G}(x) \mathbf{P}_s^\perp \right]. \quad (471)$$

Equivalently,

$$\epsilon_\perp(x; \mathbf{V}_s) = \max_{\substack{\|\mathbf{u}\|_2=1 \\ \mathbf{V}_s^\top \mathbf{u}=0}} \mathbf{u}^\top \mathbf{G}(x) \mathbf{u}. \quad (472)$$

The global neglected-direction indicator is

$$\boxed{\epsilon_\perp^{\max}(\mathbf{V}_s) = \max_{0 \leq x \leq x_{\max}} \epsilon_\perp(x; \mathbf{V}_s)}. \quad (473)$$

If

$$x_* \in \arg \max_x \epsilon_\perp(x; \mathbf{V}_s) \quad (474)$$

and

$$\mathbf{v}_* \in \arg \max_{\substack{\|\mathbf{v}\|_2=1 \\ \mathbf{V}_s^\top \mathbf{v}=0}} \mathbf{v}^\top \mathbf{G}(x_*) \mathbf{v}, \quad (475)$$

then $\mathbf{v}_*$ is the currently omitted direction with the largest normalized energy over the interval. It can be added to the basis:

$$\mathbf{V}_{s+1} = \text{orth} \begin{bmatrix} \mathbf{V}_s & \mathbf{v}_* \end{bmatrix}. \quad (476)$$

This criterion detects energetic directions lying entirely outside the current basis. It does not by itself measure cross-coupling between retained and omitted directions. A stronger matrix approximation indicator is introduced in section 9.11.

9.8 Greedy transition-oriented basis construction

A practical greedy procedure begins with one or more globally dangerous directions and adds the largest omitted transient-growth directions until a tolerance or maximum reduced rank is reached.

The maximization in Algorithm 10 should use a streamwise grid that resolves the important eigenvalue peaks. The eigenvalue function

$$x \mapsto \lambda_{\max} \left[\mathbf{P}_s^\perp \mathbf{G}(x) \mathbf{P}_s^\perp \right] \quad (477)$$

may be interpolated or adaptively maximized between stored stations, analogously to the directional maximum procedure in ??.

Several directions may be added at each greedy iteration by retaining all eigenvectors of the worst-station residual satisfying

$$\lambda_j(\mathbf{R}_{s,n_*}) \geq \varepsilon_{\text{block}} \lambda_1(\mathbf{R}_{s,n_*}). \quad (478)$$

Block enrichment is useful when the dangerous subspace is multidimensional or when the dominant eigenvalue is nearly repeated.

9.9 Hybrid transition-oriented basis

No single basis criterion is optimal in every regime. A robust construction combines complementary information:

1. Compute leading eigenvectors of the integrated matrix $\mathbf{K}_\omega$ to represent broadly energetic directions.
2. Add leading eigenvectors of $\mathbf{G}(x)$ at stations where $\lambda_{\max}[\mathbf{G}(x)]$ establishes a new streamwise record.
3. Add modes from the union of dominant eigenspaces at stations associated with distinct physical amplification mechanisms.
4. Apply the greedy complement algorithm to identify important directions not contained in the initial basis.
5. Orthonormalize the combined snapshot matrix and compress it using an SVD.
6. Increase the final rank until the transition CDF is converged.

If

$$\mathbf{Z}_{\text{hybrid}} = [\mathbf{Z}_{\text{int}} \quad \mathbf{Z}_{\text{record}} \quad \mathbf{Z}_{\text{greedy}}] \quad (479)$$

contains the collected candidate directions, then

$$\mathbf{Z}_{\text{hybrid}} = \mathbf{U}_h \boldsymbol{\Sigma}_h \mathbf{V}_h^\top \quad (480)$$

and

$$\mathbf{V}_s = \mathbf{U}_h(:, 1 : s) \quad (481)$$

defines the final fixed basis.

The singular values of $\mathbf{Z}_{\text{hybrid}}$ indicate linear dependence among the candidate directions, but they are not by themselves a transition error measure. Final rank selection should be based on the transition CDF and matrix residual indicators.

9.10 Relation to finite-horizon balanced truncation

The initial Gaussian coordinate acts as an input applied at $x = 0$, while the energy-weighted disturbance is a spatially distributed output:

$$\mathbf{y}_{\text{out}}(x) = \mathbf{D}(x)\mathbf{w}. \quad (482)$$

The matrix

$$\mathbf{K}_\omega = \int_0^{x_{\max}} \omega(x) \mathbf{D}(x)^* \mathbf{D}(x) dx \quad (483)$$

is therefore a finite-horizon observability-type Gramian in the initial stochastic coordinates.

If additional input or state reduction is needed, finite-horizon balanced truncation may be used to balance the directions excited by the initial uncertainty against those observed by the energy output. This can be useful when reduction is performed in the full OWNS state space before constructing $\mathbf{B}(x)$.

Standard infinite-horizon balanced truncation is not automatically appropriate for the present problem because:

- (i) the OWNS operator is spatially varying;
- (ii) the relevant domain is a finite streamwise interval;
- (iii) transition depends on a maximum rather than an integrated quadratic output; and
- (iv) the transition threshold can vary with x .

Finite-horizon or empirical Gramians are therefore more relevant than their infinite-horizon counterparts. Any state-space reduction should ultimately be checked against the family $\mathbf{G}(x)$ and the resulting first-transition CDF.

9.11 Matrix approximation error

The reduced matrix can be embedded in the original stochastic space as

$$\widehat{\mathbf{G}}_s(x) = \mathbf{P}_s \mathbf{G}(x) \mathbf{P}_s. \quad (484)$$

Define the residual

$$\mathbf{R}_s(x) = \mathbf{G}(x) - \widehat{\mathbf{G}}_s(x). \quad (485)$$

A uniform spectral-norm error indicator is

$$\boxed{\varepsilon_{G,s} = \max_{0 \leq x \leq x_{\max}} \|\mathbf{G}(x) - \mathbf{P}_s \mathbf{G}(x) \mathbf{P}_s\|_2.} \quad (486)$$

For every unit direction $\mathbf{u}$,

$$\left| \mathbf{u}^\top \mathbf{G}(x) \mathbf{u} - \mathbf{u}^\top \widehat{\mathbf{G}}_s(x) \mathbf{u} \right| \leq \varepsilon_{G,s}. \quad (487)$$

Consequently, the full and embedded reduced directional envelopes satisfy

$$|m_{\mathbf{u}}(x) - \widehat{m}_{s,\mathbf{u}}(x)| \leq \varepsilon_{G,s}, \quad (488)$$

where

$$\widehat{m}_{s,\mathbf{u}}(x) = \max_{0 \leq \xi \leq x} \mathbf{u}^\top \widehat{\mathbf{G}}_s(\xi) \mathbf{u}. \quad (489)$$

For a general Gaussian realization $\mathbf{w} = \mathbf{R}\mathbf{u}$,

$$|y(x; \mathbf{w}) - \widehat{y}_s(x; \mathbf{w})| \leq \varepsilon_{G,s} \|\mathbf{w}\|_2^2, \quad (490)$$

where

$$\widehat{y}_s(x; \mathbf{w}) = \mathbf{w}^\top \widehat{\mathbf{G}}_s(x) \mathbf{w}. \quad (491)$$

The same bound applies to the running maxima:

$$\left| \max_{\xi \leq x} y(\xi; \mathbf{w}) - \max_{\xi \leq x} \widehat{y}_s(\xi; \mathbf{w}) \right| \leq \varepsilon_{G,s} \|\mathbf{w}\|_2^2. \quad (492)$$

This result shows that transition classifications can differ only for realizations sufficiently close to the threshold relative to the residual scale. In particular, the symmetric difference of the full and embedded reduced transition events is contained in

$$\left\{ \mathbf{w} : \left| \max_{\xi \leq x} \widehat{y}_s(\xi; \mathbf{w}) - 1 \right| \leq \varepsilon_{G,s} \|\mathbf{w}\|_2^2 \right\}. \quad (493)$$

A small matrix residual is therefore most important when a large probability mass lies near the transition boundary.

The residual in eq. (485) includes three components:

$$\mathbf{R}_s = \mathbf{P}_s^\perp \mathbf{G} \mathbf{P}_s^\perp + \mathbf{P}_s \mathbf{G} \mathbf{P}_s^\perp + \mathbf{P}_s^\perp \mathbf{G} \mathbf{P}_s. \quad (494)$$

The first term represents energy entirely within the omitted subspace. The final two terms represent coupling between retained and omitted directions. The neglected-direction indicator $\epsilon_\perp^{\max}$ from eq. (473) measures only the first component. The full residual norm $\varepsilon_{G,s}$ is therefore a stronger diagnostic.

9.12 Conditional matrix blocks

Complete the reduced basis to an orthogonal matrix

$$\mathbf{V} = [\mathbf{V}_s \quad \mathbf{V}_\perp]. \quad (495)$$

In these coordinates,

$$\mathbf{V}^\top \mathbf{G}(x) \mathbf{V} = \begin{bmatrix} \mathbf{G}_{ss}(x) & \mathbf{G}_{s\perp}(x) \\ \mathbf{G}_{s\perp}(x)^\top & \mathbf{G}_{\perp\perp}(x) \end{bmatrix}, \quad (496)$$

where

$$\mathbf{G}_{ss}(x) = \mathbf{V}_s^\top \mathbf{G}(x) \mathbf{V}_s, \quad (497)$$

$$\mathbf{G}_{s\perp}(x) = \mathbf{V}_s^\top \mathbf{G}(x) \mathbf{V}_\perp, \quad (498)$$

$$\mathbf{G}_{\perp\perp}(x) = \mathbf{V}_\perp^\top \mathbf{G}(x) \mathbf{V}_\perp. \quad (499)$$

Useful block diagnostics are

$$\varepsilon_{\perp,s} = \max_x \|\mathbf{G}_{\perp\perp}(x)\|_2, \quad (500)$$

$$\varepsilon_{\times,s} = \max_x \|\mathbf{G}_{s\perp}(x)\|_2. \quad (501)$$

The full residual satisfies

$$\varepsilon_{G,s} \leq \varepsilon_{\perp,s} + 2\varepsilon_{\times,s}. \quad (502)$$

Thus, both omitted energy and retained–omitted coupling should be small.

If

$$\mathbf{G}_{s\perp}(x) = \mathbf{0} \quad \text{for every } x, \quad (503)$$

then the retained subspace is invariant for the entire matrix family. In this special case, the retained and omitted quadratic forms are block diagonal. In general, the matrices $\{\mathbf{G}(x)\}$ do not share a common invariant subspace, especially when non-normal amplification rotates the dangerous directions.

9.13 Probability bounds from a matrix residual

The uniform residual provides directional envelope bounds:

$$\max[\widehat{m}_{s,\mathbf{u}}(x) - \varepsilon_{G,s}, 0] \leq m_{\mathbf{u}}(x) \leq \widehat{m}_{s,\mathbf{u}}(x) + \varepsilon_{G,s}. \quad (504)$$

Since $\Psi_r(m) = \overline{F}_{\chi_r^2}(1/m)$ is nondecreasing,

$$\begin{aligned}\Psi_r(\max[\widehat{m}_{s,\mathbf{u}}(x) - \varepsilon_{G,s}, 0]) &\leq \Psi_r[m_{\mathbf{u}}(x)] \\ &\leq \Psi_r(\widehat{m}_{s,\mathbf{u}}(x) + \varepsilon_{G,s}).\end{aligned}\tag{505}$$

Averaging these inequalities over the original sphere $\mathbb{S}^{r-1}$ produces upper and lower bounds on the full transition CDF.

These bounds concern the embedded approximation $\widehat{\mathbf{G}}_s = \mathbf{P}_s \mathbf{G} \mathbf{P}_s$ under the original r -dimensional Gaussian measure. They are not identical to simply evaluating the s -dimensional reduced model in eq. (442). The latter discards the omitted Gaussian coordinates and uses a χ_s^2 radial distribution. This distinction should be maintained when interpreting reduced-order error estimates.

The full-dimensional residual bounds are most useful as diagnostics or certificates when they can be evaluated with a moderate-cost angular rule. The reduced s -dimensional calculation remains the principal low-cost approximation.

9.14 CDF convergence with reduced rank

The most direct practical error assessment is to compute the transition distribution for an increasing sequence of nested bases:

$$\text{range}(\mathbf{V}_{s_1}) \subset \text{range}(\mathbf{V}_{s_2}) \subset \cdots, \quad s_1 < s_2 < \cdots.\tag{506}$$

Define the CDF difference

$$\Delta_s^F = \max_{0 \leq n \leq N} |F_{X_{\text{tr},s}}(x_n) - F_{X_{\text{tr},s^-}}(x_n)|,\tag{507}$$

where s^- is the preceding reduced rank. Similarly, define

$$\Delta_s^p = \sum_{n=0}^N |p_{n,s} - p_{n,s^-}| + |p_{\text{censored},s} - p_{\text{censored},s^-}|,\tag{508}$$

$$\Delta_s^{x_p} = |x_{p,s} - x_{p,s^-}|.\tag{509}$$

The rank is considered adequate when

$$\Delta_s^F \leq \varepsilon_F^{\text{red}},\tag{510}$$

$$\Delta_s^p \leq \varepsilon_p^{\text{red}},\tag{511}$$

$$\Delta_s^{x_p} \leq \varepsilon_x^{\text{red}},\tag{512}$$

for the reported quantities, while also satisfying matrix residual criteria such as

$$\varepsilon_{G,s} \leq \varepsilon_G^{\text{tol}}.\tag{513}$$

Unlike streamwise and angular refinement, increasing the stochastic rank does not necessarily produce a monotone sequence of transition CDFs. Adding an independent Gaussian coordinate changes both the quadratic-form family and the radial dimension. Therefore, convergence should be judged from stabilization rather than from an assumed one-sided bound.

9.15 Sensitivity-weighted reduction criteria

Not all errors in $\mathbf{G}(x)$ have the same effect on the transition probability. For a directional maximum $m > 0$, the transition-tail sensitivity is

$$\Psi'_r(m) = \frac{f_{\chi_r^2}(1/m)}{m^2}. \quad (514)$$

A directional maximum error $\delta m_{\mathbf{u}}$ produces the first-order probability change

$$\delta \Psi_r \approx \frac{f_{\chi_r^2}(1/m_{\mathbf{u}})}{m_{\mathbf{u}}^2} \delta m_{\mathbf{u}}. \quad (515)$$

This suggests emphasizing directions and stations for which the gain is close to the transition-sensitive range of the chi-square tail, rather than selecting modes solely from the largest eigenvalues.

Given a pilot angular rule $\{(\mathbf{u}_k, \omega_k)\}$, one may define a sensitivity-weighted projection indicator

$$\mathcal{E}_s^{\text{sens}} = \sum_{k=1}^K \omega_k \frac{f_{\chi_r^2}(1/m_k)}{m_k^2} |m_k - \hat{m}_{s,k}|. \quad (516)$$

This quantity estimates the induced transition-probability error more directly than a uniform matrix norm. It is not a rigorous bound and requires a pilot approximation to the full directional maxima, but it can guide basis enrichment toward directions with the greatest probabilistic influence.

A streamwise sensitivity weight for the integrated matrix $\mathbf{K}_{\omega}$ can likewise be constructed from a pilot transition calculation. For example, $\omega(x)$ may be increased where the first-transition CDF rises rapidly or where many directions establish new gain records.

9.16 Reduced-basis construction algorithm

The following algorithm summarizes a hybrid practical procedure.

The angular and streamwise tolerances used in each reduced calculation should be tighter than the desired rank-convergence tolerance. Otherwise, changes caused by angular or streamwise discretization can be mistaken for reduced-rank error.

9.17 Recommended reduction strategy for OWNS

For the strongly non-normal OWNS problem, the following reduction strategy is recommended:

1. Construct the initial factor with enough covariance modes that potentially amplified low-variance directions have not already been discarded.
2. Propagate this factor, or apply the propagator in a matrix-free manner, to build the normalized energy family $\mathbf{G}(x)$.
3. Resolve the streamwise variation of $\lambda_{\max}[\mathbf{G}(x)]$ and its leading eigenspaces.
4. Form an initial basis from the union of dominant eigenspaces at representative and record-gain stations.
5. Supplement this basis with leading eigenvectors of a weighted integrated Gramian to preserve broadly energetic directions.

6. Apply greedy complement enrichment to detect omitted directions associated with localized transient growth.
7. Use one fixed basis over the full streamwise interval.
8. Evaluate the first-transition CDF for a sequence of increasing reduced ranks.
9. Monitor both the CDF convergence and the full matrix residual, including retained–omitted cross-coupling.
10. Validate the final reduced model against a higher-rank calculation or an independent trajectory ensemble for selected cases.

The reduced dimension should be chosen according to the transition quantity of interest rather than solely according to covariance reconstruction. A basis that accurately reproduces $\text{tr}[\mathbf{C}(x, x)]$ can still misrepresent the tails and directional maxima governing first transition.

10 Numerical Conditioning for Strongly Non-Normal Propagation

Strong non-normality can produce large transient amplification and substantial separation among the singular values of the propagated factor

$$\mathbf{B}(x) = \mathbf{M}(0, x)\mathbf{B}_0. \quad (517)$$

As a result, direct propagation and storage of $\mathbf{B}(x)$ can become numerically ill-conditioned even when the physically relevant quadratic form

$$\mathbf{G}(x) = \frac{\Re[\mathbf{B}(x)^* \mathbf{H}(x) \mathbf{B}(x)]}{e_{\text{thres}}(x)} \quad (518)$$

remains well defined.

This section discusses stable propagation, factorization, scaling, and evaluation strategies. The essential requirement is that every numerical transformation preserve the mapping between the original Gaussian coordinate $\mathbf{w}$ and the propagated state:

$$\mathbf{q}(x) = \mathbf{B}(x)\mathbf{w}. \quad (519)$$

A propagated basis may be orthogonalized or rescaled, but the associated coordinate transformation must be retained. Discarding it changes the energy history of individual realizations and therefore changes the first-transition distribution.

10.1 Conditioning induced by non-normal amplification

Let the singular-value decomposition of the propagated factor be

$$\mathbf{B}(x) = \mathbf{U}_B(x) \mathbf{\Sigma}_B(x) \mathbf{V}_B(x)^*, \quad (520)$$

where

$$\mathbf{\Sigma}_B(x) = \text{diag}[\sigma_1(x), \dots, \sigma_r(x)], \quad \sigma_1(x) \geq \dots \geq \sigma_r(x) \geq 0. \quad (521)$$

The condition number on the numerically retained range is

$$\kappa_2[\mathbf{B}(x)] = \frac{\sigma_1(x)}{\sigma_{\min,+}(x)}, \quad (522)$$

where $\sigma_{\min,+}$ is the smallest retained nonzero singular value.

A large value of $\kappa_2[\mathbf{B}(x)]$ indicates that the propagated images of the initial stochastic directions have become nearly linearly dependent in the physical state space. This can occur because the non-normal propagator strongly amplifies a small set of preferred structures while other structures remain weak or decay.

The numerical consequences include:

- (i) loss of relative accuracy in weak propagated directions;
- (ii) cancellation when evaluating linear combinations $\mathbf{B}(x)\mathbf{u}$;
- (iii) loss of symmetry or positive semidefiniteness when forming $\mathbf{B}^*\mathbf{H}\mathbf{B}$;
- (iv) overflow or underflow if amplification spans a range larger than the floating-point representation; and
- (v) inaccurate reduced-rank decisions if numerically weak directions are discarded without considering their downstream transition relevance.

The condition number of $\mathbf{B}$ is coordinate dependent and should not be interpreted as a physical observable by itself. The relevant physical amplification is measured in the energy norm through

$$\mathbf{D}(x) = \frac{\mathbf{H}(x)^{1/2}\mathbf{B}(x)}{\sqrt{e_{\text{thres}}(x)}}, \quad (523)$$

for which

$$\mathbf{G}(x) = \Re[\mathbf{D}(x)^*\mathbf{D}(x)]. \quad (524)$$

Monitoring the singular values of $\mathbf{D}(x)$ is often more informative than monitoring those of $\mathbf{B}(x)$ alone.

10.2 Separation of propagated basis and coordinate map

A stable implementation may represent the propagated factor as

$$\boxed{\mathbf{B}(x) = \mathbf{Q}(x)\mathbf{T}(x)}, \quad (525)$$

where

$$\mathbf{Q}(x)^*\mathbf{Q}(x) = \mathbf{I}_r \quad (526)$$

and $\mathbf{T}(x) \in \mathbb{C}^{r \times r}$ stores the mapping from the original Gaussian coordinates to the orthonormal propagated basis. For a thin QR factorization, $\mathbf{T}(x)$ is upper triangular.

The physical state is then

$$\mathbf{q}(x) = \mathbf{Q}(x)\mathbf{T}(x)\mathbf{w}. \quad (527)$$

The energy matrix in the original stochastic coordinates is

$$\boxed{\mathbf{A}(x) = \Re[\mathbf{T}(x)^*\mathbf{Q}(x)^*\mathbf{H}(x)\mathbf{Q}(x)\mathbf{T}(x)]}. \quad (528)$$

Thus, orthogonalizing the physical columns does not remove the non-normal amplification. It moves the amplification and coordinate coupling into $\mathbf{T}(x)$.

It would be incorrect to replace $\mathbf{B}(x)$ by $\mathbf{Q}(x)$ while continuing to use $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. That replacement would produce

$$\tilde{\mathbf{q}}(x) = \mathbf{Q}(x)\mathbf{w}, \quad (529)$$

which generally has a different covariance and a different first-transition distribution from $\mathbf{q}(x) = \mathbf{Q}(x)\mathbf{T}(x)\mathbf{w}$.

One may instead define transformed coordinates

$$\mathbf{z}(x) = \mathbf{T}(x)\mathbf{w}. \quad (530)$$

Their covariance is

$$\mathbb{E}[\mathbf{z}(x)\mathbf{z}(x)^*] = \mathbf{T}(x)\mathbf{T}(x)^*, \quad (531)$$

so $\mathbf{z}(x)$ is not generally a standard Gaussian vector. Moreover, its definition varies with x . The radial-directional method should therefore continue to operate in the original standard Gaussian coordinate $\mathbf{w}$, with all factors in eq. (528) retained.

10.3 Periodic QR stabilization

Suppose the OWNS march advances the propagated factor from x_n to x_{n+1} through the numerical map

$$\mathbf{B}_{n+1} = \mathbf{P}_{n+1,n}\mathbf{B}_n, \quad (532)$$

where $\mathbf{P}_{n+1,n}$ approximates $\mathbf{M}(x_n, x_{n+1})$. Let

$$\mathbf{B}_n = \mathbf{Q}_n\mathbf{T}_n. \quad (533)$$

Then

$$\mathbf{B}_{n+1} = (\mathbf{P}_{n+1,n}\mathbf{Q}_n)\mathbf{T}_n. \quad (534)$$

Compute a thin QR factorization

$$\mathbf{P}_{n+1,n}\mathbf{Q}_n = \mathbf{Q}_{n+1}\mathbf{R}_{n+1}. \quad (535)$$

The coordinate factor is updated as

$$\boxed{\mathbf{T}_{n+1} = \mathbf{R}_{n+1}\mathbf{T}_n}. \quad (536)$$

Therefore,

$$\mathbf{B}_{n+1} = \mathbf{Q}_{n+1}\mathbf{T}_{n+1}. \quad (537)$$

At the inlet, compute

$$\mathbf{B}_0 = \mathbf{Q}_0\mathbf{T}_0 \quad (538)$$

by a thin QR factorization. The matrix $\mathbf{T}_0$ is part of the initial map and must not be replaced by the identity unless $\mathbf{B}_0$ already has orthonormal columns.

The diagonal phases or signs of each QR factorization should be fixed consistently. For example, the diagonal entries of $\mathbf{R}_{n+1}$ may be made real and nonnegative. This convention reduces discontinuous sign or phase changes in the stored factors, although the reconstructed product $\mathbf{Q}_{n+1}\mathbf{T}_{n+1}$ is invariant to such choices.

Periodic QR stabilizes the physical basis $\mathbf{Q}_n$, but the accumulated matrix $\mathbf{T}_n$ may itself become ill-conditioned. This ill-conditioning reflects the true separation among propagated directions and cannot be removed by orthogonalization alone. It can, however, be represented using additional scaling or singular-value factorizations.

10.4 Continuous orthonormalization

Instead of discrete QR factorizations, one may continuously factor

$$\mathbf{B}(x) = \mathbf{Q}(x)\mathbf{T}(x), \quad \mathbf{Q}(x)^*\mathbf{Q}(x) = \mathbf{I}. \quad (539)$$

Differentiating gives

$$\mathbf{B}' = \mathbf{Q}'\mathbf{T} + \mathbf{Q}\mathbf{T}'. \quad (540)$$

Since

$$\mathbf{B}' = \mathbf{L}\mathbf{B} = \mathbf{L}\mathbf{Q}\mathbf{T}, \quad (541)$$

one possible orthonormalization equation is

$$\mathbf{Q}' = (\mathbf{I} - \mathbf{Q}\mathbf{Q}^*)\mathbf{L}\mathbf{Q} + \mathbf{Q}\mathbf{\Omega}, \quad (542)$$

where

$$\mathbf{\Omega}^* = -\mathbf{\Omega} \quad (543)$$

is a skew-Hermitian gauge matrix. The coordinate factor then satisfies

$$\mathbf{T}' = (\mathbf{Q}^*\mathbf{L}\mathbf{Q} - \mathbf{\Omega})\mathbf{T}. \quad (544)$$

The gauge can be chosen to impose a desired structure on $\mathbf{T}$, such as an upper-triangular continuous-QR factor. The exact form depends on the selected continuous-orthonormalization method. Discrete periodic QR is usually simpler to integrate into an existing OWNS solver, while continuous QR can provide smoother basis evolution and finite-distance Lyapunov diagnostics.

Any continuous orthonormalization must preserve

$$\mathbf{B}(x) = \mathbf{Q}(x)\mathbf{T}(x) \quad (545)$$

to the accuracy required by the first-transition calculation.

10.5 SVD reconditioning

At selected stations, the factor may be re-expressed using a thin singular-value decomposition:

$$\mathbf{B}(x) = \mathbf{U}(x)\mathbf{\Sigma}(x)\mathbf{V}(x)^*. \quad (546)$$

Define

$$\mathbf{Q}(x) = \mathbf{U}(x), \quad \mathbf{T}(x) = \mathbf{\Sigma}(x)\mathbf{V}(x)^*. \quad (547)$$

Then

$$\mathbf{B}(x) = \mathbf{Q}(x)\mathbf{T}(x) \quad (548)$$

and the original Gaussian coordinate remains unchanged.

The right singular-vector matrix $\mathbf{V}(x)$ cannot be discarded. Replacing $\mathbf{T}(x)$ by $\mathbf{\Sigma}(x)$ would rotate the original stochastic coordinates in a streamwise-dependent manner. Although the local covariance could still be represented after a corresponding coordinate rotation, the pathwise relationship between stations would be lost unless the rotation were accumulated.

The SVD is more expensive than QR but directly reveals:

- (i) the numerical rank of the propagated factor;

- (ii) the spread of the propagated singular values;
- (iii) nearly dependent physical columns; and
- (iv) candidate directions for transition-oriented rank compression.

A singular direction should not be discarded solely because $\sigma_j[\mathbf{B}(x)]$ is small at one station. It may become important at a later station, or its energy relevance may be altered by $\mathbf{H}(x)$. Any truncation should follow the transition-oriented criteria in section 9.

10.6 Scaled representation of the coordinate factor

If entries or singular values of $\mathbf{T}(x)$ become very large or small, introduce a scalar scale factor:

$$\mathbf{T}(x) = \exp[\ell(x)]\bar{\mathbf{T}}(x), \quad (549)$$

where $\ell(x) \in \mathbb{R}$ and $\bar{\mathbf{T}}(x)$ is maintained at a moderate norm. The propagated factor is

$$\mathbf{B}(x) = \exp[\ell(x)]\mathbf{Q}(x)\bar{\mathbf{T}}(x). \quad (550)$$

The normalized energy matrix becomes

$$\mathbf{G}(x) = \exp[2\ell(x)] \frac{\Re [\bar{\mathbf{T}}(x)^* \mathbf{Q}(x)^* \mathbf{H}(x) \mathbf{Q}(x) \bar{\mathbf{T}}(x)]}{e_{\text{thres}}(x)}. \quad (551)$$

A scalar scale controls overflow associated with the overall amplification but does not resolve a large condition number within $\mathbf{T}$. More general diagonal scaling may be used:

$$\mathbf{T}(x) = \bar{\mathbf{T}}(x)\mathbf{D}(x), \quad (552)$$

where

$$\mathbf{D}(x) = \text{diag}[\exp(\ell_1(x)), \dots, \exp(\ell_r(x))]. \quad (553)$$

Then

$$\mathbf{G}(x) = \mathbf{D}(x)\bar{\mathbf{G}}(x)\mathbf{D}(x), \quad (554)$$

where

$$\bar{\mathbf{G}}(x) = \frac{\Re [\bar{\mathbf{T}}(x)^* \mathbf{Q}(x)^* \mathbf{H}(x) \mathbf{Q}(x) \bar{\mathbf{T}}(x)]}{e_{\text{thres}}(x)}. \quad (555)$$

The diagonal factors are part of the mapping from the original Gaussian coordinate and must be included in every directional quadratic form:

$$a_{\mathbf{u}}(x) = [\mathbf{D}(x)\mathbf{u}]^T \bar{\mathbf{G}}(x) [\mathbf{D}(x)\mathbf{u}]. \quad (556)$$

Rescaling the columns and then treating the rescaled coefficients as independent standard Gaussians would alter the stochastic model.

10.7 Logarithmic evaluation of directional gains

When a common scale factor is used, the directional gain can be represented logarithmically:

$$\log a_{\mathbf{u}}(x) = 2\ell(x) + \log \left[\mathbf{u}^\top \overline{\mathbf{G}}(x) \mathbf{u} \right]. \quad (557)$$

This avoids explicitly forming a number outside the floating-point range.

The critical squared radius is

$$\rho_{\mathbf{u}}^2(x) = \frac{1}{m_{\mathbf{u}}(x)}, \quad (558)$$

so

$$\log \rho_{\mathbf{u}}^2(x) = -\log m_{\mathbf{u}}(x). \quad (559)$$

If the statistical library accepts a logarithmic argument or provides asymptotic chi-square tail evaluation, the transition probability can be computed without exponentiating extreme values.

In ordinary double precision, a practical implementation may use thresholds:

- if $\log m_{\mathbf{u}}$ is sufficiently negative that $1/m_{\mathbf{u}}$ lies in an extreme upper chi-square tail, evaluate the logarithmic transition probability;
- if $\log m_{\mathbf{u}}$ is sufficiently positive that $1/m_{\mathbf{u}}$ is numerically zero, the directional transition probability is effectively one; and
- otherwise, evaluate the lower or upper chi-square tail using a standard stable routine.

The tolerance for replacing a probability by zero or one should be set relative to the desired probability accuracy, not merely to machine epsilon.

10.8 Stable formation of the reduced energy matrix

The expression

$$\mathbf{A} = \Re[\mathbf{B}^* \mathbf{H} \mathbf{B}] \quad (560)$$

should be evaluated as a Gram matrix whenever possible. Suppose

$$\mathbf{H} = \mathbf{W}^* \mathbf{W}. \quad (561)$$

Define

$$\mathbf{Y} = \mathbf{W} \mathbf{B}. \quad (562)$$

Then

$$\mathbf{A} = \Re[\mathbf{Y}^* \mathbf{Y}]. \quad (563)$$

This construction makes positive semidefiniteness explicit and can be implemented using stable level-3 matrix operations.

For a quadrature-weighted energy norm, $\mathbf{W}$ may often be constructed by applying square roots of the wall-normal quadrature weights and physical variable weights. It is not necessary to form a dense matrix square root of $\mathbf{H}$ when the weighting operation is available directly.

If $\mathbf{B} = \mathbf{Q} \mathbf{T}$, first form

$$\mathbf{Y}_Q = \mathbf{W} \mathbf{Q} \quad (564)$$

and then

$$\mathbf{A} = \Re[\mathbf{T}^* \mathbf{Y}_Q^* \mathbf{Y}_Q \mathbf{T}]. \quad (565)$$

The intermediate matrix

$$\mathbf{H}_Q = \mathbf{Q}^* \mathbf{H} \mathbf{Q} = \mathbf{Y}_Q^* \mathbf{Y}_Q \quad (566)$$

is Hermitian positive semidefinite.

After forming $\mathbf{G}$, enforce symmetry:

$$\mathbf{G} \leftarrow \frac{1}{2} (\mathbf{G} + \mathbf{G}^\top). \quad (567)$$

The relative symmetry defect before correction,

$$\eta_{\text{sym}} = \frac{\|\mathbf{G} - \mathbf{G}^\top\|_F}{\|\mathbf{G}\|_F}, \quad (568)$$

should be monitored. A defect much larger than the expected floating-point or marching error can indicate inconsistent complex conventions, an inaccurate adjoint weighting, or loss of propagation accuracy.

The positive-semidefinite defect can be monitored through

$$\eta_{\text{psd}} = \frac{\max[-\lambda_{\min}(\mathbf{G}), 0]}{\max[\lambda_{\max}(\mathbf{G}), \varepsilon_{\text{scale}}]}. \quad (569)$$

Only roundoff-scale negative eigenvalues should be clipped to zero.

10.9 Stable matrix-free directional energy

If only directional gains are required, forming $\mathbf{G}$ can be avoided. For direction $\mathbf{u}$, compute

$$\mathbf{v}(x) = \mathbf{B}(x)\mathbf{u} \quad (570)$$

and

$$a_{\mathbf{u}}(x) = \frac{\mathbf{v}(x)^* \mathbf{H}(x) \mathbf{v}(x)}{e_{\text{thres}}(x)}. \quad (571)$$

If $\mathbf{H} = \mathbf{W}^* \mathbf{W}$, evaluate

$$a_{\mathbf{u}}(x) = \frac{\|\mathbf{W}(x)\mathbf{v}(x)\|_2^2}{e_{\text{thres}}(x)}. \quad (572)$$

This norm-based form avoids cancellation and produces a nonnegative result in exact arithmetic.

Using the factored representation,

$$\mathbf{c}(x) = \mathbf{T}(x)\mathbf{u}, \quad (573)$$

$$\mathbf{v}(x) = \mathbf{Q}(x)\mathbf{c}(x). \quad (574)$$

The gain is then

$$a_{\mathbf{u}}(x) = \frac{\mathbf{c}(x)^* \mathbf{Q}(x)^* \mathbf{H}(x) \mathbf{Q}(x) \mathbf{c}(x)}{e_{\text{thres}}(x)}. \quad (575)$$

For many directions, assemble

$$\mathbf{U} = [\mathbf{u}_1 \quad \cdots \quad \mathbf{u}_K] \quad (576)$$

and compute

$$\mathbf{C}(x) = \mathbf{T}(x)\mathbf{U}, \quad (577)$$

$$\mathbf{V}(x) = \mathbf{Q}(x)\mathbf{C}(x). \quad (578)$$

The directional energies are the squared weighted norms of the columns of $\mathbf{V}$. This batched calculation uses matrix-matrix operations and is generally more efficient than processing directions individually.

10.10 Direct propagation of directional states

Instead of propagating the complete factor, one may propagate selected directional states:

$$\mathbf{v}_k(0) = \mathbf{B}_0 \mathbf{u}_k, \quad (579)$$

followed by

$$\frac{d\mathbf{v}_k}{dx} = \mathbf{L}(x) \mathbf{v}_k(x). \quad (580)$$

Then

$$\mathbf{v}_k(x) = \mathbf{B}(x) \mathbf{u}_k. \quad (581)$$

This strategy is exact for the selected angular directions and can reduce storage from $n_q r$ values per station to $n_q K_b$ values for a processed block of K_b directions.

Directional propagation is attractive when:

- (i) the number of angular directions is smaller than the stochastic rank;
- (ii) the OWNS solver efficiently marches multiple right-hand sides;
- (iii) only the transition CDF is required, rather than the full matrix family $\mathbf{G}(x)$; and
- (iv) angular refinement is known in advance.

It is less attractive when the angular rule is repeatedly enriched, because each new direction requires an additional OWNS march. Forming or storing $\mathbf{G}(x)$ allows arbitrary new directions to be evaluated without further propagation.

A hybrid strategy propagates a transition-oriented reduced factor

$$\mathbf{B}_s(x) = \mathbf{B}(x) \mathbf{V}_s \quad (582)$$

and then performs angular integration in the reduced stochastic space. This often provides the best balance between OWNS cost and angular flexibility.

10.11 Propagation-error amplification

Strong non-normality can amplify numerical errors introduced by the spatial march. Let the numerical factor be

$$\tilde{\mathbf{B}}(x) = \mathbf{B}(x) + \Delta \mathbf{B}(x). \quad (583)$$

The resulting energy-matrix perturbation is

$$\begin{aligned} \Delta \mathbf{A} &= \Re \left[\tilde{\mathbf{B}}^* \mathbf{H} \tilde{\mathbf{B}} - \mathbf{B}^* \mathbf{H} \mathbf{B} \right] \\ &= \Re \left[\mathbf{B}^* \mathbf{H} \Delta \mathbf{B} + \Delta \mathbf{B}^* \mathbf{H} \mathbf{B} + \Delta \mathbf{B}^* \mathbf{H} \Delta \mathbf{B} \right]. \end{aligned} \quad (584)$$

Therefore,

$$\|\Delta \mathbf{A}\|_2 \leq 2\|\mathbf{H}\|_2 \|\mathbf{B}\|_2 \|\Delta \mathbf{B}\|_2 + \|\mathbf{H}\|_2 \|\Delta \mathbf{B}\|_2^2. \quad (585)$$

For a deterministic threshold,

$$\|\Delta \mathbf{G}\|_2 \leq \frac{2\|\mathbf{H}\|_2 \|\mathbf{B}\|_2 \|\Delta \mathbf{B}\|_2 + \|\mathbf{H}\|_2 \|\Delta \mathbf{B}\|_2^2}{e_{\text{thres}}}. \quad (586)$$

If $\mathbf{H}$ or the threshold is also approximated, their perturbations must be included. To first order,

$$\Delta \mathbf{G} \approx \frac{\Delta \mathbf{A}}{e_{\text{thres}}} - \frac{\Delta e_{\text{thres}}}{e_{\text{thres}}} \mathbf{G}. \quad (587)$$

These bounds can be conservative but show that relative errors in the propagated factor can be magnified when the factor norm is large. OWNS marching tolerances should therefore be selected according to the required accuracy in $\mathbf{G}$ or the transition CDF, rather than solely according to the state residual.

A direct convergence check is to compute

$$\eta_{\text{march}} = \max_n \frac{\|\mathbf{G}_{\text{fine}}(x_n) - \mathbf{G}_{\text{coarse}}(x_n)\|_2}{\max[\|\mathbf{G}_{\text{fine}}(x_n)\|_2, \varepsilon_{\text{scale}}]} \quad (588)$$

using two OWNS marching resolutions or tolerances. More importantly, the induced CDF difference should be evaluated:

$$\eta_{\text{march}}^F = \max_n |F_{\text{fine}}(x_n) - F_{\text{coarse}}(x_n)|. \quad (589)$$

10.12 Derivative consistency

The continuous maximum-detection method uses

$$\mathbf{G}'(x) \quad (590)$$

or directional derivatives $a'_{\mathbf{u}}(x)$. Inconsistent derivatives can create false Hermite peaks or suppress genuine ones.

A useful check compares the analytical derivative with a centered finite difference:

$$\mathbf{G}'_{\text{FD}}(x_n) = \frac{\mathbf{G}(x_{n+1}) - \mathbf{G}(x_{n-1}))}{x_{n+1} - x_{n-1}} \quad (591)$$

on a sufficiently resolved grid. Define

$$\eta_{G'} = \frac{\|\mathbf{G}'_{\text{analytic}}(x_n) - \mathbf{G}'_{\text{FD}}(x_n)\|_F}{\max[\|\mathbf{G}'_{\text{analytic}}(x_n)\|_F, \varepsilon_{\text{scale}}]}. \quad (592)$$

The finite-difference approximation has its own truncation error, so this check should be repeated under streamwise refinement.

For the factored representation $\mathbf{B} = \mathbf{Q}\mathbf{T}$, the derivative must include both factors:

$$\mathbf{B}' = \mathbf{Q}'\mathbf{T} + \mathbf{Q}\mathbf{T}'. \quad (593)$$

Differentiating only the orthonormal physical basis $\mathbf{Q}$ omits the amplification and mixing carried by $\mathbf{T}$.

10.13 Mixed precision and compensated accumulation

If the OWNS state dimension is large and the transition probability is sensitive to weak directions, higher precision may be needed for selected operations even when the main spatial march is performed in double precision. Candidate operations for increased precision include:

- (i) accumulation of $\mathbf{T}$ over many QR steps;

- (ii) formation of $\mathbf{T}^* \mathbf{H}_Q \mathbf{T}$;
- (iii) evaluation of nearly cancelling directional combinations $\mathbf{T} \mathbf{u}$;
- (iv) eigendecomposition of nearly rank-deficient $\mathbf{G}(x)$; and
- (v) accumulation of very small angular probabilities.

For angular summation,

$$F_K(x) = \sum_{k=1}^K \omega_k \Psi_x(\mathbf{u}_k), \quad (594)$$

pairwise summation or compensated summation can reduce loss of accuracy. For extremely small probabilities, the logarithmic weighted-sum procedure in section 7.9 should be used.

Higher precision cannot recover information already lost because a weak propagated direction was numerically eliminated. Stable orthogonalization and adequate marching accuracy remain necessary.

10.14 Diagnostic invariance tests

The following invariance tests are especially useful for detecting basis-handling errors.

10.14.1 Fixed orthogonal stochastic rotation

Choose a fixed random orthogonal matrix

$$\mathbf{U}^\top \mathbf{U} = \mathbf{I} \quad (595)$$

and replace

$$\mathbf{B}(x) \longrightarrow \tilde{\mathbf{B}}(x) = \mathbf{B}(x) \mathbf{U}. \quad (596)$$

Then

$$\tilde{\mathbf{G}}(x) = \mathbf{U}^\top \mathbf{G}(x) \mathbf{U}. \quad (597)$$

Using a correspondingly rotated angular rule or an independently converged isotropic angular integration, the first-transition CDF must remain unchanged.

10.14.2 Direct and factored energy agreement

At selected stations, compare

$$\mathbf{A}_{\text{direct}} = \Re[\mathbf{B}^* \mathbf{H} \mathbf{B}] \quad (598)$$

with

$$\mathbf{A}_{\text{factored}} = \Re[\mathbf{T}^* \mathbf{Q}^* \mathbf{H} \mathbf{Q} \mathbf{T}]. \quad (599)$$

The relative defect

$$\eta_{QT} = \frac{\|\mathbf{A}_{\text{direct}} - \mathbf{A}_{\text{factored}}\|_F}{\max[\|\mathbf{A}_{\text{direct}}\|_F, \varepsilon_{\text{scale}}]} \quad (600)$$

should be consistent with the factorization and floating-point tolerances.

10.14.3 Explicit and matrix-free directional agreement

For selected directions, compare

$$a_{\mathbf{u}}^{\text{explicit}} = \mathbf{u}^\top \mathbf{G} \mathbf{u} \quad (601)$$

with

$$a_{\mathbf{u}}^{\text{mf}} = \frac{[\mathbf{B}\mathbf{u}]^* \mathbf{H} [\mathbf{B}\mathbf{u}]}{\epsilon_{\text{thres}}}. \quad (602)$$

Their relative difference should remain small over the full streamwise domain:

$$\eta_{\text{gain}} = \frac{|a_{\mathbf{u}}^{\text{explicit}} - a_{\mathbf{u}}^{\text{mf}}|}{\max[a_{\mathbf{u}}^{\text{mf}}, \epsilon_{\text{scale}}]}. \quad (603)$$

10.14.4 Covariance reconstruction

The factored representation must satisfy

$$\mathbf{C}(x, x) = \mathbf{Q}(x) \mathbf{T}(x) \mathbf{T}(x)^* \mathbf{Q}(x)^*. \quad (604)$$

If an independently propagated covariance is available, compare it with this reconstruction. Agreement of the one-point covariance alone does not guarantee correct first-transition history, but disagreement immediately indicates a propagation or factorization error.

10.15 Recommended stabilized OWNS workflow

For a strongly non-normal OWNS calculation, the following workflow is recommended:

1. Construct the initial synthesis factor $\mathbf{B}_0$ using a clearly documented real or proper-complex Gaussian convention.
2. Compute a thin QR factorization $\mathbf{B}_0 = \mathbf{Q}_0 \mathbf{T}_0$.
3. March the orthonormal physical basis $\mathbf{Q}$ over short streamwise intervals using the OWNS solver.
4. Reorthogonalize periodically and accumulate every triangular factor into $\mathbf{T}$.
5. Apply scalar or diagonal scaling to $\mathbf{T}$ when needed, retaining all scale factors explicitly.
6. At each output station, compute $\mathbf{G}(x)$ from the complete factored map $\mathbf{Q}(x) \mathbf{T}(x)$, not from $\mathbf{Q}(x)$ alone.
7. Form the energy matrix through an energy-weighted Gram product whenever possible.
8. Symmetrize $\mathbf{G}$, monitor its symmetry and positive-semidefinite defects, and correct only roundoff-scale violations.
9. Compare explicit and matrix-free directional gains at representative stations and directions.
10. Verify analytical streamwise derivatives against refined finite differences before using Hermite maximum detection.
11. Repeat the OWNS march with tighter tolerances or a finer streamwise discretization and check convergence of both $\mathbf{G}(x)$ and the first-transition CDF.

12. If the propagated stochastic rank becomes numerically excessive, apply a fixed transition-oriented reduction rather than independently truncating the propagated factor at each station.

The central principle is that numerical stabilization may change the representation of the propagated basis, but not the underlying map from the original Gaussian realization to the physical disturbance. Maintaining that map is necessary for preserving the streamwise memory on which the first-transition method depends.

11 Relation to Absorbing Liouville and First-Passage Methods

The radial-directional method can be interpreted as a specialized solution of an absorbing probability-transport problem. In the physical OWNS state space, probability density is transported by the spatial dynamics and removed when it first reaches the transition boundary. In the common initial stochastic coordinates, the Gaussian density remains stationary while the transition boundary moves and deforms. The latter representation is especially advantageous for the present linear, random-initial-condition problem.

This section establishes the relation among:

- (i) the absorbing Liouville equation in the propagated physical state;
- (ii) the moving safe-set formulation in the initial Gaussian coordinates;
- (iii) the radial-directional probability formula; and
- (iv) Rice-type level-crossing formulas.

The formulations describe the same first-transition event under their respective regularity assumptions, but they differ substantially in their numerical cost and in how streamwise memory is represented.

11.1 Transition boundary in physical state space

Let the deterministic spatial dynamics be

$$\frac{d\mathbf{q}}{dx} = \mathbf{L}(x)\mathbf{q}(x), \quad (605)$$

and define the transition limit-state function

$$h(\mathbf{q}, x) = \mathbf{q}^* \mathbf{H}(x) \mathbf{q} - e_{\text{thres}}(x). \quad (606)$$

The instantaneous safe domain is

$$\mathcal{D}(x) = \{\mathbf{q} : h(\mathbf{q}, x) < 0\}, \quad (607)$$

and its boundary is

$$\partial\mathcal{D}(x) = \{\mathbf{q} : h(\mathbf{q}, x) = 0\}. \quad (608)$$

The threshold-exceeded region is

$$\mathcal{D}_{\text{tr}}(x) = \{\mathbf{q} : h(\mathbf{q}, x) \geq 0\}. \quad (609)$$

The transition boundary can vary with x through both the energy metric and the threshold. Its state-space gradient is

$$\nabla_{\mathbf{q}} h(\mathbf{q}, x) = 2\mathbf{H}(x)\mathbf{q} \quad (610)$$

for a real symmetric representation. In a realification of a complex Fourier state, the same expression applies using $\mathbf{H}_R$ and $\mathbf{q}_R$.

Along an OWNS trajectory,

$$\begin{aligned} \dot{h}(\mathbf{q}, x) &= \frac{d}{dx} h[\mathbf{q}(x), x] \\ &= \frac{\partial h}{\partial x} + \nabla_{\mathbf{q}} h^\top \mathbf{L}(x) \mathbf{q}. \end{aligned} \quad (611)$$

For the quadratic energy,

$$\dot{h}(\mathbf{q}, x) = \mathbf{q}^* [\mathbf{L}(x)^* \mathbf{H}(x) + \mathbf{H}(x) \mathbf{L}(x) + \mathbf{H}'(x)] \mathbf{q} - e'_{\text{thres}}(x). \quad (612)$$

At a regular point of the boundary, an outward crossing from the safe domain to the transitioned domain satisfies

$$h(\mathbf{q}, x) = 0, \quad \dot{h}(\mathbf{q}, x) > 0. \quad (613)$$

The quantity $\dot{h}$ includes both the velocity of the state relative to fixed coordinates and the motion of the boundary itself. It is therefore the appropriate crossing speed for a spatially varying energy metric or threshold.

11.2 Unconditional Liouville equation

Suppose initially that the random state has a density $p_0(\mathbf{q})$ with respect to volume measure in the physical state space. Under the deterministic dynamics in eq. (605), its unconditional density $p(\mathbf{q}, x)$ satisfies the Liouville equation

$$\frac{\partial p}{\partial x} + \nabla_{\mathbf{q}} \cdot [\mathbf{L}(x) \mathbf{q} p] = 0. \quad (614)$$

Expanding the divergence gives

$$\frac{\partial p}{\partial x} + [\mathbf{L}(x) \mathbf{q}]^\top \nabla_{\mathbf{q}} p + \text{tr}[\mathbf{L}(x)] p = 0 \quad (615)$$

for a real state-space representation.

Along a characteristic satisfying

$$\frac{d\mathbf{q}}{dx} = \mathbf{L}(x) \mathbf{q}, \quad (616)$$

the density evolves according to

$$\frac{dp}{dx} = -\text{tr}[\mathbf{L}(x)] p. \quad (617)$$

For an invertible propagator,

$$p(\mathbf{q}, x) = \frac{p_0[\mathbf{M}(0, x)^{-1} \mathbf{q}]}{|\det \mathbf{M}(0, x)|}. \quad (618)$$

The unconditional Liouville equation does not retain transition history. A trajectory that exits the safe domain and later re-enters continues to contribute to $p(\mathbf{q}, x)$. Integrating the unconditional density over the current safe domain gives only the local probability

$$\int_{\mathcal{D}(x)} p(\mathbf{q}, x) d\mathbf{q} = \mathbb{P}[e(x) < e_{\text{thres}}(x)], \quad (619)$$

not the survival probability $\mathbb{P}[X_{\text{tr}} > x]$.

11.3 Absorbing survivor density

Let

$$\rho(\mathbf{q}, x) \quad (620)$$

denote the unnormalized density of trajectories that have remained inside the safe domain for their complete upstream history. In the interior of the safe domain, it satisfies

$$\boxed{\frac{\partial \rho}{\partial x} + \nabla_{\mathbf{q}} \cdot [\mathbf{L}(x) \mathbf{q} \rho] = 0, \quad \mathbf{q} \in \mathcal{D}(x).} \quad (621)$$

Its initial condition is

$$\rho(\mathbf{q}, 0) = p_0(\mathbf{q}) \mathbf{1}_{\mathcal{D}(0)}(\mathbf{q}). \quad (622)$$

If threshold exceedance at the inlet counts as immediate transition, the excluded initial mass is

$$p_0^{\text{tr}} = 1 - \int_{\mathcal{D}(0)} p_0(\mathbf{q}) \, d\mathbf{q}. \quad (623)$$

The survival probability is the total remaining mass:

$$\boxed{S(x) = \int_{\mathcal{D}(x)} \rho(\mathbf{q}, x) \, d\mathbf{q}.} \quad (624)$$

The survivor density is intentionally not normalized. Its loss of mass is the cumulative probability of first transition:

$$F_{X_{\text{tr}}}(x) = 1 - S(x). \quad (625)$$

The normalized conditional density of the untransitioned population is

$$p_{\text{surv}}(\mathbf{q}, x) = p(\mathbf{q}, x \mid X_{\text{tr}} > x) = \frac{\rho(\mathbf{q}, x)}{S(x)}, \quad (626)$$

provided $S(x) > 0$. Even though the unconditional state remains Gaussian under linear dynamics, the survivor density is generally non-Gaussian because threshold-crossing trajectories have been selectively removed.

11.4 Absorbing boundary condition for deterministic transport

For a diffusion equation, an absorbing boundary is commonly imposed through a homogeneous Dirichlet condition. The Liouville equation is instead a first-order hyperbolic transport equation. Imposing

$$\rho = 0 \quad \text{on all of } \partial\mathcal{D}(x) \quad (627)$$

can overconstrain the deterministic transport problem.

The physically relevant rule is:

- (i) probability may leave the safe domain through an outward boundary crossing;
- (ii) the departing probability is permanently removed; and
- (iii) no probability is allowed to enter the safe domain from the transitioned side.

Define the outward-crossing portion of the boundary by

$$\partial\mathcal{D}_+(x) = \left\{ \mathbf{q} \in \partial\mathcal{D}(x) : \dot{h}(\mathbf{q}, x) > 0 \right\}, \quad (628)$$

and the inward-moving or re-entry portion by

$$\partial\mathcal{D}_-(x) = \left\{ \mathbf{q} \in \partial\mathcal{D}(x) : \dot{h}(\mathbf{q}, x) < 0 \right\}. \quad (629)$$

The survivor density has an outgoing trace on $\partial\mathcal{D}_+$, which determines the lost probability flux. On $\partial\mathcal{D}_-$, the incoming density from the transitioned region is set to zero. This prevents a previously removed trajectory from re-entering the survivor population.

In a characteristic interpretation, the boundary rule is especially simple: propagate each density element until the first station at which $h \geq 0$, then terminate that characteristic. This is the deterministic analogue of a killed stochastic process.

11.5 First-transition density as boundary flux

Under sufficient regularity, the loss of survivor mass is the flux through the outward transition boundary. The first-transition-location density is

$$f_{X_{\text{tr}}}(x) = -\frac{dS}{dx}. \quad (630)$$

Using the limit-state function, it can be written as

$$f_{X_{\text{tr}}}(x) = \int_{\partial\mathcal{D}(x)} \rho(\mathbf{q}, x) \frac{[\dot{h}(\mathbf{q}, x)]_+}{\|\nabla_{\mathbf{q}} h(\mathbf{q}, x)\|_2} dS_{\mathbf{q}}, \quad (631)$$

where

$$[a]_+ = \max(a, 0) \quad (632)$$

and $dS_{\mathbf{q}}$ is surface measure on $\partial\mathcal{D}(x)$.

Equivalently, using the Dirac delta distribution,

$$f_{X_{\text{tr}}}(x) = \int_{\mathbb{R}^{n_q}} \rho(\mathbf{q}, x) \delta[h(\mathbf{q}, x)] [\dot{h}(\mathbf{q}, x)]_+ d\mathbf{q}. \quad (633)$$

The equivalence follows from the coarea identity

$$\int f(\mathbf{q}) \delta[h(\mathbf{q}, x)] d\mathbf{q} = \int_{h(\mathbf{q}, x)=0} \frac{f(\mathbf{q})}{\|\nabla_{\mathbf{q}} h(\mathbf{q}, x)\|_2} dS_{\mathbf{q}}. \quad (634)$$

The flux formula counts first crossings because ρ contains only surviving trajectories. A trajectory removed at an earlier station contributes no density to a later boundary crossing.

Tangential contacts with

$$h = 0, \quad \dot{h} = 0 \quad (635)$$

do not contribute to the regular flux formula. If tangential threshold attainment is assigned transition and occurs with nonzero probability, it must be treated separately. For a smooth, nondegenerate Gaussian model, such events are typically of probability zero.

11.6 Transfer-operator formulation

The absorbing Liouville problem can also be written using Perron–Frobenius transfer operators. Let

$$\mathcal{P}_{n+1,n} \quad (636)$$

propagate a density from x_n to x_{n+1} . For an invertible linear map

$$\mathbf{q}_{n+1} = \mathbf{M}_{n+1,n} \mathbf{q}_n, \quad (637)$$

the operator acts as

$$(\mathcal{P}_{n+1,n} \rho_n)(\mathbf{q}) = \frac{\rho_n(\mathbf{M}_{n+1,n}^{-1} \mathbf{q})}{|\det \mathbf{M}_{n+1,n}|}. \quad (638)$$

Define the safe-domain restriction operator

$$(\mathcal{R}_n f)(\mathbf{q}) = \mathbf{1}_{\mathcal{D}(x_n)}(\mathbf{q}) f(\mathbf{q}). \quad (639)$$

A stationwise absorbing update is

$$\boxed{\rho_{n+1} = \mathcal{R}_{n+1} \mathcal{P}_{n+1,n} \rho_n}. \quad (640)$$

Repeated application gives

$$\rho_n = \mathcal{R}_n \mathcal{P}_{n,n-1} \cdots \mathcal{R}_1 \mathcal{P}_{1,0} \mathcal{R}_0 \rho_0. \quad (641)$$

The operator product is substochastic because each restriction can remove probability mass:

$$\int \rho_{n+1}(\mathbf{q}) \, d\mathbf{q} \leq \int \rho_n(\mathbf{q}) \, d\mathbf{q}. \quad (642)$$

The lost mass over a step is

$$p_{n+1} = \int \rho_n(\mathbf{q}) \, d\mathbf{q} - \int \rho_{n+1}(\mathbf{q}) \, d\mathbf{q}. \quad (643)$$

Checking the safe domain only at the endpoint can miss a crossing and recrossing inside the interval. A continuous-step restriction must instead remove all characteristics for which

$$\max_{x_n \leq x \leq x_{n+1}} h[\mathbf{q}(x), x] \geq 0. \quad (644)$$

This is the transfer-operator analogue of the interval maximum detection developed in ??.

11.7 Low-rank initial distributions in physical state space

A direct physical-state density formulation requires special care when the initial stochastic rank is smaller than the OWNS state dimension:

$$r < n_q. \quad (645)$$

In that case,

$$\mathbf{q}(0) = \mathbf{B}_0 \mathbf{w} \quad (646)$$

is supported on the r -dimensional range of $\mathbf{B}_0$. It does not possess a nonsingular density with respect to n_q -dimensional volume measure. Under deterministic linear propagation, the distribution remains supported on

$$\text{range}[\mathbf{B}(x)]. \quad (647)$$

The Liouville equation must then be interpreted as transport of a measure concentrated on a lower-dimensional evolving subspace, or be written in intrinsic reduced coordinates. A full-state finite-volume density solver would waste most of its computational domain and may not represent the singular measure correctly.

This issue provides an additional reason to work in the initial stochastic coordinates. The Gaussian measure in $\mathbf{w}$ -space is full rank by construction, even when the physical-state covariance is low rank.

11.8 Moving-boundary formulation in the initial coordinates

Substitute

$$\mathbf{q}(x) = \mathbf{B}(x)\mathbf{w} \quad (648)$$

into the physical limit-state function:

$$\begin{aligned} h[\mathbf{B}(x)\mathbf{w}, x] &= \mathbf{w}^\top \mathbf{A}(x)\mathbf{w} - e_{\text{thres}}(x) \\ &= e_{\text{thres}}(x) \left[\mathbf{w}^\top \mathbf{G}(x)\mathbf{w} - 1 \right]. \end{aligned} \quad (649)$$

Define the normalized limit-state function

$$g(\mathbf{w}, x) = \mathbf{w}^\top \mathbf{G}(x)\mathbf{w} - 1. \quad (650)$$

The instantaneous safe set is

$$\mathcal{E}(x) = \{\mathbf{w} : g(\mathbf{w}, x) < 0\}. \quad (651)$$

The initial coordinate does not evolve:

$$\frac{d\mathbf{w}}{dx} = \mathbf{0}. \quad (652)$$

Therefore, the standard Gaussian density

$$\phi_r(\mathbf{w}) = \frac{1}{(2\pi)^{r/2}} \exp\left(-\frac{1}{2}\|\mathbf{w}\|_2^2\right) \quad (653)$$

is stationary. All streamwise evolution appears in the moving quadratic boundary

$$g(\mathbf{w}, x) = 0. \quad (654)$$

The survivor set is

$$\mathcal{S}(x) = \bigcap_{0 \leq \xi \leq x} \{\mathbf{w} : g(\mathbf{w}, \xi) < 0\}. \quad (655)$$

The survival probability is

$$S(x) = \int_{\mathcal{S}(x)} \phi_r(\mathbf{w}) d\mathbf{w}. \quad (656)$$

The equivalent unnormalized survivor measure in $\mathbf{w}$ -space has density

$$\rho_w(\mathbf{w}, x) = \phi_r(\mathbf{w}) \mathbf{1}_{\mathcal{S}(x)}(\mathbf{w}). \quad (657)$$

Unlike the physical-state Liouville density, this density does not need to be advected. Its support simply shrinks as new portions of the initial-coordinate space are excluded.

The streamwise boundary speed in these coordinates is

$$\frac{\partial g}{\partial x} = \mathbf{w}^\top \mathbf{G}'(x) \mathbf{w}. \quad (658)$$

A first outward crossing satisfies

$$g(\mathbf{w}, x) = 0, \quad \frac{\partial g}{\partial x} > 0, \quad g(\mathbf{w}, \xi) < 0 \text{ for all } \xi < x. \quad (659)$$

11.9 Equivalence with the radial–directional method

Write

$$\mathbf{w} = R\mathbf{u}. \quad (660)$$

Along direction $\mathbf{u}$, the moving-boundary condition is

$$R^2 \mathbf{u}^\top \mathbf{G}(x) \mathbf{u} = 1. \quad (661)$$

Survival through x requires

$$R^2 \max_{0 \leq \xi \leq x} \mathbf{u}^\top \mathbf{G}(\xi) \mathbf{u} < 1. \quad (662)$$

Thus, the survivor-set radial boundary is

$$R < \frac{1}{\sqrt{m_{\mathbf{u}}(x)}}. \quad (663)$$

Integrating the stationary Gaussian measure radially gives

$$\begin{aligned} S(x) &= \mathbb{E}_{\mathbf{u}} \left[\mathbb{P} \left(R^2 < \frac{1}{m_{\mathbf{u}}(x)} \mid \mathbf{u} \right) \right] \\ &= \mathbb{E}_{\mathbf{u}} \left[F_{\chi_r^2} \left(\frac{1}{m_{\mathbf{u}}(x)} \right) \right]. \end{aligned} \quad (664)$$

Consequently,

$$F_{X_{\text{tr}}}(x) = \mathbb{E}_{\mathbf{u}} \left[\bar{F}_{\chi_r^2} \left(\frac{1}{m_{\mathbf{u}}(x)} \right) \right]. \quad (665)$$

The radial–directional method can therefore be viewed as an exact integration of the absorbing moving-boundary problem along every ray in the initial Gaussian coordinate space. It avoids:

- (i) discretizing the high-dimensional physical state space;
- (ii) transporting an increasingly anisotropic density;
- (iii) representing a potentially singular low-rank physical-state measure; and
- (iv) explicitly applying absorbing boundary conditions to a hyperbolic PDE.

The only remaining numerical approximations are associated with stochastic dimension reduction, angular integration, streamwise maximum detection, and the underlying OWNS propagation.

11.10 Exact first-crossing flux in the initial coordinates

The first-transition density can be represented directly in the initial Gaussian coordinates. Define the upstream-survival indicator immediately before x by

$$\mathcal{I}_{\text{surv}}(\mathbf{w}, x^-) = \mathbf{1} \{g(\mathbf{w}, \xi) < 0 \text{ for every } 0 \leq \xi < x\}. \quad (666)$$

Under regular-crossing assumptions,

$$f_{X_{\text{tr}}}(x) = \int_{\mathbb{R}^r} \phi_r(\mathbf{w}) \mathcal{I}_{\text{surv}}(\mathbf{w}, x^-) \delta[g(\mathbf{w}, x)] \left[\frac{\partial g}{\partial x}(\mathbf{w}, x) \right]_+ d\mathbf{w}. \quad (667)$$

Using the quadratic form,

$$f_{X_{\text{tr}}}(x) = \int_{\mathbb{R}^r} \phi_r(\mathbf{w}) \mathcal{I}_{\text{surv}}(\mathbf{w}, x^-) \delta[\mathbf{w}^\top \mathbf{G}(x) \mathbf{w} - 1] [\mathbf{w}^\top \mathbf{G}'(x) \mathbf{w}]_+ d\mathbf{w}. \quad (668)$$

The upstream-survival indicator is the term that gives this formula memory. Without it, the integral counts every positive crossing at x , including recrossings by trajectories that transitioned earlier.

Equation (668) is equivalent to differentiating the radial-directional CDF where the required regularity conditions hold. The running-maximum formulation is generally more convenient numerically because it incorporates the survival indicator implicitly.

11.11 Relation to Rice upcrossing theory

For the normalized energy process

$$y(x) = \mathbf{w}^\top \mathbf{G}(x) \mathbf{w}, \quad (669)$$

the derivative is

$$y'(x) = \mathbf{w}^\top \mathbf{G}'(x) \mathbf{w}. \quad (670)$$

A standard Rice formula gives the expected positive-crossing density of level one:

$$\nu_+(x) = \mathbb{E} [\delta[y(x) - 1] [y'(x)]_+]. \quad (671)$$

If the joint density of $y(x)$ and $y'(x)$ exists, this becomes

$$\nu_+(x) = \int_0^\infty v f_{y(x), y'(x)}(1, v) dv. \quad (672)$$

The expected number of positive crossings over $[0, x]$ is

$$\mathbb{E}[N_+(0, x)] = \int_0^x \nu_+(\xi) d\xi. \quad (673)$$

In the present quadratic-Gaussian problem,

$$y(x) = \mathbf{w}^\top \mathbf{G}(x) \mathbf{w}, \quad y'(x) = \mathbf{w}^\top \mathbf{G}'(x) \mathbf{w}. \quad (674)$$

Their joint characteristic function is available analytically. For real $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_r)$,

$$\varphi_{y, y'}(s, t) = \mathbb{E} [\exp (i s y + i t y')] = \det [\mathbf{I} - 2i (s \mathbf{G} + t \mathbf{G}')]^{-1/2}. \quad (675)$$

The joint density can, in principle, be recovered through a two-dimensional Fourier inversion and used in eq. (672).

The ordinary Rice intensity does not distinguish first upcrossings from later upcrossings:

$$\nu_+(x) = \int_{\mathbb{R}^r} \phi_r(\mathbf{w}) \delta[g(\mathbf{w}, x)] \left[\frac{\partial g}{\partial x} \right]_+ d\mathbf{w}. \quad (676)$$

Comparing with eq. (667), the difference is precisely the upstream-survival indicator:

$$f_{X_{\text{tr}}}(x) = \mathbb{E} [\delta[y(x) - 1][y'(x)]_+ \mathbf{1} \{y(\xi) < 1 \text{ for every } \xi < x\}] , \quad (677)$$

$$\nu_+(x) = \mathbb{E} [\delta[y(x) - 1][y'(x)]_+]. \quad (678)$$

Therefore,

$$f_{X_{\text{tr}}}(x) \leq \nu_+(x). \quad (679)$$

If trajectories cross the threshold at most once in the positive direction, then

$$f_{X_{\text{tr}}}(x) = \nu_+(x) \quad (680)$$

apart from any inlet mass. In a strongly non-normal system with amplification, decay, and later reamplification, multiple crossings may occur and the ordinary Rice intensity can overcount transition.

11.12 Bounds and approximations from upcrossings

Let $N_+(0, x)$ denote the number of positive crossings after the inlet. If transition at $x = 0$ is treated separately, then

$$\{X_{\text{tr}} \leq x\} \subseteq \{y(0) \geq 1\} \cup \{N_+(0, x) \geq 1\}. \quad (681)$$

Using the union bound and Markov's inequality,

$$\begin{aligned} F_{X_{\text{tr}}}(x) &\leq \mathbb{P}[y(0) \geq 1] + \mathbb{P}[N_+(0, x) \geq 1] \\ &\leq \mathbb{P}[y(0) \geq 1] + \mathbb{E}[N_+(0, x)] \\ &= \mathbb{P}[y(0) \geq 1] + \int_0^x \nu_+(\xi) d\xi. \end{aligned} \quad (682)$$

The right-hand side should be truncated at one.

If positive crossings are rare and approximately independent, a Poisson approximation gives

$$S(x) \approx \mathbb{P}[y(0) < 1] \exp \left[- \int_0^x \nu_+(\xi) d\xi \right]. \quad (683)$$

This approximation can be inaccurate when crossings cluster or when a trajectory repeatedly crosses the threshold near a broad energy peak. The exact radial running-maximum formulation does not require a Poisson assumption and is preferred when computationally feasible.

Higher-order Rice formulas can account for correlations among multiple crossings through factorial moments of N_+ . They provide useful analytical approximations and diagnostics but are generally more complex than directly evaluating the running maximum in the finite initial stochastic space.

11.13 Relation to survival analysis

The absorbing formulation naturally defines a hazard rate:

$$\lambda(x) = \frac{f_{X_{\text{tr}}}(x)}{S(x)}. \quad (684)$$

Using the survivor boundary flux,

$$\lambda(x) = \frac{\int_{\partial\mathcal{D}(x)} \rho(\mathbf{q}, x) \frac{[\dot{h}(\mathbf{q}, x)]_+}{\|\nabla_{\mathbf{q}} h(\mathbf{q}, x)\|_2} dS_{\mathbf{q}}}{\int_{\mathcal{D}(x)} \rho(\mathbf{q}, x) d\mathbf{q}}. \quad (685)$$

Thus, the hazard is the outward first-crossing flux divided by the remaining survivor mass.

In the initial-coordinate representation,

$$\lambda(x) = \frac{\int \phi_r(\mathbf{w}) \mathcal{I}_{\text{surv}}(\mathbf{w}, x^-) \delta[g(\mathbf{w}, x)] [\partial_x g(\mathbf{w}, x)]_+ d\mathbf{w}}{\int_{\mathcal{S}(x)} \phi_r(\mathbf{w}) d\mathbf{w}}. \quad (686)$$

This expression shows explicitly that the hazard is conditional on having survived all previous stations. It is not equal in general to a local threshold-exceedance probability or an unconditional Rice upcrossing rate.

11.14 When a direct Liouville method is advantageous

For the present linear OWNS problem with Gaussian initial uncertainty and a quadratic transition criterion, the moving-boundary radial-directional formulation is generally more efficient than a direct Liouville solve. A direct absorbing-density method becomes more attractive when:

- (i) a very low-dimensional physical or reduced state is available;
- (ii) the dynamics are nonlinear, so that $\mathbf{q}(x)$ is no longer a linear function of a fixed Gaussian coordinate;
- (iii) stochastic forcing is injected continuously downstream;
- (iv) the transition criterion is nonquadratic or depends on additional state variables;
- (v) the complete conditional survivor density is required; or
- (vi) post-transition and relaminarization states are represented explicitly by a multistate model.

For nonlinear deterministic dynamics,

$$\frac{d\mathbf{q}}{dx} = \mathbf{f}(\mathbf{q}, x), \quad (687)$$

the survivor density satisfies

$$\frac{\partial \rho}{\partial x} + \nabla_{\mathbf{q}} \cdot [\mathbf{f}(\mathbf{q}, x) \rho] = 0 \quad (688)$$

inside the safe domain, with the same killing rule at first exit. The safe set in the initial coordinates is no longer generally an intersection of ellipsoids, and the Gaussian radial coordinate cannot usually be integrated analytically.

If distributed stochastic forcing is added,

$$d\mathbf{q} = \mathbf{L}(x)\mathbf{q}dx + \mathbf{K}_f(x)d\mathbf{W}_x, \quad (689)$$

the density satisfies an absorbing Fokker–Planck equation:

$$\frac{\partial \rho}{\partial x} = -\nabla_{\mathbf{q}} \cdot [\mathbf{L}(x)\mathbf{q}\rho] + \frac{1}{2}\nabla_{\mathbf{q}}\nabla_{\mathbf{q}} : [\mathbf{D}_f(x)\rho], \quad (690)$$

where

$$\mathbf{D}_f(x) = \mathbf{K}_f(x)\mathbf{K}_f(x)^\top. \quad (691)$$

In that case, randomness is not confined to a fixed initial coordinate, and the radial–directional formula derived for the present problem no longer applies directly.

11.15 Interpretation for the OWNS transition problem

For the linear OWNS model considered here, the following statements summarize the relationship among the methods:

1. The unconditional propagated Gaussian distribution determines local energy probabilities but does not retain irreversible transition classification.
2. An absorbing Liouville equation retains memory by removing probability at its first outward crossing of the energy boundary.
3. Strong non-normality does not invalidate the absorbing formulation, but it can make a physical-state density highly anisotropic and numerically expensive to resolve.
4. If the initial covariance is low rank, the physical-state distribution may be singular with respect to full state-space volume, further complicating a direct density PDE.
5. The common initial coordinate $\mathbf{w}$ transforms the problem into a stationary Gaussian measure with a moving family of quadratic safe boundaries.
6. The survivor set is the intersection of all upstream safe ellipsoids in $\mathbf{w}$ -space.
7. Radial integration evaluates the Gaussian measure removed by this absorbing boundary exactly along each initial direction.
8. The directional running maximum is the geometric equivalent of irreversible probability killing.
9. Ordinary Rice theory counts all positive threshold crossings; the first-transition density additionally requires conditioning on upstream survival.
10. The radial–directional method is therefore an exact first-passage calculation within the finite-dimensional linear Gaussian model, rather than a stationwise exceedance model or an independence approximation.

12 Verification and Validation

Verification establishes that the numerical implementation correctly evaluates the mathematical first-transition model. Validation assesses whether the OWNS propagation, disturbance statistics, energy norm, and transition threshold adequately represent the physical transition problem. These are distinct tasks:

- *Numerical verification* concerns implementation, discretization, and convergence errors.
- *Model validation* concerns agreement with independent numerical, experimental, or observational transition data.

The method contains several separable sources of numerical error:

- (i) OWNS spatial-propagation and wall-normal discretization error;
- (ii) stochastic-rank truncation error;
- (iii) streamwise maximum-detection error;
- (iv) angular-integration error;
- (v) floating-point and factor-conditioning error; and
- (vi) interpolation or differentiation error in reported transition densities, hazards, and quantiles.

These errors should be studied separately whenever possible. A single comparison with trajectory Monte Carlo is useful but does not by itself identify which component is responsible for a discrepancy.

12.1 Internal consistency checks

The first set of tests follows directly from the mathematical structure of the method and should be applied to every calculation.

12.1.1 Properties of the normalized energy matrices

At every station, the normalized energy matrix should be real symmetric and positive semidefinite:

$$\mathbf{G}(x_n) = \mathbf{G}(x_n)^\top, \quad \mathbf{G}(x_n) \succeq \mathbf{0}. \quad (692)$$

Monitor the relative symmetry defect

$$\eta_{\text{sym}}(x_n) = \frac{\|\mathbf{G}(x_n) - \mathbf{G}(x_n)^\top\|_F}{\max[\|\mathbf{G}(x_n)\|_F, \varepsilon_{\text{scale}}]} \quad (693)$$

and the relative positive-semidefinite defect

$$\eta_{\text{psd}}(x_n) = \frac{\max[-\lambda_{\min}(\mathbf{G}(x_n)), 0]}{\max[\lambda_{\max}(\mathbf{G}(x_n)), \varepsilon_{\text{scale}}]}. \quad (694)$$

Values substantially larger than the expected propagation and floating-point errors indicate an inconsistency in the energy metric, complex-to-real convention, or factor propagation.

For selected directions, compare the explicit quadratic form

$$a_{\mathbf{u}}^{\text{explicit}}(x_n) = \mathbf{u}^\top \mathbf{G}(x_n) \mathbf{u} \quad (695)$$

with the physical-state evaluation

$$a_{\mathbf{u}}^{\text{state}}(x_n) = \frac{[\mathbf{B}(x_n)\mathbf{u}]^* \mathbf{H}(x_n) [\mathbf{B}(x_n)\mathbf{u}]}{e_{\text{thres}}(x_n)}. \quad (696)$$

Their relative difference,

$$\eta_a(\mathbf{u}, x_n) = \frac{|a_{\mathbf{u}}^{\text{explicit}}(x_n) - a_{\mathbf{u}}^{\text{state}}(x_n)|}{\max[a_{\mathbf{u}}^{\text{state}}(x_n), \varepsilon_{\text{scale}}]}, \quad (697)$$

should be consistent with the numerical precision.

12.1.2 Monotonicity and probability conservation

For every angular direction, the running maximum must satisfy

$$m_{k,n+1} \geq m_{k,n}. \quad (698)$$

Consequently,

$$S(x_{n+1}) \leq S(x_n), \quad (699)$$

$$F_{X_{\text{tr}}}(x_{n+1}) \geq F_{X_{\text{tr}}}(x_n). \quad (700)$$

The probability values must satisfy

$$0 \leq S(x_n) \leq 1, \quad 0 \leq F_{X_{\text{tr}}}(x_n) \leq 1, \quad (701)$$

and

$$S(x_n) + F_{X_{\text{tr}}}(x_n) = 1. \quad (702)$$

The interval probabilities

$$p_n = F_{X_{\text{tr}}}(x_n) - F_{X_{\text{tr}}}(x_{n-1}) \quad (703)$$

must be nonnegative, and total probability must be conserved:

$$p_0 + \sum_{n=1}^N p_n + p_{\text{censored}} = 1. \quad (704)$$

Material violations of monotonicity or probability conservation are implementation errors rather than properties of the underlying model.

12.1.3 Relation to the local exceedance probability

At every station,

$$p_{\text{local}}(x_n) = \mathbb{P} \left[\mathbf{w}^\top \mathbf{G}(x_n) \mathbf{w} \geq 1 \right] \quad (705)$$

must satisfy

$$\boxed{p_{\text{local}}(x_n) \leq F_{X_{\text{tr}}}(x_n)}. \quad (706)$$

The local probability can be evaluated independently using a generalized chi-square algorithm based on the eigenvalues of $\mathbf{G}(x_n)$. Failure of eq. (706) beyond numerical tolerance indicates an error in the running maximum, angular integration, or local quadratic-form calculation.

At the inlet, no upstream history exists, so

$$F_{X_{\text{tr}}}(0) = p_{\text{local}}(0). \quad (707)$$

More generally, if

$$\mathbf{G}(x_2) - \mathbf{G}(x_1) \succeq \mathbf{0} \quad \text{for every } x_2 \geq x_1, \quad (708)$$

then every normalized energy realization is nondecreasing and

$$F_{X_{\text{tr}}}(x) = p_{\text{local}}(x). \quad (709)$$

This provides a useful special-case test.

12.1.4 Basis invariance

Let

$$\mathbf{U}^\top \mathbf{U} = \mathbf{I}_r \quad (710)$$

be a fixed orthogonal matrix. Transform

$$\tilde{\mathbf{B}}(x) = \mathbf{B}(x)\mathbf{U}, \quad (711)$$

$$\tilde{\mathbf{G}}(x) = \mathbf{U}^\top \mathbf{G}(x)\mathbf{U}. \quad (712)$$

The exact first-transition distribution is invariant under this transformation:

$$\tilde{F}_{X_{\text{tr}}}(x) = F_{X_{\text{tr}}}(x). \quad (713)$$

For a finite deterministic angular rule, the two numerical values may differ because the rotated directions are not necessarily members of the original quadrature rule. Two valid tests are:

- (i) rotate the angular points consistently according to

$$\tilde{\mathbf{u}}_k = \mathbf{U}^\top \mathbf{u}_k, \quad (714)$$

in which case the directional gains should agree pointwise; or

- (ii) independently converge the angular integration for the original and transformed matrices and compare the resulting CDFs.

The first test is exact up to floating-point error and directly checks the coordinate transformations.

12.1.5 Scaling checks

If the propagated factor is scaled by a constant c ,

$$\tilde{\mathbf{B}}(x) = c\mathbf{B}(x), \quad (715)$$

then

$$\tilde{\mathbf{G}}(x) = c^2 \mathbf{G}(x). \quad (716)$$

The same result is obtained by leaving $\mathbf{B}$ unchanged and scaling the threshold according to

$$\tilde{e}_{\text{thres}}(x) = \frac{e_{\text{thres}}(x)}{c^2}. \quad (717)$$

These equivalent calculations should produce identical transition distributions.

If both the energy matrix and threshold are multiplied by the same positive scalar,

$$\widetilde{\mathbf{H}}(x) = c\mathbf{H}(x), \quad \tilde{e}_{\text{thres}}(x) = ce_{\text{thres}}(x), \quad (718)$$

then $\mathbf{G}(x)$ and the entire first-transition distribution must remain unchanged.

12.2 Analytical and semi-analytical test cases

A sequence of low-dimensional tests should be completed before applying the algorithm to the full OWNS problem. These tests isolate the probability and first-passage calculations from the spatial solver.

12.2.1 Scalar stochastic coordinate

Let $r = 1$, so that

$$y(x) = w^2 g(x), \quad w \sim \mathcal{N}(0, 1), \quad g(x) \geq 0. \quad (719)$$

Define

$$m(x) = \max_{0 \leq \xi \leq x} g(\xi). \quad (720)$$

The exact first-transition CDF is

$$F_{X_{\text{tr}}}(x) = \bar{F}_{\chi_1^2} \left[\frac{1}{m(x)} \right]. \quad (721)$$

No angular integration is required. This case tests the chi-square tail, running maximum, interval probabilities, and censoring logic.

Choose a nonmonotone gain, for example,

$$g(x) = g_0 + A \exp \left[-\frac{(x - x_p)^2}{2\sigma_x^2} \right], \quad (722)$$

to test whether a narrow transient peak is resolved. The exact maximum is available analytically when the peak lies in the interval.

12.2.2 Isotropic quadratic form

Let

$$\mathbf{G}(x) = g(x)\mathbf{I}_r. \quad (723)$$

Then every direction has the same gain:

$$a_{\mathbf{u}}(x) = g(x), \quad m_{\mathbf{u}}(x) = m(x). \quad (724)$$

The exact transition CDF is

$$F_{X_{\text{tr}}}(x) = \bar{F}_{\chi_r^2} \left[\frac{1}{m(x)} \right]. \quad (725)$$

Every valid angular rule should reproduce this result exactly up to roundoff because the integrand is constant over the sphere.

12.2.3 Separable matrix family

Let

$$\mathbf{G}(x) = g(x)\mathbf{G}_0, \quad g(x) \geq 0, \quad \mathbf{G}_0 \succeq \mathbf{0}. \quad (726)$$

Then

$$m_{\mathbf{u}}(x) = m_g(x)\mathbf{u}^\top \mathbf{G}_0 \mathbf{u}, \quad m_g(x) = \max_{\xi \leq x} g(\xi). \quad (727)$$

The first-transition event becomes

$$m_g(x)\mathbf{w}^\top \mathbf{G}_0 \mathbf{w} \geq 1. \quad (728)$$

Therefore,

$$F_{X_{\text{tr}}}(x) = \mathbb{P} \left[\mathbf{w}^\top \mathbf{G}_0 \mathbf{w} \geq \frac{1}{m_g(x)} \right]. \quad (729)$$

The right-hand side is a one-point generalized chi-square probability that can be evaluated independently using the eigenvalues of $\mathbf{G}_0$. This test compares the angular method with a completely different numerical evaluation.

12.2.4 Rank-one matrix family with fixed direction

Let

$$\mathbf{G}(x) = g(x)\mathbf{v}\mathbf{v}^\top, \quad \|\mathbf{v}\|_2 = 1. \quad (730)$$

Then

$$y(x) = g(x)(\mathbf{v}^\top \mathbf{w})^2. \quad (731)$$

Since

$$\mathbf{v}^\top \mathbf{w} \sim \mathcal{N}(0, 1), \quad (732)$$

the exact CDF is

$$F_{X_{\text{tr}}}(x) = \overline{F}_{\chi_1^2} \left[\frac{1}{\max_{\xi \leq x} g(\xi)} \right]. \quad (733)$$

This result is independent of the ambient stochastic dimension r . It provides a stringent test of the radial-angular integration because the full method uses a χ_r^2 radial variable, while angular averaging must recover the effective one-dimensional probability.

12.2.5 Commuting matrix family

Suppose all normalized energy matrices share a fixed eigenbasis:

$$\mathbf{G}(x) = \mathbf{U} \text{diag}[\gamma_1(x), \dots, \gamma_r(x)] \mathbf{U}^\top. \quad (734)$$

Then

$$y(x) = \sum_{j=1}^r \gamma_j(x) Z_j^2, \quad \mathbf{Z} = \mathbf{U}^\top \mathbf{w}. \quad (735)$$

If all eigenvalue functions are proportional,

$$\gamma_j(x) = c_j g(x), \quad (736)$$

the problem reduces to the separable test above.

If the $\gamma_j(x)$ attain maxima at different stations, the test retains nontrivial first-passage memory while avoiding rotation of the eigenvectors. High-accuracy tensor quadrature in the squared Gaussian components, or a large independent reference calculation, can be used for comparison.

12.2.6 Rotating two-dimensional quadratic form

To test sensitivity to relative eigenvector orientation, let

$$\mathbf{G}(x) = \mathbf{R}[\theta(x)] \begin{bmatrix} \gamma_1(x) & 0 \\ 0 & \gamma_2(x) \end{bmatrix} \mathbf{R}[\theta(x)]^\top, \quad (737)$$

where

$$\mathbf{R}(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}. \quad (738)$$

The angular direction can be parameterized by

$$\mathbf{u}(\varphi) = \begin{bmatrix} \cos \varphi \\ \sin \varphi \end{bmatrix}, \quad (739)$$

giving

$$a_\varphi(x) = \gamma_1(x) \cos^2[\varphi - \theta(x)] + \gamma_2(x) \sin^2[\varphi - \theta(x)]. \quad (740)$$

The exact angular integral reduces to

$$F_{X_{\text{tr}}}(x) = \frac{1}{\pi} \int_0^\pi \bar{F}_{\chi_2^2} \left[\frac{1}{\max_{\xi \leq x} a_\varphi(\xi)} \right] d\varphi. \quad (741)$$

A highly resolved periodic trapezoidal rule provides a near-exact reference. This test verifies that the method retains the rotation of dangerous directions that would be lost by treating the local eigendecompositions independently.

12.2.7 Crossing and recrossing test

Construct a gain history with two separated peaks:

$$g(x) = g_0 + A_1 \exp \left[-\frac{(x - x_1)^2}{2\sigma_1^2} \right] + A_2 \exp \left[-\frac{(x - x_2)^2}{2\sigma_2^2} \right]. \quad (742)$$

If

$$A_2 < A_1, \quad (743)$$

the second peak should add no new first-transition probability in a scalar or fixed-direction problem because it does not establish a new record. If

$$A_2 > A_1, \quad (744)$$

the second peak should increase the CDF only by the probability mass associated with radii that survived the first peak but cross at the second.

This test directly verifies the irreversible running-maximum update.

12.3 Verification of continuous maximum detection

The streamwise maximum algorithm should be tested independently of the angular integration. For prescribed smooth functions $a_{\mathbf{u}}(x)$ with known maxima, compare:

- (i) the endpoint-only maximum;

- (ii) the cubic Hermite maximum;
- (iii) the adaptively refined Hermite maximum; and
- (iv) a highly resolved reference maximum.

For direction k , define the maximum error

$$\epsilon_{m,k} = \left| \hat{m}_k(x_{\max}) - m_k^{\text{ref}}(x_{\max}) \right|. \quad (745)$$

The induced directional probability error is

$$\epsilon_{\Psi,k} = \left| \overline{F}_{\chi_r^2} \left[\frac{1}{\hat{m}_k} \right] - \overline{F}_{\chi_r^2} \left[\frac{1}{m_k^{\text{ref}}} \right] \right|. \quad (746)$$

This second measure is often more relevant because a given maximum error can have either negligible or substantial probability effect, depending on the chi-square tail sensitivity.

For OWNS data, compare the adaptively interpolated maximum with a calculation performed on a globally refined streamwise grid. Define

$$\eta_{\max}^F = \max_n |F_{\text{adaptive}}(x_n) - F_{\text{fine}}(x_n)|. \quad (747)$$

Agreement should improve according to the prescribed streamwise tolerance.

The analytical derivative

$$a'_{\mathbf{u}}(x) = \mathbf{u}^\top \mathbf{G}'(x) \mathbf{u} \quad (748)$$

should also be compared with refined finite differences:

$$a'_{\mathbf{u},\text{FD}}(x_n) = \frac{a_{\mathbf{u}}(x_n + \Delta x) - a_{\mathbf{u}}(x_n - \Delta x)}{2\Delta x}. \quad (749)$$

This comparison should be repeated for decreasing Δx until finite-difference truncation and roundoff trends are understood.

12.4 Verification of angular integration

Angular integration should first be tested using functions with known sphere averages. For a uniform direction on $\mathbb{S}^{r-1}$,

$$\mathbb{E}[\mathbf{u}\mathbf{u}^\top] = \frac{1}{r} \mathbf{I}_r. \quad (750)$$

Therefore, for any symmetric matrix $\mathbf{A}$,

$$\mathbb{E}_{\mathbf{u}} \left[\mathbf{u}^\top \mathbf{A} \mathbf{u} \right] = \frac{\text{tr}(\mathbf{A})}{r}. \quad (751)$$

A candidate angular rule should approximately satisfy

$$\sum_{k=1}^K \omega_k \mathbf{u}_k \mathbf{u}_k^\top \approx \frac{1}{r} \mathbf{I}_r. \quad (752)$$

Fourth-order moments provide a stronger test:

$$\mathbb{E}[u_i u_j u_k u_\ell] = \frac{\delta_{ij} \delta_{kl} + \delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk}}{r(r+2)}. \quad (753)$$

In particular,

$$\mathbb{E}[u_i^4] = \frac{3}{r(r+2)}, \quad (754)$$

$$\mathbb{E}[u_i^2 u_j^2] = \frac{1}{r(r+2)}, \quad i \neq j. \quad (755)$$

After these generic tests, apply the angular rule to the isotropic, rank-one, separable, and rotating-matrix cases from section 12.2. For deterministic rules, compare a sequence of increasing resolutions. For RQMC, compare:

- (i) the empirical variation among independent scramblings;
- (ii) the change when the number of points per scrambling is doubled; and
- (iii) the reported standard error against the observed error in analytical test cases.

The same angular directions must be reused at all streamwise stations. Regenerating independent angular points at each station can produce a nonmonotone numerical CDF even though the exact directional integrands are monotone.

12.5 Comparison with trajectory Monte Carlo

Direct trajectory Monte Carlo provides an independent reference for the complete first-transition calculation. It should be used for verification rather than as the primary production method.

Draw

$$\mathbf{w}^{(j)} \stackrel{\text{iid}}{\sim} \mathcal{N}(\mathbf{0}, \mathbf{I}_r), \quad j = 1, \dots, N_{\text{MC}}. \quad (756)$$

For each draw, evaluate

$$y^{(j)}(x) = [\mathbf{w}^{(j)}]^\top \mathbf{G}(x) \mathbf{w}^{(j)} \quad (757)$$

and determine

$$X_{\text{tr}}^{(j)} = \inf \left\{ x : y^{(j)}(x) \geq 1 \right\}. \quad (758)$$

The Monte Carlo CDF estimator is

$$\hat{F}_{\text{MC}}(x) = \frac{1}{N_{\text{MC}}} \sum_{j=1}^{N_{\text{MC}}} \mathbf{1} \left\{ X_{\text{tr}}^{(j)} \leq x \right\}. \quad (759)$$

At a fixed station, the standard error is

$$\text{SE} \left[\hat{F}_{\text{MC}}(x) \right] = \sqrt{\frac{F_{X_{\text{tr}}}(x)[1 - F_{X_{\text{tr}}}(x)]}{N_{\text{MC}}}}. \quad (760)$$

A plug-in estimate replaces the unknown probability with $\hat{F}_{\text{MC}}(x)$.

For pointwise confidence intervals, a binomial interval such as the Wilson interval is preferable to a normal approximation when the probability is close to zero or one. If simultaneous confidence over the complete CDF is desired, the Dvoretzky–Kiefer–Wolfowitz inequality gives

$$\mathbb{P} \left[\sup_x \left| \hat{F}_{\text{MC}}(x) - F_{X_{\text{tr}}}(x) \right| > \epsilon \right] \leq 2 \exp(-2N_{\text{MC}}\epsilon^2). \quad (761)$$

Thus, a simultaneous band with confidence at least $1 - \alpha$ is obtained using

$$\epsilon_{\text{DKW}} = \sqrt{\frac{\log(2/\alpha)}{2N_{\text{MC}}}}. \quad (762)$$

The deterministic or RQMC radial–directional CDF should lie within the Monte Carlo uncertainty band after accounting for its own streamwise, angular, rank, and propagation errors.

Algorithm 13 uses only stored stations. For a fair comparison with a continuous-maximum radial–directional result, the Monte Carlo trajectories must use the same continuous interpolation or a sufficiently refined common streamwise grid. Otherwise, the comparison confounds probability integration error with maximum-detection error.

12.5.1 Coupled radial–directional verification

A particularly useful diagnostic uses the same sampled directions in both a direct trajectory estimate and the radial-integrated estimate. Generate

$$\mathbf{w}^{(j)} = R_j \mathbf{u}_j, \quad (763)$$

and compare the binary trajectory indicator

$$I_j(x) = \mathbf{1} \{R_j^2 m_{\mathbf{u}_j}(x) \geq 1\} \quad (764)$$

with the conditional radial probability

$$\Psi_j(x) = \bar{F}_{\chi_r^2} \left[\frac{1}{m_{\mathbf{u}_j}(x)} \right]. \quad (765)$$

The sample means of both quantities converge to the same CDF, while the variance of the radial-integrated estimator should be no larger. This directly verifies the Rao–Blackwell interpretation developed in section 8.6.

12.6 Rare-event verification

Direct Monte Carlo becomes inefficient when

$$F_{X_{\text{tr}}}(x) \ll 1. \quad (766)$$

The expected number of observed transitions is

$$N_{\text{event}} = N_{\text{MC}} F_{X_{\text{tr}}}(x). \quad (767)$$

A meaningful relative-error estimate generally requires N_{event} to be substantially larger than one.

For a small probability p , the approximate relative standard error of direct Monte Carlo is

$$\frac{\text{SE}(\hat{p})}{p} \approx \frac{1}{\sqrt{N_{\text{MC}} p}}. \quad (768)$$

Therefore, achieving relative error ϵ_{rel} requires approximately

$$N_{\text{MC}} \gtrsim \frac{1}{p \epsilon_{\text{rel}}^2}. \quad (769)$$

For rare-event verification, possible alternatives include:

- directional importance sampling on the sphere;
- importance sampling of the Gaussian vector around dominant transition directions;
- subset simulation;
- FORM- or SORM-guided importance sampling; and
- threshold continuation from a less rare reference problem.

A useful continuation test varies the threshold by a scalar factor:

$$e_{\text{thres}}^{(\alpha)}(x) = \alpha e_{\text{thres}}(x). \quad (770)$$

As α increases, transition becomes less likely. The computed CDF should vary smoothly and monotonically:

$$\alpha_1 < \alpha_2 \implies F_{X_{\text{tr}}}^{(\alpha_1)}(x) \geq F_{X_{\text{tr}}}^{(\alpha_2)}(x). \quad (771)$$

12.7 Convergence studies

A converged transition prediction requires independent refinement of the spatial, stochastic, and probabilistic discretizations.

12.7.1 OWNS and wall-normal convergence

Repeat the calculation while varying:

- wall-normal grid resolution;
- streamwise marching step;
- OWNS projection or recursion parameters;
- linear-solver tolerances;
- boundary-condition placement; and
- base-flow interpolation resolution.

For two resolutions labeled $j - 1$ and j , monitor

$$\Delta_j^G = \max_n \frac{\|\mathbf{G}^{(j)}(x_n) - \mathbf{G}^{(j-1)}(x_n)\|_2}{\max[\|\mathbf{G}^{(j)}(x_n)\|_2, \varepsilon_{\text{scale}}]}, \quad (772)$$

$$\Delta_j^F = \max_n \left| F^{(j)}(x_n) - F^{(j-1)}(x_n) \right|. \quad (773)$$

The CDF difference is the more directly relevant quantity, but the matrix difference helps diagnose the source of an observed change.

12.7.2 Streamwise maximum convergence

Compare endpoint-only, Hermite-interpolated, adaptively refined, and globally refined streamwise calculations. Define

$$\Delta_{\text{stream}}^F = \max_n |F_{\text{refined}}(x_n) - F_{\text{coarse}}(x_n)|. \quad (774)$$

The endpoint-only CDF should approach the continuous result from below under nested streamwise refinement, provided the same angular rule and matrix family are used.

12.7.3 Angular convergence

For increasing angular resolution K_j , monitor

$$\Delta_{K_j}^F = \max_n |F_{K_j}(x_n) - F_{K_{j-1}}(x_n)|. \quad (775)$$

Also compare interval probabilities and quantiles:

$$\Delta_{K_j}^p = \sum_n |p_n^{(K_j)} - p_n^{(K_{j-1})}|, \quad (776)$$

$$\Delta_{K_j}^{x_p} = |x_p^{(K_j)} - x_p^{(K_{j-1})}|. \quad (777)$$

For RQMC, point-set refinement should be accompanied by replicate-based standard errors.

12.7.4 Stochastic-rank convergence

For a nested sequence of transition-oriented bases, monitor

$$\Delta_s^F = \max_n |F_s(x_n) - F_{s-}(x_n)| \quad (778)$$

and the matrix residual

$$\varepsilon_{G,s} = \max_x \|\mathbf{G}(x) - \mathbf{P}_s \mathbf{G}(x) \mathbf{P}_s\|_2. \quad (779)$$

Rank convergence need not be monotone because changing the stochastic dimension also changes the radial chi-square distribution.

12.7.5 Factorization and precision convergence

Compare:

- direct propagation of $\mathbf{B}$;
- QR-stabilized propagation;
- explicit- $\mathbf{G}$ and matrix-free directional evaluation; and
- standard and increased floating-point precision at selected cases.

All representations should converge to the same CDF when their coordinate transformations are retained correctly.

12.8 Error-budget summary

Let the computed CDF be denoted by

$$\hat{F}(x). \quad (780)$$

A practical error budget may be summarized schematically as

$$\left| \hat{F}(x) - F(x) \right| \lesssim \epsilon_{\text{OWNS}} + \epsilon_{\text{rank}} + \epsilon_{\text{stream}} + \epsilon_{\text{angular}} + \epsilon_{\text{roundoff}}, \quad (781)$$

where the terms denote estimated effects on the CDF rather than raw state errors.

This additive expression is a practical bookkeeping approximation; the errors can interact and are not necessarily statistically independent. Each component can be estimated by a targeted refinement:

$$\epsilon_{\text{OWNS}} \approx \max_n |F_{\text{fine OWNS}}(x_n) - F_{\text{baseline}}(x_n)|, \quad (782)$$

$$\epsilon_{\text{rank}} \approx \max_n |F_{s+}(x_n) - F_s(x_n)|, \quad (783)$$

$$\epsilon_{\text{stream}} \approx \max_n |F_{\text{refined stream}}(x_n) - F_{\text{baseline}}(x_n)|, \quad (784)$$

$$\epsilon_{\text{angular}} \approx \max_n |F_{K+}(x_n) - F_K(x_n)|. \quad (785)$$

The refinement used to estimate one error component should hold the others fixed at sufficiently accurate values.

12.9 Physical model validation

After numerical verification, the predicted transition distribution should be compared with independent physical evidence. Depending on the application, validation data may include:

- direct numerical simulation;
- large-eddy simulation with resolved transition;
- wind-tunnel measurements;
- flight-test measurements;
- experimentally measured intermittency distributions; or
- independently calibrated transition-location statistics.

The model predicts a threshold-defined first-transition location. The quantity extracted from the validation data must use a compatible definition. Possible physical transition indicators include:

- disturbance-energy exceedance;
- skin-friction departure from the laminar value;
- wall-shear intermittency;
- spectral broadbanding;
- heat-transfer increase; or
- sustained turbulent kinetic energy.

Different indicators can identify different points within the transition process. Agreement should not be expected unless the relationship between the model threshold and the observed transition criterion is specified.

If validation data provide transition locations

$$\left\{ X_{\text{tr}}^{(j)} \right\}_{j=1}^{N_{\text{obs}}}, \quad (786)$$

possibly with right-censored observations, compare:

- (i) empirical and predicted survival curves;
- (ii) transition-location quantiles;
- (iii) interval transition probabilities;
- (iv) predicted and observed censoring fractions; and
- (v) hazard or intermittency trends, when available.

For censored physical data, survival-analysis estimators such as the Kaplan–Meier estimator can be used to estimate the empirical survival function without treating all censored runs as transitions at the end of the domain.

12.10 Threshold calibration and out-of-sample validation

If the transition threshold is calibrated from data, calibration and validation should use separate cases. Let the threshold depend on parameters $\boldsymbol{\theta}$:

$$e_{\text{thres}}(x; \boldsymbol{\theta}). \quad (787)$$

A likelihood for observed exact transition locations can be constructed from

$$f_{X_{\text{tr}}}(x; \boldsymbol{\theta}), \quad (788)$$

while a right-censored observation at x_c contributes

$$S(x_c; \boldsymbol{\theta}). \quad (789)$$

For interval-censored transition in $(x_{n-1}, x_n]$, the contribution is

$$F_{X_{\text{tr}}}(x_n; \boldsymbol{\theta}) - F_{X_{\text{tr}}}(x_{n-1}; \boldsymbol{\theta}). \quad (790)$$

For observations indexed by j , a generic log-likelihood is

$$\begin{aligned} \log \mathcal{L}(\boldsymbol{\theta}) = & \sum_{j \in \mathcal{E}} \log f_{X_{\text{tr}}}(x_j; \boldsymbol{\theta}) \\ & + \sum_{j \in \mathcal{I}} \log [F_{X_{\text{tr}}}(x_{j,R}; \boldsymbol{\theta}) - F_{X_{\text{tr}}}(x_{j,L}; \boldsymbol{\theta})] \\ & + \sum_{j \in \mathcal{C}} \log S(x_{j,c}; \boldsymbol{\theta}), \end{aligned} \quad (791)$$

where $\mathcal{E}$, $\mathcal{I}$, and $\mathcal{C}$ denote exact, interval-censored, and right-censored observations, respectively.

After calibration, the model should be evaluated on conditions not used to determine $\boldsymbol{\theta}$, such as different frequencies, spanwise wavenumbers, free-stream disturbance levels, Reynolds numbers, pressure gradients, or surface conditions. Out-of-sample performance is necessary to distinguish a transferable transition model from a case-specific threshold fit.

12.11 Recommended verification sequence

A practical verification and validation sequence is:

1. Verify the Gaussian and complex-to-real conventions using covariance reconstruction.
2. Verify $\mathbf{G}(x)$ using explicit and matrix-free energy evaluations.
3. Apply scalar, isotropic, separable, rank-one, and rotating analytical test cases.
4. Verify the cubic Hermite and adaptive streamwise maximum algorithms against functions with known maxima.
5. Verify the angular rule using exact spherical moments and analytical transition test cases.
6. Test fixed-basis invariance under a random orthogonal transformation.
7. Establish convergence with respect to streamwise maximum resolution and angular integration.
8. Establish stochastic-rank convergence using a transition-oriented nested basis.
9. Establish OWNS and wall-normal discretization convergence.
10. Compare the complete CDF with independent trajectory Monte Carlo using a common streamwise crossing treatment.
11. For rare-event cases, use importance sampling or another independent rare-event method.
12. Validate the threshold-defined transition distribution against DNS or experimental data with a compatible transition definition.
13. If the threshold is calibrated, reserve independent cases for out-of-sample validation.

Successful completion of these tests supports the conclusion that the computed transition CDF reflects the specified OWNS–Gaussian first-passage model rather than numerical artifacts. Physical agreement then depends on the adequacy of the underlying linear propagation, initial disturbance statistics, energy metric, and transition criterion.

13 Computational Complexity

The computational cost of the first-transition method consists of three principal components:

- (i) propagation of the stochastic factor or selected directional states through the OWNS equations;
- (ii) evaluation of the directional energy histories and their streamwise running maxima; and
- (iii) angular integration of the analytically marginalized radial probabilities.

The relative cost of these components depends on the physical-state dimension, stochastic rank, angular resolution, and availability of matrix-free OWNS operations. This section compares explicit reduced- matrix, matrix-free, and direct trajectory approaches.

13.1 Dimensions and notation

Let

$$n_q = \text{dimension of the discretized OWNS state}, \quad (792)$$

$$r = \text{rank of the initial Gaussian model}, \quad (793)$$

$$s = \text{transition-oriented reduced stochastic rank}, \quad (794)$$

$$N + 1 = \text{number of stored streamwise stations}, \quad (795)$$

$$K = \text{number of angular directions}, \quad (796)$$

$$J = \text{number of RQMC randomizations}, \quad (797)$$

and let n_H denote the operation count required to apply the energy matrix $\mathbf{H}(x)$ to one physical-state vector. For a diagonal or block-diagonal quadrature-weighted energy norm,

$$n_H = \mathcal{O}(n_q). \quad (798)$$

For a general dense matrix,

$$n_H = \mathcal{O}(n_q^2), \quad (799)$$

although such a dense physical energy matrix is usually unnecessary.

Let

$$\mathcal{C}_{\text{OWNS}}(p) \quad (800)$$

denote the cost of marching p right-hand sides through the complete streamwise domain. The exact scaling depends on the OWNS formulation, wall-normal discretization, projection method, linear solver, and degree to which multiple right-hand sides can share matrix factorizations.

If one operator factorization is reused across a block of right-hand sides, the marginal cost of increasing p can be substantially less than p independent marches. It is therefore useful to distinguish between:

- (i) operator setup or factorization cost; and
- (ii) application or triangular-solve cost per right-hand side.

A generic per-step model is

$$\mathcal{C}_{\text{step}}(p) = \mathcal{C}_{\text{setup}} + p\mathcal{C}_{\text{rhs}}, \quad (801)$$

so that

$$\mathcal{C}_{\text{OWNS}}(p) \approx N_{\text{march}} (\mathcal{C}_{\text{setup}} + p\mathcal{C}_{\text{rhs}}), \quad (802)$$

where N_{march} is the number of internal marching steps, which can exceed the number N of stored output intervals.

13.2 Cost of propagating the stochastic factor

Propagating the complete initial factor

$$\mathbf{B}_0 \in \mathbb{C}^{n_q \times r} \quad (803)$$

requires marching r right-hand sides:

$$\boxed{\mathcal{C}_B = \mathcal{C}_{\text{OWNS}}(r)}. \quad (804)$$

If a fixed reduced basis $\mathbf{V}_s \in \mathbb{R}^{r \times s}$ is available before propagation, one may instead march

$$\mathbf{B}_{0,s} = \mathbf{B}_0 \mathbf{V}_s \in \mathbb{C}^{n_q \times s}, \quad (805)$$

at cost

$$\boxed{\mathcal{C}_{B_s} = \mathcal{C}_{\text{OWNS}}(s)}. \quad (806)$$

When $s \ll r$, this can significantly reduce the OWNS propagation cost.

A transition-oriented basis often requires information from an initial higher-rank propagation. Two-stage workflows are possible:

- (i) perform one high-rank offline march to construct $\mathbf{G}(x)$ and the reduced basis, then use the reduced factor for parameter studies; or
- (ii) construct the basis using matrix-free dominant-eigenvector or randomized range-finding methods that require only selected forward and adjoint propagator actions.

The first approach is simpler when a single high-rank factor march is affordable. The second can reduce memory and propagation cost but requires additional operator infrastructure.

13.3 QR-stabilization cost

A thin QR factorization of an $n_q \times r$ complex matrix has leading cost

$$\mathcal{O}(n_q r^2) \quad (807)$$

when $n_q \gg r$. If QR stabilization is applied at N_{QR} stations, its total cost is

$$\mathcal{C}_{\text{QR}} = \mathcal{O}(N_{\text{QR}} n_q r^2). \quad (808)$$

For reduced rank s , this becomes

$$\mathcal{C}_{\text{QR},s} = \mathcal{O}(N_{\text{QR}} n_q s^2). \quad (809)$$

Updating the accumulated triangular coordinate factor,

$$\mathbf{T}_{n+1} = \mathbf{R}_{n+1} \mathbf{T}_n, \quad (810)$$

requires

$$\mathcal{O}(r^3) \quad (811)$$

operations per reorthogonalization in a straightforward dense implementation. Since n_q is typically much larger than r , the QR operation on the physical basis often dominates this update.

The stabilization frequency should be selected according to loss of orthogonality and conditioning rather than applied at every output station automatically. If

$$\eta_Q = \|\mathbf{Q}^* \mathbf{Q} - \mathbf{I}\|_F \quad (812)$$

exceeds a prescribed tolerance, reorthogonalization is required.

13.4 Explicit formation of the normalized energy matrices

Suppose

$$\mathbf{B}_n \in \mathbb{C}^{n_q \times r}. \quad (813)$$

To form

$$\mathbf{G}_n = \frac{\Re[\mathbf{B}_n^* \mathbf{H}_n \mathbf{B}_n]}{e_{\text{thres},n}}, \quad (814)$$

first apply the energy weighting to all columns. The cost is approximately

$$\mathcal{O}(rn_H). \quad (815)$$

The subsequent Gram product costs

$$\mathcal{O}(n_q r^2). \quad (816)$$

Thus, over all stored stations,

$$\boxed{\mathcal{C}_G = \mathcal{O}[N(rn_H + n_q r^2)]}. \quad (817)$$

For a diagonal or sparse energy metric,

$$\mathcal{C}_G = \mathcal{O}(Nn_q r^2). \quad (818)$$

After transition-oriented reduction, forming $\mathbf{G}_s(x)$ directly from $\mathbf{B}_s(x) \in \mathbb{C}^{n_q \times s}$ costs

$$\boxed{\mathcal{C}_{G_s} = \mathcal{O}[N(sn_H + n_q s^2)]}. \quad (819)$$

If the full matrices $\mathbf{G}(x_n)$ have already been formed, their projection into a reduced basis,

$$\mathbf{G}_s(x_n) = \mathbf{V}_s^\top \mathbf{G}(x_n) \mathbf{V}_s, \quad (820)$$

costs

$$\mathcal{O}(Nr^2 s) \quad (821)$$

using dense matrix operations, with an additional $\mathcal{O}(Nrs^2)$ term that is lower order when $s < r$.

13.5 Cost of explicit directional quadratic forms

Given $\mathbf{G}_n \in \mathbb{R}^{r \times r}$, evaluating

$$a_{k,n} = \mathbf{u}_k^\top \mathbf{G}_n \mathbf{u}_k \quad (822)$$

for one direction costs

$$\mathcal{O}(r^2). \quad (823)$$

For all directions and stations,

$$\boxed{\mathcal{C}_{\text{ang},G} = \mathcal{O}(NKr^2)}. \quad (824)$$

The running-maximum and chi-square-tail updates add only

$$\mathcal{O}(NK) \quad (825)$$

scalar operations.

The directional forms can be evaluated in a batch. Let

$$\mathbf{U} = [\mathbf{u}_1 \quad \cdots \quad \mathbf{u}_K] \in \mathbb{R}^{r \times K}. \quad (826)$$

Compute

$$\mathbf{Z}_n = \mathbf{G}_n \mathbf{U} \quad (827)$$

at cost

$$\mathcal{O}(r^2 K), \quad (828)$$

and obtain the gains from columnwise inner products:

$$a_{k,n} = \mathbf{u}_k^\top \mathbf{z}_{k,n}. \quad (829)$$

This does not change the asymptotic operation count but substantially improves computational efficiency through matrix–matrix operations.

In reduced dimension s , the cost becomes

$$\boxed{\mathcal{C}_{\text{ang}, G_s} = \mathcal{O}(NKs^2)}. \quad (830)$$

13.6 Matrix-free angular-evaluation cost

Without explicitly forming $\mathbf{G}_n$, the directional state is

$$\mathbf{v}_{k,n} = \mathbf{B}_n \mathbf{u}_k. \quad (831)$$

For one direction, this multiplication costs

$$\mathcal{O}(n_q r). \quad (832)$$

Applying the energy metric and evaluating the weighted norm adds approximately

$$\mathcal{O}(n_H + n_q). \quad (833)$$

For all directions and stations,

$$\boxed{\mathcal{C}_{\text{ang}, B} = \mathcal{O}[NK(n_q r + n_H)]}. \quad (834)$$

For a sparse energy metric,

$$\mathcal{C}_{\text{ang}, B} = \mathcal{O}(NK n_q r). \quad (835)$$

After stochastic reduction,

$$\mathbf{B}_{s,n} \in \mathbb{C}^{n_q \times s}, \quad (836)$$

so

$$\mathcal{C}_{\text{ang}, B_s} = \mathcal{O}(NK n_q s). \quad (837)$$

Comparing eq. (824) and eq. (834), explicit formation of $\mathbf{G}$ is generally preferable when:

$$Kr^2 \ll K n_q r \quad (838)$$

and the one-time Gram-matrix formation is affordable. Since $n_q \gg r$ is common, the reduced matrices can make repeated angular evaluation much cheaper than repeatedly applying $\mathbf{B}$.

Matrix-free evaluation is preferable when:

- K is small;
- $\mathbf{G}$ cannot be stored;
- only selected directions are needed;
- the factor itself is available only as an operator; or
- a small block of directional states can be marched directly.

13.7 Direct propagation of angular directions

Instead of propagating the factor and subsequently multiplying by the angular directions, initialize

$$\mathbf{V}_0 = \mathbf{B}_0 \mathbf{U} \in \mathbb{C}^{n_q \times K} \quad (839)$$

and propagate these K physical states directly. The cost is

$$\boxed{\mathcal{C}_{\text{dir march}} = \mathcal{C}_{\text{OWNS}}(K)}. \quad (840)$$

The energy histories then cost

$$\mathcal{O}(NK n_H) \quad (841)$$

to evaluate.

This approach is advantageous if

$$K < r \quad (842)$$

and if no subsequent angular refinement or alternative stochastic postprocessing is required. In most moderate- or high-dimensional angular integrations, however, K can be much larger than r . Then propagating the factor once and evaluating many directions in the small stochastic space is preferable:

$$\mathcal{C}_{\text{OWNS}}(r) + \mathcal{O}(NK r^2) \quad \text{versus} \quad \mathcal{C}_{\text{OWNS}}(K). \quad (843)$$

The break-even point depends strongly on multi-right-hand-side reuse inside the OWNS solver and should be measured for the specific implementation.

13.8 Cost of continuous streamwise maximum detection

Endpoint-only running maxima add

$$\mathcal{O}(NK) \quad (844)$$

scalar comparisons after the directional gains are known.

Cubic Hermite interval maximization requires, for each direction and interval:

- (i) formation of four cubic coefficients;
- (ii) solution of a quadratic stationary-point equation; and
- (iii) evaluation at at most four candidate points.

The scalar cost remains

$$\mathcal{O}(NK). \quad (845)$$

If $\mathbf{G}'(x_n)$ is explicitly available, evaluating all directional derivatives adds

$$\mathcal{O}(NK r^2) \quad (846)$$

for the full rank, or

$$\mathcal{O}(NK s^2) \quad (847)$$

after reduction.

Adaptive streamwise refinement introduces new OWNS or matrix evaluations. Let

$$N_{\text{add}} \quad (848)$$

be the number of additional shared streamwise stations. If the factor can be evaluated at those points through dense output or stored marching data, the incremental cost is dominated by forming or evaluating the corresponding energy matrices:

$$\mathcal{O} [N_{\text{add}} (n_q r^2 + K r^2)] \quad (849)$$

for an explicit full-rank implementation.

If new OWNS substeps or remarks are required, the additional cost is problem dependent and may dominate the probability calculation. Shared refinement across all directions is preferable to independent direction-by-direction refinement because one new $\mathbf{G}(x)$ evaluation serves the entire angular rule.

13.9 Memory requirements

Storing the full propagated factors at every station requires

$$\boxed{\mathcal{M}_B = \mathcal{O}(N n_q r)} \quad (850)$$

real or complex floating-point values. This can be the dominant memory cost.

Storing only the reduced normalized energy matrices requires

$$\boxed{\mathcal{M}_G = \mathcal{O}(N r^2)}. \quad (851)$$

Since the matrices are symmetric, packed storage can reduce this to

$$\frac{N r (r + 1)}{2} \quad (852)$$

real values. If derivatives are also stored, the matrix-storage cost approximately doubles.

For a reduced stochastic dimension s ,

$$\mathcal{M}_{G_s} = \mathcal{O}(N s^2). \quad (853)$$

This is often sufficiently small that the entire matrix family can be retained in memory.

Storing all directional gain histories requires

$$\mathcal{M}_a = \mathcal{O}(N K). \quad (854)$$

This storage is unnecessary if only the CDF and final running maxima are required. A streaming implementation stores:

- the current matrix $\mathbf{G}_n$;
- the current running maxima $\{m_k\}_{k=1}^K$;
- the current CDF value; and
- endpoint values and derivatives needed for the next Hermite interval.

Its angular-memory requirement is only

$$\mathcal{O}(Kr + K) \quad (855)$$

for directions and running maxima, in addition to the current factor or matrix.

If the transition CDF must be recomputed for many different thresholds, storing the unnormalized matrices

$$\mathbf{A}(x_n) = \Re[\mathbf{B}(x_n)^* \mathbf{H}(x_n) \mathbf{B}(x_n)] \quad (856)$$

is preferable to storing only $\mathbf{G}$, since

$$\mathbf{G}(x_n) = \frac{\mathbf{A}(x_n)}{e_{\text{thres}}(x_n)} \quad (857)$$

can then be reconstructed for each threshold model without repeating the OWNS propagation.

13.10 Comparison with direct trajectory Monte Carlo

Let

$$N_{\text{MC}} \quad (858)$$

be the number of direct Gaussian trajectories. If $\mathbf{G}(x_n)$ has already been formed, evaluating all trajectory energies costs

$$\boxed{\mathcal{C}_{\text{MC},G} = \mathcal{O}(NN_{\text{MC}}r^2)}. \quad (859)$$

This has the same algebraic form as angular evaluation with $K = N_{\text{MC}}$, but each Monte Carlo trajectory produces a binary transition indicator. Each radial-directional point instead integrates the complete Gaussian radial distribution and generally provides lower variance.

If the physical trajectories are propagated separately through OWNS, the cost is

$$\boxed{\mathcal{C}_{\text{MC},\text{OWNS}} = \mathcal{C}_{\text{OWNS}}(N_{\text{MC}})}. \quad (860)$$

This is typically much more expensive than propagating the rank- r factor once when

$$N_{\text{MC}} \gg r. \quad (861)$$

For a transition probability $p \ll 1$, direct Monte Carlo requires approximately

$$N_{\text{MC}} = \mathcal{O}\left(\frac{1}{p\epsilon_{\text{rel}}^2}\right) \quad (862)$$

samples to achieve relative statistical error ϵ_{rel} . The radial-directional method does not incur this radial rare-event sampling cost. Its difficulty is instead controlled by angular localization of the dangerous directions.

The two methods can have similar costs if the transition integrand is extremely localized on the sphere and requires a very large angular rule. In that regime, adaptive angular importance sampling or FORM-guided methods may be preferable to uniform cubature.

Table 2: Leading computational and storage costs for alternative first-transition implementations.

Method	Leading computational cost	Leading storage
Full factor and explicit $\mathbf{G}$	$\mathcal{C}_{\text{OWNS}}(r) + \mathcal{O}(Nn_q r^2) + \mathcal{O}(NKr^2)$	$\mathcal{O}(Nr^2)$ if only $\mathbf{G}$ is retained
Reduced factor and explicit $\mathbf{G}_s$	$\mathcal{C}_{\text{OWNS}}(s) + \mathcal{O}(Nn_q s^2) + \mathcal{O}(NKs^2)$	$\mathcal{O}(Ns^2)$
Full factor, matrix-free angular evaluation	$\mathcal{C}_{\text{OWNS}}(r) + \mathcal{O}(NKn_q r)$	$\mathcal{O}(n_q r + Kr)$ in a streaming implementation
Direct propagation of angular directions	$\mathcal{C}_{\text{OWNS}}(K) + \mathcal{O}(NKn_q)$	$\mathcal{O}(n_q K)$ per processed block
Direct trajectory Monte Carlo from explicit $\mathbf{G}$	$\mathcal{O}(NN_{\text{MC}} r^2)$ after forming $\mathbf{G}$	$\mathcal{O}(N_{\text{MC}} r)$ or less in blocks

13.11 Comparison of principal implementation strategies

The leading costs of several implementation strategies are summarized in table 2. The table omits constants and assumes a sparse or diagonal energy metric.

The reduced explicit- $\mathbf{G}_s$ strategy is generally most attractive when a reliable transition-oriented basis with

$$s \ll \min(r, n_q) \quad (863)$$

is available. It separates the expensive physical propagation from a small stochastic-space probability calculation.

13.12 Parallelism

The method offers parallelism at several levels.

13.12.1 Parallelism over frequency and wavenumber

If different Fourier pairs (ω, β) are treated independently, their OWNS propagations and transition calculations can be distributed across separate processors or compute nodes. Any later combination across spectral components must preserve the assumed joint stochastic model and should not automatically assume independence.

13.12.2 Parallelism over factor columns

The columns of $\mathbf{B}(x)$ satisfy the same linear OWNS equations with different initial conditions. Multiple right-hand sides can be processed in blocks, sharing operator assembly and factorization.

QR stabilization requires communication among the columns, but the march between orthogonalization points can often be parallelized.

13.12.3 Parallelism over angular directions

Once $\mathbf{G}(x)$ is available, each directional history

$$a_{\mathbf{u}_k}(x) = \mathbf{u}_k^\top \mathbf{G}(x) \mathbf{u}_k \quad (864)$$

can be evaluated independently. Parallel reduction is needed only to sum the weighted directional probabilities:

$$F_K(x_n) = \sum_{k=1}^K \omega_k \Psi_{k,n}. \quad (865)$$

RQMC randomizations are also independent and can be assigned to separate processor groups.

13.12.4 Parallelism over parameter cases

Base-flow parameters, threshold models, initial covariance models, and reduced ranks can be evaluated independently. If only the threshold changes, the propagated matrices $\mathbf{A}(x)$ can be reused, making threshold-sensitivity studies particularly inexpensive.

13.13 Offline–online decomposition

The calculation naturally separates into an expensive offline stage and a relatively inexpensive online probability stage.

13.13.1 Offline stage

The offline stage includes:

- (i) construction of the base flow and OWNS operators;
- (ii) factorization of the initial covariance;
- (iii) propagation of $\mathbf{B}(x)$;
- (iv) QR or SVD stabilization;
- (v) construction of $\mathbf{A}(x) = \Re[\mathbf{B}^* \mathbf{H} \mathbf{B}]$;
- (vi) construction of a transition-oriented reduced basis; and
- (vii) storage of the reduced matrix family $\mathbf{A}_s(x)$.

The principal offline cost is approximately

$$\mathcal{C}_{\text{offline}} \approx \mathcal{C}_{\text{OWNS}}(r) + \mathcal{O}(N n_q r^2), \quad (866)$$

before any reduced rerun.

13.13.2 Online stage

For a selected threshold,

$$\mathbf{G}_s(x_n) = \frac{\mathbf{A}_s(x_n)}{e_{\text{thres}}(x_n)}, \quad (867)$$

and the online stage consists of:

- (i) threshold normalization;
- (ii) directional quadratic-form evaluation;

- (iii) streamwise maximum detection;
- (iv) chi-square-tail evaluation; and
- (v) angular summation.

Its leading cost is

$$\boxed{\mathcal{C}_{\text{online}} = \mathcal{O}(NKs^2)} \quad (868)$$

for an explicit reduced matrix family, plus any adaptive interpolation work.

This decomposition is useful for:

- threshold calibration;
- uncertainty and sensitivity studies;
- comparison of alternative transition criteria;
- repeated angular convergence tests; and
- generation of many scalar draws from the final X_{tr} distribution.

Once the CDF has been computed, generating each scalar transition location by inverse transform requires

$$\mathcal{O}(\log N) \quad (869)$$

operations using a binary search on the streamwise CDF grid.

13.14 Complexity of transition-oriented basis construction

If the full matrices $\mathbf{G}(x_n) \in \mathbb{R}^{r \times r}$ are available, a dense full eigendecomposition at every station costs

$$\mathcal{O}(Nr^3). \quad (870)$$

This can be excessive when r is large. If only the first p eigenvectors are needed, iterative symmetric eigensolvers reduce the cost to approximately

$$\mathcal{O}(Np\mathcal{C}_{Gv}), \quad (871)$$

where $\mathcal{C}_{Gv}$ is the cost of one matrix–vector product with $\mathbf{G}(x_n)$. For explicit dense matrices,

$$\mathcal{C}_{Gv} = \mathcal{O}(r^2). \quad (872)$$

The integrated matrix

$$\mathbf{K}_\omega = \sum_n \nu_n \omega_n \mathbf{G}(x_n) \quad (873)$$

costs

$$\mathcal{O}(Nr^2) \quad (874)$$

to assemble, followed by

$$\mathcal{O}(r^3) \quad (875)$$

for a dense eigendecomposition or less for a partial iterative solve.

For the greedy complement method, each iteration requires dominant eigenvalue calculations over all streamwise stations. If s iterations are used and each dominant eigenpair requires n_{it} matrix–vector products, a rough cost is

$$\mathcal{O}(sNn_{\text{it}}r^2) \quad (876)$$

for explicit dense matrices. Warm-starting the eigensolver from nearby stations or previous greedy iterations can reduce this cost.

Basis construction is ordinarily an offline expense and can be amortized over many threshold or parameter evaluations.

13.15 Practical break-even considerations

Asymptotic operation counts do not capture several important practical effects:

- (i) OWNS operator factorizations may be reusable across many right-hand sides.
- (ii) Dense reduced-space operations can attain much higher hardware efficiency than sparse physical-space solves.
- (iii) Storing $\mathbf{B}(x)$ can incur substantial memory-traffic and input/output costs.
- (iv) Adaptive angular refinement favors storing $\mathbf{G}(x)$, because new directions can be evaluated without a new OWNS march.
- (v) Adaptive streamwise refinement favors retaining dense-output marching information or check-pointed factors.
- (vi) GPU acceleration is particularly suitable for batched evaluations of $\mathbf{G}_n\mathbf{U}$ and columnwise energy norms.
- (vii) For very small transition probabilities, stable special functions and logarithmic summation can cost more than elementary arithmetic but remain negligible relative to OWNS propagation.

The break-even point between explicit and matrix-free strategies should therefore be established through timing measurements using representative values of (n_q, r, s, N, K) .

13.16 Recommended computational workflow

For a typical OWNS application with

$$n_q \gg r \quad (877)$$

and repeated probability evaluations, the recommended workflow is:

1. Propagate the initial stochastic factor in blocks, reusing OWNS operator setup across columns.
2. Apply periodic QR stabilization while retaining the complete coordinate map.
3. Form and store the reduced matrices $\mathbf{A}(x_n)$ or $\mathbf{G}(x_n)$, rather than storing every full physical factor, unless physical-state reconstruction is required.
4. Construct a fixed transition-oriented basis and reduce the stochastic dimension from r to s .

5. Perform angular integration using batched dense matrix–matrix operations in the s -dimensional space.
6. Process angular directions in blocks if K is too large to store simultaneously.
7. Share adaptive streamwise refinement points across all angular directions.
8. Parallelize over angular blocks, RQMC randomizations, spectral components, or parameter cases.
9. Reuse the offline matrix family for threshold calibration and sensitivity studies.
10. Report wall-clock time, peak memory, and the values of n_q , r , s , N , and K , so that computational performance can be interpreted and reproduced.

For this workflow, the expensive physical propagation is performed once per OWNS and initial-covariance configuration. The subsequent first-transition calculation occurs in a much smaller stochastic space and can be repeated efficiently for alternative thresholds and probability-resolution requirements.

14 Boundary-Layer Application

This section applies the first-transition method to disturbances propagated through a boundary layer using the linear One-Way Navier–Stokes equations. The purpose is to demonstrate how the propagated stochastic factor, energy threshold, streamwise maximum calculation, and angular integration are assembled for a specific flow configuration.

Quantities that depend on the particular numerical experiment are marked by bracketed placeholders and should be replaced by the values used in the final calculation.

14.1 Base flow and computational configuration

The base state is denoted by

$$\bar{\mathbf{q}}(x, y) = [\bar{\rho} \quad \bar{u} \quad \bar{v} \quad \bar{w} \quad \bar{T}]^T, \quad (878)$$

with the set of variables modified as appropriate for the incompressible or compressible formulation. The base flow is assumed steady and homogeneous in the spanwise coordinate. Its streamwise variation is retained by the OWNS operator.

The boundary-layer configuration considered here is zero-pressure-gradient flat-plate boundary layer. The principal flow parameters are summarized in table 3.

If the physical streamwise coordinate is not initialized at zero, introduce the shifted coordinate

$$\tilde{x} = x - x_0. \quad (879)$$

The derivations in the preceding sections use $\tilde{x} = 0$ as the stochastic inlet. For notational simplicity, the tilde may be omitted after this change of variables.

The base flow is obtained from the compressible Blasius self-similarity solution. It is interpolated onto the OWNS streamwise and wall-normal grids using *[state the interpolation method]*. The interpolation is selected to preserve sufficient smoothness for evaluating the spatial operator and, when used, the derivative of the normalized energy matrix.

The wall-normal coordinate is discretized using *[Chebyshev collocation, finite differences, finite elements, or another method]*. Boundary conditions are *[state the wall and free-stream disturbance*

Table 3: Base-flow and disturbance parameters for the boundary-layer application. Replace the bracketed entries with the values and nondimensional conventions used in the calculation.

Parameter	Value or definition
Flow configuration	$[configuration]$
Mach number	$[value \text{ or not applicable}]$
Reynolds-number definition	$[definition]$
Inlet Reynolds number	$[value]$
Streamwise domain	$x \in [[value], [value]]$
Inlet location for stochastic propagation	$x_0 = [value]$
Temporal frequency	$\omega = [value]$
Spanwise wavenumber	$\beta = [value]$
Wall-normal domain	$y \in [0, [value]]$
Number of wall-normal degrees of freedom	$N_y = [value]$
Number of stored streamwise stations	$N_x = [value]$
OWNS formulation	$[linear \text{ OWNS variant}]$
OWNS projection parameters	$[values]$

boundary conditions]. The outer wall-normal boundary is placed sufficiently far from the boundary layer that the transition distribution is insensitive to further increases in the domain height.

The discrete Fourier disturbance state at fixed (ω, β) is

$$\mathbf{q}(x) = [\mathbf{q}_1(x)^\top \cdots \mathbf{q}_{n_v}(x)^\top]^\top \in \mathbb{C}^{n_q}, \quad (880)$$

where n_v is the number of retained disturbance variables and each block contains the wall-normal degrees of freedom. The resulting state dimension is

$$n_q = [state \text{ the dimension or its formula}]. \quad (881)$$

After application of the OWNS projection, the disturbance satisfies

$$\frac{d\mathbf{q}}{dx} = \mathbf{L}(x; \omega, \beta) \mathbf{q}(x). \quad (882)$$

The propagator is not formed explicitly. Instead, the columns of the initial correlation factor, or a transition-oriented reduction of those columns, are marched as multiple right-hand sides:

$$\frac{d\mathbf{B}}{dx} = \mathbf{L}(x; \omega, \beta) \mathbf{B}(x), \quad \mathbf{B}(0) = \mathbf{B}_0. \quad (883)$$

The marching scheme is *[state the integration scheme and order]*, with a nominal streamwise step

$$\Delta x = [value]. \quad (884)$$

The internal OWNS marching grid may be finer than the set of stations at which $\mathbf{B}(x)$, $\mathbf{A}(x)$, or $\mathbf{G}(x)$ is stored.

14.2 Spectral component under consideration

The present calculation is performed for the Fourier pair

$$(\omega, \beta) = ([frequency], [spanwise \text{ wavenumber}]). \quad (885)$$

The corresponding dimensional or nondimensional frequency parameters are *[state Strouhal number, frequency parameter, or other conventions]*. The selected pair represents *[describe the relevant instability or nonmodal disturbance mechanism]*.

If several Fourier components are analyzed, the procedure described below is applied separately to each pair unless a joint stochastic model is constructed. Transition probabilities from separate frequencies or wavenumbers cannot generally be combined according to

$$1 - \prod_j [1 - F_j(x)] \quad (886)$$

without an explicit independence assumption. A broadband disturbance model should instead define a joint Gaussian coordinate and a total energy quadratic form containing all retained spectral components.

For independent Fourier components with a total energy defined by the sum of their component energies, the factors can be assembled into a block representation. If

$$\mathbf{q}_{\text{all}}(x) = \begin{bmatrix} \mathbf{q}_1(x) \\ \vdots \\ \mathbf{q}_{n_f}(x) \end{bmatrix}, \quad (887)$$

then a joint factor may be written as

$$\mathbf{B}_{\text{all}}(x) = \text{blockdiag} [\mathbf{B}_1(x), \dots, \mathbf{B}_{n_f}(x)] \quad (888)$$

under an independent-coordinate model. The subsequent first-transition formulation applies to the resulting joint stochastic dimension. Cross-spectral correlations require nonzero off-diagonal covariance blocks and a corresponding joint factorization.

14.3 Initial disturbance statistics

The inlet disturbance is modeled as a zero-mean Gaussian random vector:

$$\mathbf{q}(0) = \mathbf{B}_0 \mathbf{w}, \quad \mathbf{w} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_r). \quad (889)$$

Its correlation matrix is

$$\mathbf{C}_0 = \mathbb{E}[\mathbf{q}(0)\mathbf{q}(0)^*] = \mathbf{B}_0 \mathbf{B}_0^*. \quad (890)$$

The initial correlation is constructed from *[describe the physical, experimental, simulation-based, or parametric covariance model]*. For example, if the covariance is defined by a wall-normal correlation kernel $\mathbf{K}(y, y')$, its discrete representation is

$$\mathbf{C}_0 = [\mathbf{K}(y_i, y_j)]_{i,j=1}^{N_y}, \quad (891)$$

with the appropriate variable blocks, quadrature scaling, and cross-variable correlations.

If the initial statistics are obtained from data, the covariance estimate is

$$\hat{\mathbf{C}}_0 = \frac{1}{N_s - 1} \sum_{j=1}^{N_s} \left(\mathbf{q}_0^{(j)} - \bar{\mathbf{q}}_0 \right) \left(\mathbf{q}_0^{(j)} - \bar{\mathbf{q}}_0 \right)^*. \quad (892)$$

Any regularization, shrinkage, filtering, or symmetry enforcement applied to this estimate should be stated explicitly.

The covariance eigendecomposition is

$$\mathbf{C}_0 = \mathbf{V}_C \mathbf{\Lambda}_C \mathbf{V}_C^*. \quad (893)$$

An initial rank r_0 is retained:

$$\mathbf{B}_0 = \mathbf{V}_{C,r_0} \mathbf{\Lambda}_{C,r_0}^{1/2}. \quad (894)$$

The retained covariance fraction is

$$\eta_C(r_0) = \frac{\sum_{j=1}^{r_0} \lambda_j}{\sum_j \lambda_j} = [\text{value}]. \quad (895)$$

Because initial covariance energy alone does not guarantee transition accuracy under non-normal propagation, the value r_0 is selected large enough to support the transition-oriented convergence study in section 14.6. The final probabilistic calculation uses a fixed transition-oriented rank

$$s = [\text{value}], \quad (896)$$

obtained from the procedure described in section 9.

The total inlet disturbance level is normalized according to *[state the physical normalization]*. One possible convention is

$$\mathbb{E}[e(0)] = \text{tr} [\mathbf{B}_0^* \mathbf{H}(0) \mathbf{B}_0] = e_0, \quad (897)$$

where

$$e_0 = [\text{value}]. \quad (898)$$

If an amplitude parameter ϵ scales the inlet factor,

$$\mathbf{B}_0(\epsilon) = \epsilon \mathbf{B}_{0,\text{ref}}, \quad (899)$$

then the propagated normalized energy matrix scales as

$$\mathbf{G}(x; \epsilon) = \epsilon^2 \mathbf{G}_{\text{ref}}(x). \quad (900)$$

This identity permits disturbance-level studies without repeating the linear OWNS propagation.

14.4 Energy norm

The disturbance energy is

$$e(x) = \mathbf{q}(x)^* \mathbf{H}(x) \mathbf{q}(x). \quad (901)$$

For the present application, the continuous energy functional is defined as

$$e(x) = \int_0^{y_{\max}} \mathcal{E} [\hat{\mathbf{q}}(x, y); \bar{\mathbf{q}}(x, y)] \, dy, \quad (902)$$

where $\mathcal{E}$ is *[state the kinetic-energy, compressible-energy, Chu-energy, or other energy density]*.

For example, an incompressible kinetic-energy norm may be written as

$$e_{\text{kin}}(x) = \frac{1}{2} \int_0^{y_{\max}} (|\hat{u}|^2 + |\hat{v}|^2 + |\hat{w}|^2) \, dy. \quad (903)$$

This expression is illustrative and should be replaced if a different energy definition is used.

After wall-normal discretization,

$$\mathbf{H}(x) = \mathbf{H}_{\text{phys}}(x)\mathbf{H}_{\text{quad}}, \quad (904)$$

with the product interpreted in a symmetrized or factored form that preserves Hermitian positive semidefiniteness. Here, $\mathbf{H}_{\text{quad}}$ contains wall-normal quadrature weights and $\mathbf{H}_{\text{phys}}$ contains the physical variable weights.

The energy norm is checked by comparing its discrete value with direct quadrature of representative disturbance fields. Under wall-normal refinement, the relative energy difference

$$\eta_H = \frac{|e_{N_y^+} - e_{N_y}|}{\max(e_{N_y^+}, \varepsilon_{\text{scale}})} \quad (905)$$

is required to be below *[state the tolerance]*.

14.5 Transition criterion

Transition is defined operationally as the first station at which the disturbance energy reaches a prescribed threshold:

$$X_{\text{tr}} = \inf \{x : e(x) \geq e_{\text{thres}}(x)\}. \quad (906)$$

The threshold used in the present application is

$$e_{\text{thres}}(x) = [\textit{state the threshold model}]. \quad (907)$$

Possible threshold conventions include:

- (i) a constant absolute disturbance energy,

$$e_{\text{thres}}(x) = e_c; \quad (908)$$

- (ii) a constant energy normalized by a local base-flow scale,

$$\frac{e_{\text{thres}}(x)}{e_{\text{base}}(x)} = \eta_c; \quad (909)$$

- (iii) an empirically calibrated streamwise threshold,

$$e_{\text{thres}}(x) = e_c(x; \boldsymbol{\theta}); \quad (910)$$

or

- (iv) a threshold selected to correspond to an independently observed physical transition indicator.

The selected criterion must be interpreted as a first-passage definition. The linear OWNS model is used only up to the first threshold crossing; it is not assumed to model the post-transition turbulent state.

The normalized energy matrix is

$$\mathbf{G}(x) = \frac{\Re[\mathbf{B}(x)^* \mathbf{H}(x) \mathbf{B}(x)]}{e_{\text{thres}}(x)}. \quad (911)$$

For a reduced stochastic basis $\mathbf{V}_s$, the matrix used in the probability calculation is

$$\mathbf{G}_s(x) = \mathbf{V}_s^\top \mathbf{G}(x) \mathbf{V}_s. \quad (912)$$

Threshold uncertainty is assessed by introducing a multiplicative factor α :

$$e_{\text{thres}}^{(\alpha)}(x) = \alpha e_{\text{thres}}(x). \quad (913)$$

The values

$$\alpha \in \{[list\ the\ tested\ factors]\} \quad (914)$$

are used to quantify the sensitivity of the predicted transition distribution to the threshold definition.

14.6 Transition-oriented stochastic reduction

The initial covariance factor has rank

$$r_0 = [value]. \quad (915)$$

A fixed transition-oriented basis is constructed from *[state the selected method: union of leading eigenspaces, integrated Gramian, greedy enrichment, or a hybrid method]*.

For the integrated component of the basis, the matrix

$$\mathbf{K}_\omega = \int_0^{x_{\max}} \omega(x) \mathbf{G}(x) dx \quad (916)$$

uses the weight

$$\omega(x) = [state\ the\ weight]. \quad (917)$$

The basis is supplemented by *[state the number and selection criterion]* leading eigenvectors from streamwise record-gain stations.

The candidate reduced ranks are

$$s \in \{[list\ tested\ ranks]\}. \quad (918)$$

Rank convergence is evaluated using

$$\Delta_s^F = \max_n |F_{X_{\text{tr},s}}(x_n) - F_{X_{\text{tr},s^-}}(x_n)| \quad (919)$$

and

$$\varepsilon_{G,s} = \max_n \|\mathbf{G}(x_n) - \mathbf{P}_s \mathbf{G}(x_n) \mathbf{P}_s\|_2. \quad (920)$$

The final rank is selected as

$$s = [value], \quad (921)$$

for which

$$\Delta_s^F = [value], \quad \varepsilon_{G,s} = [value]. \quad (922)$$

Table 4: Convergence of the first-transition calculation with stochastic rank. Replace the entries with computed values.

Rank s	$F_{X_{\text{tr}}}(x_{\text{max}})$	Δ_s^F	p_{censored}
[rank 1]	[value]	[value]	[value]
[rank 2]	[value]	[value]	[value]
[rank 3]	[value]	[value]	[value]

14.7 Streamwise maximum and angular integration settings

For each angular direction $\mathbf{u}_k \in \mathbb{S}^{s-1}$, the directional gain is

$$a_k(x) = \mathbf{u}_k^\top \mathbf{G}_s(x) \mathbf{u}_k. \quad (923)$$

The continuous running maximum is approximated by

$$m_k(x_n) = \max_{0 \leq \xi \leq x_n} a_k(\xi). \quad (924)$$

The streamwise maximum is resolved using [cubic Hermite interpolation, adaptive midpoint refinement, dense OWNS output, or another method]. The absolute and relative maximum tolerances are

$$\varepsilon_{\text{abs}} = [\text{value}], \quad \varepsilon_{\text{rel}} = [\text{value}]. \quad (925)$$

Adaptive refinement adds

$$N_{\text{add}} = [\text{value}] \quad (926)$$

streamwise stations to the initial grid.

The angular expectation is evaluated using [state the deterministic spherical quadrature, spherical design, RQMC, or importance-sampling method]. The principal angular parameters are listed in table 5.

Table 5: Numerical parameters for the first-transition probability calculation.

Parameter	Value
Reduced stochastic dimension s	[value]
Angular integration method	[method]
Number of angular points K	[value]
Number of RQMC randomizations J	[value or not applicable]
Initial streamwise stations	[value]
Added adaptive stations	[value]
Maximum absolute tolerance	[value]
Maximum relative tolerance	[value]
Angular CDF tolerance	[value]
Chi-square-tail implementation	[library or method]

The transition CDF is computed as

$$F_{X_{\text{tr}}}(x_n) \approx \sum_{k=1}^K \omega_k \bar{F}_{\chi_s^2} \left[\frac{1}{m_k(x_n)} \right]. \quad (927)$$

The same angular points are used at every streamwise station to preserve the monotonicity of the computed CDF.

14.8 Non-normal amplification diagnostics

Before evaluating transition probability, the non-normal amplification encoded in the matrix family is characterized through

$$\gamma_{\max}(x) = \lambda_{\max}[\mathbf{G}_s(x)]. \quad (928)$$

This quantity is the largest normalized energy attained at x by a unit vector in the reduced stochastic-coordinate space.

The corresponding dominant direction is

$$\mathbf{v}_1(x) \in \arg \max_{\|\mathbf{v}\|_2=1} \mathbf{v}^\top \mathbf{G}_s(x) \mathbf{v}. \quad (929)$$

Rotation of the dominant direction is measured by the principal-angle correlation

$$c_1(x_a, x_b) = \left| \mathbf{v}_1(x_a)^\top \mathbf{v}_1(x_b) \right|. \quad (930)$$

Values close to one indicate nearly aligned directions, whereas values close to zero indicate substantial rotation.

For a multidimensional dominant subspace $\mathbf{V}_p(x)$, the principal angles between stations are obtained from the singular values

$$\sigma_j \left[\mathbf{V}_p(x_a)^\top \mathbf{V}_p(x_b) \right] = \cos \theta_j. \quad (931)$$

These diagnostics help determine whether transition is dominated by a single persistent mechanism or by several streamwise-varying directions.

The largest normalized gain over the domain is

$$\Gamma_{\max} = \max_{0 \leq x \leq x_{\max}} \lambda_{\max}[\mathbf{G}_s(x)], \quad (932)$$

attained at

$$x_\Gamma \in \arg \max_x \lambda_{\max}[\mathbf{G}_s(x)]. \quad (933)$$

14.9 Local and first-transition probabilities

The local exceedance probability is

$$p_{\text{local}}(x) = \mathbb{P} \left[\mathbf{w}^\top \mathbf{G}_s(x) \mathbf{w} \geq 1 \right]. \quad (934)$$

It is evaluated independently from the generalized chi-square distribution determined by the eigenvalues of $\mathbf{G}_s(x)$.

The memory-aware transition probability is

$$F_{X_{\text{tr}}}(x) = \mathbb{P} \left[\max_{0 \leq \xi \leq x} \mathbf{w}^\top \mathbf{G}_s(\xi) \mathbf{w} \geq 1 \right]. \quad (935)$$

The difference

$$\Delta_{\text{memory}}(x) = F_{X_{\text{tr}}}(x) - p_{\text{local}}(x) \geq 0 \quad (936)$$

quantifies the probability mass associated with realizations that crossed upstream but are below the threshold at station x .

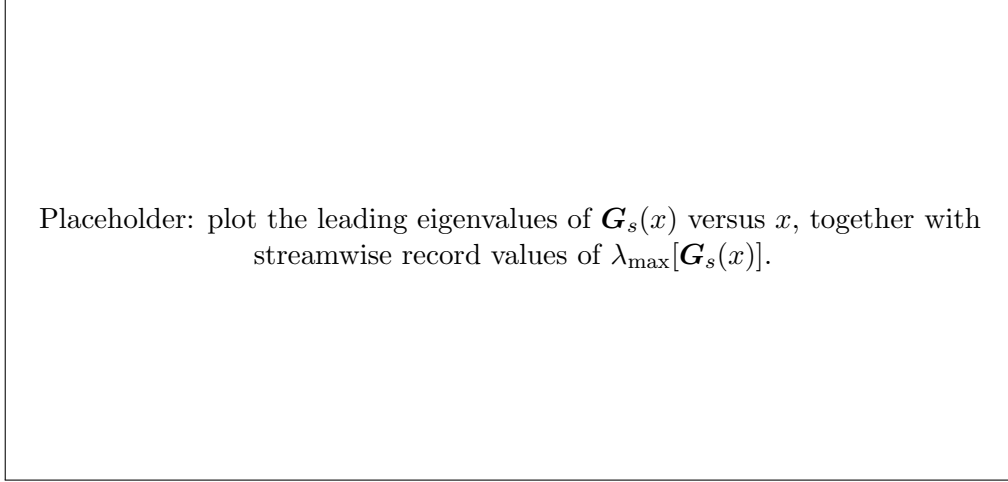


Figure 1: Leading eigenvalues of the normalized propagated energy matrix. The largest eigenvalue gives the maximum normalized energy at each station for a unit initial stochastic direction.

A relative memory measure may be defined as

$$R_{\text{memory}}(x) = \frac{F_{X_{\text{tr}}}(x) - p_{\text{local}}(x)}{\max[F_{X_{\text{tr}}}(x), \varepsilon_{\text{prob}}]}. \quad (937)$$

This quantity lies between zero and one when the denominator is not regularized. A value near zero indicates that most transitioned realizations remain locally above the threshold. A larger value indicates substantial downstream decay or recrossing.

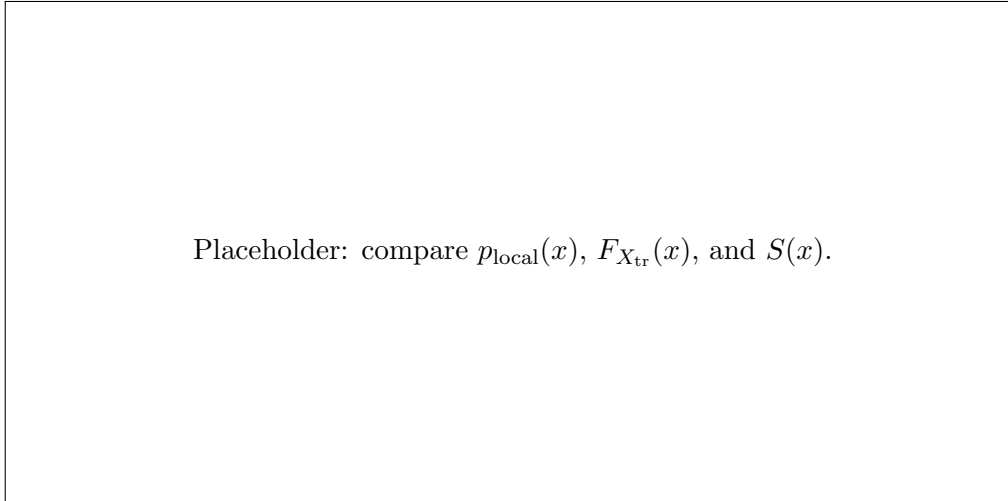


Figure 2: Comparison of the stationwise energy-exceedance probability and the cumulative first-transition probability. The difference between the two curves represents realizations that exceeded the threshold upstream but are locally below it.

The principal numerical values are reported as

$$F_{X_{\text{tr}}}(x_{\text{max}}) = [\text{value}], \quad (938)$$

$$S(x_{\text{max}}) = [\text{value}], \quad (939)$$

$$\max_x \Delta_{\text{memory}}(x) = [\text{value}], \quad (940)$$

$$\arg \max_x \Delta_{\text{memory}}(x) = [\text{value}]. \quad (941)$$

14.10 Transition-location distribution

The probability of first transition in the interval $(x_{n-1}, x_n]$ is

$$p_n = F_{X_{\text{tr}}}(x_n) - F_{X_{\text{tr}}}(x_{n-1}). \quad (942)$$

A piecewise-constant transition-location density may be reported as

$$\hat{f}_{X_{\text{tr}}}(x) = \frac{p_n}{x_n - x_{n-1}}, \quad x \in (x_{n-1}, x_n]. \quad (943)$$

This conservative interval representation is preferable to directly differentiating a noisy CDF.

The interval-averaged hazard is

$$\bar{\lambda}_n = -\frac{\log S(x_n) - \log S(x_{n-1})}{x_n - x_{n-1}}. \quad (944)$$

The hazard is defined only while $S(x_{n-1}) > 0$.

The transition quantiles are

$$x_p = \inf \{x : F_{X_{\text{tr}}}(x) \geq p\}. \quad (945)$$

Representative values are listed in table 6.

Table 6: Predicted first-transition-location quantiles. A quantile is reported as right-censored if the terminal CDF does not reach the requested probability.

Probability p	Transition quantile x_p
0.05	<i>[value or censored]</i>
0.10	<i>[value or censored]</i>
0.25	<i>[value or censored]</i>
0.50	<i>[value or censored]</i>
0.75	<i>[value or censored]</i>
0.90	<i>[value or censored]</i>
0.95	<i>[value or censored]</i>

If

$$F_{X_{\text{tr}}}(x_{\text{max}}) < p, \quad (946)$$

the unconditional p -quantile lies beyond the modeled domain and should not be assigned x_{max} . A conditional quantile among realizations that transition by x_{max} may instead be computed from

$$F_{\text{cond}}(x) = \frac{F_{X_{\text{tr}}}(x)}{F_{X_{\text{tr}}}(x_{\text{max}})}. \quad (947)$$

Placeholder: plot the transition CDF, interval transition density, and hazard as functions of x .

Figure 3: First-transition-location statistics. The transition density is computed from conservative CDF differences, and the hazard is conditioned on survival to the beginning of each interval.

14.11 Directional contributions to transition

For angular direction $\mathbf{u}_k$, define the terminal directional transition probability

$$\Psi_k(x_{\max}) = \bar{F}_{\chi_s^2} \left[\frac{1}{m_k(x_{\max})} \right]. \quad (948)$$

Its weighted contribution to the angular integral is

$$c_k = \omega_k \Psi_k(x_{\max}). \quad (949)$$

The contributions satisfy

$$\sum_{k=1}^K c_k = F_{X_{\text{tr}}}(x_{\max}). \quad (950)$$

The active record station for direction k is

$$x_k^* \in \arg \max_{0 \leq x \leq x_{\max}} a_k(x). \quad (951)$$

A weighted distribution of record stations can be formed as

$$\mu_{\text{record}}(I_n) = \sum_{k: x_k^* \in I_n} c_k, \quad (952)$$

where I_n is a streamwise interval. This quantity identifies where the directional mechanisms contributing to terminal transition reach their largest normalized energy. It is not identical to the first-transition-location distribution, because a realization with a large radius may cross before its direction reaches its maximum.

The conditional angular distribution among realizations transitioned by x has discrete weights

$$\tilde{\omega}_k^{\text{tr}}(x) = \frac{\omega_k \Psi_k(x)}{F_{X_{\text{tr}}}(x)}, \quad F_{X_{\text{tr}}}(x) > 0. \quad (953)$$

These weights can be used to calculate transition-conditioned directional summaries.

For example, the projective second-moment matrix of transition-producing directions is

$$\mathbf{P}_{\text{tr}}(x) = \sum_{k=1}^K \tilde{\omega}_k^{\text{tr}}(x) \mathbf{u}_k \mathbf{u}_k^{\text{T}}. \quad (954)$$

Its leading eigenvectors identify broad initial stochastic subspaces associated with transition. Since $\mathbf{u}$ and $-\mathbf{u}$ represent the same quadratic-energy direction, the second moment is more meaningful than a signed angular mean.

A physical initial disturbance associated with a representative direction is

$$\mathbf{q}_{0,k} = \mathbf{B}_{0,s} \mathbf{u}_k, \quad \mathbf{B}_{0,s} = \mathbf{B}_0 \mathbf{V}_s. \quad (955)$$

Its propagated structure is

$$\mathbf{q}_k(x) = \mathbf{B}_s(x) \mathbf{u}_k. \quad (956)$$

Plotting these fields for the directions with the largest c_k provides a physical interpretation of the initial structures that contribute most strongly to transition probability.

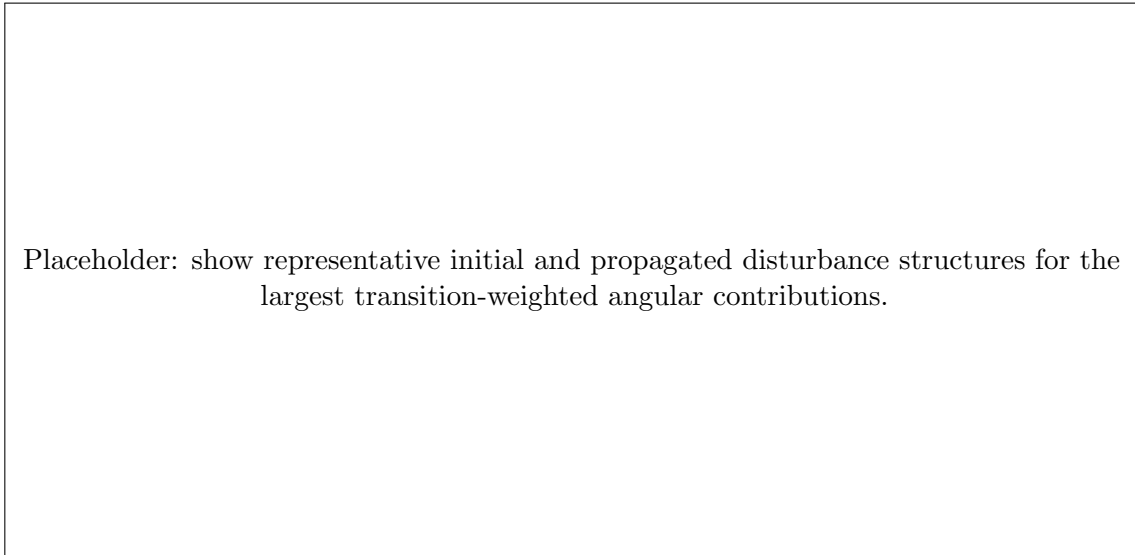


Figure 4: Representative transition-producing disturbance directions. Directions are ranked by their weighted contributions $c_k = \omega_k \Psi_k(x_{\text{max}})$, rather than only by their maximum deterministic gain.

14.12 Sensitivity to disturbance amplitude and threshold

If the inlet covariance factor is scaled by ϵ , the transition CDF becomes

$$F_{X_{\text{tr}}}(x; \epsilon) = \mathbb{E}_{\mathbf{u}} \left[\bar{F}_{\chi_s^2} \left(\frac{1}{\epsilon^2 m_{\mathbf{u}, \text{ref}}(x)} \right) \right]. \quad (957)$$

This relation permits an inexpensive disturbance-amplitude sweep after the reference directional maxima have been computed.

The tested amplitudes are

$$\epsilon \in \{[list \text{ amplitudes}]\}. \quad (958)$$

For each amplitude, report:

$$F_{X_{\text{tr}}}(x_{\text{max}}; \epsilon), \quad p_{\text{censored}}(\epsilon), \quad x_{0.5}(\epsilon), \quad (959)$$

with the median reported only when it lies within the computational domain.

Similarly, multiplying the threshold by α gives

$$\mathbf{G}^{(\alpha)}(x) = \frac{1}{\alpha} \mathbf{G}(x), \quad (960)$$

and

$$F_{X_{\text{tr}}}^{(\alpha)}(x) = \mathbb{E}_{\mathbf{u}} \left[\bar{F}_{\chi_s^2} \left(\frac{\alpha}{m_{\mathbf{u}}(x)} \right) \right]. \quad (961)$$

The transition probability must decrease monotonically with α :

$$\alpha_1 < \alpha_2 \implies F_{X_{\text{tr}}}^{(\alpha_1)}(x) \geq F_{X_{\text{tr}}}^{(\alpha_2)}(x). \quad (962)$$

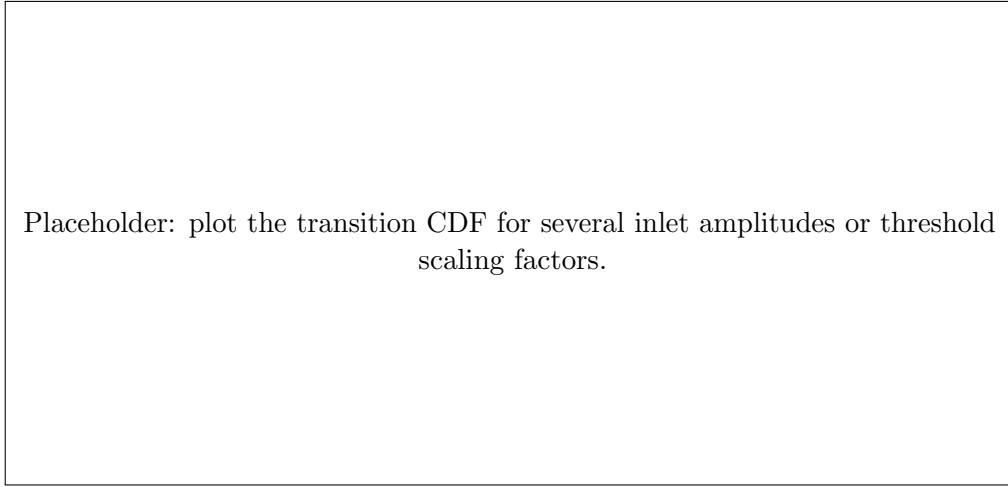


Figure 5: Sensitivity of the first-transition distribution to inlet disturbance amplitude or transition-threshold scaling. The linear OWNS propagation is reused across all curves.

14.13 Numerical convergence for the application

The application-specific convergence study varies the streamwise resolution, angular resolution, stochastic rank, and OWNS discretization. The baseline and refined settings are summarized in table 7.

For refinement parameter j , define the complete-CDF difference

$$\Delta_j^F = \max_{0 \leq n \leq N} \left| F^{(j)}(x_n) - F^{(\text{ref})}(x_n) \right|. \quad (963)$$

The reported convergence values are given in table 8.

The final numerical uncertainty assigned to the CDF is

$$\varepsilon_F^{\text{num}} = [\text{value or estimated range}]. \quad (964)$$

This value combines the observed effects of the separate refinement studies and should not be interpreted as uncertainty in the physical transition model.

Table 7: Numerical refinement settings for the boundary-layer application.

Discretization	Coarse	Baseline	Refined
Wall-normal points N_y	[value]	[value]	[value]
Streamwise output stations N_x	[value]	[value]	[value]
Stochastic rank s	[value]	[value]	[value]
Angular points K	[value]	[value]	[value]
RQMC replicates J	[value]	[value]	[value]

Table 8: Application-specific convergence of the first-transition CDF.

Refinement	Δ^F	Change in $x_{0.5}$
Wall-normal resolution	[value]	[value or not applicable]
Streamwise maximum resolution	[value]	[value or not applicable]
Stochastic rank	[value]	[value or not applicable]
Angular resolution	[value]	[value or not applicable]
OWNS parameters	[value]	[value or not applicable]

14.14 Comparison with trajectory Monte Carlo

An independent trajectory Monte Carlo calculation is performed using

$$N_{\text{MC}} = [\text{value}] \quad (965)$$

Gaussian realizations. Each realization uses the same reduced matrix family and continuous streamwise maximum treatment as the radial-directional calculation.

The empirical CDF is

$$\hat{F}_{\text{MC}}(x) = \frac{1}{N_{\text{MC}}} \sum_{j=1}^{N_{\text{MC}}} \mathbf{1} \left\{ X_{\text{tr}}^{(j)} \leq x \right\}. \quad (966)$$

A simultaneous confidence band is constructed using

$$\epsilon_{\text{DKW}} = \sqrt{\frac{\log(2/\alpha)}{2N_{\text{MC}}}}, \quad (967)$$

with confidence level

$$1 - \alpha = [\text{value}]. \quad (968)$$

The maximum difference between the radial-directional and empirical CDFs is

$$\Delta_{\text{MC}}^F = \max_n \left| F_{X_{\text{tr}}}(x_n) - \hat{F}_{\text{MC}}(x_n) \right| = [\text{value}]. \quad (969)$$

This difference is compared with both ϵ_{DKW} and the estimated numerical error of the radial-directional calculation.

14.15 Summary of application results

The principal outputs of the boundary-layer calculation are summarized in table 9. The entries should be completed after the numerical convergence and validation studies have been performed.

The application should be interpreted through three distinct quantities:

Placeholder: compare the radial-directional CDF with trajectory Monte Carlo and its confidence band.

Figure 6: Verification of the first-transition CDF against trajectory Monte Carlo. Both methods use the same transition threshold and streamwise crossing treatment.

Table 9: Summary of the boundary-layer first-transition results.

Quantity	Result
Terminal transition probability $F_{X_{tr}}(x_{\max})$	<i>[value]</i>
Right-censoring probability $S(x_{\max})$	<i>[value]</i>
Median transition location $x_{0.5}$	<i>[value or censored]</i>
Modal transition interval	<i>[value]</i>
Maximum local exceedance probability	<i>[value]</i>
Maximum memory correction	<i>[value]</i>
Location of maximum memory correction	<i>[value]</i>
Most dangerous deterministic-gain station x_{Γ}	<i>[value]</i>
Final stochastic rank	<i>[value]</i>
Angular resolution	<i>[value]</i>
Estimated numerical CDF uncertainty	<i>[value]</i>

- (i) the local exceedance probability, which describes the ensemble at one station;
- (ii) the cumulative first-transition probability, which retains all upstream threshold crossings; and
- (iii) the directional contribution distribution, which identifies the initial stochastic structures responsible for the transition probability.

Together, these quantities distinguish transient amplification from irreversible transition classification and expose the role of non-normal disturbance growth in the predicted transition-location distribution.

15 Discussion

The first-transition formulation changes the probabilistic quantity used to describe boundary-layer transition. A local exceedance probability asks whether a realization is above the energy threshold at one streamwise station. The first-transition CDF instead asks whether that realization has reached

the threshold at any upstream station. This distinction is essential when the disturbance dynamics are non-normal and the energy can amplify, decay, and later reamplify.

The present method exploits the special structure of the linear OWNS model with Gaussian initial uncertainty. All downstream states are linear functions of the same finite-dimensional Gaussian coordinate, and the energy history is therefore a family of correlated quadratic forms. This structure makes it possible to retain streamwise memory without solving a probability-density equation in the full physical state space or reconstructing a large ensemble of Gaussian trajectories.

15.1 Interpretation of the memory correction

The local exceedance probability and first-transition CDF are

$$p_{\text{local}}(x) = \mathbb{P}[e(x) \geq e_{\text{thres}}(x)], \quad (970)$$

$$F_{X_{\text{tr}}}(x) = \mathbb{P}\left[\max_{0 \leq \xi \leq x} \frac{e(\xi)}{e_{\text{thres}}(\xi)} \geq 1\right]. \quad (971)$$

Their difference,

$$\Delta_{\text{memory}}(x) = F_{X_{\text{tr}}}(x) - p_{\text{local}}(x), \quad (972)$$

is the probability of the event

$$\left\{\max_{0 \leq \xi < x} \frac{e(\xi)}{e_{\text{thres}}(\xi)} \geq 1\right\} \cap \left\{\frac{e(x)}{e_{\text{thres}}(x)} < 1\right\}. \quad (973)$$

Thus, the memory correction has a direct physical interpretation: it is the fraction of realizations that have exceeded the transition threshold upstream but are locally below it.

A small value of $\Delta_{\text{memory}}(x)$ can arise in several distinct regimes:

- (i) transition probability is small at all upstream stations;
- (ii) normalized energy is approximately monotone for the transition-producing realizations; or
- (iii) an upstream peak has occurred, but most realizations that crossed it remain above the threshold downstream.

A large memory correction indicates that the local ensemble no longer contains an obvious signature of all previous threshold crossings. In that regime, interpreting $p_{\text{local}}(x)$ as the probability of having transitioned can substantially underestimate cumulative transition.

The correction should not automatically be interpreted as physical relaminarization. Within the present model, a realization that drops below the energy threshold after first crossing is still classified as transitioned. The downstream decrease merely demonstrates that a local linear-energy criterion is not itself an irreversible state variable. Whether the corresponding physical flow becomes turbulent and remains turbulent is part of the threshold-modeling assumption.

15.2 Role of non-normality

Strong non-normality is often viewed as a complication for probability-density propagation because it produces anisotropic amplification, rotating disturbance structures, and severe numerical conditioning. In the common initial coordinates, however, non-normality is represented directly by

$$\mathbf{G}(x) = \frac{\Re[\mathbf{B}_0^* \mathbf{M}(0, x)^* \mathbf{H}(x) \mathbf{M}(0, x) \mathbf{B}_0]}{e_{\text{thres}}(x)}. \quad (974)$$

The full matrix family records both the magnitude and orientation of the transiently amplified stochastic directions.

The dominant eigenvalue

$$\lambda_{\max}[\mathbf{G}(x)] \quad (975)$$

identifies the largest normalized energy at a specified station for a unit initial stochastic direction. It does not, by itself, determine the transition probability. Probability also depends on:

- (i) the angular width of the high-gain region;
- (ii) the subdominant eigenvalues;
- (iii) rotation of the eigenspaces with x ;
- (iv) competition among peaks at different streamwise stations; and
- (v) the Gaussian radial probability associated with the critical amplitude for each direction.

A narrow direction with extremely large gain can dominate an optimal growth calculation but contribute relatively little total transition probability if its angular neighborhood has small measure. Conversely, a broader collection of moderately amplified directions can dominate the probabilistic result. Deterministic optimal growth and stochastic transition prediction are therefore related but not equivalent.

Non-normality also makes local eigendecomposition insufficient for pathwise prediction. At each station, the eigenvalues of $\mathbf{G}(x)$ determine the marginal energy distribution, but the relative orientations of the matrices determine how the same initial realization evolves downstream. Two matrix families can possess identical eigenvalues at every station and nevertheless produce different first-transition distributions if their eigenvectors rotate differently.

15.3 The common stochastic basis as the source of memory

The key modeling object is the propagated synthesis factor

$$\mathbf{B}(x) = \mathbf{M}(0, x)\mathbf{B}_0, \quad (976)$$

which gives

$$\mathbf{q}(x; \mathbf{w}) = \mathbf{B}(x)\mathbf{w}. \quad (977)$$

The coefficient vector $\mathbf{w}$ labels one realization across the entire streamwise interval. This labeling, rather than Gaussianity alone, supplies the dependence required for first-passage prediction.

The one-point covariance

$$\mathbf{C}(x, x) = \mathbf{B}(x)\mathbf{B}(x)^* \quad (978)$$

does not uniquely determine the joint streamwise process. Independent factorizations of $\mathbf{C}(x, x)$ can differ by streamwise-dependent orthogonal or unitary transformations. These transformations preserve the local energy PDF but generally alter the implied relationship between realizations at different stations.

The method consequently requires either:

- (i) the propagated factor in a common initial coordinate system; or
- (ii) an equivalent representation of the full two-point dependence that permits consistent trajectory construction.

This requirement is also relevant to numerical stabilization. QR and SVD factorizations may change the representation of the propagated columns, but all accumulated coordinate transformations must remain part of the map from $\mathbf{w}$ to $\mathbf{q}(x)$.

15.4 Interpretation of the radial–directional reduction

The Gaussian decomposition

$$\mathbf{w} = R\mathbf{u} \quad (979)$$

separates two physically distinct aspects of the inlet uncertainty:

- $\mathbf{u}$ describes the structure or mixture of the initial stochastic modes; and
- R describes the overall stochastic amplitude along that structure.

For each initial direction, the OWNS equations produce the deterministic normalized gain history

$$a_{\mathbf{u}}(x) = \frac{[\mathbf{B}(x)\mathbf{u}]^* \mathbf{H}(x) [\mathbf{B}(x)\mathbf{u}]}{e_{\text{thres}}(x)}. \quad (980)$$

Its running maximum determines the smallest Gaussian radius that would have transitioned by x :

$$R_{\text{crit}}(\mathbf{u}, x) = \frac{1}{\sqrt{m_{\mathbf{u}}(x)}}. \quad (981)$$

The conditional transition probability is the Gaussian radial tail beyond this value.

This interpretation provides more than a numerical shortcut. It partitions the transition problem into:

- (i) deterministic amplification physics for each initial structure; and
- (ii) a known probability law for the amplitude of that structure.

Analytically integrating the radius is a Rao–Blackwellization of trajectory Monte Carlo. It removes variance associated with randomly sampling the Gaussian amplitude and replaces each binary trajectory outcome with an exact conditional probability.

The remaining angular integral can still be difficult if the dangerous directions occupy a small part of the sphere. This difficulty is a physical feature of the model rather than an artifact of radial sampling: it indicates that transition is controlled by a highly specific set of initial disturbance structures.

15.5 First transition versus all threshold crossings

The directional running maximum counts a realization once, at its first threshold crossing. This differs from an ordinary upcrossing calculation, which can count the same realization multiple times.

Let

$$N_+(0, x) \quad (982)$$

be the number of positive crossings of the threshold. Then

$$F_{X_{\text{tr}}}(x) = \mathbb{P}[y(0) \geq 1 \text{ or } N_+(0, x) \geq 1], \quad (983)$$

whereas a Rice calculation of

$$\mathbb{E}[N_+(0, x)] \quad (984)$$

counts all positive crossings. The two agree only when multiple upcrossings are negligible.

For strongly transient systems, repeated threshold crossings can occur if the normalized energy contains multiple separated peaks. The running-maximum formulation remains exact in this setting. A later peak adds transition probability only if it exceeds the previous directional record. This record interpretation is a useful diagnostic: streamwise regions in which no direction establishes a new probability-relevant record cannot increase the transition CDF.

15.6 Interpretation of the transition hazard

The hazard

$$\lambda(x) = \frac{f_{X_{\text{tr}}}(x)}{S(x)} \quad (985)$$

is the first-transition probability per unit downstream distance conditioned on survival to x . It separates two effects:

- (i) the number of realizations still available to transition, represented by $S(x)$; and
- (ii) the rate at which the surviving population encounters new threshold-producing amplification, represented by $f_{X_{\text{tr}}}(x)$.

A small transition density downstream does not necessarily imply weak instantaneous amplification. It can also occur because most susceptible realizations transitioned upstream. Conversely, a large hazard indicates that the current streamwise region is effective at removing probability from the population that has survived to that point.

The hazard should not be identified with the local exceedance probability or its derivative. It is explicitly conditional on the complete upstream safe history.

15.7 Survivor selection and conditional disturbance statistics

Although the unconditional state is Gaussian under the linear OWNS model, the population conditioned on survival is generally non-Gaussian. The survival event preferentially excludes initial directions and amplitudes that experience large normalized gain.

In initial coordinates, the conditional survivor distribution is

$$p(\mathbf{w} \mid X_{\text{tr}} > x) = \frac{\phi_r(\mathbf{w}) \mathbf{1}_{\mathcal{S}(x)}(\mathbf{w})}{S(x)}. \quad (986)$$

Its mean remains zero because the safe set is centrally symmetric:

$$\mathbb{E}[\mathbf{w} \mid X_{\text{tr}} > x] = \mathbf{0}. \quad (987)$$

Its covariance, however, changes:

$$\boldsymbol{\Sigma}_{w, \text{surv}}(x) = \mathbb{E}[\mathbf{w} \mathbf{w}^\top \mid X_{\text{tr}} > x]. \quad (988)$$

The corresponding conditional physical-state covariance is

$$\mathbf{C}_{\text{surv}}(x, x) = \mathbf{B}(x) \boldsymbol{\Sigma}_{w, \text{surv}}(x) \mathbf{B}(x)^*. \quad (989)$$

These statistics quantify how transition selectively removes disturbance structures from the surviving ensemble. For example, a direction that undergoes strong lift-up amplification may be strongly depleted among downstream survivors even if it was prominent in the unconditional inlet covariance.

The radial–directional quadrature can be extended to calculate survivor moments. Conditional on direction $\mathbf{u}$, survival truncates the squared radius at

$$R^2 < \frac{1}{m_{\mathbf{u}}(x)}. \quad (990)$$

Hence,

$$\mathbb{E} \left[\mathbf{w} \mathbf{w}^\top \mathbf{1}_{\{X_{\text{tr}} > x\}} \right] = \mathbb{E}_{\mathbf{u}} \left[\mathbf{u} \mathbf{u}^\top \mathbb{E}_R \left[R^2 \mathbf{1}_{\left\{ R^2 < \frac{1}{m_{\mathbf{u}}(x)} \right\}} \right] \right]. \quad (991)$$

The inner truncated radial moment is available from incomplete gamma functions. Thus, conditional survivor statistics can be recovered without solving an absorbing Liouville PDE.

15.8 Transition-producing directions

Several definitions of a “dominant” disturbance direction are possible, and they need not agree:

- (i) the direction maximizing energy at a selected station,

$$\arg \max_{\|\mathbf{u}\|=1} \mathbf{u}^\top \mathbf{G}(x) \mathbf{u}; \quad (992)$$

- (ii) the direction maximizing energy over the complete interval,

$$\arg \max_{\|\mathbf{u}\|=1} m_{\mathbf{u}}(x_{\text{max}}); \quad (993)$$

- (iii) the direction maximizing the conditional transition probability $\Psi_x(\mathbf{u})$, which is equivalent to maximizing $m_{\mathbf{u}}(x)$; and
- (iv) the angular region contributing the most integrated probability after accounting for its spherical measure.

The first three are pointwise optimization concepts. The fourth is a probability-mass concept and can favor a broader subspace rather than a single optimal vector.

For a numerical angular rule, the weighted contribution

$$c_k(x) = \omega_k \overline{F}_{\chi_r^2} \left[\frac{1}{m_k(x)} \right] \quad (994)$$

includes both gain and represented angular measure. Ranking directions by c_k is therefore more informative for probability interpretation than ranking them by m_k alone. For nonuniform or adaptive angular rules, the weights must be included; otherwise, densely sampled regions would appear artificially important.

15.9 Dimension reduction and omitted transition mechanisms

Transition-oriented dimension reduction is necessary for computational efficiency but introduces a potentially important model error. A basis selected from the largest initial covariance eigenvalues can omit low-variance directions that undergo strong downstream amplification.

The relevant family is not the inlet covariance alone but

$$\{\mathbf{G}(x) : 0 \leq x \leq x_{\max}\}. \quad (995)$$

A suitable fixed basis should represent:

- dominant eigenspaces at multiple stations;
- rotating non-normal amplification directions;
- narrow streamwise gain peaks;
- retained–omitted cross-coupling; and
- directions near the probability-sensitive transition boundary.

A small error in mean energy does not guarantee a small error in the transition CDF. First-passage probabilities depend on tails and maxima, which can be more sensitive to omitted directions than low-order moments.

Increasing the reduced stochastic rank does not necessarily yield a monotone sequence of transition probabilities. The reduced Gaussian model changes dimension, radial distribution, and covariance content as modes are added. Rank adequacy should therefore be established by convergence of the reported CDF and by residual diagnostics, not by assuming that reduced probabilities approach the full value from one side.

15.10 Numerical resolution of non-normal peaks

The continuous first-transition event depends on the maximum over all streamwise locations. An endpoint-only calculation has a systematic one-sided bias:

$$F_{\text{grid}}(x) \leq F_{\text{continuous}}(x). \quad (996)$$

This property is useful because it identifies the direction of the error, but it does not indicate its magnitude.

The error can be significant when the non-normal propagator produces a narrow amplification peak. The streamwise resolution needed for the first-passage calculation can therefore be more demanding than that needed to represent the state accurately at selected output stations.

Cubic Hermite interpolation uses both the gain and its derivative to locate interior stationary points efficiently. It is nevertheless an interpolation model rather than a guaranteed upper envelope. Adaptive midpoint refinement or derivative-based bounds remain necessary when high confidence in the continuous maximum is required.

Streamwise and angular resolution interact. A narrow streamwise peak may occur only for a narrow set of angular directions. Missing either feature can underestimate transition. Convergence studies should vary the streamwise and angular resolutions independently and then jointly confirm the final result.

15.11 Choice and interpretation of the transition threshold

The probability method is exact for the event defined by the selected energy threshold. The physical meaning of that event depends entirely on the adequacy of

$$e(x) \geq e_{\text{thres}}(x) \quad (997)$$

as a transition criterion.

Several interpretations are possible:

- (i) threshold crossing is a proxy for the onset of nonlinear effects;
- (ii) threshold crossing identifies a disturbance amplitude associated empirically with turbulent breakdown;
- (iii) threshold crossing marks loss of validity of the linear OWNS approximation; or
- (iv) threshold crossing defines an operational engineering transition location without claiming to resolve the complete breakdown process.

These interpretations should not be conflated. In particular, a linear energy threshold does not derive turbulent breakdown from the Navier–Stokes equations. It maps a linear stochastic amplification model to a transition event.

The predicted CDF may be highly sensitive to the threshold because the Gaussian tail can change rapidly with the critical radius. If the threshold is uncertain, reporting one transition curve can give a misleading impression of precision. A threshold-sensitivity family, or a calibrated threshold distribution, is preferable.

If the threshold is estimated from transition data, uncertainty in its parameters should be propagated to the CDF. For threshold parameters θ , the predictive transition probability is

$$F_{\text{pred}}(x) = \int F_{X_{\text{tr}}}(x \mid \theta) p(\theta \mid \mathcal{D}) d\theta, \quad (998)$$

in a Bayesian formulation, where $\mathcal{D}$ denotes calibration data. A frequentist alternative reports confidence intervals obtained from parameter uncertainty or bootstrap resampling.

15.12 Limitations of linear OWNS as a transition model

The linear OWNS model provides a spatially evolving disturbance field before transition. Its use in the present first-passage calculation has several limitations.

15.12.1 Neglect of nonlinear disturbance interactions

Once disturbances become sufficiently large, nonlinear interactions can alter their growth, generate new frequencies and wavenumbers, and change the transition pathway. The linear relation

$$\mathbf{q}(x) = \mathbf{B}(x)\mathbf{w} \quad (999)$$

then ceases to be exact. The energy is no longer a quadratic form in a fixed Gaussian vector, and the radial integration generally loses its closed form.

A threshold can be interpreted as terminating the linear calculation before these nonlinear effects must be modeled. The credibility of that interpretation depends on whether the chosen threshold lies near the limit of linear validity and has been connected to observed transition behavior.

15.12.2 Randomness only at the inlet

The current model assumes that all stochasticity is contained in the initial condition. It excludes continuously distributed uncertainty such as:

- downstream free-stream forcing;
- surface roughness distributed over x ;
- acoustic or vortical forcing entering through the outer boundary;
- uncertain base-flow evolution; and
- model error accumulated during propagation.

With distributed Gaussian forcing, the state can remain Gaussian, but it is no longer a deterministic function of one finite inlet vector unless the entire forcing field is parameterized by an enlarged Gaussian coordinate. A finite Karhunen–Loève representation of the distributed forcing could restore a common-coordinate formulation, at the cost of a larger stochastic dimension. Alternatively, an absorbing Fokker–Planck or stochastic first-passage method may be required.

15.12.3 No post-threshold evolution

The first-transition method deliberately does not model the flow after threshold crossing. It cannot predict:

- the detailed nonlinear breakdown sequence;
- turbulent spot formation and growth;
- intermittency generated by spot convection;
- post-transition mean-flow modification; or
- physical relaminarization.

If these quantities are required, the first-transition distribution can serve as an input to a downstream intermittency or turbulent-spot model rather than as a complete transition-zone model.

15.12.4 Dependence on the selected energy norm

Different positive-semidefinite energy metrics can rank disturbance structures differently. For example, kinetic energy, compressible disturbance energy, wall-shear energy, and measurement-weighted output norms need not identify the same dangerous directions.

The transition threshold and energy norm should be treated as a pair. Changing $\mathbf{H}(x)$ without recalibrating $e_{\text{thres}}(x)$ changes the physical event being modeled.

15.13 Extension to broadband disturbances

The development has been stated for a fixed temporal frequency and spanwise wavenumber. A physical transition criterion may depend on the combined energy of many spectral components.

Let the joint Gaussian coordinate be

$$\mathbf{w}_{\text{all}} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_{r_{\text{all}}}), \quad (1000)$$

and let the total normalized energy be

$$y_{\text{all}}(x) = \mathbf{w}_{\text{all}}^\top \mathbf{G}_{\text{all}}(x) \mathbf{w}_{\text{all}}. \quad (1001)$$

The same first-transition and radial-directional formulas apply with dimension r_{all} .

If the spectral components are independent and the total energy is additive, $\mathbf{G}_{\text{all}}$ is block diagonal. Cross-spectral correlations create off-diagonal blocks. In either case, the angular dimension can become large, making transition-oriented reduction especially important.

Computing separate transition CDFs for each component and then combining them as independent failure events is generally not equivalent to thresholding total disturbance energy. The physical transition criterion should determine whether the relevant event is:

- (i) exceedance by any individual component;
- (ii) exceedance by the sum of component energies; or
- (iii) exceedance by a nonlinear or measurement-based combination.

Each choice defines a different first-passage problem.

15.14 Extension to nonlinear propagation

For nonlinear spatial dynamics,

$$\frac{d\mathbf{q}}{dx} = \mathbf{f}[\mathbf{q}(x), x], \quad \mathbf{q}(0) = \mathbf{B}_0 \mathbf{w}, \quad (1002)$$

the first-transition location remains

$$X_{\text{tr}}(\mathbf{w}) = \inf \{x : e[\mathbf{q}(x; \mathbf{w}), x] \geq e_{\text{thres}}(x)\}. \quad (1003)$$

However,

$$e[\mathbf{q}(x; \mathbf{w}), x] \quad (1004)$$

is not generally homogeneous of degree two in $\mathbf{w}$. The critical radius can depend nonlinearly on both radius and direction, and the radial probability can no longer be integrated from one directional maximum.

Potential extensions include:

- sparse-grid collocation in the initial Gaussian coordinates;
- adaptive polynomial chaos with explicit first-passage detection;
- transfer-operator approximations;
- absorbing Liouville methods in a nonlinear reduced state;

- rare-event importance sampling; and
- surrogate models for $X_{\text{tr}}(\mathbf{w})$.

The linear radial–directional solution remains useful as a baseline, proposal distribution, or control variate for these nonlinear methods.

15.15 Extension to uncertain model parameters

The present derivation conditions on a deterministic base flow, propagator, energy metric, and threshold. Let $\boldsymbol{\theta}$ represent uncertain parameters in these quantities. Then the predictive CDF is

$$F_{\text{pred}}(x) = \mathbb{E}_{\boldsymbol{\theta}} [F_{X_{\text{tr}}}(x \mid \boldsymbol{\theta})]. \quad (1005)$$

Conditional on $\boldsymbol{\theta}$, the radial–directional method can be applied if the state remains linear in the Gaussian initial coordinate.

A nested calculation over both $\boldsymbol{\theta}$ and $\mathbf{u}$ may be expensive. Possible reductions include:

- surrogate modeling of the reduced matrix family $\mathbf{G}(x; \boldsymbol{\theta})$;
- polynomial or low-rank parameter expansions;
- multifidelity OWNS calculations;
- common angular quadrature across parameter samples; and
- sensitivity-based parameter screening.

Aleatory disturbance uncertainty and epistemic model uncertainty should be reported separately when possible. The former defines the conditional transition distribution, while the latter expresses uncertainty in that predicted distribution.

15.16 Connection to reliability analysis

The transition event can be viewed as failure of a time-varying reliability system, with x replacing time. The limit-state function through station x is

$$g_x(\mathbf{w}) = 1 - \max_{0 \leq \xi \leq x} \mathbf{w}^\top \mathbf{G}(\xi) \mathbf{w}. \quad (1006)$$

Transition occurs when

$$g_x(\mathbf{w}) \leq 0. \quad (1007)$$

For rare transition, the most probable failure point solves

$$\min_{\mathbf{w}} \frac{1}{2} \|\mathbf{w}\|_2^2 \quad \text{subject to} \quad \max_{\xi \leq x} \mathbf{w}^\top \mathbf{G}(\xi) \mathbf{w} \geq 1. \quad (1008)$$

The smallest squared norm satisfying the constraint is

$$\beta_{\text{FORM}}^2(x) = \frac{1}{\max_{\xi \leq x} \lambda_{\max}[\mathbf{G}(\xi)]}. \quad (1009)$$

The corresponding direction is a dominant eigenvector at a station attaining the largest eigenvalue.

This design point identifies the most probable individual transition-causing realization in the asymptotic rare-event sense. It does not account for the total angular volume of competing directions. The radial–directional method includes that volume exactly within its angular-integration accuracy. FORM and SORM can nevertheless be useful for:

- diagnosing rare-event directions;
- constructing angular importance densities;
- initializing reduced bases; and
- validating asymptotic trends at high thresholds.

If several stations or disturbance mechanisms have comparable design points, a multiple-design-point reliability treatment is required.

15.17 Relationship to intermittency and transition zones

The cumulative first-transition probability

$$F_{X_{\text{tr}}}(x) \tag{1010}$$

resembles an intermittency curve in that it is nondecreasing and represents the fraction of realizations that have initiated transition. It should not automatically be equated with local temporal or spatial turbulent intermittency.

Classical intermittency describes the fraction of time or area occupied by turbulent flow at a specified location, often accounting for the birth, convection, and growth of turbulent spots. The present CDF describes the probability that the linear disturbance energy first reached a threshold upstream. To connect the two, one may require an additional model for:

- (i) delay between threshold crossing and turbulent breakdown;
- (ii) convection from the first-crossing location to the observation location;
- (iii) turbulent-spot growth and merging; and
- (iv) possible variation of these processes across realizations.

If X_{tr} is interpreted as a spot-birth or breakdown location, its distribution can provide a stochastic source term for a downstream intermittency model. This is a promising connection but is outside the present linear first-passage calculation.

15.18 Recommended interpretation of reported results

A complete result should report more than one terminal transition probability. At minimum, the following quantities are recommended:

1. The local exceedance probability $p_{\text{local}}(x)$.
2. The cumulative first-transition CDF $F_{X_{\text{tr}}}(x)$.
3. The survival probability $S(x)$.
4. Interval transition probabilities or a conservatively differentiated transition-location density.
5. The right-censoring probability at the end of the modeled domain.
6. Transition quantiles, with unreached quantiles explicitly marked as censored.

7. Sensitivity to the transition threshold and inlet disturbance amplitude.
8. Streamwise, angular, stochastic-rank, and OWNS convergence measures.
9. Diagnostics of the leading normalized-gain directions and their streamwise rotation.
10. Transition-weighted directional structures, including angular weights.
11. A comparison with independent trajectory Monte Carlo or another reference method.
12. A clear statement that the prediction is conditional on the chosen linear OWNS model, Gaussian inlet statistics, energy norm, and threshold definition.

Reporting these quantities distinguishes numerical probability accuracy from physical-model uncertainty and makes clear what aspect of transition is being predicted.

15.19 Principal strengths and limitations

The principal strengths of the method are:

- it gives an irreversible, history-aware transition probability;
- it exactly retains the joint streamwise dependence generated by the linear propagator;
- it accommodates strongly non-normal and nonmonotone amplification;
- it integrates the Gaussian radial uncertainty analytically;
- it avoids a high-dimensional physical-state Liouville equation;
- it naturally handles right censoring;
- it supports transition-conditioned interpretation of initial disturbance directions; and
- its expensive OWNS propagation can be reused for threshold and amplitude studies.

Its principal limitations are:

- accuracy depends on the physical validity of a linear energy-threshold transition criterion;
- angular integration can remain expensive at high stochastic dimension;
- a transition-oriented basis is needed when the inlet covariance rank is large;
- narrow streamwise and angular peaks require adaptive resolution;
- nonlinear interactions and distributed stochastic forcing are not represented in the basic formulation;
- the method predicts first threshold crossing, not the post-transition turbulent state; and
- uncertainty in the base flow, covariance model, energy norm, and threshold can exceed the numerical integration error.

Within these limitations, the method provides a direct way to use the history already contained in the propagated factor $\mathbf{B}(x)$. Rather than replacing the local energy PDF with an empirical memory model, it derives the first-transition distribution from the common Gaussian realization and the complete non-normal OWNS propagator.

16 Conclusions

This work developed a history-aware probabilistic formulation for predicting the first transition location of Gaussian disturbances propagated through a boundary layer by the linear One-Way Navier–Stokes equations. The method replaces the stationwise energy-exceedance probability

$$p_{\text{local}}(x) = \mathbb{P}[e(x) \geq e_{\text{thres}}(x)] \quad (1011)$$

with the cumulative first-transition probability

$$F_{X_{\text{tr}}}(x) = \mathbb{P}[X_{\text{tr}} \leq x], \quad (1012)$$

where

$$X_{\text{tr}} = \inf \{x : e(x) \geq e_{\text{thres}}(x)\}. \quad (1013)$$

The first-transition CDF classifies a realization as transitioned at all stations downstream of its first threshold encounter, even if its energy subsequently falls below the threshold.

The initial disturbance is represented as

$$\mathbf{q}(0) = \mathbf{B}_0 \mathbf{w}, \quad \mathbf{w} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_r). \quad (1014)$$

Under linear OWNS propagation,

$$\mathbf{q}(x) = \mathbf{M}(0, x) \mathbf{B}_0 \mathbf{w} = \mathbf{B}(x) \mathbf{w}. \quad (1015)$$

The same Gaussian vector $\mathbf{w}$ labels a realization at every streamwise station. This common stochastic basis retains the streamwise dependence and relative rotation of propagated disturbance directions generated by the strongly non-normal OWNS propagator.

For a quadratic disturbance-energy norm,

$$e(x) = \mathbf{q}(x)^* \mathbf{H}(x) \mathbf{q}(x), \quad (1016)$$

the normalized energy is a quadratic form in the initial Gaussian coordinate:

$$\frac{e(x)}{e_{\text{thres}}(x)} = \mathbf{w}^\top \mathbf{G}(x) \mathbf{w}, \quad (1017)$$

where, for real stochastic coordinates,

$$\mathbf{G}(x) = \frac{\Re[\mathbf{B}(x)^* \mathbf{H}(x) \mathbf{B}(x)]}{e_{\text{thres}}(x)}. \quad (1018)$$

The desired transition probability is consequently

$$F_{X_{\text{tr}}}(x) = \mathbb{P} \left[\sup_{0 \leq \xi \leq x} \mathbf{w}^\top \mathbf{G}(\xi) \mathbf{w} \geq 1 \right]. \quad (1019)$$

This is a first-passage problem for a finite-dimensional Gaussian quadratic-form process.

Writing the Gaussian vector as

$$\mathbf{w} = \mathbf{R} \mathbf{u}, \quad R^2 \sim \chi_r^2, \quad \mathbf{u} \sim \text{Unif}(\mathbb{S}^{r-1}), \quad (1020)$$

permits the radial probability to be evaluated analytically. For each initial direction, define

$$m_{\mathbf{u}}(x) = \max_{0 \leq \xi \leq x} \mathbf{u}^\top \mathbf{G}(\xi) \mathbf{u}. \quad (1021)$$

The exact finite-dimensional transition CDF is then

$$F_{X_{\text{tr}}}(x) = \mathbb{E}_{\mathbf{u}} \left[\bar{F}_{\chi_r^2} \left(\frac{1}{m_{\mathbf{u}}(x)} \right) \right]. \quad (1022)$$

The corresponding survival function is

$$S(x) = \mathbb{E}_{\mathbf{u}} \left[F_{\chi_r^2} \left(\frac{1}{m_{\mathbf{u}}(x)} \right) \right]. \quad (1023)$$

The directional running maximum is the mechanism that supplies streamwise memory. A downstream gain peak increases the transition probability only when it establishes a new record for the corresponding initial direction. Threshold recrossings and later sub-record peaks do not count the same realization again.

The radial–directional formulation is equivalent to an absorbing Liouville problem. In the physical OWNS state space, the survivor density is transported by the spatial dynamics and removed at its first encounter with the transition boundary. In the initial stochastic coordinates, the Gaussian density is stationary and the safe set is the evolving intersection

$$\mathcal{S}(x) = \bigcap_{0 \leq \xi \leq x} \left\{ \mathbf{w} : \mathbf{w}^\top \mathbf{G}(\xi) \mathbf{w} < 1 \right\}. \quad (1024)$$

The radial–directional method evaluates the Gaussian probability content of this set without solving a high-dimensional density PDE.

The principal methodological conclusions are as follows:

1. **The propagated factor contains the required history.** The common-basis factor $\mathbf{B}(x) = \mathbf{M}(0, x) \mathbf{B}_0$ preserves the identity of each initial Gaussian realization. Independently diagonalizing the one-point covariance at every station preserves local energy PDFs but generally discards the dependence needed for first-transition prediction.
2. **Strong non-normality does not invalidate the first-passage formulation.** Non-normal amplification and rotation are retained in the full matrix family $\mathbf{G}(x)$. They complicate numerical propagation, streamwise maximum detection, and stochastic reduction, but the first-transition event remains well defined.
3. **The Gaussian radius can be removed analytically.** Each angular evaluation accounts for every Gaussian amplitude along one initial direction. This is a Rao–Blackwellization of direct trajectory Monte Carlo and avoids sampling a binary transition outcome for every random amplitude.
4. **Streamwise maxima must be resolved continuously.** Using only stored OWNS stations generally underestimates the transition probability. Cubic Hermite interpolation, adaptive streamwise refinement, or derivative-based bounds are required when transient amplification peaks are narrow.

5. **Angular resolution controls the remaining probability error.** Deterministic spherical quadrature is appropriate at low stochastic dimension, while randomized quasi-Monte Carlo, stratification, or angular importance sampling is more suitable at moderate or high dimension.
6. **Dimension reduction must be transition oriented.** Initial covariance energy alone is not a sufficient truncation criterion for a non-normal propagator. A suitable fixed basis should preserve the dominant and rotating eigenspaces of $\mathbf{G}(x)$, including localized transient-growth directions and retained–omitted coupling.
7. **Numerical stabilization must preserve the original Gaussian coordinates.** QR, SVD, and scaling transformations may be used to stabilize propagation, but all accumulated coordinate maps must be retained. Replacing the propagated factor by an independently normalized basis changes the stochastic process and its first-transition distribution.
8. **The output is a censored transition-location distribution.** The method provides the survival function, transition CDF, interval probabilities, hazard, quantiles, and probability of no transition within the computational domain. Unreached quantiles should be reported as censored rather than assigned to the terminal station.
9. **The method predicts threshold crossing, not the complete turbulent transition process.** Its physical validity depends on the linear OWNS approximation, initial covariance model, selected energy norm, and transition threshold. The post-threshold turbulent state is not modeled.

For practical OWNS calculations, the recommended procedure is:

1. factor the inlet covariance and define a consistent real or proper-complex Gaussian convention;
2. propagate the factor while preserving the mapping to the original stochastic coordinate;
3. construct the normalized energy matrices $\mathbf{G}(x)$;
4. form a fixed transition-oriented reduced basis if necessary;
5. evaluate directional gain histories using spherical quadrature or RQMC directions;
6. resolve their continuous streamwise running maxima;
7. analytically evaluate the corresponding chi-square radial tails;
8. angularly average the conditional probabilities to obtain $S(x)$ and $F_{X_{\text{tr}}}(x)$;
9. compute interval transition probabilities and the right-censoring mass; and
10. verify convergence with respect to OWNS resolution, stochastic rank, streamwise maxima, and angular integration.

The resulting calculation uses the history already present in the propagated stochastic factor rather than introducing an empirical station-to-station memory model. Within the linear Gaussian energy-threshold framework, it provides a direct first-passage distribution that remains valid under nonmonotone, strongly non-normal disturbance amplification.

Future work should address:

- calibration and uncertainty quantification of the transition threshold using censored transition-location data;
- broadband and cross-spectrally correlated disturbances;
- distributed downstream stochastic forcing;
- nonlinear OWNS or reduced nonlinear propagation;
- coupling of the first-transition distribution to turbulent-spot and intermittency models;
- adaptive angular importance sampling guided by reliability design points;
- rigorous streamwise and stochastic-reduction probability bounds; and
- validation against DNS, experiments, and flight data across multiple boundary-layer configurations.

These extensions would retain the central principle of the present approach: transition should be modeled as a first event along a coherently propagated realization, rather than as a collection of independent local exceedance probabilities.

A Derivation of the Radial–Directional Formula

This appendix gives a complete derivation of the radial–directional representation of the first-transition probability. The derivation is stated for a real standard Gaussian vector

$$\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_r), \quad (1025)$$

and a continuous family of real symmetric positive-semidefinite matrices

$$\mathbf{G}(x) = \mathbf{G}(x)^\top \succeq \mathbf{0}, \quad x \in [0, x_{\max}]. \quad (1026)$$

The corresponding extension to proper complex Gaussian coordinates is given in section B.

A.1 Statement of the result

Define the normalized energy process

$$y(x; \mathbf{w}) = \mathbf{w}^\top \mathbf{G}(x) \mathbf{w}, \quad (1027)$$

and the first-transition location

$$X_{\text{tr}}(\mathbf{w}) = \inf \{x \in [0, x_{\max}] : y(x; \mathbf{w}) \geq 1\}, \quad (1028)$$

with the convention

$$\inf \emptyset = \infty. \quad (1029)$$

For a unit direction

$$\mathbf{u} \in \mathbb{S}^{r-1}, \quad (1030)$$

define the directional gain and its running maximum by

$$a_{\mathbf{u}}(x) = \mathbf{u}^\top \mathbf{G}(x) \mathbf{u}, \quad (1031)$$

$$m_{\mathbf{u}}(x) = \max_{0 \leq \xi \leq x} a_{\mathbf{u}}(\xi). \quad (1032)$$

The maximum exists because $\mathbf{G}(\xi)$ is continuous and $[0, x]$ is compact.

The principal result is

$$S(x) = \mathbb{P}(X_{\text{tr}} > x) = \mathbb{E}_{\mathbf{u}} \left[F_{\chi_r^2} \left(\frac{1}{m_{\mathbf{u}}(x)} \right) \right], \quad (1033)$$

and

$$F_{X_{\text{tr}}}(x) = \mathbb{P}(X_{\text{tr}} \leq x) = \mathbb{E}_{\mathbf{u}} \left[\bar{F}_{\chi_r^2} \left(\frac{1}{m_{\mathbf{u}}(x)} \right) \right], \quad (1034)$$

where

$$\mathbf{u} \sim \text{Unif}(\mathbb{S}^{r-1}). \quad (1035)$$

When $m_{\mathbf{u}}(x) = 0$, the convention

$$\frac{1}{m_{\mathbf{u}}(x)} = \infty \quad (1036)$$

is used, so that

$$F_{\chi_r^2}(\infty) = 1, \quad (1037)$$

$$\bar{F}_{\chi_r^2}(\infty) = 0. \quad (1038)$$

A.2 Polar decomposition of a standard Gaussian vector

The standard Gaussian density on $\mathbb{R}^r$ is

$$\phi_r(\mathbf{w}) = \frac{1}{(2\pi)^{r/2}} \exp \left(-\frac{1}{2} \|\mathbf{w}\|_2^2 \right). \quad (1039)$$

For $\mathbf{w} \neq \mathbf{0}$, introduce polar coordinates

$$\mathbf{w} = R\mathbf{u}, \quad R = \|\mathbf{w}\|_2, \quad \mathbf{u} = \frac{\mathbf{w}}{\|\mathbf{w}\|_2}. \quad (1040)$$

Here,

$$R \in (0, \infty), \quad \mathbf{u} \in \mathbb{S}^{r-1}. \quad (1041)$$

The point $\mathbf{w} = \mathbf{0}$ has Gaussian probability zero and does not affect any probability integral.

The Euclidean volume element becomes

$$d\mathbf{w} = R^{r-1} dR d\sigma(\mathbf{u}), \quad (1042)$$

where $d\sigma$ is surface measure on the unit sphere. Therefore,

$$\phi_r(\mathbf{w}) d\mathbf{w} = \frac{1}{(2\pi)^{r/2}} \exp \left(-\frac{R^2}{2} \right) R^{r-1} dR d\sigma(\mathbf{u}). \quad (1043)$$

The surface area of the unit sphere is

$$|\mathbb{S}^{r-1}| = \frac{2\pi^{r/2}}{\Gamma(r/2)}. \quad (1044)$$

Using this identity, the Gaussian measure factors as

$$\phi_r(\mathbf{w}) d\mathbf{w} = f_R(R) dR \frac{d\sigma(\mathbf{u})}{|\mathbb{S}^{r-1}|}, \quad (1045)$$

where

$$f_R(R) = \frac{2^{1-r/2}}{\Gamma(r/2)} R^{r-1} \exp\left(-\frac{R^2}{2}\right), \quad R > 0. \quad (1046)$$

This is the density of a chi-distributed random variable with r degrees of freedom.

Equation (1045) shows that:

- (i) R and $\mathbf{u}$ are independent;
- (ii) $\mathbf{u}$ is uniformly distributed on $\mathbb{S}^{r-1}$; and
- (iii) R^2 has a chi-square distribution with r degrees of freedom.

To verify the final statement, let

$$Z = R^2. \quad (1047)$$

Then

$$R = \sqrt{Z}, \quad dR = \frac{1}{2\sqrt{Z}} dZ. \quad (1048)$$

Substitution into eq. (1046) gives

$$f_Z(z) = \frac{z^{r/2-1} \exp(-z/2)}{2^{r/2} \Gamma(r/2)}, \quad z > 0, \quad (1049)$$

which is the density of

$$Z = R^2 \sim \chi_r^2. \quad (1050)$$

A.3 First-transition event along a fixed ray

For a realization $\mathbf{w} = R\mathbf{u}$, the normalized energy is

$$\begin{aligned} y(x; R, \mathbf{u}) &= (R\mathbf{u})^\top \mathbf{G}(x) (R\mathbf{u}) \\ &= R^2 \mathbf{u}^\top \mathbf{G}(x) \mathbf{u} \\ &= R^2 a_{\mathbf{u}}(x). \end{aligned} \quad (1051)$$

The maximum normalized energy attained through station x is

$$\begin{aligned} \max_{0 \leq \xi \leq x} y(\xi; R, \mathbf{u}) &= \max_{0 \leq \xi \leq x} [R^2 a_{\mathbf{u}}(\xi)] \\ &= R^2 \max_{0 \leq \xi \leq x} a_{\mathbf{u}}(\xi) \\ &= R^2 m_{\mathbf{u}}(x). \end{aligned} \quad (1052)$$

The factor R^2 can be taken outside the maximum because it is nonnegative and independent of ξ .

It follows that the transition event through x is

$$\begin{aligned}\{X_{\text{tr}} \leq x\} &= \left\{ \max_{0 \leq \xi \leq x} y(\xi; R, \mathbf{u}) \geq 1 \right\} \\ &= \{R^2 m_{\mathbf{u}}(x) \geq 1\}.\end{aligned}\tag{1053}$$

Similarly, the survival event is

$$\{X_{\text{tr}} > x\} = \{R^2 m_{\mathbf{u}}(x) < 1\}.\tag{1054}$$

If $m_{\mathbf{u}}(x) > 0$, define the critical squared radius

$$z_{\text{crit}}(\mathbf{u}, x) = \frac{1}{m_{\mathbf{u}}(x)}.\tag{1055}$$

Then

$$X_{\text{tr}} > x \iff R^2 < z_{\text{crit}}(\mathbf{u}, x),\tag{1056}$$

$$X_{\text{tr}} \leq x \iff R^2 \geq z_{\text{crit}}(\mathbf{u}, x).\tag{1057}$$

If $m_{\mathbf{u}}(x) = 0$, then

$$a_{\mathbf{u}}(\xi) = 0 \quad \text{for every } \xi \in [0, x],\tag{1058}$$

because each directional gain is nonnegative. Hence,

$$y(\xi; R, \mathbf{u}) = 0 \quad \text{for every finite } R,\tag{1059}$$

and every realization on that ray survives through x . This case is represented by

$$z_{\text{crit}}(\mathbf{u}, x) = \infty.\tag{1060}$$

A.4 Conditional survival probability

Condition on a fixed direction $\mathbf{u}$. Since R^2 is independent of $\mathbf{u}$,

$$\begin{aligned}S(x \mid \mathbf{u}) &= \mathbb{P}(X_{\text{tr}} > x \mid \mathbf{u}) \\ &= \mathbb{P}[R^2 m_{\mathbf{u}}(x) < 1 \mid \mathbf{u}].\end{aligned}\tag{1061}$$

For $m_{\mathbf{u}}(x) > 0$,

$$\begin{aligned}S(x \mid \mathbf{u}) &= \mathbb{P}\left[R^2 < \frac{1}{m_{\mathbf{u}}(x)}\right] \\ &= F_{\chi_r^2}\left[\frac{1}{m_{\mathbf{u}}(x)}\right].\end{aligned}\tag{1062}$$

The same expression remains valid for $m_{\mathbf{u}}(x) = 0$ under the infinity convention.

The conditional transition probability is the complementary radial tail:

$$\begin{aligned}F_{X_{\text{tr}}}(x \mid \mathbf{u}) &= \mathbb{P}(X_{\text{tr}} \leq x \mid \mathbf{u}) \\ &= \mathbb{P}\left[R^2 \geq \frac{1}{m_{\mathbf{u}}(x)}\right] \\ &= \overline{F}_{\chi_r^2}\left[\frac{1}{m_{\mathbf{u}}(x)}\right].\end{aligned}\tag{1063}$$

Whether equality is assigned to transition has no effect on this conditional probability because R^2 has a continuous distribution:

$$\mathbb{P} \left[R^2 = \frac{1}{m_{\mathbf{u}}(x)} \right] = 0 \quad (1064)$$

for every direction with finite positive $m_{\mathbf{u}}(x)$.

A.5 Angular averaging

The law of total probability gives

$$\begin{aligned} S(x) &= \mathbb{E}_{\mathbf{u}} [\mathbb{P}(X_{\text{tr}} > x \mid \mathbf{u})] \\ &= \mathbb{E}_{\mathbf{u}} \left[F_{\chi_r^2} \left(\frac{1}{m_{\mathbf{u}}(x)} \right) \right]. \end{aligned} \quad (1065)$$

This proves eq. (1033).

Likewise,

$$\begin{aligned} F_{X_{\text{tr}}}(x) &= \mathbb{E}_{\mathbf{u}} [\mathbb{P}(X_{\text{tr}} \leq x \mid \mathbf{u})] \\ &= \mathbb{E}_{\mathbf{u}} \left[\bar{F}_{\chi_r^2} \left(\frac{1}{m_{\mathbf{u}}(x)} \right) \right], \end{aligned} \quad (1066)$$

which proves eq. (1034).

Because the angular distribution is uniform,

$$S(x) = \frac{1}{|\mathbb{S}^{r-1}|} \int_{\mathbb{S}^{r-1}} F_{\chi_r^2} \left[\frac{1}{m_{\mathbf{u}}(x)} \right] d\sigma(\mathbf{u}), \quad (1067)$$

and

$$F_{X_{\text{tr}}}(x) = \frac{1}{|\mathbb{S}^{r-1}|} \int_{\mathbb{S}^{r-1}} \bar{F}_{\chi_r^2} \left[\frac{1}{m_{\mathbf{u}}(x)} \right] d\sigma(\mathbf{u}). \quad (1068)$$

Since

$$F_{\chi_r^2}(z) + \bar{F}_{\chi_r^2}(z) = 1, \quad (1069)$$

the formulas satisfy

$$S(x) + F_{X_{\text{tr}}}(x) = 1. \quad (1070)$$

A.6 Direct derivation from the survivor-set integral

The same result can be obtained directly from the Gaussian measure of the survivor set. Recall that

$$\mathcal{S}(x) = \bigcap_{0 \leq \xi \leq x} \left\{ \mathbf{w} : \mathbf{w}^\top \mathbf{G}(\xi) \mathbf{w} < 1 \right\}. \quad (1071)$$

Substitute $\mathbf{w} = R\mathbf{u}$:

$$\begin{aligned} \mathbf{w} \in \mathcal{S}(x) &\iff R^2 \mathbf{u}^\top \mathbf{G}(\xi) \mathbf{u} < 1 \quad \text{for every } \xi \leq x \\ &\iff R^2 m_{\mathbf{u}}(x) < 1. \end{aligned} \quad (1072)$$

Therefore,

$$\mathcal{S}(x) = \left\{ R\mathbf{u} : \mathbf{u} \in \mathbb{S}^{r-1}, \quad 0 \leq R < \rho_{\mathbf{u}}(x) \right\}, \quad (1073)$$

where

$$\rho_{\mathbf{u}}(x) = \frac{1}{\sqrt{m_{\mathbf{u}}(x)}}. \quad (1074)$$

When $m_{\mathbf{u}}(x) = 0$, $\rho_{\mathbf{u}}(x) = \infty$.

The survival probability is

$$S(x) = \int_{\mathcal{S}(x)} \phi_r(\mathbf{w}) \, \mathrm{d}\mathbf{w}. \quad (1075)$$

Using polar coordinates and the radial description of the safe set,

$$S(x) = \frac{1}{(2\pi)^{r/2}} \int_{\mathbb{S}^{r-1}} \int_0^{\rho_{\mathbf{u}}(x)} \exp\left(-\frac{R^2}{2}\right) R^{r-1} \, \mathrm{d}R \, \mathrm{d}\sigma(\mathbf{u}). \quad (1076)$$

Using eq. (1045),

$$S(x) = \frac{1}{|\mathbb{S}^{r-1}|} \int_{\mathbb{S}^{r-1}} \left[\int_0^{\rho_{\mathbf{u}}(x)} f_R(R) \, \mathrm{d}R \right] \, \mathrm{d}\sigma(\mathbf{u}). \quad (1077)$$

The inner integral is

$$\begin{aligned} \int_0^{\rho_{\mathbf{u}}(x)} f_R(R) \, \mathrm{d}R &= \mathbb{P}[R \leq \rho_{\mathbf{u}}(x)] \\ &= \mathbb{P}[R^2 \leq \rho_{\mathbf{u}}^2(x)] \\ &= F_{\chi_r^2} \left[\frac{1}{m_{\mathbf{u}}(x)} \right]. \end{aligned} \quad (1078)$$

Substitution into eq. (1077) again yields eq. (1033).

This derivation shows geometrically that the method integrates the Gaussian measure of the intersection of all upstream safe ellipsoids. The intersection need not itself be an ellipsoid, but it is star-shaped with respect to the origin and therefore has one radial boundary value for each direction.

A.7 Convexity and radial representation of the survivor set

For each station ξ , the instantaneous safe set is

$$\mathcal{E}(\xi) = \left\{ \mathbf{w} : \mathbf{w}^\top \mathbf{G}(\xi) \mathbf{w} < 1 \right\}. \quad (1079)$$

Because $\mathbf{G}(\xi) \succeq \mathbf{0}$, the function

$$\mathbf{w} \mapsto \mathbf{w}^\top \mathbf{G}(\xi) \mathbf{w} \quad (1080)$$

is convex. Hence, each $\mathcal{E}(\xi)$ is convex, and their intersection $\mathcal{S}(x)$ is convex.

Each instantaneous set is also centrally symmetric:

$$\mathbf{w} \in \mathcal{E}(\xi) \iff -\mathbf{w} \in \mathcal{E}(\xi). \quad (1081)$$

Therefore,

$$\mathbf{w} \in \mathcal{S}(x) \iff -\mathbf{w} \in \mathcal{S}(x). \quad (1082)$$

The origin belongs to every instantaneous safe set because

$$\mathbf{0}^\top \mathbf{G}(\xi) \mathbf{0} = 0 < 1. \quad (1083)$$

Convexity and inclusion of the origin imply that the survivor set is star-shaped with respect to the origin. Along a fixed ray $\{R\mathbf{u} : R \geq 0\}$, its intersection is therefore an interval of the form

$$[0, \rho_{\mathbf{u}}(x)) \quad (1084)$$

or the complete ray if $\rho_{\mathbf{u}}(x) = \infty$.

The instantaneous radial boundary in direction $\mathbf{u}$ is

$$\rho_{\mathbf{u}}^{(\xi)} = \frac{1}{\sqrt{a_{\mathbf{u}}(\xi)}}. \quad (1085)$$

The intersection selects the smallest upstream radius:

$$\begin{aligned} \rho_{\mathbf{u}}(x) &= \inf_{0 \leq \xi \leq x} \rho_{\mathbf{u}}^{(\xi)} \\ &= \inf_{0 \leq \xi \leq x} \frac{1}{\sqrt{a_{\mathbf{u}}(\xi)}} \\ &= \frac{1}{\sqrt{\sup_{0 \leq \xi \leq x} a_{\mathbf{u}}(\xi)}} \\ &= \frac{1}{\sqrt{m_{\mathbf{u}}(x)}}. \end{aligned} \quad (1086)$$

This is the geometric origin of the directional running maximum.

A.8 Antipodal symmetry and projective directions

Because the energy is quadratic,

$$a_{-\mathbf{u}}(x) = (-\mathbf{u})^T \mathbf{G}(x) (-\mathbf{u}) = a_{\mathbf{u}}(x). \quad (1087)$$

Therefore,

$$m_{-\mathbf{u}}(x) = m_{\mathbf{u}}(x), \quad (1088)$$

$$\rho_{-\mathbf{u}}(x) = \rho_{\mathbf{u}}(x), \quad (1089)$$

$$F_{X_{\text{tr}}}(x \mid -\mathbf{u}) = F_{X_{\text{tr}}}(x \mid \mathbf{u}). \quad (1090)$$

Thus, the angular integrand is naturally defined on the real projective space

$$\mathbb{RP}^{r-1} = \mathbb{S}^{r-1} / \{\mathbf{u} \sim -\mathbf{u}\}. \quad (1091)$$

A quadrature rule containing both members of each antipodal pair can combine their weights without changing the probability integral. This symmetry is exact and does not rely on any symmetry of the OWNS operator.

A.9 Monotonicity of the transition distribution

For $0 \leq x_1 \leq x_2$,

$$[0, x_1] \subseteq [0, x_2], \quad (1092)$$

so

$$m_{\mathbf{u}}(x_1) \leq m_{\mathbf{u}}(x_2). \quad (1093)$$

Since

$$m \mapsto \overline{F}_{\chi_r^2}(1/m) \quad (1094)$$

is nondecreasing for $m > 0$,

$$F_{X_{\text{tr}}}(x_1 \mid \mathbf{u}) \leq F_{X_{\text{tr}}}(x_2 \mid \mathbf{u}). \quad (1095)$$

Angular averaging preserves this inequality:

$$F_{X_{\text{tr}}}(x_1) \leq F_{X_{\text{tr}}}(x_2). \quad (1096)$$

Similarly,

$$S(x_1) \geq S(x_2). \quad (1097)$$

This proof does not require the instantaneous energy $a_{\mathbf{u}}(x)$ to be monotone. Only its running maximum must be monotone, which follows directly from its definition.

A.10 Inlet probability mass and right censoring

At the inlet,

$$m_{\mathbf{u}}(0) = a_{\mathbf{u}}(0) = \mathbf{u}^{\text{T}} \mathbf{G}(0) \mathbf{u}. \quad (1098)$$

The immediate-transition probability is therefore

$$\mathbb{P}(X_{\text{tr}} = 0) = \mathbb{E}_{\mathbf{u}} \left[\overline{F}_{\chi_r^2} \left(\frac{1}{\mathbf{u}^{\text{T}} \mathbf{G}(0) \mathbf{u}} \right) \right]. \quad (1099)$$

This is equal to the local exceedance probability at the inlet:

$$\mathbb{P}(X_{\text{tr}} = 0) = \mathbb{P} \left[\mathbf{w}^{\text{T}} \mathbf{G}(0) \mathbf{w} \geq 1 \right]. \quad (1100)$$

The probability that no transition occurs within the modeled domain is

$$\mathbb{P}(X_{\text{tr}} = \infty) = S(x_{\text{max}}) \quad (1101)$$

under the convention that every realization surviving to x_{max} is assigned an infinite transition location. Equivalently, this is the right-censoring probability:

$$p_{\text{censored}} = \mathbb{E}_{\mathbf{u}} \left[F_{\chi_r^2} \left(\frac{1}{m_{\mathbf{u}}(x_{\text{max}})} \right) \right]. \quad (1102)$$

The complete probability measure of X_{tr} can therefore contain:

- (i) a point mass at $x = 0$;
- (ii) a continuous or mixed distribution over $0 < x \leq x_{\text{max}}$; and
- (iii) a point mass at $X_{\text{tr}} = \infty$.

A.11 Discrete streamwise approximation

Let

$$0 = x_0 < x_1 < \cdots < x_N = x_{\max} \quad (1103)$$

be a discrete streamwise grid. Define

$$m_{\mathbf{u}}^{(N)}(x_n) = \max_{0 \leq j \leq n} \mathbf{u}^\top \mathbf{G}(x_j) \mathbf{u}. \quad (1104)$$

The discrete first-transition CDF is

$$F_{X_{\text{tr}}^{(N)}}(x_n) = \mathbb{E}_{\mathbf{u}} \left[\bar{F}_{\chi_r^2} \left(\frac{1}{m_{\mathbf{u}}^{(N)}(x_n)} \right) \right]. \quad (1105)$$

Since

$$m_{\mathbf{u}}^{(N)}(x_n) \leq m_{\mathbf{u}}(x_n), \quad (1106)$$

it follows that

$$F_{X_{\text{tr}}^{(N)}}(x_n) \leq F_{X_{\text{tr}}}(x_n). \quad (1107)$$

Thus, endpoint-only streamwise sampling gives a lower estimate of the continuous transition probability.

For a nested sequence of grids whose union is dense in $[0, x]$, continuity of $a_{\mathbf{u}}(\xi)$ gives

$$m_{\mathbf{u}}^{(N)}(x) \nearrow m_{\mathbf{u}}(x) \quad (1108)$$

for every direction. Since the directional transition integrand is bounded between zero and one, the bounded convergence theorem yields

$$F_{X_{\text{tr}}^{(N)}}(x) \longrightarrow F_{X_{\text{tr}}}(x). \quad (1109)$$

This establishes convergence of the stationwise approximation under streamwise refinement.

A.12 Differentiation with respect to the terminal station

Define

$$\Psi_r(m) = \bar{F}_{\chi_r^2}(1/m), \quad m > 0. \quad (1110)$$

Its derivative is

$$\begin{aligned} \Psi_r'(m) &= -f_{\chi_r^2}(1/m) \frac{d}{dm} \left(\frac{1}{m} \right) \\ &= \frac{f_{\chi_r^2}(1/m)}{m^2}. \end{aligned} \quad (1111)$$

Where $m_{\mathbf{u}}(x) > 0$ is differentiable,

$$\frac{\partial}{\partial x} \Psi_r[m_{\mathbf{u}}(x)] = f_{\chi_r^2} \left[\frac{1}{m_{\mathbf{u}}(x)} \right] \frac{m'_{\mathbf{u}}(x)}{m_{\mathbf{u}}(x)^2}. \quad (1112)$$

If differentiation can be interchanged with angular expectation, the first-transition density is

$$f_{X_{\text{tr}}}(x) = \mathbb{E}_{\mathbf{u}} \left[f_{\chi_r^2} \left(\frac{1}{m_{\mathbf{u}}(x)} \right) \frac{m'_{\mathbf{u}}(x)}{m_{\mathbf{u}}(x)^2} \right]. \quad (1113)$$

The derivative of the running maximum is zero whenever the current gain lies below its previous record:

$$a_{\mathbf{u}}(x) < m_{\mathbf{u}}(x^-) \implies m'_{\mathbf{u}}(x) = 0. \quad (1114)$$

At regular points where the current gain establishes a new record,

$$a_{\mathbf{u}}(x) = m_{\mathbf{u}}(x), \quad a'_{\mathbf{u}}(x) > 0, \quad (1115)$$

one has

$$m'_{\mathbf{u}}(x) = a'_{\mathbf{u}}(x). \quad (1116)$$

The running maximum can fail to be differentiable where a new record begins or where equal maxima exchange activity. Consequently, eq. (1113) is best interpreted almost everywhere or in a weak sense. Numerically, conservative differences of the monotone CDF are generally more robust than direct evaluation of this derivative.

A.13 Conditions underlying the derivation

The radial–directional result relies on the following assumptions:

- (i) **Fixed finite stochastic dimension.** The initial uncertainty is represented by $\mathbf{w} \in \mathbb{R}^r$ for finite r .
- (ii) **Standard isotropic Gaussian coordinates.** The coordinate satisfies $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_r)$. A general nonsingular Gaussian covariance can be transformed to this form by whitening.
- (iii) **Zero mean.** The energy is a central homogeneous quadratic form in $\mathbf{w}$. A nonzero deterministic mean introduces linear and constant terms that generally prevent the same radial separation.
- (iv) **Linear state propagation.** The physical state is $\mathbf{q}(x) = \mathbf{B}(x)\mathbf{w}$.
- (v) **Quadratic threshold statistic.** The normalized transition statistic is $\mathbf{w}^\top \mathbf{G}(x) \mathbf{w}$.
- (vi) **Positive-semidefinite energy matrices.** The matrices satisfy $\mathbf{G}(x) \succeq \mathbf{0}$, so directional gains and their radial thresholds are nonnegative.
- (vii) **A common stochastic basis.** The same realization of $\mathbf{w}$ is used at every streamwise station.
- (viii) **First threshold attainment is irreversible.** Once the threshold is reached, the realization is classified as transitioned at all downstream stations.
- (ix) **Sufficient streamwise regularity.** Continuity of $\mathbf{G}(x)$ ensures that the directional maximum exists on each compact interval. The probability formulas can also be written with a supremum under weaker regularity.

The derivation does not require:

- normality of the OWNS propagator;
- monotonicity of the disturbance energy;
- statistical independence between streamwise stations;

- a Markov model for the energy;
- a Poisson approximation for threshold crossings; or
- diagonalizability of the entire matrix family in a common basis.

These observations are important for the boundary-layer application: strong non-normality changes the matrix family $\mathbf{G}(x)$, but it does not alter the validity of the radial–directional derivation.

A.14 Summary

The derivation can be summarized by the sequence

$$\mathbf{w} = R\mathbf{u}, \quad R^2 \sim \chi_r^2, \quad R \perp \mathbf{u}, \quad (1117)$$

$$\begin{aligned} \max_{\xi \leq x} \mathbf{w}^\top \mathbf{G}(\xi) \mathbf{w} &= R^2 \max_{\xi \leq x} \mathbf{u}^\top \mathbf{G}(\xi) \mathbf{u} \\ &= R^2 m_{\mathbf{u}}(x), \end{aligned} \quad (1118)$$

$$X_{\text{tr}} \leq x \iff R^2 \geq \frac{1}{m_{\mathbf{u}}(x)}, \quad (1119)$$

$$\mathbb{P}(X_{\text{tr}} \leq x \mid \mathbf{u}) = \overline{F}_{\chi_r^2} \left[\frac{1}{m_{\mathbf{u}}(x)} \right], \quad (1120)$$

$$F_{X_{\text{tr}}}(x) = \mathbb{E}_{\mathbf{u}} \left[\overline{F}_{\chi_r^2} \left(\frac{1}{m_{\mathbf{u}}(x)} \right) \right]. \quad (1121)$$

The streamwise first-passage memory is carried entirely by the directional running maximum, while the complete Gaussian amplitude distribution is incorporated analytically through the chi-square tail.

B Complex Gaussian Convention

The main derivation uses a real standard Gaussian coordinate,

$$\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_r), \quad (1122)$$

even when the Fourier-transformed physical state is complex. An alternative is to model the stochastic coefficients themselves as a proper complex Gaussian vector. This appendix defines that convention, derives the corresponding radial–directional formula, and establishes its equivalence to a $2r$ -dimensional realification.

The distinction between real and proper complex stochastic coefficients is important. The two models generally describe different ensembles and lead to different radial distributions:

$$\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_r) \implies \|\mathbf{w}\|_2^2 \sim \chi_r^2, \quad (1123)$$

$$\mathbf{z} \sim \mathcal{CN}(\mathbf{0}, \mathbf{I}_r) \implies \|\mathbf{z}\|_2^2 \sim \text{Gamma}(r, 1) = \frac{1}{2} \chi_{2r}^2. \quad (1124)$$

A calculation must use one convention consistently when constructing the correlation factor, energy matrix, radial distribution, and angular integration rule.

B.1 Covariance and pseudo-covariance of a complex random vector

Let

$$\mathbf{q} \in \mathbb{C}^{n_q} \quad (1125)$$

be a complex random vector with mean

$$\boldsymbol{\mu} = \mathbb{E}[\mathbf{q}]. \quad (1126)$$

Its covariance is

$$\mathbf{C} = \mathbb{E}[(\mathbf{q} - \boldsymbol{\mu})(\mathbf{q} - \boldsymbol{\mu})^*], \quad (1127)$$

while its pseudo-covariance, or relation matrix, is

$$\mathbf{P} = \mathbb{E}[(\mathbf{q} - \boldsymbol{\mu})(\mathbf{q} - \boldsymbol{\mu})^\top]. \quad (1128)$$

The covariance is Hermitian positive semidefinite:

$$\mathbf{C} = \mathbf{C}^*, \quad \mathbf{C} \succeq \mathbf{0}, \quad (1129)$$

whereas the pseudo-covariance is complex symmetric:

$$\mathbf{P} = \mathbf{P}^\top. \quad (1130)$$

A complex Gaussian vector is called *proper* when

$$\mathbf{P} = \mathbf{0}. \quad (1131)$$

For a zero-mean complex Gaussian vector, properness is equivalent to circular symmetry:

$$\mathbf{q} \stackrel{d}{=} \exp(i\vartheta)\mathbf{q} \quad \text{for every } \vartheta \in [0, 2\pi). \quad (1132)$$

The covariance matrix alone completely specifies a zero-mean proper complex Gaussian distribution. For an improper complex Gaussian vector, both $\mathbf{C}$ and $\mathbf{P}$ are required. Therefore, a factorization of only $\mathbf{C}$ does not in general specify an improper complex stochastic model.

B.2 Standard proper complex Gaussian convention

A standard proper complex Gaussian vector is denoted by

$$\boxed{\mathbf{z} \sim \mathcal{CN}(\mathbf{0}, \mathbf{I}_r).} \quad (1133)$$

The convention adopted here is

$$\mathbb{E}[\mathbf{z}] = \mathbf{0}, \quad (1134)$$

$$\mathbb{E}[\mathbf{z}\mathbf{z}^*] = \mathbf{I}_r, \quad (1135)$$

$$\mathbb{E}[\mathbf{z}\mathbf{z}^\top] = \mathbf{0}. \quad (1136)$$

Its density with respect to $2r$ -dimensional Lebesgue measure is

$$\boxed{\phi_{\mathbb{C},r}(\mathbf{z}) = \frac{1}{\pi^r} \exp(-\mathbf{z}^*\mathbf{z}).} \quad (1137)$$

The real and imaginary parts satisfy

$$\Re \mathbf{z} \sim \mathcal{N}\left(\mathbf{0}, \frac{1}{2} \mathbf{I}_r\right), \quad (1138)$$

$$\Im \mathbf{z} \sim \mathcal{N}\left(\mathbf{0}, \frac{1}{2} \mathbf{I}_r\right), \quad (1139)$$

and are independent. Equivalently,

$$\boxed{\mathbf{z} = \frac{1}{\sqrt{2}} (\mathbf{w}_R + i\mathbf{w}_I)}, \quad (1140)$$

where

$$\mathbf{w}_R, \mathbf{w}_I \stackrel{\text{ind}}{\sim} \mathcal{N}(\mathbf{0}, \mathbf{I}_r). \quad (1141)$$

Some software libraries use a different complex-normal convention in which each real and imaginary component has unit variance. Under that alternative convention, $\mathbb{E}[\mathbf{z}\mathbf{z}^*] = 2\mathbf{I}$. The factor of $1/\sqrt{2}$ in eq. (1140) fixes the convention used throughout this appendix and should not be omitted.

B.3 Complex correlation factor and OWNS propagation

Let the inlet Fourier state be modeled as

$$\mathbf{q}(0) = \mathbf{B}_0 \mathbf{z}, \quad \mathbf{z} \sim \mathcal{CN}(\mathbf{0}, \mathbf{I}_r), \quad (1142)$$

where

$$\mathbf{B}_0 \in \mathbb{C}^{n_q \times r}. \quad (1143)$$

Then

$$\mathbb{E}[\mathbf{q}(0)\mathbf{q}(0)^*] = \mathbf{B}_0 \mathbf{B}_0^*, \quad (1144)$$

$$\mathbb{E}[\mathbf{q}(0)\mathbf{q}(0)^\top] = \mathbf{0}. \quad (1145)$$

Thus, $\mathbf{q}(0)$ is proper on the range of $\mathbf{B}_0$.

Under linear OWNS propagation,

$$\mathbf{q}(x) = \mathbf{M}(0, x) \mathbf{B}_0 \mathbf{z} = \mathbf{B}(x) \mathbf{z}, \quad (1146)$$

where

$$\mathbf{B}(x) = \mathbf{M}(0, x) \mathbf{B}_0. \quad (1147)$$

The same proper complex coordinate $\mathbf{z}$ labels a realization at every station. The propagated covariance and pseudo-covariance are

$$\mathbf{C}(x, x) = \mathbf{B}(x) \mathbf{B}(x)^*, \quad (1148)$$

$$\mathbf{P}(x, x) = \mathbf{0}. \quad (1149)$$

Properness is preserved by deterministic complex-linear propagation.

The two-point covariance is

$$\mathbf{C}(x_1, x_2) = \mathbf{B}(x_1) \mathbf{B}(x_2)^*, \quad (1150)$$

and the two-point pseudo-covariance remains zero:

$$\mathbf{P}(x_1, x_2) = \mathbb{E}[\mathbf{q}(x_1)\mathbf{q}(x_2)^\top] = \mathbf{0}. \quad (1151)$$

B.4 Whitening a general proper complex Gaussian coordinate

Suppose the stochastic coordinate has a general proper covariance:

$$\zeta \sim \mathcal{CN}(\mathbf{0}, \Sigma), \quad \Sigma \succeq \mathbf{0}. \quad (1152)$$

Let

$$\Sigma = U_{\Sigma} \Lambda_{\Sigma} U_{\Sigma}^* \quad (1153)$$

and define

$$\zeta = \Sigma^{1/2} z, \quad z \sim \mathcal{CN}(\mathbf{0}, \mathbf{I}). \quad (1154)$$

If the physical state is originally written as

$$\mathbf{q}(x) = \tilde{\mathbf{B}}(x) \zeta, \quad (1155)$$

then the standard-coordinate factor is

$$\mathbf{B}(x) = \tilde{\mathbf{B}}(x) \Sigma^{1/2}, \quad (1156)$$

so that

$$\mathbf{q}(x) = \mathbf{B}(x) z. \quad (1157)$$

The radial–directional formula should be applied after this whitening step.

B.5 Quadratic energy for proper complex coordinates

Let the disturbance energy be

$$e(x) = \mathbf{q}(x)^* \mathbf{H}(x) \mathbf{q}(x), \quad (1158)$$

where

$$\mathbf{H}(x) = \mathbf{H}(x)^* \succeq \mathbf{0}. \quad (1159)$$

Substituting $\mathbf{q}(x) = \mathbf{B}(x) z$ gives

$$e(x) = z^* \mathbf{A}_{\mathbb{C}}(x) z, \quad (1160)$$

where

$$\boxed{\mathbf{A}_{\mathbb{C}}(x) = \mathbf{B}(x)^* \mathbf{H}(x) \mathbf{B}(x).} \quad (1161)$$

Unlike the real-coordinate convention, no real-part operation is needed. The matrix $\mathbf{A}_{\mathbb{C}}(x)$ is Hermitian positive semidefinite, and

$$z^* \mathbf{A}_{\mathbb{C}}(x) z \in \mathbb{R}_{\geq 0}. \quad (1162)$$

For a positive transition threshold, define

$$\boxed{\mathbf{G}_{\mathbb{C}}(x) = \frac{\mathbf{A}_{\mathbb{C}}(x)}{e_{\text{thres}}(x).} \quad (1163)$$

The normalized energy is

$$y(x) = z^* \mathbf{G}_{\mathbb{C}}(x) z. \quad (1164)$$

The first-transition location is

$$X_{\text{tr}}(z) = \inf \{x : z^* \mathbf{G}_{\mathbb{C}}(x) z \geq 1\}. \quad (1165)$$

B.6 One-point distribution and moments

At a fixed station, diagonalize

$$\mathbf{A}_{\mathbb{C}}(x) = \mathbf{U}(x) \text{diag}[\alpha_1(x), \dots, \alpha_r(x)] \mathbf{U}(x)^*. \quad (1166)$$

By unitary invariance of the standard proper complex Gaussian,

$$\tilde{\mathbf{z}} = \mathbf{U}(x)^* \mathbf{z} \sim \mathcal{CN}(\mathbf{0}, \mathbf{I}_r). \quad (1167)$$

Therefore,

$$e(x) \stackrel{d}{=} \sum_{j=1}^r \alpha_j(x) |\tilde{z}_j|^2. \quad (1168)$$

Under the adopted convention,

$$|\tilde{z}_j|^2 \sim \text{Exponential}(1), \quad (1169)$$

independently. Thus, the one-point energy is a weighted sum of independent unit-mean exponential variables.

For a proper complex standard Gaussian,

$$\mathbb{E}[e(x)] = \text{tr}[\mathbf{A}_{\mathbb{C}}(x)], \quad (1170)$$

$$\text{Var}[e(x)] = \text{tr}[\mathbf{A}_{\mathbb{C}}(x)^2]. \quad (1171)$$

For two stations,

$$\boxed{\text{Cov}[e(x_1), e(x_2)] = \text{tr}[\mathbf{A}_{\mathbb{C}}(x_1) \mathbf{A}_{\mathbb{C}}(x_2)]}. \quad (1172)$$

These expressions differ from the real-Gaussian identities by a factor of two:

$$\text{Cov}_{\mathbb{R}}[\mathbf{w}^{\top} \mathbf{A}_1 \mathbf{w}, \mathbf{w}^{\top} \mathbf{A}_2 \mathbf{w}] = 2 \text{tr}(\mathbf{A}_1 \mathbf{A}_2). \quad (1173)$$

The complex and real conventions can therefore produce the same one-point mean for suitably matched matrices but different variances and tails. The stochastic convention must be selected according to the intended physical ensemble, not merely for computational convenience.

B.7 Complex radial decomposition

Write

$$\boxed{\mathbf{z} = R_{\mathbb{C}} \mathbf{u}_{\mathbb{C}}}, \quad (1174)$$

where

$$R_{\mathbb{C}} = \|\mathbf{z}\|_2 \quad (1175)$$

and

$$\mathbf{u}_{\mathbb{C}} = \frac{\mathbf{z}}{\|\mathbf{z}\|_2} \in \mathbb{S}_{\mathbb{C}}^{r-1}. \quad (1176)$$

The complex unit sphere is

$$\mathbb{S}_{\mathbb{C}}^{r-1} = \{\mathbf{u} \in \mathbb{C}^r : \mathbf{u}^* \mathbf{u} = 1\}. \quad (1177)$$

As a real manifold, this sphere has dimension

$$2r - 1. \quad (1178)$$

Identifying $\mathbb{C}^r$ with $\mathbb{R}^{2r}$, the volume element in complex polar coordinates is

$$d\mathbf{z} = R_{\mathbb{C}}^{2r-1} dR_{\mathbb{C}} d\sigma_{\mathbb{C}}(\mathbf{u}_{\mathbb{C}}). \quad (1179)$$

The surface area is

$$|\mathbb{S}_{\mathbb{C}}^{r-1}| = \frac{2\pi^r}{\Gamma(r)}. \quad (1180)$$

Substituting eq. (1137) gives the radial density

$$f_{R_{\mathbb{C}}}(R) = \frac{2}{\Gamma(r)} R^{2r-1} \exp(-R^2), \quad R > 0. \quad (1181)$$

The radius and direction are independent:

$$R_{\mathbb{C}} \perp \mathbf{u}_{\mathbb{C}}, \quad (1182)$$

and

$$\mathbf{u}_{\mathbb{C}} \sim \text{Unif}(\mathbb{S}_{\mathbb{C}}^{r-1}). \quad (1183)$$

Let

$$Z_{\mathbb{C}} = R_{\mathbb{C}}^2. \quad (1184)$$

The change of variables gives

$$f_{Z_{\mathbb{C}}}(z) = \frac{z^{r-1} \exp(-z)}{\Gamma(r)}, \quad z > 0. \quad (1185)$$

Hence,

$$\boxed{R_{\mathbb{C}}^2 \sim \text{Gamma}(r, 1).} \quad (1186)$$

Equivalently,

$$2R_{\mathbb{C}}^2 \sim \chi_{2r}^2. \quad (1187)$$

B.8 Complex directional gain and running maximum

For a complex unit direction, define

$$a_{\mathbf{u}_{\mathbb{C}}}(x) = \mathbf{u}_{\mathbb{C}}^* \mathbf{G}_{\mathbb{C}}(x) \mathbf{u}_{\mathbb{C}}. \quad (1188)$$

Because $\mathbf{G}_{\mathbb{C}}(x) \succeq \mathbf{0}$,

$$a_{\mathbf{u}_{\mathbb{C}}}(x) \in \mathbb{R}_{\geq 0}. \quad (1189)$$

The normalized energy is

$$\begin{aligned} y(x) &= (R_{\mathbb{C}} \mathbf{u}_{\mathbb{C}})^* \mathbf{G}_{\mathbb{C}}(x) (R_{\mathbb{C}} \mathbf{u}_{\mathbb{C}}) \\ &= R_{\mathbb{C}}^2 a_{\mathbf{u}_{\mathbb{C}}}(x). \end{aligned} \quad (1190)$$

Define the directional running maximum

$$\boxed{m_{\mathbf{u}_{\mathbb{C}}}(x) = \max_{0 \leq \xi \leq x} \mathbf{u}_{\mathbb{C}}^* \mathbf{G}_{\mathbb{C}}(\xi) \mathbf{u}_{\mathbb{C}}.} \quad (1191)$$

Then

$$X_{\text{tr}} \leq x \iff R_{\mathbb{C}}^2 m_{\mathbf{u}_{\mathbb{C}}}(x) \geq 1. \quad (1192)$$

For $m_{\mathbf{u}_{\mathbb{C}}}(x) > 0$, the critical squared complex radius is

$$R_{\text{crit}, \mathbb{C}}^2(\mathbf{u}_{\mathbb{C}}, x) = \frac{1}{m_{\mathbf{u}_{\mathbb{C}}}(x)}. \quad (1193)$$

B.9 Complex radial–directional first-transition formula

Conditioning on the complex direction gives

$$\begin{aligned} S(x \mid \mathbf{u}_\mathbb{C}) &= \mathbb{P} \left[R_\mathbb{C}^2 < \frac{1}{m_{\mathbf{u}_\mathbb{C}}(x)} \right] \\ &= F_{\text{Gamma}(r,1)} \left[\frac{1}{m_{\mathbf{u}_\mathbb{C}}(x)} \right]. \end{aligned} \quad (1194)$$

The conditional transition probability is

$$F_{X_{\text{tr}}}(x \mid \mathbf{u}_\mathbb{C}) = \overline{F}_{\text{Gamma}(r,1)} \left[\frac{1}{m_{\mathbf{u}_\mathbb{C}}(x)} \right]. \quad (1195)$$

Averaging over the uniformly distributed complex direction yields

$$S(x) = \mathbb{E}_{\mathbf{u}_\mathbb{C}} \left[F_{\text{Gamma}(r,1)} \left(\frac{1}{m_{\mathbf{u}_\mathbb{C}}(x)} \right) \right], \quad (1196)$$

and

$$F_{X_{\text{tr}}}(x) = \mathbb{E}_{\mathbf{u}_\mathbb{C}} \left[\overline{F}_{\text{Gamma}(r,1)} \left(\frac{1}{m_{\mathbf{u}_\mathbb{C}}(x)} \right) \right]. \quad (1197)$$

Since

$$2R_\mathbb{C}^2 \sim \chi_{2r}^2, \quad (1198)$$

the same formulas can be written as

$$F_{X_{\text{tr}}}(x) = \mathbb{E}_{\mathbf{u}_\mathbb{C}} \left[\overline{F}_{\chi_{2r}^2} \left(\frac{2}{m_{\mathbf{u}_\mathbb{C}}(x)} \right) \right]. \quad (1199)$$

Similarly,

$$S(x) = \mathbb{E}_{\mathbf{u}_\mathbb{C}} \left[F_{\chi_{2r}^2} \left(\frac{2}{m_{\mathbf{u}_\mathbb{C}}(x)} \right) \right]. \quad (1200)$$

For integer r , the gamma survival function has the finite-sum representation

$$\overline{F}_{\text{Gamma}(r,1)}(z) = \exp(-z) \sum_{j=0}^{r-1} \frac{z^j}{j!}. \quad (1201)$$

Thus, the directional transition probability can be evaluated as

$$F_{X_{\text{tr}}}(x \mid \mathbf{u}_\mathbb{C}) = \exp \left[-\frac{1}{m_{\mathbf{u}_\mathbb{C}}(x)} \right] \sum_{j=0}^{r-1} \frac{m_{\mathbf{u}_\mathbb{C}}(x)^{-j}}{j!}. \quad (1202)$$

For extreme tails, a dedicated logarithmic gamma-survival routine is preferable to direct evaluation of this sum.

B.10 Phase symmetry and complex projective directions

For any global phase $\vartheta \in [0, 2\pi)$,

$$\begin{aligned} a_{\exp(i\vartheta)\mathbf{u}_\mathbb{C}}(x) &= [\exp(i\vartheta)\mathbf{u}_\mathbb{C}]^* \mathbf{G}_\mathbb{C}(x) [\exp(i\vartheta)\mathbf{u}_\mathbb{C}] \\ &= a_{\mathbf{u}_\mathbb{C}}(x). \end{aligned} \quad (1203)$$

Therefore,

$$m_{\exp(i\vartheta)\mathbf{u}_\mathbb{C}}(x) = m_{\mathbf{u}_\mathbb{C}}(x), \quad (1204)$$

$$F_{X_{\text{tr}}}(x \mid \exp(i\vartheta)\mathbf{u}_\mathbb{C}) = F_{X_{\text{tr}}}(x \mid \mathbf{u}_\mathbb{C}). \quad (1205)$$

The complex angular integrand is therefore naturally defined on the complex projective space

$$\mathbb{CP}^{r-1} = \mathbb{S}_\mathbb{C}^{r-1} / \{\mathbf{u} \sim \exp(i\vartheta)\mathbf{u}\}. \quad (1206)$$

This symmetry is stronger than the antipodal symmetry of the real problem. In the real case, only $\mathbf{u}$ and $-\mathbf{u}$ are equivalent. In the proper complex case, every global phase represents the same quadratic-energy direction.

A numerical angular rule may integrate over the complete complex unit sphere without explicitly quotienting out this phase. The redundant global phase does not bias the result, although a projective parameterization can reduce the effective angular dimension by one.

B.11 Realification of a complex state

For any complex physical state

$$\mathbf{q} = \mathbf{q}_\text{R} + i\mathbf{q}_\text{I}, \quad (1207)$$

define its realification by

$$\mathcal{R}(\mathbf{q}) = \begin{bmatrix} \Re \mathbf{q} \\ \Im \mathbf{q} \end{bmatrix} \in \mathbb{R}^{2n_q}. \quad (1208)$$

For a complex matrix $\mathbf{K} \in \mathbb{C}^{m \times n}$, define

$$\boxed{\mathcal{R}(\mathbf{K}) = \begin{bmatrix} \Re \mathbf{K} & -\Im \mathbf{K} \\ \Im \mathbf{K} & \Re \mathbf{K} \end{bmatrix}}. \quad (1209)$$

This mapping satisfies

$$\mathcal{R}(\mathbf{K}\mathbf{q}) = \mathcal{R}(\mathbf{K})\mathcal{R}(\mathbf{q}), \quad (1210)$$

and

$$\mathcal{R}(\mathbf{K}_1\mathbf{K}_2) = \mathcal{R}(\mathbf{K}_1)\mathcal{R}(\mathbf{K}_2). \quad (1211)$$

If $\mathbf{H}$ is Hermitian, then

$$\mathcal{R}(\mathbf{H}) = \begin{bmatrix} \Re \mathbf{H} & -\Im \mathbf{H} \\ \Im \mathbf{H} & \Re \mathbf{H} \end{bmatrix} \quad (1212)$$

is real symmetric. Moreover,

$$\boxed{\mathbf{q}^* \mathbf{H} \mathbf{q} = \mathcal{R}(\mathbf{q})^\top \mathcal{R}(\mathbf{H}) \mathcal{R}(\mathbf{q})}. \quad (1213)$$

If the complex OWNS equation is

$$\frac{d\mathbf{q}}{dx} = \mathbf{L}(x)\mathbf{q}, \quad (1214)$$

then the realified state satisfies

$$\frac{d}{dx} \mathcal{R}(\mathbf{q}) = \mathcal{R}[\mathbf{L}(x)] \mathcal{R}(\mathbf{q}). \quad (1215)$$

Thus, realification preserves the deterministic spatial dynamics as well as the quadratic energy.

B.12 Realification of proper complex Gaussian coordinates

Define the $2r$ -dimensional real standard Gaussian vector

$$\tilde{\mathbf{w}} = \begin{bmatrix} \mathbf{w}_R \\ \mathbf{w}_I \end{bmatrix} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_{2r}). \quad (1216)$$

From eq. (1140),

$$\mathcal{R}(z) = \frac{1}{\sqrt{2}} \tilde{\mathbf{w}}. \quad (1217)$$

If

$$\mathbf{q}(x) = \mathbf{B}(x)\mathbf{z}, \quad (1218)$$

then

$$\begin{aligned} \mathcal{R}[\mathbf{q}(x)] &= \mathcal{R}[\mathbf{B}(x)]\mathcal{R}(z) \\ &= \frac{1}{\sqrt{2}} \mathcal{R}[\mathbf{B}(x)] \tilde{\mathbf{w}}. \end{aligned} \quad (1219)$$

Define the realified synthesis factor

$$\tilde{\mathbf{B}}_R(x) = \frac{1}{\sqrt{2}} \mathcal{R}[\mathbf{B}(x)] = \frac{1}{\sqrt{2}} \begin{bmatrix} \Re \mathbf{B}(x) & -\Im \mathbf{B}(x) \\ \Im \mathbf{B}(x) & \Re \mathbf{B}(x) \end{bmatrix}. \quad (1220)$$

Then

$$\mathcal{R}[\mathbf{q}(x)] = \tilde{\mathbf{B}}_R(x) \tilde{\mathbf{w}}. \quad (1221)$$

The corresponding real normalized energy matrix is

$$\tilde{\mathbf{G}}_R(x) = \frac{\tilde{\mathbf{B}}_R(x)^\top \mathcal{R}[\mathbf{H}(x)] \tilde{\mathbf{B}}_R(x)}{e_{\text{thres}}(x)}. \quad (1222)$$

Using the realification identities gives

$$\tilde{\mathbf{G}}_R(x) = \frac{1}{2} \mathcal{R}[\mathbf{G}_\mathbb{C}(x)]. \quad (1223)$$

The normalized energies are identical:

$$\mathbf{z}^* \mathbf{G}_\mathbb{C}(x) \mathbf{z} = \tilde{\mathbf{w}}^\top \tilde{\mathbf{G}}_R(x) \tilde{\mathbf{w}}. \quad (1224)$$

The factor of $1/2$ in eq. (1223) is essential. It follows from the convention $\mathbf{z} = (\mathbf{w}_R + i\mathbf{w}_I)/\sqrt{2}$. Omitting this factor doubles the modeled energy.

B.13 Equivalence of the complex and real radial formulas

Write the real Gaussian coordinate as

$$\tilde{\mathbf{w}} = R_R \mathbf{u}_R, \quad R_R^2 \sim \chi_{2r}^2, \quad (1225)$$

where

$$\mathbf{u}_R = \begin{bmatrix} \mathbf{u}_R \\ \mathbf{u}_I \end{bmatrix} \in \mathbb{S}^{2r-1}. \quad (1226)$$

Associate with it the complex unit direction

$$\mathbf{u}_{\mathbb{C}} = \mathbf{u}_{\mathbb{R}} + i\mathbf{u}_{\mathbb{I}}. \quad (1227)$$

Since

$$\|\mathbf{u}_{\mathbb{C}}\|_2^2 = \|\mathbf{u}_{\mathbb{R}}\|_2^2 + \|\mathbf{u}_{\mathbb{I}}\|_2^2 = 1, \quad (1228)$$

this is a point on the complex unit sphere.

The real and complex radii are related by

$$R_{\mathbb{C}} = \frac{R_{\mathbb{R}}}{\sqrt{2}}. \quad (1229)$$

For corresponding directions,

$$\mathbf{u}_{\mathbb{R}}^T \tilde{\mathbf{G}}_{\mathbb{R}}(x) \mathbf{u}_{\mathbb{R}} = \frac{1}{2} \mathbf{u}_{\mathbb{C}}^* \mathbf{G}_{\mathbb{C}}(x) \mathbf{u}_{\mathbb{C}}. \quad (1230)$$

Therefore, the real and complex directional running maxima satisfy

$$m_{\mathbf{u}_{\mathbb{R}}}^{\mathbb{R}}(x) = \frac{1}{2} m_{\mathbf{u}_{\mathbb{C}}}^{\mathbb{C}}(x). \quad (1231)$$

The real radial formula in dimension $2r$ gives

$$\begin{aligned} F_{X_{\text{tr}}}(x) &= \mathbb{E}_{\mathbf{u}_{\mathbb{R}}} \left[\bar{F}_{\chi_{2r}^2} \left(\frac{1}{m_{\mathbf{u}_{\mathbb{R}}}^{\mathbb{R}}(x)} \right) \right] \\ &= \mathbb{E}_{\mathbf{u}_{\mathbb{C}}} \left[\bar{F}_{\chi_{2r}^2} \left(\frac{2}{m_{\mathbf{u}_{\mathbb{C}}}^{\mathbb{C}}(x)} \right) \right], \end{aligned} \quad (1232)$$

which is exactly eq. (1199).

Thus, a proper complex Gaussian problem can be implemented in either of two equivalent ways:

- (i) use complex unit directions and a $\text{Gamma}(r, 1)$ radial distribution; or
- (ii) realify the problem, use real directions on $\mathbb{S}^{2r-1}$, and use a χ_{2r}^2 radial distribution.

The two implementations should agree to within their angular and streamwise numerical errors and provide a useful verification test.

B.14 Eigenvalue structure after realification

Let

$$\mathbf{G}_{\mathbb{C}} = \mathbf{U} \text{diag}(\gamma_1, \dots, \gamma_r) \mathbf{U}^*, \quad \gamma_j \geq 0. \quad (1233)$$

The realification $\mathcal{R}(\mathbf{G}_{\mathbb{C}})$ has each eigenvalue twice:

$$\text{spec}[\mathcal{R}(\mathbf{G}_{\mathbb{C}})] = \{\gamma_1, \gamma_1, \dots, \gamma_r, \gamma_r\}. \quad (1234)$$

Because the realified normalized matrix is $\tilde{\mathbf{G}}_{\mathbb{R}} = \frac{1}{2} \mathcal{R}(\mathbf{G}_{\mathbb{C}})$, its eigenvalues are

$$\text{spec}(\tilde{\mathbf{G}}_{\mathbb{R}}) = \left\{ \frac{\gamma_1}{2}, \frac{\gamma_1}{2}, \dots, \frac{\gamma_r}{2}, \frac{\gamma_r}{2} \right\}. \quad (1235)$$

The real generalized chi-square representation is consequently

$$y \stackrel{d}{=} \sum_{j=1}^r \frac{\gamma_j}{2} (W_{j,R}^2 + W_{j,I}^2). \quad (1236)$$

Since

$$\frac{1}{2} (W_{j,R}^2 + W_{j,I}^2) \sim \text{Exponential}(1), \quad (1237)$$

this agrees with the complex representation

$$y \stackrel{d}{=} \sum_{j=1}^r \gamma_j |z_j|^2. \quad (1238)$$

B.15 Generating uniform complex directions

A uniform direction on the complex unit sphere can be generated by drawing a proper complex Gaussian vector and normalizing it. Draw

$$\boldsymbol{\eta} = \frac{1}{\sqrt{2}} (\mathbf{g}_R + i\mathbf{g}_I), \quad (1239)$$

where

$$\mathbf{g}_R, \mathbf{g}_I \stackrel{\text{ind}}{\sim} \mathcal{N}(\mathbf{0}, \mathbf{I}_r). \quad (1240)$$

Then

$$\boxed{\mathbf{u}_C = \frac{\boldsymbol{\eta}}{\|\boldsymbol{\eta}\|_2}} \quad (1241)$$

is uniformly distributed on $\mathbb{S}_C^{r-1}$.

Equivalently, draw

$$\mathbf{g} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_{2r}), \quad (1242)$$

normalize it on $\mathbb{S}^{2r-1}$, and combine its first and second r components into the real and imaginary parts of $\mathbf{u}_C$.

For randomized quasi-Monte Carlo, generate a low-discrepancy point

$$\mathbf{v}_k \in (0, 1)^{2r}, \quad (1243)$$

apply the inverse normal CDF componentwise, normalize the resulting $2r$ -dimensional real vector, and convert it to a complex direction. Using $2r$ real coordinates is necessary; applying the inverse-normal map to only r coordinates generates real rather than proper complex directions.

The factor $1/\sqrt{2}$ is unnecessary in Algorithm 14 because normalization removes any common radial scale. It remains necessary when generating an unnormalized proper complex Gaussian vector with covariance $\mathbf{I}$.

B.16 Complex angular-integration algorithm

Let

$$\{(\mathbf{u}_{C,k}, \omega_k)\}_{k=1}^K \quad (1244)$$

be a normalized integration rule on the complex unit sphere:

$$\mathbf{u}_{\mathbb{C},k}^* \mathbf{u}_{\mathbb{C},k} = 1, \quad \omega_k \geq 0, \quad \sum_{k=1}^K \omega_k = 1. \quad (1245)$$

For each direction,

$$a_{k,n} = \mathbf{u}_{\mathbb{C},k}^* \mathbf{G}_{\mathbb{C}}(x_n) \mathbf{u}_{\mathbb{C},k}. \quad (1246)$$

The running maximum is

$$m_{k,n} = \max_{0 \leq j \leq n} a_{k,j}, \quad (1247)$$

or its continuously resolved streamwise counterpart.

As in the real formulation, a stable upper-gamma-tail routine should be used when the transition probability is small. The real part in the directional quadratic form removes only roundoff-scale imaginary components; a material imaginary value indicates that $\mathbf{G}_{\mathbb{C}}$ is not numerically Hermitian.

B.17 Complex derivative and streamwise maximum detection

If

$$\mathbf{B}'(x) = \mathbf{L}(x) \mathbf{B}(x), \quad (1248)$$

then

$$\mathbf{A}'_{\mathbb{C}}(x) = \mathbf{B}(x)^* [\mathbf{L}(x)^* \mathbf{H}(x) + \mathbf{H}(x) \mathbf{L}(x) + \mathbf{H}'(x)] \mathbf{B}(x). \quad (1249)$$

The normalized matrix derivative is

$$\mathbf{G}'_{\mathbb{C}}(x) = \frac{\mathbf{A}'_{\mathbb{C}}(x)}{e_{\text{thres}}(x)} - \frac{e'_{\text{thres}}(x)}{e_{\text{thres}}(x)} \mathbf{G}_{\mathbb{C}}(x). \quad (1250)$$

For a fixed complex direction,

$$a'_{\mathbf{u}_{\mathbb{C}}}(x) = \mathbf{u}_{\mathbb{C}}^* \mathbf{G}'_{\mathbb{C}}(x) \mathbf{u}_{\mathbb{C}}. \quad (1251)$$

This quantity is real in exact arithmetic. The cubic Hermite and adaptive streamwise maximum procedures from ?? apply without modification after replacing real transposes by conjugate transposes in the directional quadratic forms.

B.18 Real coefficients multiplying complex Fourier structures

A different model uses a real Gaussian coefficient vector with a complex Fourier-state factor:

$$\mathbf{q}(x) = \mathbf{B}(x) \mathbf{w}, \quad \mathbf{w} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_r). \quad (1252)$$

Its covariance is

$$\mathbf{C}(x, x) = \mathbf{B}(x) \mathbf{B}(x)^*, \quad (1253)$$

but its pseudo-covariance is generally nonzero:

$$\mathbf{P}(x, x) = \mathbf{B}(x) \mathbf{B}(x)^{\top}. \quad (1254)$$

Thus, this ensemble is generally improper as a complex random vector.

The energy is

$$\begin{aligned} e(x) &= \mathbf{w}^{\top} \mathbf{B}(x)^* \mathbf{H}(x) \mathbf{B}(x) \mathbf{w} \\ &= \mathbf{w}^{\top} \Re [\mathbf{B}(x)^* \mathbf{H}(x) \mathbf{B}(x)] \mathbf{w}. \end{aligned} \quad (1255)$$

The imaginary skew-symmetric part makes no contribution for real $\mathbf{w}$. Therefore, the correct radial distribution is

$$\|\mathbf{w}\|_2^2 \sim \chi_r^2, \quad (1256)$$

and the real formula from section A applies.

This model must not be evaluated using the $\text{Gamma}(r, 1)$ radial distribution merely because the physical Fourier state is complex. The radial law is determined by the stochastic coordinate, not by the data type of the propagated state.

B.19 Improper complex Gaussian coordinates

If the stochastic coordinate is complex Gaussian but improper,

$$\mathbb{E}[\mathbf{z}\mathbf{z}^\top] \neq \mathbf{0}, \quad (1257)$$

its distribution is not circularly symmetric. The decomposition into an independent scalar radius and uniform complex direction generally does not hold in the original complex coordinates.

An improper complex Gaussian vector should be represented by its augmented covariance

$$\mathbf{C}_{\text{aug}} = \begin{bmatrix} \mathbf{C} & \mathbf{P} \\ \mathbf{P}^* & \mathbf{C}^* \end{bmatrix} \quad (1258)$$

or, equivalently, by the covariance of its realification

$$\mathbf{z}_R = \begin{bmatrix} \Re \mathbf{z} \\ \Im \mathbf{z} \end{bmatrix}. \quad (1259)$$

The real covariance is

$$\mathbf{\Sigma}_R = \frac{1}{2} \begin{bmatrix} \Re(\mathbf{C} + \mathbf{P}) & -\Im(\mathbf{C} - \mathbf{P}) \\ \Im(\mathbf{C} + \mathbf{P}) & \Re(\mathbf{C} - \mathbf{P}) \end{bmatrix}. \quad (1260)$$

After factorizing

$$\mathbf{\Sigma}_R = \mathbf{L}_R \mathbf{L}_R^\top, \quad (1261)$$

write

$$\mathbf{z}_R = \mathbf{L}_R \mathbf{w}, \quad \mathbf{w} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_{r_R}). \quad (1262)$$

The problem is then treated with the real radial-directional method using the whitened real coordinate $\mathbf{w}$.

The dimension r_R is the rank of $\mathbf{\Sigma}_R$ and can lie between r and $2r$, depending on the degree of impropriety and any rank deficiency.

B.20 Fourier conjugacy and physical-field normalization

A complex Fourier coefficient represents a real physical disturbance only after the conjugate spectral component is included. Two common conventions are

$$\mathbf{q}'(x, y, z, t) = \Re \{ \hat{\mathbf{q}}(x, y) \exp[i\beta z - i\omega t] \}, \quad (1263)$$

$$\mathbf{q}'(x, y, z, t) = \hat{\mathbf{q}}(x, y) \exp[i\beta z - i\omega t] + \text{c.c.} \quad (1264)$$

These expressions differ by a factor of two if the same $\hat{\mathbf{q}}$ is used.

For a scalar coefficient $\hat{q}$, the temporal average under the real-part convention is

$$\left\langle |\Re[\hat{q} \exp(-i\omega t)]|^2 \right\rangle_t = \frac{1}{2} |\hat{q}|^2. \quad (1265)$$

Under the explicit conjugate-pair convention,

$$\left\langle |\hat{q} \exp(-i\omega t) + \hat{q}^* \exp(i\omega t)|^2 \right\rangle_t = 2|\hat{q}|^2. \quad (1266)$$

These factors should be incorporated into the energy matrix $\mathbf{H}$, the Fourier amplitude, or the transition threshold. The first-transition method is unchanged as long as the normalized energy

$$\frac{\mathbf{q}^* \mathbf{H} \mathbf{q}}{e_{\text{thres}}} \quad (1267)$$

uses one consistent convention.

If both positive and negative frequencies or spanwise wavenumbers are included explicitly in a real physical-field model, their Fourier coefficients are related by conjugacy and should not be treated as independent Gaussian coordinates. Double-counting conjugate components would alter the covariance, radial dimension, and energy distribution.

B.21 Choosing between real and proper complex coefficients

The appropriate stochastic convention depends on the physical meaning of the Fourier coefficients.

A real Gaussian coordinate multiplying complex structures is suitable when:

- each stochastic mode has a fixed complex phase structure;
- randomness changes only the signed amplitude of that mode;
- the correlation factor was constructed from real random coefficients; or
- the complex state is a numerical representation of a real-coordinate stochastic model.

A proper complex Gaussian coordinate is suitable when:

- the phase of each stochastic coefficient is itself uniformly random;
- real and imaginary coefficient components are independent with equal variance;
- the modeled Fourier coefficient is circularly symmetric; or
- the estimated pseudo-covariance is negligible relative to the covariance.

An improper complex Gaussian model is required when:

- the real and imaginary coefficient variances differ;
- they are correlated;
- a preferred phase is present; or
- conjugacy or other physical constraints produce a nonzero pseudo-covariance.

A useful empirical diagnostic is

$$\eta_{\text{proper}} = \frac{\|\hat{\mathbf{P}}\|_F}{\max[\|\hat{\mathbf{C}}\|_F, \varepsilon_{\text{scale}}]}. \quad (1268)$$

A small value supports a proper-complex approximation, subject to sampling uncertainty. A material value indicates that the pseudo-covariance should not be discarded.

B.22 Verification tests for the complex convention

The following tests are recommended.

B.22.1 Complex-to-real energy agreement

For randomly selected proper complex vectors $\mathbf{z}^{(j)}$, construct

$$\tilde{\mathbf{w}}^{(j)} = \sqrt{2} \begin{bmatrix} \Re \mathbf{z}^{(j)} \\ \Im \mathbf{z}^{(j)} \end{bmatrix}. \quad (1269)$$

Verify

$$[\mathbf{z}^{(j)}]^* \mathbf{G}_C(x) \mathbf{z}^{(j)} = [\tilde{\mathbf{w}}^{(j)}]^\top \tilde{\mathbf{G}}_R(x) \tilde{\mathbf{w}}^{(j)} \quad (1270)$$

at representative stations.

B.22.2 Radial-distribution test

Generate a large set of proper complex standard Gaussian samples and verify

$$\mathbb{E}[\|\mathbf{z}\|_2^2] = r, \quad (1271)$$

$$\text{Var}[\|\mathbf{z}\|_2^2] = r. \quad (1272)$$

A distributional test should compare $\|\mathbf{z}\|_2^2$ with $\text{Gamma}(r, 1)$, or $2\|\mathbf{z}\|_2^2$ with χ_{2r}^2 .

B.22.3 Complex and realified CDF agreement

Compute the first-transition CDF using:

- (i) the complex formula with $\text{Gamma}(r, 1)$ radial tails; and
- (ii) the realified formula with χ_{2r}^2 radial tails.

Using corresponding angular points, the directional probabilities should agree pointwise:

$$\bar{F}_{\text{Gamma}(r,1)}\left(\frac{1}{m_{\mathbb{C}}}\right) = \bar{F}_{\chi_{2r}^2}\left(\frac{1}{m_{\mathbb{R}}}\right), \quad m_{\mathbb{R}} = \frac{m_{\mathbb{C}}}{2}. \quad (1273)$$

B.22.4 Unitary invariance

For a fixed unitary matrix

$$\mathbf{U}^* \mathbf{U} = \mathbf{I}, \quad (1274)$$

transform

$$\tilde{\mathbf{B}}(x) = \mathbf{B}(x) \mathbf{U}, \quad (1275)$$

$$\tilde{\mathbf{G}}_C(x) = \mathbf{U}^* \mathbf{G}_C(x) \mathbf{U}. \quad (1276)$$

The proper complex Gaussian distribution is unitarily invariant, so the exact first-transition CDF must be unchanged. With angular directions transformed consistently as

$$\tilde{\mathbf{u}}_k = \mathbf{U}^* \mathbf{u}_k, \quad (1277)$$

the directional gain histories should agree to floating-point accuracy.

B.22.5 Moment test

For selected matrices, compare sample moments with

$$\mathbb{E}[\mathbf{z}^* \mathbf{A} \mathbf{z}] = \text{tr}(\mathbf{A}), \quad (1278)$$

$$\text{Var}[\mathbf{z}^* \mathbf{A} \mathbf{z}] = \text{tr}(\mathbf{A}^2). \quad (1279)$$

A factor-of-two discrepancy in the variance commonly indicates that a real-Gaussian and proper-complex convention have been mixed.

B.23 Summary of conventions

The three principal cases are summarized in table 10.

Table 10: Summary of Gaussian-coordinate conventions for a complex Fourier state.

Model	State representation	Radial distribution	Energy matrix
Real coefficients	$\mathbf{q} = \mathbf{B}\mathbf{w}$, $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_r)$	$\ \mathbf{w}\ ^2 \sim \chi_r^2$	$\Re[\mathbf{B}^* \mathbf{H} \mathbf{B}]$
Proper complex coefficients	$\mathbf{q} = \mathbf{B}\mathbf{z}$, $\mathbf{z} \sim \mathcal{CN}(\mathbf{0}, \mathbf{I}_r)$	$\ \mathbf{z}\ ^2 \sim \text{Gamma}(r, 1) = \frac{1}{2}\chi_{2r}^2$	$\mathbf{B}^* \mathbf{H} \mathbf{B}$
Realified proper complex	$\mathcal{R}(\mathbf{q}) = \tilde{\mathbf{B}}_R \tilde{\mathbf{w}}$, $\tilde{\mathbf{w}} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_{2r})$	$\ \tilde{\mathbf{w}}\ ^2 \sim \chi_{2r}^2$	$\frac{1}{2}\mathcal{R}[\mathbf{B}^* \mathbf{H} \mathbf{B}]$

The corresponding first-transition formulas are:

$$F_{X_{\text{tr}}}^{\text{real}}(x) = \mathbb{E}_{\mathbf{u} \in \mathbb{S}^{r-1}} \left[\bar{F}_{\chi_r^2} \left(\frac{1}{m_{\mathbf{u}}(x)} \right) \right], \quad (1280)$$

$$F_{X_{\text{tr}}}^{\text{proper}}(x) = \mathbb{E}_{\mathbf{u}_{\mathbb{C}}} \left[\bar{F}_{\text{Gamma}(r,1)} \left(\frac{1}{m_{\mathbf{u}_{\mathbb{C}}}(x)} \right) \right], \quad (1281)$$

$$F_{X_{\text{tr}}}^{\text{realified}}(x) = \mathbb{E}_{\mathbf{u}_{\mathbb{R}} \in \mathbb{S}^{2r-1}} \left[\bar{F}_{\chi_{2r}^2} \left(\frac{1}{m_{\mathbf{u}_{\mathbb{R}}}(x)} \right) \right]. \quad (1282)$$

The final two expressions are equivalent when

$$\tilde{\mathbf{G}}_R = \frac{1}{2} \mathcal{R}(\mathbf{G}_{\mathbb{C}}). \quad (1283)$$

The appropriate formula is determined by the stochastic model for the Fourier coefficients. The fact that the OWNS state is stored as a complex vector does not, by itself, imply a proper complex Gaussian coordinate.

C Continuous Maximum-Finding Details

This appendix provides implementation details for approximating the continuous directional running maximum

$$m_{\mathbf{u}}(x) = \max_{0 \leq \xi \leq x} a_{\mathbf{u}}(\xi), \quad a_{\mathbf{u}}(x) = \mathbf{u}^{\top} \mathbf{G}(x) \mathbf{u}. \quad (1284)$$

The objective is to detect interior amplification peaks between stored OWNS stations without reconstructing random energy trajectories.

The principal method uses cubic Hermite interpolation of each directional gain. Adaptive midpoint refinement is then used to assess and reduce interpolation error. The same procedure applies to real and proper-complex Gaussian coordinates after replacing the transpose by a conjugate transpose where appropriate.

C.1 Directional values and derivatives

Let

$$0 = x_0 < x_1 < \dots < x_N = x_{\max} \quad (1285)$$

be the current streamwise grid. For direction $\mathbf{u}_k$, define

$$a_{k,n} = \mathbf{u}_k^\top \mathbf{G}(x_n) \mathbf{u}_k, \quad (1286)$$

$$d_{k,n} = \mathbf{u}_k^\top \mathbf{G}'(x_n) \mathbf{u}_k. \quad (1287)$$

The exact interval maximum is

$$m_{k,n}^I = \max_{x \in [x_n, x_{n+1}]} a_{\mathbf{u}_k}(x). \quad (1288)$$

If the full matrix derivatives are unavailable, the directional derivative can be calculated from the propagated directional state. Define

$$\mathbf{v}_k(x) = \mathbf{B}(x) \mathbf{u}_k. \quad (1289)$$

Then

$$a_{\mathbf{u}_k}(x) = \frac{\mathbf{v}_k(x)^* \mathbf{H}(x) \mathbf{v}_k(x)}{e_{\text{thres}}(x)}. \quad (1290)$$

If

$$\mathbf{v}'_k(x) = \mathbf{L}(x) \mathbf{v}_k(x), \quad (1291)$$

then

$$\begin{aligned} a'_{\mathbf{u}_k}(x) &= \frac{2\Re[\mathbf{v}_k(x)^* \mathbf{H}(x) \mathbf{v}'_k(x)] + \mathbf{v}_k(x)^* \mathbf{H}'(x) \mathbf{v}_k(x)}{e_{\text{thres}}(x)} \\ &\quad - \frac{e'_{\text{thres}}(x)}{e_{\text{thres}}(x)} a_{\mathbf{u}_k}(x). \end{aligned} \quad (1292)$$

The derivative should be computed using the projected OWNS evolution actually used by the marching solver. If the generalized OWNS formulation contains algebraic constraints, explicitly forming $\mathbf{L} = \mathbf{E}^{-1} \mathbf{K}$ may be invalid or poorly conditioned. The derivative returned by the consistent marching equations is preferred.

C.2 Cubic Hermite interpolant

Consider one direction on the interval

$$I_n = [x_n, x_{n+1}], \quad h_n = x_{n+1} - x_n. \quad (1293)$$

Introduce

$$s = \frac{x - x_n}{h_n}, \quad s \in [0, 1]. \quad (1294)$$

For compactness, denote

$$a_0 = a_{k,n}, \quad a_1 = a_{k,n+1}, \quad (1295)$$

$$d_0 = d_{k,n}, \quad d_1 = d_{k,n+1}. \quad (1296)$$

The cubic Hermite interpolant is

$$\hat{a}(s) = h_{00}(s)a_0 + h_{10}(s)h_nd_0 + h_{01}(s)a_1 + h_{11}(s)h_nd_1, \quad (1297)$$

where

$$h_{00}(s) = 2s^3 - 3s^2 + 1, \quad (1298)$$

$$h_{10}(s) = s^3 - 2s^2 + s, \quad (1299)$$

$$h_{01}(s) = -2s^3 + 3s^2, \quad (1300)$$

$$h_{11}(s) = s^3 - s^2. \quad (1301)$$

It satisfies

$$\hat{a}(0) = a_0, \quad \hat{a}(1) = a_1, \quad (1302)$$

$$\frac{1}{h_n}\hat{a}'(0) = d_0, \quad \frac{1}{h_n}\hat{a}'(1) = d_1, \quad (1303)$$

where the prime on $\hat{a}$ in eq. (1303) denotes differentiation with respect to s .

In power form,

$$\hat{a}(s) = c_3s^3 + c_2s^2 + c_1s + c_0, \quad (1304)$$

with

$$c_0 = a_0, \quad (1305)$$

$$c_1 = h_nd_0, \quad (1306)$$

$$c_2 = 3(a_1 - a_0) - h_n(2d_0 + d_1), \quad (1307)$$

$$c_3 = 2(a_0 - a_1) + h_n(d_0 + d_1). \quad (1308)$$

C.3 Stationary points of the interpolant

The derivative with respect to s is

$$\hat{a}'(s) = 3c_3s^2 + 2c_2s + c_1. \quad (1309)$$

Every maximum of the cubic on the closed interval occurs at an endpoint or an interior root of $\hat{a}'(s) = 0$.

If

$$|c_3| > \varepsilon_{\text{poly}}, \quad (1310)$$

the discriminant is

$$\Delta = 4c_2^2 - 12c_3c_1. \quad (1311)$$

For $\Delta \geq 0$, the formal roots are

$$s_{\pm} = \frac{-2c_2 \pm \sqrt{\Delta}}{6c_3}. \quad (1312)$$

Only roots in the open interval $(0, 1)$ are retained.

For improved numerical stability, solve the equivalent quadratic

$$As^2 + Bs + C = 0, \quad A = 3c_3, \quad B = 2c_2, \quad C = c_1, \quad (1313)$$

using

$$q = -\frac{1}{2} \left[B + \text{sign}(B) \sqrt{B^2 - 4AC} \right]. \quad (1314)$$

When $q \neq 0$, the roots are

$$s_1 = \frac{q}{A}, \quad s_2 = \frac{C}{q}. \quad (1315)$$

If $B = 0$, either sign may be used consistently in eq. (1314).

If

$$|c_3| \leq \varepsilon_{\text{poly}}, \quad |c_2| > \varepsilon_{\text{poly}}, \quad (1316)$$

the interpolant is effectively quadratic and its stationary point is

$$s_* = -\frac{c_1}{2c_2}. \quad (1317)$$

If both c_3 and c_2 are negligible, the interpolant is effectively linear or constant and has no isolated interior stationary point.

Define the candidate set

$$\mathcal{C} = \{0, 1\} \cup \{s \in (0, 1) : \hat{a}'(s) = 0\}. \quad (1318)$$

The interpolated interval maximum is

$$\boxed{\hat{m}^I = \max_{s \in \mathcal{C}} \hat{a}(s)}. \quad (1319)$$

Its estimated location is

$$\hat{x}^* = x_n + h_n \arg \max_{s \in \mathcal{C}} \hat{a}(s). \quad (1320)$$

C.4 Robust interval-maximum routine

The following routine includes scale-aware polynomial tolerances, stable quadratic roots, and endpoint merging.

The clipping of $\hat{m}^I$ to zero is intended only to remove roundoff-scale negativity. A substantially negative interpolated gain indicates inaccurate endpoint data or an inadequately resolved interval.

C.5 A posteriori interpolation indicator

A single cubic interpolant does not provide a rigorous maximum bound. An a posteriori indicator is obtained by comparing it with two interpolants on half intervals.

Let

$$x_m = \frac{x_0 + x_1}{2}. \quad (1321)$$

Evaluate

$$a_m = a_{\mathbf{u}}(x_m), \quad d_m = a'_{\mathbf{u}}(x_m). \quad (1322)$$

Let

$$\hat{m}_C^I = \text{maximum of the cubic on } [x_0, x_1], \quad (1323)$$

$$\hat{m}_L^I = \text{maximum of the cubic on } [x_0, x_m], \quad (1324)$$

$$\hat{m}_R^I = \text{maximum of the cubic on } [x_m, x_1]. \quad (1325)$$

Define

$$\hat{m}_S^I = \max(\hat{m}_L^I, \hat{m}_R^I). \quad (1326)$$

The maximum-difference indicator is

$$\eta_m = |\hat{m}_S^I - \hat{m}_C^I|. \quad (1327)$$

The midpoint interpolation residual is

$$\eta_a = \left| a_m - \hat{a}_C \left(\frac{1}{2} \right) \right|. \quad (1328)$$

If the coarse derivative at the midpoint is also desired, define

$$\eta_d = h \left| d_m - \frac{1}{h} \hat{a}'_C \left(\frac{1}{2} \right) \right|. \quad (1329)$$

Multiplication by h gives η_d the same units as the gain.

A combined indicator is

$$\boxed{\eta_I = \max(\eta_m, \eta_a, \eta_d)}. \quad (1330)$$

If derivative values are not sufficiently accurate for a meaningful η_d , the first two terms may be used alone.

The interval is accepted when

$$\eta_I \leq \varepsilon_{\text{abs}} + \varepsilon_{\text{rel}} \max(\hat{m}_S^I, m_{\text{scale}}), \quad (1331)$$

where m_{scale} is a small reference gain that prevents the relative tolerance from vanishing near zero.

C.6 Probability-oriented refinement indicator

The ultimate objective is accuracy in the transition CDF rather than uniform accuracy in every directional maximum. Define

$$\Psi_r(m) = \overline{F}_{\chi_r^2}(1/m), \quad m > 0. \quad (1332)$$

For $m = 0$, set

$$\Psi_r(0) = 0. \quad (1333)$$

A direct probability indicator is

$$\boxed{\eta_\Psi = |\Psi_r(\hat{m}_S^I) - \Psi_r(\hat{m}_C^I)|}. \quad (1334)$$

This indicator automatically accounts for the nonlinear sensitivity of the chi-square tail.

For angular direction k with weight ω_k , its estimated CDF contribution error is

$$\eta_{F,k} = \omega_k \eta_{\Psi,k}. \quad (1335)$$

An aggregate indicator for interval I_n is

$$\eta_{F,n} = \sum_{k=1}^K \eta_{F,k,n}. \quad (1336)$$

The direct probability indicator is preferable to the linearized estimate

$$\eta_{\Psi} \approx \frac{f_{\chi_r^2}(1/m)}{m^2} \eta_m \quad (1337)$$

when the maximum change is not small.

Only a new directional record affects the transition CDF. Let m_k^{up} be the running maximum before the current interval. Define the coarse and split updated records:

$$m_{k,C}^{\text{new}} = \max(m_k^{\text{up}}, \hat{m}_{k,C}^I), \quad (1338)$$

$$m_{k,S}^{\text{new}} = \max(m_k^{\text{up}}, \hat{m}_{k,S}^I). \quad (1339)$$

The record-aware probability indicator is

$$\boxed{\eta_{\Psi,k}^{\text{record}} = |\Psi_r(m_{k,S}^{\text{new}}) - \Psi_r(m_{k,C}^{\text{new}})|}. \quad (1340)$$

If both interval estimates lie below the previous record, this indicator is exactly zero. The interval does not need refinement for the current first-transition CDF unless it is being retained for other diagnostics or later threshold studies.

Care is required when reusing an adaptively refined grid for a lower threshold or larger disturbance amplitude. A peak that is probability-insensitive for one parameter choice may become important for another. A parameter-independent gain tolerance is safer when the same gain histories will support many threshold values.

C.7 Recursive adaptive interval routine

The following routine recursively resolves the maximum for one direction over one interval. It accepts an interval only when both the gain-based and probability-based criteria are satisfied.

The previous running maximum passed to the right half interval includes the resolved maximum from the left half. This ordering is important for record-aware probability refinement. A peak in the right half that does not exceed the left-half record cannot add first-transition probability.

For reusable gain data or threshold studies, it can be preferable to disable the record-aware shortcut and resolve every interval to a specified gain tolerance independently of m^{up} .

C.8 Shared adaptive grid for all angular directions

Independent recursion for every angular direction can require different streamwise evaluation points and repeated OWNS work. A shared-grid algorithm instead refines an interval whenever its aggregated error over the angular rule exceeds tolerance.

At every current station x_n , store

$$\mathbf{G}_n = \mathbf{G}(x_n), \quad \mathbf{G}'_n = \mathbf{G}'(x_n). \quad (1341)$$

For angular matrix

$$\mathbf{U} = [\mathbf{u}_1 \quad \cdots \quad \mathbf{u}_K], \quad (1342)$$

all endpoint gains can be evaluated in a batch:

$$\mathbf{Z}_n = \mathbf{G}_n \mathbf{U}, \quad (1343)$$

$$a_{k,n} = \mathbf{u}_k^\top \mathbf{z}_{k,n}. \quad (1344)$$

Similarly,

$$\mathbf{Z}'_n = \mathbf{G}'_n \mathbf{U}, \quad (1345)$$

$$d_{k,n} = \mathbf{u}_k^\top \mathbf{z}'_{k,n}. \quad (1346)$$

For interval I_n , let $\eta_{I,k,n}$ and $\eta_{\Psi,k,n}^{\text{record}}$ be the direction-specific gain and probability indicators. Possible aggregate criteria are

$$\eta_{I,n}^{\max} = \max_k \eta_{I,k,n}, \quad (1347)$$

$$\eta_{F,n} = \sum_{k=1}^K \omega_k \eta_{\Psi,k,n}^{\text{record}}, \quad (1348)$$

$$\eta_{\Psi,n}^{\max} = \max_k \eta_{\Psi,k,n}^{\text{record}}. \quad (1349)$$

An interval can be marked for refinement when

$$\eta_{I,n}^{\max} > \tau_{I,n} \quad (1350)$$

or

$$\eta_{F,n} > \varepsilon_{F,n}^{\text{stream}}. \quad (1351)$$

The gain criterion protects directions with small angular weight and supports later parameter reuse. The weighted CDF criterion focuses work on the current probability output. Using both is generally safer.

In practice, the full left-to-right running-maximum update should be repeated after each refinement pass, because insertion of new points in an upstream interval can alter the record passed to all downstream intervals.

C.9 Dense output from the OWNS marching scheme

If the OWNS integrator provides dense output, the propagated factor can be evaluated inside a marching step without restarting the spatial march. For an interval $[x_n, x_{n+1}]$, suppose the dense-output representation is

$$\mathbf{B}(x_n + \theta h_n) \approx \sum_{j=0}^p \mathbf{D}_{j,n} \theta^j, \quad 0 \leq \theta \leq 1. \quad (1352)$$

The directional state is

$$\mathbf{v}_{\mathbf{u}}(x_n + \theta h_n) \approx \sum_{j=0}^p \mathbf{D}_{j,n} \mathbf{u} \theta^j. \quad (1353)$$

The gain can then be evaluated at arbitrary θ using

$$a_{\mathbf{u}}(x) = \frac{\mathbf{v}_{\mathbf{u}}(x)^* \mathbf{H}(x) \mathbf{v}_{\mathbf{u}}(x)}{e_{\text{thres}}(x)}. \quad (1354)$$

Dense output is particularly useful for:

- midpoint and quarter-point refinement;
- verification of Hermite maxima;
- scalar optimization of selected directional gains; and
- root finding for threshold crossings after a radius has been specified.

The order and error tolerance of the dense-output formula should be consistent with the main OWNS marching method. Interpolating $\mathbf{B}(x)$ at high order does not improve an under-resolved or low-order spatial march.

C.10 Direct bounded optimization of directional gain

For selected directions, the interval maximum can be evaluated by a one-dimensional bounded optimization routine:

$$m_{\mathbf{u},n}^I = \max_{x \in [x_n, x_{n+1}]} a_{\mathbf{u}}(x). \quad (1355)$$

Derivative-free methods such as Brent's method can be used after bracketing an interior maximum. If a reliable derivative is available, one can instead find roots of

$$a'_{\mathbf{u}}(x) = 0 \quad (1356)$$

and evaluate the gain at every root and endpoint.

A single unimodal bounded maximization is not sufficient if the directional gain can have multiple extrema inside one interval. Possible safeguards are:

- (i) subdivide the interval using derivative sign changes;
- (ii) sample a coarse set of interior points and create separate candidate brackets;
- (iii) use adaptive Chebyshev interpolation to identify all stationary points; or
- (iv) refine the interval until at most one stationary point is resolved in each subinterval.

Direct optimization is more expensive than cubic interpolation but provides a valuable verification method for directions with large transition contributions.

C.11 Chebyshev interpolation alternative

For an interval with complicated directional gain variation, map $x \in [x_n, x_{n+1}]$ to $\zeta \in [-1, 1]$ and sample the gain at Chebyshev–Lobatto points:

$$\zeta_j = \cos\left(\frac{j\pi}{p}\right), \quad j = 0, \dots, p. \quad (1357)$$

The physical locations are

$$x_j = \frac{x_n + x_{n+1}}{2} + \frac{x_{n+1} - x_n}{2} \zeta_j. \quad (1358)$$

Construct the polynomial interpolant

$$\hat{a}_p(\zeta) = \sum_{j=0}^p \alpha_j T_j(\zeta), \quad (1359)$$

where T_j are Chebyshev polynomials.

The interpolant maximum is found by evaluating the endpoints and all real roots of

$$\frac{d\hat{a}_p}{d\zeta} = 0 \quad (1360)$$

inside $[-1, 1]$. The interpolation order is increased until the maximum and the trailing coefficients stabilize.

Chebyshev interpolation is useful when:

- dense output allows inexpensive interior evaluations;
- several extrema occur within one original OWNS interval;
- endpoint derivatives are unreliable; or
- a high-accuracy reference maximum is needed.

Cubic Hermite interpolation remains preferable for routine production calculations because it uses data already available at the interval endpoints and requires only quadratic stationary-point solves.

C.12 Conservative Lipschitz bounds

Suppose

$$\|\mathbf{G}'(x)\|_2 \leq L_n \quad \text{for every } x \in [x_n, x_{n+1}]. \quad (1361)$$

For a unit direction,

$$|a'_{\mathbf{u}}(x)| = |\mathbf{u}^\top \mathbf{G}'(x) \mathbf{u}| \leq L_n. \quad (1362)$$

Let

$$a_0 = a_{\mathbf{u}}(x_n), \quad a_1 = a_{\mathbf{u}}(x_{n+1}), \quad h = x_{n+1} - x_n. \quad (1363)$$

Then, for $t = x - x_n \in [0, h]$,

$$a_{\mathbf{u}}(x) \leq \min(a_0 + L_n t, a_1 + L_n(h - t)). \quad (1364)$$

The intersection of the two affine upper bounds occurs at

$$t_* = \frac{a_1 - a_0 + L_n h}{2L_n} \quad (1365)$$

when $L_n > 0$. If $t_* \in [0, h]$, the maximum of the envelope is

$$U_{\mathbf{u},n} = \frac{a_0 + a_1 + L_n h}{2}. \quad (1366)$$

If the intersection lies outside the interval, the tighter endpoint bound gives

$$U_{\mathbf{u},n} = \max(a_0, a_1). \quad (1367)$$

A universally valid, slightly simpler expression is

$$\boxed{U_{\mathbf{u},n} = \max(a_0, a_1) + \frac{L_n h}{2}}. \quad (1368)$$

The endpoint lower bound is

$$L_{\mathbf{u},n} = \max(a_0, a_1). \quad (1369)$$

Thus,

$$L_{\mathbf{u},n} \leq m_{\mathbf{u},n}^I \leq U_{\mathbf{u},n}. \quad (1370)$$

The interval can be certified to have negligible CDF uncertainty for direction k if

$$\omega_k \left| \Psi_r(\max[m_k^{\text{up}}, U_{k,n}]) - \Psi_r(\max[m_k^{\text{up}}, L_{k,n}]) \right| \leq \varepsilon_{F,k,n}. \quad (1371)$$

Summing over directions gives a certified bound for the represented angular rule:

$$F_n^L \leq F_n \leq F_n^U. \quad (1372)$$

The main difficulty is obtaining a reliable bound L_n . Possible choices include:

- a conservative analytical bound on $\|\mathbf{G}'(x)\|_2$;
- the maximum of $\|\mathbf{G}'(x)\|_2$ over a sufficiently refined set of interior points, multiplied by a safety factor;
- interval arithmetic or validated numerics; or
- a direction-specific bound on $|a'_{\mathbf{u}}(x)|$.

A sampled derivative maximum is an error indicator rather than a rigorous bound unless additional regularity information is available.

C.13 Threshold-crossing location for a specified radius

The CDF calculation requires directional running maxima but not an individual crossing location for each Gaussian radius. If a conditional realization $(R, \mathbf{u})$ is given, however, its first crossing solves

$$R^2 a_{\mathbf{u}}(x) = 1. \quad (1373)$$

Define

$$f_{R,\mathbf{u}}(x) = R^2 a_{\mathbf{u}}(x) - 1. \quad (1374)$$

Search streamwise intervals from upstream to downstream. The first interval containing either:

$$f_{R,\mathbf{u}}(x_n) < 0, \quad f_{R,\mathbf{u}}(x_{n+1}) \geq 0 \quad (1375)$$

or an interior Hermite maximum satisfying

$$R^2 \hat{m}_{\mathbf{u},n}^I \geq 1 \quad (1376)$$

contains the first crossing.

If both endpoints are below threshold but the interior maximum is above threshold, the Hermite polynomial can have two roots. The smaller root in $s \in (0, 1)$ is the first crossing. Solve

$$R^2 \hat{a}(s) - 1 = 0 \quad (1377)$$

using a bracketed root finder on the increasing portion preceding the first threshold-exceeding maximum. A bracketed method is preferable to an unconstrained cubic formula because it preserves the upstream root selection and is more robust near tangential contact.

This algorithm is useful for trajectory verification and conditional sampling. It is not needed for evaluating the marginal transition CDF, for which radial integration is more efficient.

C.14 Handling nonsmooth or discontinuous matrix data

Cubic Hermite interpolation assumes sufficient smoothness of the directional gain. Nonsmoothness can enter through:

- piecewise interpolation of the base flow;
- abrupt changes in the OWNS projection parameters;
- remeshing or wall-normal coordinate changes;
- a piecewise-defined transition threshold;
- discontinuous energy normalization; or
- numerical factor resets that are not transformed consistently.

If $\mathbf{G}(x)$ is continuous but its derivative jumps at a known station x_c , split the streamwise domain at x_c and construct separate interpolants on either side. Do not use one Hermite interval that spans the derivative discontinuity.

If $\mathbf{G}(x)$ itself has a physical or modeling discontinuity, the first-transition event should use the appropriate one-sided value at the discontinuity. Store x_c explicitly and evaluate both sides if needed. The transition CDF can then contain a jump if a positive Gaussian probability mass is moved instantaneously across the threshold by the discontinuity.

A discontinuity caused only by inconsistent numerical coordinates is not physical and should be corrected before maximum detection.

C.15 Near-degenerate and flat maxima

A broad or flat maximum can satisfy

$$a'_{\mathbf{u}}(x_*) \approx 0, \quad a''_{\mathbf{u}}(x_*) \approx 0. \quad (1378)$$

Its location may be poorly determined even when its maximum value is accurate. Since the transition CDF depends on the maximum value rather than the maximizing location, refinement should focus primarily on

$$|\hat{m}^I - m^I|. \quad (1379)$$

A large change in the estimated maximizer with a negligible change in the maximum value does not necessarily require further refinement for the CDF.

If a transition-location density is inferred from record locations, flat maxima require more care. They can distribute threshold crossings over an extended streamwise region depending on radius, even though the directional maximum value is nearly constant. In this case, the CDF should be resolved directly as a function of x rather than assigning all directional contribution to a single maximizer.

C.16 Streamwise CDF construction inside an interval

The running maximum at stored endpoints gives $F_{X_{\text{tr}}}(x_n)$, but a smooth CDF inside each interval can also be constructed from the directional interpolants.

For $x \in [x_n, x_{n+1}]$, define

$$m_k(x) = \max \left[m_k(x_n), \max_{\xi \in [x_n, x]} \hat{a}_{k,n}(\xi) \right]. \quad (1380)$$

Then

$$\hat{F}_{X_{\text{tr}}}(x) = \sum_{k=1}^K \omega_k \bar{F}_{\chi_r^2} \left[\frac{1}{m_k(x)} \right]. \quad (1381)$$

This construction is monotone because every $m_k(x)$ is a running maximum.

A generic cubic interpolation applied directly to the endpoint CDF values does not automatically preserve the first-passage structure. For plotting or inverse-CDF sampling, either use eq. (1381) or a monotone scalar interpolant such as a piecewise cubic Hermite interpolating polynomial with monotonicity-preserving slopes.

For conservative reporting, interval probabilities

$$p_n = F_{X_{\text{tr}}}(x_{n+1}) - F_{X_{\text{tr}}}(x_n) \quad (1382)$$

remain preferable to a pointwise differentiated density.

C.17 Recommended tolerance hierarchy

The streamwise maximum tolerance should be chosen relative to the desired CDF accuracy. Let

$$\varepsilon_F^{\text{target}} \quad (1383)$$

be the total desired CDF error. A possible allocation is

$$\varepsilon_F^{\text{target}} = \varepsilon_F^{\text{OWNS}} + \varepsilon_F^{\text{red}} + \varepsilon_F^{\text{stream}} + \varepsilon_F^{\text{angular}} + \varepsilon_F^{\text{roundoff}}. \quad (1384)$$

The streamwise refinement should target

$$\varepsilon_F^{\text{stream}} < \varepsilon_F^{\text{target}} \quad (1385)$$

with a margin sufficient to distinguish it from angular and OWNS errors.

For N_I streamwise intervals, a simple local allocation is

$$\varepsilon_{F,n}^{\text{stream}} = \frac{\varepsilon_F^{\text{stream}}}{N_I}. \quad (1386)$$

This allocation can be overly conservative. An adaptive allocation based on interval transition contribution is

$$\varepsilon_{F,n}^{\text{stream}} = \varepsilon_F^{\text{stream}} \frac{\max(p_n^{\text{pilot}}, p_{\min})}{\sum_j \max(p_j^{\text{pilot}}, p_{\min})}, \quad (1387)$$

where p_n^{pilot} is a pilot interval probability and $p_{\min} > 0$ prevents any interval from receiving zero tolerance.

A gain tolerance should also be imposed even when the probability indicator is small:

$$\eta_{I,k,n} \leq \varepsilon_m^{\text{abs}} + \varepsilon_m^{\text{rel}} \max(m_{k,n}, m_{\text{scale}}). \quad (1388)$$

This protects parameter reuse and prevents a numerically unresolved gain history from appearing converged merely because the current threshold places it in an insensitive probability regime.

C.18 Verification tests for maximum detection

The maximum-finding implementation should be verified using functions with known extrema.

C.18.1 Exactly cubic gain

Let

$$a(x) = c_3x^3 + c_2x^2 + c_1x + c_0. \quad (1389)$$

A cubic Hermite interpolant constructed from exact endpoint values and derivatives must reproduce $a(x)$ exactly. The computed interval maximum should agree with the analytical stationary-point result to floating-point accuracy.

C.18.2 Narrow Gaussian peak

Let

$$a(x) = a_0 + A \exp \left[-\frac{(x - x_p)^2}{2\sigma^2} \right]. \quad (1390)$$

The exact maximum is

$$a(x_p) = a_0 + A \quad (1391)$$

when x_p lies within the interval. Vary σ/h and the offset of x_p from the midpoint to test adaptive detection of narrow and off-center peaks.

C.18.3 Two-peak record test

Use

$$a(x) = a_0 + A_1 \exp \left[-\frac{(x - x_1)^2}{2\sigma_1^2} \right] + A_2 \exp \left[-\frac{(x - x_2)^2}{2\sigma_2^2} \right]. \quad (1392)$$

If the second peak is lower than the first, it should not change the running maximum. If it is higher, only the increment in the directional radial tail should contribute to downstream first-transition probability.

C.18.4 Flat maximum test

Let

$$a(x) = a_{\max} - c(x - x_p)^4. \quad (1393)$$

This tests whether the maximum value converges even when the second derivative vanishes at the maximum.

C.18.5 Oscillatory test

Let

$$a(x) = a_0 + A \sin^2(\kappa x). \quad (1394)$$

Choose κ so that several extrema occur in one initial interval. This verifies that adaptive subdivision is required when one cubic cannot represent all interior peaks.

For each test, record:

$$\epsilon_m = |\hat{m} - m_{\text{exact}}|, \quad (1395)$$

$$\epsilon_x = |\hat{x}^* - x_{\text{exact}}^*|, \quad (1396)$$

$$\epsilon_\Psi = |\Psi_r(\hat{m}) - \Psi_r(m_{\text{exact}})|. \quad (1397)$$

The maximum-value and probability errors are more relevant to the transition CDF than the maximizing-location error.

C.19 Recommended production procedure

For the OWNS first-transition calculation, the recommended production procedure is:

1. Evaluate $\mathbf{G}(x_n)$ and, where reliable, $\mathbf{G}'(x_n)$ on the initial OWNS output grid.
2. Verify directional derivatives against refined finite differences at selected stations and directions.
3. Construct cubic Hermite interpolants for every angular direction and streamwise interval.
4. Evaluate endpoints and all interior stationary points using a stable quadratic-root routine.
5. Evaluate midpoint data and compare coarse and split interpolants.
6. Refine a shared streamwise grid using both gain-based and probability-based indicators.
7. Recompute all upstream running maxima after each shared-grid refinement pass.
8. Retain a gain-based tolerance even when using record-aware probability refinement, especially if the matrix family will be reused for multiple thresholds or disturbance amplitudes.
9. Verify high-contribution directions using dense output, Chebyshev interpolation, or direct bounded optimization.
10. Compare the final adaptive CDF with a globally refined streamwise-grid result.
11. If one-sided probability guarantees are required, supplement the interpolation with conservative derivative or Lipschitz bounds.
12. Report the number of added streamwise points, maximum recursion depth, unresolved-interval warnings, and the observed change in the complete transition CDF.

The continuous maximum calculation should be regarded as part of the first-passage discretization, not merely as postprocessing. An unresolved transient peak changes which realizations are classified as transitioned and therefore introduces a systematic underestimate of the transition probability.

D Stabilized Propagation of the Correlation Factor

This appendix describes stabilized procedures for propagating the initial correlation factor through a strongly non-normal OWNS system. The objective is to preserve the exact map

$$\mathbf{q}(x) = \mathbf{B}(x)\mathbf{w}, \quad \mathbf{B}(x) = \mathbf{M}(0, x)\mathbf{B}_0, \quad (1398)$$

while preventing the physical columns of $\mathbf{B}(x)$ from becoming numerically linearly dependent.

The central principle is that orthogonalization, singular-value factorization, and scaling may change the numerical representation of $\mathbf{B}(x)$, but they must not change the product mapping the original Gaussian coordinate $\mathbf{w}$ to the physical state. Every basis transformation must therefore be retained in an accumulated coordinate factor.

D.1 Discrete OWNS propagation

Let

$$0 = x_0 < x_1 < \cdots < x_N = x_{\max} \quad (1399)$$

be a sequence of stabilization or output stations. Denote the numerical OWNS propagation map over one interval by

$$\mathbf{P}_{n+1,n} \approx \mathbf{M}(x_n, x_{n+1}). \quad (1400)$$

The propagated factor satisfies

$$\mathbf{B}_{n+1} = \mathbf{P}_{n+1,n} \mathbf{B}_n, \quad \mathbf{B}_0 = \mathbf{B}(0). \quad (1401)$$

Repeated substitution gives

$$\mathbf{B}_n = \mathbf{P}_{n,n-1} \cdots \mathbf{P}_{1,0} \mathbf{B}_0. \quad (1402)$$

In a strongly non-normal system, the columns of $\mathbf{B}_n$ can become nearly linearly dependent. Directly marching these columns can then lose information about weak but potentially transition-relevant combinations. Stabilized propagation replaces the direct factor by

$$\boxed{\mathbf{B}_n = \mathbf{Q}_n \mathbf{T}_n}, \quad (1403)$$

where

$$\mathbf{Q}_n^* \mathbf{Q}_n = \mathbf{I}_r \quad (1404)$$

and $\mathbf{T}_n \in \mathbb{C}^{r \times r}$ maps the original stochastic coordinate into the current orthonormal physical basis.

The physical state is

$$\mathbf{q}(x_n) = \mathbf{Q}_n \mathbf{T}_n \mathbf{w}. \quad (1405)$$

The matrix $\mathbf{T}_n$ must be retained even though it can be ill-conditioned. Its growth and mixing contain the transient amplification generated by the non-normal propagator.

D.2 Initialization

Compute a thin QR factorization of the initial synthesis factor:

$$\mathbf{B}_0 = \mathbf{Q}_0 \mathbf{T}_0, \quad (1406)$$

where

$$\mathbf{Q}_0 \in \mathbb{C}^{n_q \times r}, \quad \mathbf{T}_0 \in \mathbb{C}^{r \times r}. \quad (1407)$$

If $\mathbf{B}_0$ has full column rank, $\mathbf{T}_0$ is nonsingular and upper triangular.

The initial state becomes

$$\mathbf{q}(0) = \mathbf{Q}_0 \mathbf{T}_0 \mathbf{w}. \quad (1408)$$

It is incorrect to replace this by

$$\mathbf{q}(0) = \mathbf{Q}_0 \mathbf{w} \quad (1409)$$

unless $\mathbf{T}_0 = \mathbf{I}$. Discarding $\mathbf{T}_0$ removes the amplitudes and correlations encoded in the initial covariance factor.

If the initial factor is numerically rank deficient, compute a rank-revealing QR factorization or singular-value decomposition:

$$\mathbf{B}_0 = \mathbf{U}_0 \mathbf{\Sigma}_0 \mathbf{V}_0^*. \quad (1410)$$

A rank should be removed only according to a documented covariance and transition-oriented tolerance. After retaining rank r , one may set

$$\mathbf{Q}_0 = \mathbf{U}_0(:, 1:r), \quad \mathbf{T}_0 = \mathbf{\Sigma}_0(1:r, 1:r) \mathbf{V}_0(:, 1:r)^*. \quad (1411)$$

D.3 Periodic QR update

Assume that

$$\mathbf{B}_n = \mathbf{Q}_n \mathbf{T}_n. \quad (1412)$$

Propagate only the orthonormal physical basis over the next interval:

$$\hat{\mathbf{Q}}_{n+1} = \mathbf{P}_{n+1,n} \mathbf{Q}_n. \quad (1413)$$

Then

$$\begin{aligned} \mathbf{B}_{n+1} &= \mathbf{P}_{n+1,n} \mathbf{B}_n \\ &= \hat{\mathbf{Q}}_{n+1} \mathbf{T}_n. \end{aligned} \quad (1414)$$

Compute the thin QR factorization

$$\hat{\mathbf{Q}}_{n+1} = \mathbf{Q}_{n+1} \mathbf{R}_{n+1}. \quad (1415)$$

The accumulated coordinate factor is updated as

$$\boxed{\mathbf{T}_{n+1} = \mathbf{R}_{n+1} \mathbf{T}_n.} \quad (1416)$$

Therefore,

$$\begin{aligned} \mathbf{Q}_{n+1} \mathbf{T}_{n+1} &= \mathbf{Q}_{n+1} \mathbf{R}_{n+1} \mathbf{T}_n \\ &= \hat{\mathbf{Q}}_{n+1} \mathbf{T}_n \\ &= \mathbf{P}_{n+1,n} \mathbf{Q}_n \mathbf{T}_n \\ &= \mathbf{P}_{n+1,n} \mathbf{B}_n = \mathbf{B}_{n+1}. \end{aligned} \quad (1417)$$

Thus, the QR update changes only the numerical representation, not the propagated factor.

After n stabilization intervals,

$$\mathbf{T}_n = \mathbf{R}_n \mathbf{R}_{n-1} \cdots \mathbf{R}_1 \mathbf{T}_0. \quad (1418)$$

The ordering of these factors is essential. Since the matrices do not generally commute, reversing or independently diagonalizing the factors changes the map from $\mathbf{w}$ to the physical state.

D.4 Phase and sign convention for QR factors

A QR factorization is not unique. For a diagonal unitary matrix

$$D = \text{diag}(d_1, \dots, d_r), \quad |d_j| = 1, \quad (1419)$$

the factors

$$\tilde{Q} = QD, \quad \tilde{R} = D^*R \quad (1420)$$

produce the same product.

For reproducibility and smoother factor histories, choose the diagonal entries of R to be real and nonnegative. If the raw QR factorization gives diagonal entry $R_{jj} \neq 0$, set

$$d_j = \frac{R_{jj}}{|R_{jj}|}, \quad (1421)$$

and apply

$$Q \leftarrow QD, \quad R \leftarrow D^*R. \quad (1422)$$

For a real factorization, this reduces to choosing positive diagonal entries of R .

The transition calculation is invariant to this convention if the coordinate factor is updated consistently. The convention improves diagnostics by preventing arbitrary sign or phase jumps.

D.5 Modified Gram–Schmidt and Householder QR

Householder QR is generally preferred for a dense block of propagated columns because it provides good numerical orthogonality:

$$\|Q^*Q - I\|_2 = \mathcal{O}(\epsilon_{\text{mach}}) \quad (1423)$$

up to dimension-dependent factors.

Modified Gram–Schmidt can be easier to integrate into a columnwise march but may require reorthogonalization. A two-pass procedure gives

$$Q = \text{MGS2}(\hat{Q}) \quad (1424)$$

and generally provides much better orthogonality than a single pass.

Monitor

$$\eta_Q = \|Q^*Q - I\|_F. \quad (1425)$$

If

$$\eta_Q > \varepsilon_Q, \quad (1426)$$

perform an additional orthogonalization step and multiply its triangular factor into the accumulated coordinate map.

If a second QR factorization gives

$$Q^{(1)} = Q^{(2)}R^{(2)}, \quad (1427)$$

then the total local triangular factor is

$$R_{\text{total}} = R^{(2)}R^{(1)}. \quad (1428)$$

Both factors must be accumulated.

D.6 Adaptive stabilization frequency

QR stabilization need not coincide with every stored output station. A short interval reduces the chance that propagated columns become numerically dependent before reorthogonalization, while a long interval reduces factorization cost.

Possible triggers include:

- (i) loss of orthogonality of a previously orthonormal block;
- (ii) growth in the estimated condition number of the propagated block;
- (iii) excessive spread in column norms;
- (iv) failure of the physical-state solver to maintain relative accuracy in weak columns; or
- (v) a prescribed maximum streamwise distance between orthogonalizations.

Before reorthogonalization, define the block Gram matrix

$$\mathbf{K}_Q = \hat{\mathbf{Q}}^* \hat{\mathbf{Q}}. \quad (1429)$$

An estimated condition number is

$$\hat{\kappa}(\hat{\mathbf{Q}}) = \sqrt{\frac{\lambda_{\max}(\mathbf{K}_Q)}{\lambda_{\min}(\mathbf{K}_Q)}}. \quad (1430)$$

A stabilization can be triggered when

$$\hat{\kappa}(\hat{\mathbf{Q}}) > \kappa_{QR}. \quad (1431)$$

The threshold κ_{QR} should remain well below the reciprocal of machine precision. Waiting until the block is numerically singular cannot recover directions that have already been lost.

D.7 Scaled QR propagation

The accumulated factor $\mathbf{T}_n$ can overflow or underflow even though $\mathbf{Q}_n$ remains well conditioned. Introduce a scalar logarithmic scale:

$$\mathbf{T}_n = \exp(\ell_n) \bar{\mathbf{T}}_n. \quad (1432)$$

After the QR update,

$$\hat{\mathbf{T}}_{n+1} = \mathbf{R}_{n+1} \bar{\mathbf{T}}_n. \quad (1433)$$

Choose a positive normalization

$$s_{n+1} = \|\hat{\mathbf{T}}_{n+1}\|_{\text{scale}}, \quad (1434)$$

where the scale norm can be the Frobenius norm, spectral norm estimate, or maximum absolute entry. Update

$$\bar{\mathbf{T}}_{n+1} = \frac{\hat{\mathbf{T}}_{n+1}}{s_{n+1}}, \quad (1435)$$

$$\ell_{n+1} = \ell_n + \log s_{n+1}. \quad (1436)$$

Then

$$\mathbf{B}_{n+1} = \exp(\ell_{n+1}) \mathbf{Q}_{n+1} \bar{\mathbf{T}}_{n+1}. \quad (1437)$$

The normalized energy matrix is

$$\mathbf{G}_{n+1} = \exp(2\ell_{n+1})\overline{\mathbf{G}}_{n+1}, \quad (1438)$$

where

$$\overline{\mathbf{G}}_{n+1} = \frac{\Re \left[\overline{\mathbf{T}}_{n+1}^* \mathbf{Q}_{n+1}^* \mathbf{H}_{n+1} \mathbf{Q}_{n+1} \overline{\mathbf{T}}_{n+1} \right]}{e_{\text{thres},n+1}}. \quad (1439)$$

For a direction $\mathbf{u}$,

$$\log a_{\mathbf{u}}(x_{n+1}) = 2\ell_{n+1} + \log \left[\mathbf{u}^\top \overline{\mathbf{G}}_{n+1} \mathbf{u} \right]. \quad (1440)$$

This expression permits running maxima to be maintained in logarithmic form when the physical gain exceeds the floating-point range.

A scalar scale controls overall growth but not the internal condition number of $\mathbf{T}_n$. If the singular values of $\mathbf{T}_n$ span many orders of magnitude, an SVD-based reconditioning or higher precision may still be required.

D.8 SVD reconditioning without loss of history

Suppose the current factor is

$$\mathbf{B}_n = \mathbf{Q}_n \mathbf{T}_n. \quad (1441)$$

Compute an SVD of the small coordinate matrix:

$$\mathbf{T}_n = \mathbf{U}_{T,n} \boldsymbol{\Sigma}_{T,n} \mathbf{V}_{T,n}^*. \quad (1442)$$

Since $\mathbf{U}_{T,n}$ is unitary, define

$$\tilde{\mathbf{Q}}_n = \mathbf{Q}_n \mathbf{U}_{T,n}, \quad (1443)$$

$$\tilde{\mathbf{T}}_n = \boldsymbol{\Sigma}_{T,n} \mathbf{V}_{T,n}^*. \quad (1444)$$

Then

$$\tilde{\mathbf{Q}}_n \tilde{\mathbf{T}}_n = \mathbf{Q}_n \mathbf{T}_n = \mathbf{B}_n. \quad (1445)$$

The physical basis remains orthonormal:

$$\tilde{\mathbf{Q}}_n^* \tilde{\mathbf{Q}}_n = \mathbf{I}. \quad (1446)$$

The right singular vectors $\mathbf{V}_{T,n}$ are part of the coordinate map and must remain in $\tilde{\mathbf{T}}_n$. Replacing $\tilde{\mathbf{T}}_n$ by $\boldsymbol{\Sigma}_{T,n}$ would perform an untracked streamwise-dependent rotation of the Gaussian coordinate.

The energy matrix after reconditioning is

$$\mathbf{A}_n = \Re \left[\tilde{\mathbf{T}}_n^* \tilde{\mathbf{Q}}_n^* \mathbf{H}_n \tilde{\mathbf{Q}}_n \tilde{\mathbf{T}}_n \right], \quad (1447)$$

which is identical to its value before reconditioning.

D.9 Rank truncation during propagation

An SVD exposes small singular values but does not by itself justify discarding them. If a truncation is applied at station x_n , write

$$\mathbf{T}_n = \mathbf{U}_{T,n} \begin{bmatrix} \boldsymbol{\Sigma}_{s,n} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{\Sigma}_{\perp,n} \end{bmatrix} \mathbf{V}_{T,n}^*. \quad (1448)$$

A local rank- s approximation is

$$\mathbf{B}_{n,s}^{\text{local}} = \mathbf{Q}_n \mathbf{U}_{T,n}(:, 1:s) \boldsymbol{\Sigma}_{s,n} \mathbf{V}_{T,n}(:, 1:s)^*. \quad (1449)$$

Although this approximation is optimal for the local factor in the Euclidean Frobenius or spectral norm, it is generally unsuitable for first-transition prediction if repeated independently at different stations. The retained right-coordinate subspace can change with x_n , so the approximate realization map is no longer generated by one fixed initial Gaussian coordinate.

A safe reduction strategy chooses one fixed initial-coordinate basis

$$\mathbf{V}_s \in \mathbb{R}^{r \times s} \quad (1450)$$

and propagates

$$\mathbf{B}_s(x) = \mathbf{B}(x) \mathbf{V}_s. \quad (1451)$$

Such a basis should be selected using the transition-oriented methods in section 9.

A local singular value may be used as a diagnostic, but a direction should be removed only after demonstrating that its contribution to the complete family $\{\mathbf{G}(x)\}$ and the transition CDF is negligible.

D.10 Continuous orthonormalization

For a continuously varying factor,

$$\mathbf{B}(x) = \mathbf{Q}(x) \mathbf{T}(x), \quad \mathbf{Q}(x)^* \mathbf{Q}(x) = \mathbf{I}, \quad (1452)$$

differentiation gives

$$\mathbf{B}' = \mathbf{Q}' \mathbf{T} + \mathbf{Q} \mathbf{T}'. \quad (1453)$$

Using

$$\mathbf{B}' = \mathbf{L} \mathbf{B} = \mathbf{L} \mathbf{Q} \mathbf{T}, \quad (1454)$$

and assuming $\mathbf{T}$ is nonsingular gives

$$\mathbf{Q}' + \mathbf{Q} \mathbf{T}' \mathbf{T}^{-1} = \mathbf{L} \mathbf{Q}. \quad (1455)$$

Let

$$\mathbf{S}(x) = \mathbf{Q}(x)^* \mathbf{L}(x) \mathbf{Q}(x). \quad (1456)$$

Choose a gauge matrix $\boldsymbol{\Omega}(x)$ satisfying

$$\boldsymbol{\Omega}^* = -\boldsymbol{\Omega}. \quad (1457)$$

One orthonormal evolution is

$$\boxed{\mathbf{Q}' = (\mathbf{I} - \mathbf{Q} \mathbf{Q}^*) \mathbf{L} \mathbf{Q} + \mathbf{Q} \boldsymbol{\Omega}.} \quad (1458)$$

The coordinate factor then satisfies

$$\boxed{\mathbf{T}' = (\mathbf{S} - \boldsymbol{\Omega}) \mathbf{T}.} \quad (1459)$$

To verify orthogonality preservation,

$$\begin{aligned}\frac{d}{dx}(Q^*Q) &= Q'^*Q + Q^*Q' \\ &= \Omega^* + \Omega = \mathbf{0}.\end{aligned}\tag{1460}$$

The gauge may be selected so that $S - \Omega$ is upper triangular, yielding a continuous QR factorization. Let

$$S = S_L + S_D + S_U,\tag{1461}$$

where the subscripts denote strictly lower, diagonal, and strictly upper parts. One possible gauge is

$$\Omega = S_L - S_L^*.\tag{1462}$$

Then

$$S - \Omega\tag{1463}$$

has zero strictly lower part and is upper triangular. Other gauge choices are possible.

Continuous orthonormalization provides smooth basis histories but is more intrusive to implement than periodic QR. Periodic Householder QR is generally the recommended default for an existing OWNS solver.

D.11 Stabilized formation of the energy matrix

Given

$$B = QT,\tag{1464}$$

define the energy metric projected onto the orthonormal physical basis:

$$H_Q = Q^* H Q.\tag{1465}$$

Then

$$\boxed{A = \Re [T^* H_Q T]}.\tag{1466}$$

If

$$H = W^* W,\tag{1467}$$

form

$$Y_Q = W Q,\tag{1468}$$

so that

$$H_Q = Y_Q^* Y_Q.\tag{1469}$$

This Gram construction makes positive semidefiniteness explicit.

A Cholesky factorization of H_Q can be used when it is positive definite:

$$H_Q = L_Q^* L_Q.\tag{1470}$$

Then

$$A = \Re [(L_Q T)^* (L_Q T)].\tag{1471}$$

After formation, apply

$$A \leftarrow \frac{1}{2} (A + A^\top).\tag{1472}$$

Only roundoff-scale negative eigenvalues should be projected to zero.

The normalized energy matrix is

$$G(x) = \frac{A(x)}{e_{\text{thres}}(x)}.\tag{1473}$$

D.12 Directional gain without reconstructing the factor

The complete physical factor need not be explicitly reconstructed. For a direction $\mathbf{u}$, first compute

$$\mathbf{c} = \mathbf{T}\mathbf{u}. \quad (1474)$$

The physical directional state is

$$\mathbf{v} = \mathbf{Q}\mathbf{c}, \quad (1475)$$

and the normalized gain is

$$a_{\mathbf{u}}(x) = \frac{\mathbf{c}^* \mathbf{H}_Q(x) \mathbf{c}}{e_{\text{thres}}(x)}. \quad (1476)$$

If $\mathbf{H}_Q = \mathbf{L}_Q^* \mathbf{L}_Q$,

$$a_{\mathbf{u}}(x) = \frac{\|\mathbf{L}_Q(x) \mathbf{T}(x) \mathbf{u}\|_2^2}{e_{\text{thres}}(x)}. \quad (1477)$$

This calculation occurs entirely in the r -dimensional propagated basis after $\mathbf{H}_Q$ has been formed.

For a block of angular directions

$$\mathbf{U} = [\mathbf{u}_1 \quad \cdots \quad \mathbf{u}_K], \quad (1478)$$

compute

$$\mathbf{C} = \mathbf{T}\mathbf{U} \quad (1479)$$

and obtain all gains from the columnwise quadratic forms of $\mathbf{H}_Q$.

D.13 Derivative of the factored map

For streamwise maximum detection, the derivative of the physical factor is

$$\mathbf{B}' = \mathbf{Q}'\mathbf{T} + \mathbf{Q}\mathbf{T}'. \quad (1480)$$

Using only $\mathbf{Q}'$ omits the amplification and mixing carried by $\mathbf{T}'$.

The energy matrix derivative can be obtained by differentiating eq. (1466):

$$\mathbf{A}' = \Re [\mathbf{T}'^* \mathbf{H}_Q \mathbf{T} + \mathbf{T}^* \mathbf{H}'_Q \mathbf{T} + \mathbf{T}^* \mathbf{H}_Q \mathbf{T}'], \quad (1481)$$

where

$$\mathbf{H}'_Q = \mathbf{Q}'^* \mathbf{H}_Q + \mathbf{Q}^* \mathbf{H}'_Q + \mathbf{Q}^* \mathbf{H}_Q \mathbf{Q}'. \quad (1482)$$

This expression is algebraically equivalent to

$$\mathbf{A}' = \Re \{ \mathbf{B}^* [\mathbf{L}^* \mathbf{H} + \mathbf{H} \mathbf{L} + \mathbf{H}'] \mathbf{B} \}. \quad (1483)$$

For periodic QR, $\mathbf{Q}(x)$ and $\mathbf{T}(x)$ can have representation changes at stabilization stations even though their product remains continuous. Derivatives used for interpolation should be computed from the physical evolution equation or from the complete product, not by finite-differencing $\mathbf{Q}$ or $\mathbf{T}$ separately across a QR reset.

D.14 Blockwise propagation

If the stochastic rank is too large to march all columns simultaneously, partition the initial factor:

$$\mathbf{B}_0 = \begin{bmatrix} \mathbf{B}_0^{(1)} & \cdots & \mathbf{B}_0^{(n_b)} \end{bmatrix}. \quad (1484)$$

Each block can be propagated independently:

$$\mathbf{B}^{(j)}(x) = \mathbf{M}(0, x) \mathbf{B}_0^{(j)}. \quad (1485)$$

The full energy matrix contains both diagonal and cross-block terms:

$$\mathbf{A}_{ij}(x) = \Re \left[\mathbf{B}^{(i)}(x)^* \mathbf{H}(x) \mathbf{B}^{(j)}(x) \right]. \quad (1486)$$

Thus,

$$\mathbf{A}(x) = \begin{bmatrix} \mathbf{A}_{11} & \cdots & \mathbf{A}_{1n_b} \\ \vdots & \ddots & \vdots \\ \mathbf{A}_{n_b 1} & \cdots & \mathbf{A}_{n_b n_b} \end{bmatrix}. \quad (1487)$$

Discarding the cross-block matrices changes the directional quadratic forms unless the corresponding subspaces are energy orthogonal at every station.

Independent QR stabilization within each block preserves each block's map, but does not orthogonalize the union of all propagated columns. If global rank or conditioning diagnostics are required, perform a global small-matrix factorization after assembling the energy-weighted cross products, or periodically orthogonalize the full block when memory permits.

D.15 Checkpointing and storage

The complete factor need not be stored at every internal marching step. Possible storage strategies include:

- (i) store $(\mathbf{Q}_n, \mathbf{T}_n)$ at every probability-output station;
- (ii) store only the reduced matrices $\mathbf{A}(x_n)$ and discard the physical factor after the march;
- (iii) store checkpoints of $(\mathbf{Q}, \mathbf{T})$ and remarch locally for adaptive streamwise refinement;
- (iv) use dense output inside each checkpoint interval; or
- (v) store a transition-oriented reduced factor $\mathbf{B}_s(x_n)$.

If multiple transition thresholds will be studied, store $\mathbf{A}(x_n)$ rather than only $\mathbf{G}(x_n)$, since

$$\mathbf{G}(x_n) = \frac{\mathbf{A}(x_n)}{e_{\text{thres}}(x_n)}. \quad (1488)$$

If physical transition-producing states will later be reconstructed, retain enough information to apply

$$\mathbf{B}(x_n) \mathbf{u} = \mathbf{Q}_n \mathbf{T}_n \mathbf{u}. \quad (1489)$$

Storing only $\mathbf{A}(x_n)$ preserves transition probabilities but not the physical disturbance fields.

D.16 Verification of stabilized propagation

Several tests should be applied before using the stabilized factor in a first-transition calculation.

D.16.1 Product reconstruction

For a case in which direct propagation remains numerically feasible, compare

$$\mathbf{B}_n^{\text{direct}} \quad (1490)$$

with

$$\mathbf{B}_n^{\text{QR}} = \mathbf{Q}_n \mathbf{T}_n. \quad (1491)$$

The relative defect is

$$\eta_B(x_n) = \frac{\|\mathbf{B}_n^{\text{direct}} - \mathbf{Q}_n \mathbf{T}_n\|_F}{\max[\|\mathbf{B}_n^{\text{direct}}\|_F, \varepsilon_{\text{scale}}]}. \quad (1492)$$

Direct factor comparison can become misleading after the direct propagation itself loses accuracy. Energy matrices and directional states then provide more robust comparisons.

D.16.2 Energy-matrix invariance

Compare

$$\mathbf{A}_{\text{direct}} = \Re[\mathbf{B}^* \mathbf{H} \mathbf{B}] \quad (1493)$$

with

$$\mathbf{A}_{QT} = \Re[\mathbf{T}^* \mathbf{Q}^* \mathbf{H} \mathbf{Q} \mathbf{T}]. \quad (1494)$$

The defect

$$\eta_A = \frac{\|\mathbf{A}_{\text{direct}} - \mathbf{A}_{QT}\|_F}{\max[\|\mathbf{A}_{\text{direct}}\|_F, \varepsilon_{\text{scale}}]} \quad (1495)$$

should be consistent with numerical precision.

D.16.3 Directional state invariance

For random unit directions $\mathbf{u}^{(j)}$, verify

$$\mathbf{B}(x_n) \mathbf{u}^{(j)} = \mathbf{Q}_n \mathbf{T}_n \mathbf{u}^{(j)}. \quad (1496)$$

Compare the corresponding energy gains rather than only state-vector norms.

D.16.4 QR-frequency invariance

Repeat the propagation with different stabilization intervals:

$$\Delta x_{\text{QR}}^{(1)}, \quad \Delta x_{\text{QR}}^{(2)}, \quad \Delta x_{\text{QR}}^{(3)}. \quad (1497)$$

After convergence, the energy matrices and transition CDF should be independent of the QR frequency:

$$\max_n \left| F^{(i)}(x_n) - F^{(j)}(x_n) \right| \leq \varepsilon_F^{\text{QR}}. \quad (1498)$$

D.16.5 Fixed-coordinate rotation

Apply a fixed orthogonal or unitary transformation to the initial stochastic coordinates:

$$\tilde{\mathbf{B}}_0 = \mathbf{B}_0 \mathbf{U}. \quad (1499)$$

Propagate and stabilize this transformed factor independently. The result must satisfy

$$\tilde{\mathbf{B}}(x) = \mathbf{B}(x) \mathbf{U}, \quad (1500)$$

and

$$\tilde{\mathbf{G}}(x) = \mathbf{U}^* \mathbf{G}(x) \mathbf{U} \quad (1501)$$

with the transpose convention adjusted for real coordinates.

D.16.6 Covariance reconstruction

The stabilized factors imply

$$\mathbf{C}(x, x) = \mathbf{Q}(x) \mathbf{T}(x) \mathbf{T}(x)^* \mathbf{Q}(x)^*. \quad (1502)$$

Compare this matrix, or selected matrix actions, with an independently propagated covariance when feasible. Agreement of covariance is a necessary but not sufficient first-passage test; energy-history invariance must also be checked.

D.17 Common stabilization errors

The following implementation errors alter the first-transition model.

Discarding the triangular factor. Replacing $\mathbf{B} = \mathbf{Q}\mathbf{T}$ by $\mathbf{Q}$ removes the accumulated amplification and initial covariance scaling.

Resetting the Gaussian coordinate after every QR step. Treating the current orthogonal basis as a new synthesis factor with a fresh standard Gaussian vector preserves neither the propagated covariance nor pathwise identity unless its coordinate covariance is also transformed.

Discarding right singular vectors. After $\mathbf{T} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*$, retaining only $\mathbf{\Sigma}$ performs an untracked coordinate rotation.

Reversing factor order. The update is

$$\mathbf{T}_{n+1} = \mathbf{R}_{n+1} \mathbf{T}_n, \quad (1503)$$

not $\mathbf{T}_n \mathbf{R}_{n+1}$.

Independent local rank truncation. Truncating a different propagated SVD subspace at each station destroys the common initial stochastic basis.

Ignoring scale factors. Column or matrix rescalings must be retained in the coordinate map and included quadratically in the energy.

Differentiating across factor resets. The factors $\mathbf{Q}$ and $\mathbf{T}$ can change representation at a QR station. Derivatives must be based on the continuous product or the physical OWNS equations.

Dropping cross-block energy terms. When the factor is propagated in separate blocks, the energy matrix contains cross-block inner products unless the blocks are energy orthogonal.

Using local covariance eigenvectors as persistent coordinates. An eigendecomposition of $\mathbf{C}(x, x)$ at each station provides a local covariance factor but does not preserve the pathwise relationship to upstream stations.

D.18 Recommended stabilized propagation algorithm

The complete recommended procedure is summarized below.

If a fixed transition-oriented basis $\mathbf{V}_s$ has already been selected, the same algorithm can be applied to

$$\mathbf{B}_{0,s} = \mathbf{B}_0 \mathbf{V}_s. \quad (1504)$$

The resulting matrices have dimension $s \times s$, reducing both the OWNS block size and probability-calculation cost.

D.19 Summary

A stabilized representation of the propagated factor has the form

$$\mathbf{B}(x) = \mathbf{Q}(x) \mathbf{T}(x), \quad (1505)$$

or, with logarithmic scaling,

$$\mathbf{B}(x) = \exp[\ell(x)] \mathbf{Q}(x) \bar{\mathbf{T}}(x). \quad (1506)$$

The corresponding normalized energy matrix is

$$\mathbf{G}(x) = \frac{\Re [\mathbf{T}(x)^* \mathbf{Q}(x)^* \mathbf{H}(x) \mathbf{Q}(x) \mathbf{T}(x)]}{e_{\text{thres}}(x)}. \quad (1507)$$

Periodic QR prevents the physical basis from collapsing numerically, while the accumulated coordinate factor retains the true non-normal amplification. SVD and scaling can reorganize this representation but must preserve the complete product. A fixed transition-oriented projection may reduce the rank, whereas independent downstream truncations generally do not preserve first-passage history.

The numerical representation is valid for first-transition analysis only if

$$\mathbf{Q}(x) \mathbf{T}(x) \mathbf{w} = \mathbf{M}(0, x) \mathbf{B}_0 \mathbf{w} \quad (1508)$$

for the same original Gaussian realization at every station. This identity is the defining requirement for all stabilization procedures used in the method.

E Additional Algorithms

This appendix collects supplementary algorithms that extend the core first-transition calculation. The procedures address:

- (i) end-to-end computation of the transition distribution;
- (ii) batched matrix-free evaluation of directional gains;
- (iii) adaptive angular convergence using randomized quasi-Monte Carlo;
- (iv) construction of a fixed transition-oriented basis;
- (v) computation of survivor-conditioned covariance statistics;
- (vi) conditional sampling of transition-producing directions and amplitudes;
- (vii) sampling realizations that first transition within a specified streamwise interval; and
- (viii) combined numerical error control.

Unless otherwise stated, the algorithms assume real standard Gaussian coordinates,

$$\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_r), \quad (1509)$$

and the normalized energy matrices

$$\mathbf{G}(x) = \frac{\Re[\mathbf{B}(x)^* \mathbf{H}(x) \mathbf{B}(x)]}{e_{\text{thres}}(x)}. \quad (1510)$$

For proper complex Gaussian coordinates, the corresponding gamma radial distribution and complex directional quadratic forms from section B should be used.

E.1 End-to-end first-transition workflow

The complete calculation consists of an offline OWNS stage followed by an online probability stage. The offline stage constructs the propagated quadratic-form family. The online stage resolves directional running maxima and performs the angular integration.

The algorithm below assumes that the propagated synthesis factor is available at the streamwise stations. If the stored implementation uses a row-oriented factor satisfying $\mathbf{C} = \mathbf{B}_{\text{row}}^* \mathbf{B}_{\text{row}}$, it should first be converted to the synthesis orientation:

$$\mathbf{B} = \mathbf{B}_{\text{row}}^*. \quad (1511)$$

If the propagated matrices $\mathbf{A}_n$ are stored rather than $\mathbf{G}_n$, the online calculation can be repeated for many threshold models without repeating the OWNS march.

E.2 Batched explicit evaluation of directional gains

Suppose the reduced matrices

$$\mathbf{G}_{s,n} \in \mathbb{R}^{s \times s} \quad (1512)$$

and angular directions

$$\mathbf{U} = [\mathbf{u}_1 \ \cdots \ \mathbf{u}_K] \in \mathbb{R}^{s \times K} \quad (1513)$$

are available. All directional quadratic forms at one station can be evaluated as

$$\mathbf{Z}_n = \mathbf{G}_{s,n} \mathbf{U}, \quad (1514)$$

followed by the columnwise dot products

$$a_{k,n} = \sum_{j=1}^s U_{j,k} Z_{j,k,n}. \quad (1515)$$

Equivalently,

$$\mathbf{a}_n = \text{colsum}[\mathbf{U} \odot \mathbf{Z}_n], \quad (1516)$$

where $\odot$ denotes elementwise multiplication.

For a continuously resolved streamwise calculation, the stationwise gain vector in this algorithm is replaced by the vector of interval maxima returned by the shared refinement procedure.

E.3 Batched matrix-free directional evaluation

If the reduced matrices are not formed explicitly, process the angular directions in blocks. Let

$$\mathbf{U}^{(b)} \in \mathbb{R}^{r \times K_b} \quad (1517)$$

contain one block of K_b directions. The propagated directional states are

$$\mathbf{V}_n^{(b)} = \mathbf{B}(x_n) \mathbf{U}^{(b)} \in \mathbb{C}^{n_q \times K_b}. \quad (1518)$$

If

$$\mathbf{H}_n = \mathbf{W}_n^* \mathbf{W}_n, \quad (1519)$$

then

$$\mathbf{Y}_n^{(b)} = \mathbf{W}_n \mathbf{V}_n^{(b)} \quad (1520)$$

and

$$a_{k,n} = \frac{\|\mathbf{y}_{k,n}^{(b)}\|_2^2}{e_{\text{thres}}(x_n)}. \quad (1521)$$

The algorithm is shown with stationwise maxima. Continuous interval maxima can be incorporated by storing the endpoint directional states and derivatives for each block and applying Algorithm 17.

E.4 Adaptive RQMC angular convergence

For moderate or high stochastic dimension, a practical angular convergence strategy doubles the number of points in nested scrambled Sobol' sequences. Independent scramblings provide replicate-based uncertainty estimates.

Let

$$K_j = 2^j K_0, \quad j = 0, 1, \dots, \quad (1522)$$

and let

$$\widehat{F}_{K_j, J}(x_n) \quad (1523)$$

be the average over J independent scramblings. Define

$$\Delta_j^F = \max_n \left| \widehat{F}_{K_j, J}(x_n) - \widehat{F}_{K_{j-1}, J}(x_n) \right| \quad (1524)$$

and

$$\text{SE}_{j, \max} = \max_n \widehat{\text{SE}} \left[\widehat{F}_{K_j, J}(x_n) \right]. \quad (1525)$$

The same scrambling should be extended when K is doubled rather than regenerated from scratch. This nested construction reduces noise in the difference between successive angular resolutions.

A convergence check based only on $F(x_{\max})$ can miss errors in the shape of the transition-location distribution. The maximum over all reported streamwise stations is therefore used.

E.5 Cell-based adaptive angular quadrature in two dimensions

For stochastic dimension $r = 2$, antipodal symmetry reduces the angular domain to

$$\theta \in [0, \pi), \quad \mathbf{u}(\theta) = \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix}. \quad (1526)$$

The transition CDF is

$$F_{X_{\text{tr}}}(x) = \frac{1}{\pi} \int_0^\pi \Psi_x(\theta) d\theta. \quad (1527)$$

A simple adaptive rule compares the midpoint estimate on an angular interval with the estimates from its two halves. For $\Theta = [\theta_L, \theta_R]$, define the midpoint

$$\theta_M = \frac{\theta_L + \theta_R}{2}. \quad (1528)$$

A coarse midpoint estimate is

$$I_C(x) = \frac{\theta_R - \theta_L}{\pi} \Psi_x(\theta_M). \quad (1529)$$

The split estimate uses quarter points:

$$\theta_{LM} = \frac{\theta_L + \theta_M}{2}, \quad \theta_{MR} = \frac{\theta_M + \theta_R}{2}, \quad (1530)$$

$$I_S(x) = \frac{\theta_R - \theta_L}{2\pi} [\Psi_x(\theta_{LM}) + \Psi_x(\theta_{MR})]. \quad (1531)$$

The cell error indicator is

$$\eta_\Theta = \max_n |I_S(x_n) - I_C(x_n)|. \quad (1532)$$

A similar method can be implemented on $\mathbb{S}^2$ using spherical triangles. In higher dimensions, RQMC is usually simpler than constructing and maintaining a spherical tessellation.

E.6 Complete transition-oriented basis algorithm

This subsection consolidates the integrated-Gramian, record-eigenspace, and greedy-complement basis constructions into one implementation.

Let

$$\mathbf{K}_\omega = \sum_{n=0}^N \nu_n \omega(x_n) \mathbf{G}(x_n) \quad (1533)$$

be the weighted aggregate matrix. Let

$$\gamma_1(x_n) = \lambda_{\max}[\mathbf{G}(x_n)]. \quad (1534)$$

A record station satisfies

$$\gamma_1(x_n) > \max_{j < n} \gamma_1(x_j) + \varepsilon_{\text{record}}. \quad (1535)$$

The complement criterion in the algorithm detects omitted energy within the orthogonal complement. The complete matrix residual

$$\max_n \|\mathbf{G}(x_n) - \mathbf{P}_s \mathbf{G}(x_n) \mathbf{P}_s\|_2 \quad (1536)$$

should also be monitored because it includes retained–omitted cross-coupling.

E.7 Survivor-conditioned stochastic covariance

The radial–directional method can calculate statistics of the population that has not transitioned by x . Define

$$\boldsymbol{\Sigma}_{w,\text{surv}}(x) = \mathbb{E} \left[\mathbf{w} \mathbf{w}^\top \mid X_{\text{tr}} > x \right]. \quad (1537)$$

For $\mathbf{w} = R\mathbf{u}$, survival requires

$$R^2 < c_{\mathbf{u}}(x), \quad c_{\mathbf{u}}(x) = \frac{1}{m_{\mathbf{u}}(x)}. \quad (1538)$$

For $Z = R^2 \sim \chi_r^2$, the truncated first moment is

$$\boxed{\mathbb{E}[Z \mathbf{1}\{Z < c\}] = r F_{\chi_{r+2}^2}(c).} \quad (1539)$$

Therefore,

$$\begin{aligned} \mathbf{M}_{w,\text{surv}}(x) &= \mathbb{E} \left[\mathbf{w} \mathbf{w}^\top \mathbf{1}\{X_{\text{tr}} > x\} \right] \\ &= r \mathbb{E}_{\mathbf{u}} \left[\mathbf{u} \mathbf{u}^\top F_{\chi_{r+2}^2} \left(\frac{1}{m_{\mathbf{u}}(x)} \right) \right]. \end{aligned} \quad (1540)$$

The conditional covariance is

$$\boxed{\boldsymbol{\Sigma}_{w,\text{surv}}(x) = \frac{\mathbf{M}_{w,\text{surv}}(x)}{S(x)}.} \quad (1541)$$

The conditional mean remains zero because the survivor set is centrally symmetric.

Under an angular quadrature rule,

$$\mathbf{M}_{w,\text{surv}}(x) \approx r \sum_{k=1}^K \omega_k \mathbf{u}_k \mathbf{u}_k^\top F_{\chi_{r+2}^2} \left[\frac{1}{m_k(x)} \right]. \quad (1542)$$

As a verification check, at a station before any probability has been removed,

$$\boldsymbol{\Sigma}_{w,\text{surv}}(x) = \mathbf{I}_r. \quad (1543)$$

At later stations, its eigenvalues reveal which initial stochastic directions have been preferentially removed by transition.

E.8 Transition-conditioned angular distribution

The probability that a direction $\mathbf{u}$ has transitioned by x , after integrating over its radius, is

$$\Psi_x(\mathbf{u}) = \overline{F}_{\chi_r^2} \left[\frac{1}{m_{\mathbf{u}}(x)} \right]. \quad (1544)$$

The conditional angular density among transitioned realizations is

$$p(\mathbf{u} \mid X_{\text{tr}} \leq x) = \frac{\Psi_x(\mathbf{u})}{F_{X_{\text{tr}}}(x)} \frac{1}{|\mathbb{S}^{r-1}|}, \quad (1545)$$

provided $F_{X_{\text{tr}}}(x) > 0$.

Under an angular rule, the discrete conditional weights are

$$\pi_k^{\text{tr}}(x) = \frac{\omega_k \Psi_x(\mathbf{u}_k)}{\sum_{\ell=1}^K \omega_{\ell} \Psi_x(\mathbf{u}_{\ell})}. \quad (1546)$$

These weights can be used to:

- rank transition-producing directions;
- reconstruct representative initial disturbances;
- form transition-conditioned directional moments; or
- sample directions conditional on transition.

The projective second moment is

$$\mathbf{P}_{\text{tr}}(x) = \sum_{k=1}^K \pi_k^{\text{tr}}(x) \mathbf{u}_k \mathbf{u}_k^{\top}. \quad (1547)$$

Its leading eigenvectors identify broad transition-producing subspaces without being affected by the sign symmetry $\mathbf{u} \sim -\mathbf{u}$.

E.9 Sampling a realization conditional on transition by x

A sample of X_{tr} alone can be drawn from the CDF. Sometimes a corresponding initial Gaussian realization is also required. To sample

$$\mathbf{w} \mid X_{\text{tr}} \leq x, \quad (1548)$$

one can first sample a direction from the transition-conditioned angular distribution and then sample its squared radius from a truncated chi-square distribution.

For direction $\mathbf{u}_k$, transition by x requires

$$Z = R^2 \geq c_k(x), \quad c_k(x) = \frac{1}{m_k(x)}. \quad (1549)$$

The conditional radial CDF is

$$\mathbb{P}(Z \leq z \mid Z \geq c_k) = \frac{F_{\chi_r^2}(z) - F_{\chi_r^2}(c_k)}{\bar{F}_{\chi_r^2}(c_k)}, \quad z \geq c_k. \quad (1550)$$

If $U \sim \text{Unif}(0, 1)$, a conditional radius can be drawn as

$$Z = F_{\chi_r^2}^{-1} [F_{\chi_r^2}(c_k) + U\bar{F}_{\chi_r^2}(c_k)]. \quad (1551)$$

For extreme tails, this inversion should use a survival-function inverse:

$$Z = \bar{F}_{\chi_r^2}^{-1} [U\bar{F}_{\chi_r^2}(c_k)], \quad (1552)$$

with the convention of the selected inverse-survival routine.

This algorithm samples from the discrete angular approximation, not from the exact continuous angular distribution. Its accuracy therefore depends on angular convergence. If the rule explicitly contains both antipodal directions, the additional random sign is unnecessary.

Once $\mathbf{w}$ has been generated, its actual first-transition location can be found by solving

$$\mathbf{w}^\top \mathbf{G}(x) \mathbf{w} = 1 \quad (1553)$$

using Algorithm 19.

E.10 Sampling a realization conditional on transition in an interval

Consider the event

$$x_a < X_{\text{tr}} \leq x_b, \quad x_a < x_b. \quad (1554)$$

For direction $\mathbf{u}$, let

$$m_a = m_{\mathbf{u}}(x_a), \quad (1555)$$

$$m_b = m_{\mathbf{u}}(x_b), \quad m_b \geq m_a. \quad (1556)$$

The realization survives through x_a and transitions by x_b when

$$Zm_a < 1, \quad Zm_b \geq 1. \quad (1557)$$

For $m_a > 0$, this is equivalent to

$$\boxed{\frac{1}{m_b} \leq Z < \frac{1}{m_a}}. \quad (1558)$$

If $m_a = 0$, the upper endpoint is infinity.

The directional interval probability is

$$\begin{aligned} \Delta\Psi_{\mathbf{u}}(x_a, x_b) &= \mathbb{P}(x_a < X_{\text{tr}} \leq x_b \mid \mathbf{u}) \\ &= F_{\chi_r^2}\left(\frac{1}{m_a}\right) - F_{\chi_r^2}\left(\frac{1}{m_b}\right) \end{aligned} \quad (1559)$$

$$= \bar{F}_{\chi_r^2}\left(\frac{1}{m_b}\right) - \bar{F}_{\chi_r^2}\left(\frac{1}{m_a}\right). \quad (1560)$$

If $m_b = m_a$, the interval contributes no new transition probability.

For a discrete angular rule, the conditional direction weights are

$$\pi_k^{(a,b)} = \frac{\omega_k \Delta \Psi_k(x_a, x_b)}{\sum_{\ell=1}^K \omega_\ell \Delta \Psi_\ell(x_a, x_b)}. \quad (1561)$$

For a selected direction, define

$$z_L = \frac{1}{m_b}, \quad z_U = \frac{1}{m_a}. \quad (1562)$$

A squared radius conditioned on $z_L \leq Z < z_U$ can be generated as

$$Z = F_{\chi_r^2}^{-1} [F_{\chi_r^2}(z_L) + U (F_{\chi_r^2}(z_U) - F_{\chi_r^2}(z_L))], \quad (1563)$$

where $U \sim \text{Unif}(0, 1)$. Tail-based inversion should be used when both truncation points lie far into the upper tail.

This algorithm is useful for reconstructing representative initial and propagated structures associated with transition in a selected streamwise region.

E.11 Sampling the first-transition location and a compatible realization

A joint sample of the transition location and a compatible initial realization can be generated in two stages:

- (i) draw a transition interval from the computed interval probabilities; and
- (ii) conditionally sample $\mathbf{w}$ within that interval using Algorithm 30.

Let

$$p_0, \quad p_1, \dots, p_N, \quad p_{\text{censored}} \quad (1564)$$

denote the inlet, interval, and censored probabilities.

Sampling from the survivor distribution requires an analogous direction–radius decomposition. The conditional angular density among survivors is proportional to

$$F_{\chi_r^2} \left[\frac{1}{m_{\mathbf{u}}(x)} \right], \quad (1565)$$

and, after selecting a direction, the squared radius is sampled from χ_r^2 truncated above at $1/m_{\mathbf{u}}(x)$.

E.12 Sampling a survivor realization

For a discrete angular rule, define survivor-conditioned direction weights

$$\pi_k^{\text{surv}}(x) = \frac{\omega_k F_{\chi_r^2} [1/m_k(x)]}{S(x)}. \quad (1566)$$

For a selected direction, survival requires

$$0 \leq Z < c_k, \quad c_k = \frac{1}{m_k(x)}. \quad (1567)$$

The conditional radius can be generated as

$$Z = F_{\chi_r^2}^{-1} [UF_{\chi_r^2}(c_k)]. \quad (1568)$$

The sampled survivor can be propagated physically using

$$\mathbf{q}(x) = \mathbf{B}(x)\mathbf{w}. \quad (1569)$$

E.13 Threshold and inlet-amplitude sweep

Suppose a reference calculation provides the unnormalized reduced energy matrices

$$\mathbf{A}_s(x) \quad (1570)$$

and the inlet factor is scaled by amplitude ϵ :

$$\mathbf{B}_{0,s}(\epsilon) = \epsilon \mathbf{B}_{0,s}^{\text{ref}}. \quad (1571)$$

Under linear propagation,

$$\mathbf{A}_s(x; \epsilon) = \epsilon^2 \mathbf{A}_s^{\text{ref}}(x). \quad (1572)$$

For threshold scale α ,

$$e_{\text{thres}}^{(\alpha)}(x) = \alpha e_{\text{thres}}^{\text{ref}}(x). \quad (1573)$$

Therefore,

$$\mathbf{G}_s(x; \epsilon, \alpha) = \frac{\epsilon^2}{\alpha} \mathbf{G}_s^{\text{ref}}(x). \quad (1574)$$

The directional maxima scale identically:

$$m_k(x; \epsilon, \alpha) = \frac{\epsilon^2}{\alpha} m_k^{\text{ref}}(x). \quad (1575)$$

Thus, no new OWNS propagation or streamwise maximum search is required for a purely multiplicative amplitude or threshold sweep.

For a threshold whose streamwise shape changes rather than only its overall scale, the directional gain histories and maxima must be recomputed.

E.14 Local exceedance probability for consistency checks

At each station, diagonalize

$$\mathbf{G}(x_n) = \mathbf{V}_n \text{diag}(\gamma_{1,n}, \dots, \gamma_{r,n}) \mathbf{V}_n^{\text{T}}. \quad (1576)$$

The local normalized energy is

$$y(x_n) \stackrel{d}{=} \sum_{j=1}^r \gamma_{j,n} Z_j^2, \quad Z_j \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1). \quad (1577)$$

A generalized chi-square routine can evaluate

$$p_{\text{local}}(x_n) = \mathbb{P}[y(x_n) \geq 1]. \quad (1578)$$

This check uses a numerical method that is independent of the angular integration and is therefore valuable for detecting implementation errors.

E.15 Combined numerical error-control driver

The total numerical error contains interacting contributions from the OWNS discretization, stochastic reduction, streamwise maximum resolution, and angular integration. A practical calculation refines these components in an ordered sequence.

Let the target CDF error be

$$\varepsilon_F^{\text{target}}. \quad (1579)$$

Allocate tolerances

$$\varepsilon_F^{\text{OWNS}} + \varepsilon_F^{\text{red}} + \varepsilon_F^{\text{stream}} + \varepsilon_F^{\text{ang}} \leq \varepsilon_F^{\text{target}}. \quad (1580)$$

Because the error sources interact, one final refinement cycle should be performed after every component separately satisfies its tolerance.

E.16 Recommended data structures

A practical implementation can store the reduced streamwise data as:

$$\mathbf{x} \in \mathbb{R}^{N+1}, \quad (1581)$$

$$\mathbf{A} \in \mathbb{R}^{(N+1) \times s \times s}, \quad (1582)$$

$$\mathbf{G} \in \mathbb{R}^{(N+1) \times s \times s}, \quad (1583)$$

$$\mathbf{U} \in \mathbb{R}^{s \times K}, \quad (1584)$$

$$\mathbf{weights} \in \mathbb{R}^K, \quad (1585)$$

$$\mathbf{running_max} \in \mathbb{R}^K, \quad (1586)$$

$$\mathbf{CDF, survival} \in \mathbb{R}^{N+1}. \quad (1587)$$

If derivatives are retained,

$$\mathbf{Gprime} \in \mathbb{R}^{(N+1) \times s \times s}. \quad (1588)$$

Symmetric packed storage can reduce the matrix memory, although dense storage can be faster for batched matrix–matrix operations at moderate s .

For an RQMC calculation with J replicates, the replicate CDF array is

$$\mathbf{CDF_rep} \in \mathbb{R}^{J \times (N+1)}. \quad (1589)$$

The complete directional histories need not be stored unless they are required for diagnostics or conditional sampling.

E.17 Minimal implementation tests

Before applying the algorithms to OWNS data, the following unit tests should pass:

1. **Isotropic test.** For $\mathbf{G}(x) = g(x)\mathbf{I}$, every angular direction must produce the same running maximum and the quadrature must reproduce

$$F(x) = \overline{F}_{\chi_r^2} \left[1 / \max_{\xi \leq x} g(\xi) \right].$$

2. **Rank-one test.** For $\mathbf{G}(x) = g(x)\mathbf{v}\mathbf{v}^\top$, the result must agree with a χ_1^2 tail independently of the ambient dimension.

3. **Basis-rotation test.** A fixed orthogonal transformation of all matrices and angular directions must leave every directional gain and CDF value unchanged.
4. **Record test.** A second directional peak lower than the first must add no new first-transition probability.
5. **Threshold-scaling test.** Scaling $\mathbf{B}$ by ϵ must agree with scaling $\mathbf{G}$ by ϵ^2 .
6. **Survivor covariance test.** If no transition is possible, the computed conditional coordinate covariance must equal $\mathbf{I}_r$.
7. **Conditional sampling test.** Samples generated conditional on transition by x must satisfy

$$\max_{\xi \leq x} \mathbf{w}^\top \mathbf{G}(\xi) \mathbf{w} \geq 1$$

up to the crossing tolerance.

8. **Interval sampling test.** Samples generated conditional on $x_a < X_{\text{tr}} \leq x_b$ must survive through x_a and cross by x_b .
9. **Probability conservation test.** The inlet, interval, and censored probabilities must sum to one.
10. **Monte Carlo test.** For a low-dimensional example, the computed CDF must agree with an independent trajectory Monte Carlo estimate within its statistical uncertainty.

E.18 Summary

The supplementary algorithms extend the core transition CDF into a more complete computational framework. In particular, they permit:

- efficient blocked evaluation in either the reduced stochastic space or the physical state space;
- adaptive control of angular error;
- construction and validation of a fixed transition-oriented basis;
- calculation of conditional survivor statistics;
- reconstruction of transition-producing initial disturbances;
- sampling of realizations conditioned on transition in a specified streamwise region;
- rapid inlet-amplitude and threshold studies; and
- systematic separation of the principal numerical error sources.

The conditional sampling algorithms remain approximations to the continuous angular distribution when a finite angular rule is used. Their accuracy should therefore be assessed using the same angular, streamwise, and stochastic-rank convergence procedures as the transition CDF itself.

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Algorithm 1 Discrete-streamwise first-transition distribution from $\mathbf{B}(x)$

Require: Stations $\{x_n\}_{n=0}^N$
Require: Propagated factors $\{\mathbf{B}_n\}_{n=0}^N$
Require: Energy matrices $\{\mathbf{H}_n\}_{n=0}^N$
Require: Thresholds $\{e_{\text{thres},n}\}_{n=0}^N$
Require: Angular rule $\{(\mathbf{u}_k, \omega_k)\}_{k=1}^K$
Ensure: Survival probabilities $\{S_n\}_{n=0}^N$
Ensure: Transition CDF $\{F_n\}_{n=0}^N$

```

1: for  $n = 0, \dots, N$  do
2:    $\mathbf{G}_n \leftarrow \frac{\Re(\mathbf{B}_n^* \mathbf{H}_n \mathbf{B}_n)}{e_{\text{thres},n}}$ 
3:    $\mathbf{G}_n \leftarrow \frac{1}{2} (\mathbf{G}_n + \mathbf{G}_n^\top)$ 
4: end for
5: for  $k = 1, \dots, K$  do
6:    $\mathbf{u}_k \leftarrow \mathbf{u}_k / \|\mathbf{u}_k\|_2$ 
7:    $m_k \leftarrow 0$ 
8: end for
9: for  $n = 0, \dots, N$  do
10:   $S_n \leftarrow 0$ 
11:  for  $k = 1, \dots, K$  do
12:     $a_{k,n} \leftarrow \mathbf{u}_k^\top \mathbf{G}_n \mathbf{u}_k$ 
13:     $a_{k,n} \leftarrow \max(a_{k,n}, 0)$ 
14:     $m_k \leftarrow \max(m_k, a_{k,n})$ 
15:    if  $m_k > 0$  then
16:       $s_{k,n} \leftarrow F_{\chi_r^2}(1/m_k)$ 
17:    else
18:       $s_{k,n} \leftarrow 1$ 
19:    end if
20:     $S_n \leftarrow S_n + \omega_k s_{k,n}$ 
21:  end for
22:   $S_n \leftarrow \min(1, \max(0, S_n))$ 
23:   $F_n \leftarrow 1 - S_n$ 
24: end for
25: return  $\{S_n, F_n\}_{n=0}^N$ 

```

▷ Remove roundoff-scale negativity

Algorithm 2 Matrix-free directional gain and running-maximum update

Require: Factor $\mathbf{B}_n$, energy matrix $\mathbf{H}_n$, and threshold $e_{\text{thres},n}$

Require: Directions $\{\mathbf{u}_k\}_{k=1}^K$

Require: Previous running maxima $\{m_k\}_{k=1}^K$

Ensure: Updated running maxima $\{m_k\}_{k=1}^K$

```
1: for  $k = 1, \dots, K$  do
2:    $\mathbf{v}_{k,n} \leftarrow \mathbf{B}_n \mathbf{u}_k$ 
3:    $a_{k,n} \leftarrow \frac{\Re(\mathbf{v}_{k,n}^* \mathbf{H}_n \mathbf{v}_{k,n})}{e_{\text{thres},n}}$ 
4:    $a_{k,n} \leftarrow \max(a_{k,n}, 0)$ 
5:    $m_k \leftarrow \max(m_k, a_{k,n})$ 
6: end for
7: return  $\{m_k\}_{k=1}^K$ 
```

Algorithm 3 First-transition distribution with interval maximum detection

Require: Streamwise data and angular rule from Algorithm 5

Require: Directional gains and, optionally, derivatives at interval endpoints

Ensure: Continuous-streamwise approximation $\{S_n, F_n\}_{n=0}^N$

```
1: for  $k = 1, \dots, K$  do
2:    $\hat{a}_{k,0} \leftarrow a_{\mathbf{u}_k}(x_0)$ 
3:    $m_k \leftarrow \max(\hat{a}_{k,0}, 0)$ 
4: end for
5: Compute  $S_0$  and  $F_0 = 1 - S_0$  from  $\{m_k\}$ 
6: for  $n = 1, \dots, N$  do
7:   for  $k = 1, \dots, K$  do
8:     Construct or evaluate  $a_{\mathbf{u}_k}(x)$  on  $[x_{n-1}, x_n]$ 
9:      $\hat{a}_{k,n} \leftarrow \max_{x \in [x_{n-1}, x_n]} a_{\mathbf{u}_k}(x)$ 
10:     $m_k \leftarrow \max(m_k, \hat{a}_{k,n})$ 
11:   end for
12:    $S_n \leftarrow \sum_{k=1}^K \omega_k F_{\chi_r^2}(1/m_k)$ , using  $F_{\chi_r^2}(\infty) = 1$ 
13:    $F_n \leftarrow 1 - S_n$ 
14: end for
15: return  $\{S_n, F_n\}_{n=0}^N$ 
```

Algorithm 4 Draw a first-transition location from a computed CDF

Require: Grid $\{x_n\}_{n=0}^N$

Require: Monotone transition CDF $\{F_n\}_{n=0}^N$

Ensure: A transition location X_{tr} or a censored outcome

```

1: Draw  $U \sim \text{Unif}(0, 1)$ 
2: if  $U > F_N$  then
3:    $X_{\text{tr}} \leftarrow \infty$                                      ▷ Right-censored at  $x_N$ 
4: else if  $U \leq F_0$  then
5:    $X_{\text{tr}} \leftarrow x_0$                                      ▷ Immediate transition
6: else
7:   Find the smallest  $n \in \{1, \dots, N\}$  such that  $F_n \geq U$ 
8:   if  $F_n > F_{n-1}$  then
9:      $X_{\text{tr}} \leftarrow x_{n-1} + (x_n - x_{n-1}) \frac{U - F_{n-1}}{F_n - F_{n-1}}$    ▷ Piecewise-linear inverse
10:  else
11:     $X_{\text{tr}} \leftarrow x_n$ 
12:  end if
13: end if
14: return  $X_{\text{tr}}$ 

```

Algorithm 5 Discrete-streamwise first-transition distribution from $\mathbf{B}(x)$

Require: Stations $\{x_n\}_{n=0}^N$
Require: Propagated factors $\{\mathbf{B}_n\}_{n=0}^N$
Require: Energy matrices $\{\mathbf{H}_n\}_{n=0}^N$
Require: Thresholds $\{e_{\text{thres},n}\}_{n=0}^N$
Require: Angular rule $\{(\mathbf{u}_k, \omega_k)\}_{k=1}^K$
Ensure: Survival probabilities $\{S_n\}_{n=0}^N$
Ensure: Transition CDF $\{F_n\}_{n=0}^N$

```

1: for  $n = 0, \dots, N$  do
2:    $\mathbf{G}_n \leftarrow \frac{\Re(\mathbf{B}_n^* \mathbf{H}_n \mathbf{B}_n)}{e_{\text{thres},n}}$ 
3:    $\mathbf{G}_n \leftarrow \frac{1}{2} (\mathbf{G}_n + \mathbf{G}_n^\top)$ 
4: end for
5: for  $k = 1, \dots, K$  do
6:    $\mathbf{u}_k \leftarrow \mathbf{u}_k / \|\mathbf{u}_k\|_2$ 
7:    $m_k \leftarrow 0$ 
8: end for
9: for  $n = 0, \dots, N$  do
10:   $S_n \leftarrow 0$ 
11:  for  $k = 1, \dots, K$  do
12:     $a_{k,n} \leftarrow \mathbf{u}_k^\top \mathbf{G}_n \mathbf{u}_k$ 
13:     $a_{k,n} \leftarrow \max(a_{k,n}, 0)$ 
14:     $m_k \leftarrow \max(m_k, a_{k,n})$ 
15:    if  $m_k > 0$  then
16:       $s_{k,n} \leftarrow F_{\chi_r^2}(1/m_k)$ 
17:    else
18:       $s_{k,n} \leftarrow 1$ 
19:    end if
20:     $S_n \leftarrow S_n + \omega_k s_{k,n}$ 
21:  end for
22:   $S_n \leftarrow \min(1, \max(0, S_n))$ 
23:   $F_n \leftarrow 1 - S_n$ 
24: end for
25: return  $\{S_n, F_n\}_{n=0}^N$ 

```

▷ Remove roundoff-scale negativity

Algorithm 6 Matrix-free directional gain and running-maximum update

Require: Factor $\mathbf{B}_n$, energy matrix $\mathbf{H}_n$, and threshold $e_{\text{thres},n}$

Require: Directions $\{\mathbf{u}_k\}_{k=1}^K$

Require: Previous running maxima $\{m_k\}_{k=1}^K$

Ensure: Updated running maxima $\{m_k\}_{k=1}^K$

```
1: for  $k = 1, \dots, K$  do
2:    $\mathbf{v}_{k,n} \leftarrow \mathbf{B}_n \mathbf{u}_k$ 
3:    $a_{k,n} \leftarrow \frac{\Re(\mathbf{v}_{k,n}^* \mathbf{H}_n \mathbf{v}_{k,n})}{e_{\text{thres},n}}$ 
4:    $a_{k,n} \leftarrow \max(a_{k,n}, 0)$ 
5:    $m_k \leftarrow \max(m_k, a_{k,n})$ 
6: end for
7: return  $\{m_k\}_{k=1}^K$ 
```

Algorithm 7 First-transition distribution with interval maximum detection

Require: Streamwise data and angular rule from Algorithm 5

Require: Directional gains and, optionally, derivatives at interval endpoints

Ensure: Continuous-streamwise approximation $\{S_n, F_n\}_{n=0}^N$

```
1: for  $k = 1, \dots, K$  do
2:    $\hat{a}_{k,0} \leftarrow a_{\mathbf{u}_k}(x_0)$ 
3:    $m_k \leftarrow \max(\hat{a}_{k,0}, 0)$ 
4: end for
5: Compute  $S_0$  and  $F_0 = 1 - S_0$  from  $\{m_k\}$ 
6: for  $n = 1, \dots, N$  do
7:   for  $k = 1, \dots, K$  do
8:     Construct or evaluate  $a_{\mathbf{u}_k}(x)$  on  $[x_{n-1}, x_n]$ 
9:      $\hat{a}_{k,n} \leftarrow \max_{x \in [x_{n-1}, x_n]} a_{\mathbf{u}_k}(x)$ 
10:     $m_k \leftarrow \max(m_k, \hat{a}_{k,n})$ 
11:   end for
12:    $S_n \leftarrow \sum_{k=1}^K \omega_k F_{\chi_r^2}(1/m_k)$ , using  $F_{\chi_r^2}(\infty) = 1$ 
13:    $F_n \leftarrow 1 - S_n$ 
14: end for
15: return  $\{S_n, F_n\}_{n=0}^N$ 
```

Algorithm 8 Draw a first-transition location from a computed CDF

Require: Grid $\{x_n\}_{n=0}^N$ **Require:** Monotone transition CDF $\{F_n\}_{n=0}^N$ **Ensure:** A transition location X_{tr} or a censored outcome

```
1: Draw  $U \sim \text{Unif}(0, 1)$ 
2: if  $U > F_N$  then
3:    $X_{\text{tr}} \leftarrow \infty$  ▷ Right-censored at  $x_N$ 
4: else if  $U \leq F_0$  then
5:    $X_{\text{tr}} \leftarrow x_0$  ▷ Immediate transition
6: else
7:   Find the smallest  $n \in \{1, \dots, N\}$  such that  $F_n \geq U$ 
8:   if  $F_n > F_{n-1}$  then
9:      $X_{\text{tr}} \leftarrow x_{n-1} + (x_n - x_{n-1}) \frac{U - F_{n-1}}{F_n - F_{n-1}}$  ▷ Piecewise-linear inverse
10:  else
11:     $X_{\text{tr}} \leftarrow x_n$ 
12:  end if
13: end if
14: return  $X_{\text{tr}}$ 
```

Algorithm 9 Randomized quasi-Monte Carlo angular integration

Require: Stochastic dimension r

Require: Points per replicate K

Require: Number of independent randomizations J

Require: Streamwise matrices or directional-gain evaluator

Ensure: Estimated CDF $\hat{F}_{K,J}(x_n)$

Ensure: Replicate-based standard errors

```
1: for  $j = 1, \dots, J$  do
2:   Generate an independently scrambled low-discrepancy set  $\{\mathbf{v}_k^{(j)}\}_{k=1}^K \subset (0, 1)^r$ 
3:   for  $k = 1, \dots, K$  do
4:     for  $\ell = 1, \dots, r$  do
5:        $z_{k,\ell}^{(j)} \leftarrow \Phi^{-1}(v_{k,\ell}^{(j)})$ 
6:     end for
7:      $\mathbf{u}_k^{(j)} \leftarrow \mathbf{z}_k^{(j)} / \|\mathbf{z}_k^{(j)}\|_2$ 
8:     Compute  $m_k^{(j)}(x_n)$  for all  $n$  using the streamwise maximum procedure
9:      $\Psi_{k,n}^{(j)} \leftarrow \overline{F}_{\chi_r^2} \left[ 1/m_k^{(j)}(x_n) \right]$ 
10:   end for
11:   for  $n = 0, \dots, N$  do
12:      $\hat{F}_K^{(j)}(x_n) \leftarrow \frac{1}{K} \sum_{k=1}^K \Psi_{k,n}^{(j)}$ 
13:   end for
14: end for
15: for  $n = 0, \dots, N$  do
16:   Compute  $\hat{F}_{K,J}(x_n)$  using eq. (387)
17:   Compute the standard error using eq. (388)
18: end for
19: return the estimated CDF and standard errors
```

Algorithm 10 Greedy transition-oriented stochastic basis

Require: Streamwise matrices $\{\mathbf{G}(x_n)\}_{n=0}^N$

Require: Maximum reduced rank $s_{\max}$

Require: Neglected-gain tolerance $\varepsilon_{\perp}^{\text{tol}}$

Ensure: Fixed orthonormal basis $\mathbf{V}_s$

```

1: Find  $n_* \in \arg \max_{0 \leq n \leq N} \lambda_{\max}[\mathbf{G}(x_n)]$ 
2: Let  $\mathbf{v}_*$  be a normalized dominant eigenvector of  $\mathbf{G}(x_{n_*})$ 
3:  $\mathbf{V}_1 \leftarrow \mathbf{v}_*$ 
4:  $s \leftarrow 1$ 
5: while  $s < s_{\max}$  do
6:    $\mathbf{P}_s^{\perp} \leftarrow \mathbf{I} - \mathbf{V}_s \mathbf{V}_s^{\top}$ 
7:   for  $n = 0, \dots, N$  do
8:      $\mathbf{R}_{s,n} \leftarrow \mathbf{P}_s^{\perp} \mathbf{G}(x_n) \mathbf{P}_s^{\perp}$ 
9:      $\epsilon_{s,n} \leftarrow \lambda_{\max}(\mathbf{R}_{s,n})$ 
10:  end for
11:   $n_* \leftarrow \arg \max_n \epsilon_{s,n}$ 
12:   $\epsilon_s^{\max} \leftarrow \epsilon_{s,n_*}$ 
13:  if  $\epsilon_s^{\max} \leq \varepsilon_{\perp}^{\text{tol}}$  then
14:    break
15:  end if
16:  Let  $\mathbf{v}_*$  be a normalized dominant eigenvector of  $\mathbf{R}_{s,n_*}$ 
17:   $\mathbf{v}_* \leftarrow \mathbf{v}_* - \mathbf{V}_s(\mathbf{V}_s^{\top} \mathbf{v}_*)$ 
18:   $\mathbf{v}_* \leftarrow \mathbf{v}_* / \|\mathbf{v}_*\|_2$ 
19:   $\mathbf{V}_{s+1} \leftarrow [\mathbf{V}_s \ \mathbf{v}_*]$ 
20:   $s \leftarrow s + 1$ 
21: end while
22: return  $\mathbf{V}_s$ 

```

Algorithm 11 Construction and validation of a transition-oriented reduced basis

Require: Resolved matrix family $\{\mathbf{G}(x_n)\}_{n=0}^N$

Require: Candidate rank sequence $s_1 < s_2 < \dots < s_J$

Require: Reduction tolerances $\varepsilon_F^{\text{red}}$ and $\varepsilon_G^{\text{tol}}$

Ensure: Fixed reduced basis $\mathbf{V}_s$

- 1: Construct the weighted aggregate $\mathbf{K}_\omega \leftarrow \sum_n \nu_n \omega_n \mathbf{G}(x_n)$
 - 2: Add leading eigenvectors of $\mathbf{K}_\omega$ to a candidate snapshot matrix $\mathbf{Z}$
 - 3: Identify stations where $\lambda_{\max}[\mathbf{G}(x_n)]$ establishes a new streamwise record
 - 4: Add leading eigenvectors of $\mathbf{G}(x_n)$ at the record stations to $\mathbf{Z}$
 - 5: Orthonormalize $\mathbf{Z}$ to obtain an initial basis
 - 6: Apply greedy complement enrichment using Algorithm 10
 - 7: Compress the combined candidate basis using an SVD
 - 8: **for** $j = 1, \dots, J$ **do**
 - 9: $\mathbf{V}_{s_j} \leftarrow$ first s_j left singular vectors of the candidate matrix
 - 10: **for** $n = 0, \dots, N$ **do**
 - 11: $\mathbf{G}_{s_j}(x_n) \leftarrow \mathbf{V}_{s_j}^\top \mathbf{G}(x_n) \mathbf{V}_{s_j}$
 - 12: **end for**
 - 13: Compute $F_{X_{\text{tr}}, s_j}(x_n)$ using the s_j -dimensional radial-directional method
 - 14: Compute the matrix residual ε_{G, s_j}
 - 15: **if** $j > 1$ **then**
 - 16: Compute $\Delta_{s_j}^F$ using eq. (507)
 - 17: **if** $\Delta_{s_j}^F \leq \varepsilon_F^{\text{red}}$ **and** $\varepsilon_{G, s_j} \leq \varepsilon_G^{\text{tol}}$ **then**
 - 18: **return** $\mathbf{V}_{s_j}$
 - 19: **end if**
 - 20: **end if**
 - 21: **end for**
 - 22: Issue a rank-convergence warning
 - 23: **return** the largest tested basis
-

Algorithm 12 Periodic QR propagation preserving the initial Gaussian coordinates

Require: Initial synthesis factor $\mathbf{B}_0$

Require: OWNS marching maps or marching operator $\{\mathbf{P}_{n+1, n}\}_{n=0}^{N-1}$

Ensure: Factored propagated maps $\mathbf{B}_n = \mathbf{Q}_n \mathbf{T}_n$

- 1: Compute the thin QR factorization $\mathbf{B}_0 = \mathbf{Q}_0 \mathbf{T}_0$
 - 2: **for** $n = 0, \dots, N-1$ **do**
 - 3: $\hat{\mathbf{Q}}_{n+1} \leftarrow \mathbf{P}_{n+1, n} \mathbf{Q}_n$
 - 4: Compute the thin QR factorization $\hat{\mathbf{Q}}_{n+1} = \mathbf{Q}_{n+1} \mathbf{R}_{n+1}$
 - 5: $\mathbf{T}_{n+1} \leftarrow \mathbf{R}_{n+1} \mathbf{T}_n$
 - 6: Store or evaluate required quantities using $\mathbf{B}_{n+1} = \mathbf{Q}_{n+1} \mathbf{T}_{n+1}$
 - 7: **end for**
 - 8: **return** $\{(\mathbf{Q}_n, \mathbf{T}_n)\}_{n=0}^N$
-

Algorithm 13 Trajectory Monte Carlo verification of the first-transition CDF

Require: Matrix family $\{\mathbf{G}(x_n)\}_{n=0}^N$

Require: Number of trajectories N_{MC}

Ensure: Empirical first-transition CDF $\hat{F}_{\text{MC}}(x_n)$

```
1: for  $j = 1, \dots, N_{\text{MC}}$  do
2:   Draw  $\mathbf{w}^{(j)} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_r)$ 
3:    $m^{(j)} \leftarrow 0$ 
4:    $X_{\text{tr}}^{(j)} \leftarrow \infty$ 
5:   for  $n = 0, \dots, N$  do
6:      $y_n^{(j)} \leftarrow [\mathbf{w}^{(j)}]^\top \mathbf{G}(x_n) \mathbf{w}^{(j)}$ 
7:      $m^{(j)} \leftarrow \max(m^{(j)}, y_n^{(j)})$ 
8:     if  $X_{\text{tr}}^{(j)} = \infty$  and  $m^{(j)} \geq 1$  then
9:        $X_{\text{tr}}^{(j)} \leftarrow x_n$ 
10:    end if
11:  end for
12: end for
13: for  $n = 0, \dots, N$  do
14:    $\hat{F}_{\text{MC}}(x_n) \leftarrow \frac{1}{N_{\text{MC}}} \sum_{j=1}^{N_{\text{MC}}} \mathbf{1}\{X_{\text{tr}}^{(j)} \leq x_n\}$ 
15: end for
16: return  $\{\hat{F}_{\text{MC}}(x_n)\}_{n=0}^N$ 
```

Algorithm 14 Generation of proper-complex angular directions

Require: Complex stochastic dimension r

Require: Number of directions K

Ensure: Uniform directions $\{\mathbf{u}_{\mathbb{C},k}\}_{k=1}^K$

```
1: for  $k = 1, \dots, K$  do
2:   Draw or quasi-randomly generate  $\mathbf{g}_{\text{R},k} \in \mathbb{R}^r$  and  $\mathbf{g}_{\text{I},k} \in \mathbb{R}^r$  with standard-normal marginals
3:    $\boldsymbol{\eta}_k \leftarrow \mathbf{g}_{\text{R},k} + \mathbf{i}\mathbf{g}_{\text{I},k}$ 
4:    $\mathbf{u}_{\mathbb{C},k} \leftarrow \boldsymbol{\eta}_k / \|\boldsymbol{\eta}_k\|_2$ 
5: end for
6: return  $\{\mathbf{u}_{\mathbb{C},k}\}_{k=1}^K$ 
```

Algorithm 15 First-transition CDF for proper complex Gaussian coordinates

Require: Stations $\{x_n\}_{n=0}^N$
Require: Hermitian matrices $\{\mathbf{G}_{\mathbb{C}}(x_n)\}_{n=0}^N$
Require: Complex angular rule $\{(\mathbf{u}_{\mathbb{C},k}, \omega_k)\}_{k=1}^K$
Require: Complex stochastic dimension r
Ensure: Survival probabilities $\{S_n\}$
Ensure: Transition CDF $\{F_n\}$

```

1: for  $k = 1, \dots, K$  do
2:    $\mathbf{u}_{\mathbb{C},k} \leftarrow \mathbf{u}_{\mathbb{C},k} / \|\mathbf{u}_{\mathbb{C},k}\|_2$ 
3:    $m_k \leftarrow 0$ 
4: end for
5: for  $n = 0, \dots, N$  do
6:    $S_n \leftarrow 0$ 
7:   for  $k = 1, \dots, K$  do
8:      $a_{k,n} \leftarrow \Re[\mathbf{u}_{\mathbb{C},k}^* \mathbf{G}_{\mathbb{C}}(x_n) \mathbf{u}_{\mathbb{C},k}]$ 
9:      $a_{k,n} \leftarrow \max(a_{k,n}, 0)$ 
10:     $m_k \leftarrow \max(m_k, a_{k,n})$ 
11:    if  $m_k > 0$  then
12:       $s_{k,n} \leftarrow F_{\text{Gamma}(r,1)}(1/m_k)$ 
13:    else
14:       $s_{k,n} \leftarrow 1$ 
15:    end if
16:     $S_n \leftarrow S_n + \omega_k s_{k,n}$ 
17:  end for
18:   $F_n \leftarrow 1 - S_n$ 
19: end for
20: return  $\{S_n, F_n\}_{n=0}^N$ 

```

Algorithm 16 Robust maximum of a cubic Hermite interpolant

Require: $x_0 < x_1$ **Require:** Values a_0, a_1 **Require:** Derivatives d_0, d_1 **Require:** Relative polynomial tolerance $\varepsilon_{\text{poly}}$ **Require:** Root tolerance ε_s **Ensure:** Maximum $\hat{m}^I$ **Ensure:** Maximizer $\hat{x}^*$

```
1:  $h \leftarrow x_1 - x_0$ 
2:  $c_0 \leftarrow a_0$ 
3:  $c_1 \leftarrow h d_0$ 
4:  $c_2 \leftarrow 3(a_1 - a_0) - h(2d_0 + d_1)$ 
5:  $c_3 \leftarrow 2(a_0 - a_1) + h(d_0 + d_1)$ 
6:  $c_{\text{scale}} \leftarrow \max(|c_0|, |c_1|, |c_2|, |c_3|, 1)$ 
7:  $\tau_{\text{poly}} \leftarrow \varepsilon_{\text{poly}} c_{\text{scale}}$ 
8:  $\mathcal{C} \leftarrow \{0, 1\}$ 
9: if  $|c_3| > \tau_{\text{poly}}$  then
10:    $A \leftarrow 3c_3$ 
11:    $B \leftarrow 2c_2$ 
12:    $C \leftarrow c_1$ 
13:    $D \leftarrow B^2 - 4AC$ 
14:    $\tau_D \leftarrow \varepsilon_{\text{poly}} \max(B^2, |4AC|, 1)$ 
15:   if  $D \geq -\tau_D$  then
16:      $D \leftarrow \max(D, 0)$ 
17:     if  $B \geq 0$  then
18:        $q \leftarrow -\frac{1}{2}(B + \sqrt{D})$ 
19:     else
20:        $q \leftarrow -\frac{1}{2}(B - \sqrt{D})$ 
21:     end if
22:     if  $|q| > \tau_{\text{poly}}$  then
23:        $s_1 \leftarrow q/A$ 
24:        $s_2 \leftarrow C/q$ 
25:     else
26:        $s_1 \leftarrow -B/(2A)$ 
27:        $s_2 \leftarrow s_1$ 
28:     end if
29:     for  $s \in \{s_1, s_2\}$  do
30:       if  $\varepsilon_s < s < 1 - \varepsilon_s$  then
31:          $\mathcal{C} \leftarrow \mathcal{C} \cup \{s\}$ 
32:       else if  $|s| \leq \varepsilon_s$  then
33:          $\mathcal{C} \leftarrow \mathcal{C} \cup \{0\}$ 
34:       else if  $|s - 1| \leq \varepsilon_s$  then
35:          $\mathcal{C} \leftarrow \mathcal{C} \cup \{1\}$ 
36:       end if
37:     end for
38:   end if
39: else if  $|c_2| > \tau_{\text{poly}}$  then
40:    $s_* \leftarrow -c_1/(2c_2)$ 
41:   if  $\varepsilon_s < s_* < 1 - \varepsilon_s$  then
42:      $\mathcal{C} \leftarrow \mathcal{C} \cup \{s_*\}$ 
43:   end if
44: end if
45:  $s_{\text{max}} \leftarrow \arg \max(c_3 s^3 + c_2 s^2 + c_1 s + c_0)$ 
```

Algorithm 17 Adaptive continuous directional maximum on one interval

Require: Direction $\mathbf{u}$

Require: Endpoint tuples (x_0, a_0, d_0) and (x_1, a_1, d_1)

Require: Previous running maximum m^{up}

Require: Gain tolerances $\varepsilon_{\text{abs}}, \varepsilon_{\text{rel}}$

Require: Probability tolerance ε_{Ψ}

Require: Remaining depth J

Ensure: Resolved interval maximum m^I

```
1:  $(m_C, x_C) \leftarrow \text{HERMITEMAXIMUM}(x_0, x_1, a_0, a_1, d_0, d_1)$ 
2:  $x_m \leftarrow (x_0 + x_1)/2$ 
3:  $(a_m, d_m) \leftarrow \text{EVALUATEDIRECTIONALDATA}(x_m, \mathbf{u})$ 
4:  $(m_L, x_L) \leftarrow \text{HERMITEMAXIMUM}(x_0, x_m, a_0, a_m, d_0, d_m)$ 
5:  $(m_R, x_R) \leftarrow \text{HERMITEMAXIMUM}(x_m, x_1, a_m, a_1, d_m, d_1)$ 
6:  $m_S \leftarrow \max(m_L, m_R)$ 
7:  $\eta_m \leftarrow |m_S - m_C|$ 
8:  $\eta_a \leftarrow |a_m - \hat{a}_C(1/2)|$ 
9:  $\eta_d \leftarrow (x_1 - x_0) |d_m - \hat{d}'_C(1/2)/(x_1 - x_0)|$ 
10:  $\eta_I \leftarrow \max(\eta_m, \eta_a, \eta_d)$ 
11:  $\tau_I \leftarrow \varepsilon_{\text{abs}} + \varepsilon_{\text{rel}} \max(m_S, m^{\text{up}}, 1)$ 
12:  $m_C^{\text{new}} \leftarrow \max(m^{\text{up}}, m_C)$ 
13:  $m_S^{\text{new}} \leftarrow \max(m^{\text{up}}, m_S)$ 
14:  $\eta_{\Psi} \leftarrow |\Psi_r(m_S^{\text{new}}) - \Psi_r(m_C^{\text{new}})|$ 
15: if  $\eta_I \leq \tau_I$  and  $\eta_{\Psi} \leq \varepsilon_{\Psi}$  then
16:   return  $m_S$ 
17: else if  $J = 0$  then
18:   Issue a maximum-resolution warning
19:   return  $\max(m_C, m_S, a_m)$ 
20: else
21:    $m_L^{\text{ref}} \leftarrow \text{ADAPTIVEDIRECTIONALMAXIMUM}(\mathbf{u}, (x_0, a_0, d_0), (x_m, a_m, d_m), m^{\text{up}}, J - 1)$ 
22:    $m^{\text{mid}} \leftarrow \max(m^{\text{up}}, m_L^{\text{ref}})$ 
23:    $m_R^{\text{ref}} \leftarrow \text{ADAPTIVEDIRECTIONALMAXIMUM}(\mathbf{u}, (x_m, a_m, d_m), (x_1, a_1, d_1), m^{\text{mid}}, J - 1)$ 
24:   return  $\max(m_L^{\text{ref}}, m_R^{\text{ref}})$ 
25: end if
```

Algorithm 18 Shared streamwise refinement for an angular rule

Require: Initial streamwise grid $\mathcal{X} = \{x_n\}$

Require: Angular rule $\{(\mathbf{u}_k, \omega_k)\}_{k=1}^K$

Require: Streamwise gain and CDF tolerances

Require: Maximum refinement level $J_{\max}$

Ensure: Refined shared grid $\mathcal{X}$

Ensure: Resolved directional interval maxima

```
1: Evaluate  $\mathbf{G}(x)$  and  $\mathbf{G}'(x)$  for every  $x \in \mathcal{X}$ 
2: for  $j = 0, \dots, J_{\max}$  do
3:   Sort  $\mathcal{X}$  in increasing order
4:   Compute directional endpoint values and derivatives for all  $k$  and all  $x \in \mathcal{X}$ 
5:   Initialize marked set  $\mathcal{M} \leftarrow \emptyset$ 
6:   Initialize each upstream running maximum from the inlet
7:   for each interval  $I_n = [x_n, x_{n+1}]$  in increasing  $x$  do
8:      $x_m \leftarrow (x_n + x_{n+1})/2$ 
9:     if data at  $x_m$  are not cached then
10:      Evaluate  $\mathbf{G}(x_m)$  and  $\mathbf{G}'(x_m)$ 
11:     end if
12:     for  $k = 1, \dots, K$  do
13:       Compute coarse and split Hermite maxima  $m_{C,k,n}^I$  and  $m_{S,k,n}^I$ 
14:       Compute  $\eta_{I,k,n}$ 
15:       Compute the record-aware probability indicator  $\eta_{\Psi,k,n}^{\text{record}}$ 
16:     end for
17:     Aggregate the gain and probability indicators
18:     if either aggregate criterion fails then
19:        $\mathcal{M} \leftarrow \mathcal{M} \cup \{x_m\}$ 
20:     else
21:       Accept the split interval maxima and update all directional running maxima
22:     end if
23:   end for
24:   if  $\mathcal{M} = \emptyset$  then
25:     return  $\mathcal{X}$  and the accepted maxima
26:   end if
27:    $\mathcal{X} \leftarrow \mathcal{X} \cup \mathcal{M}$ 
28: end for
29: Issue a streamwise-refinement warning
30: return the finest available grid and maxima
```

Algorithm 19 First threshold crossing for a specified radius and direction

Require: Radius R

Require: Direction $\mathbf{u}$

Require: Resolved interval representations of $a_{\mathbf{u}}(x)$

Ensure: X_{tr} or a censored outcome

```
1: for  $n = 0, \dots, N - 1$  do
2:   Determine the resolved interval maximum  $m_{\mathbf{u},n}^I$ 
3:   if  $R^2 m_{\mathbf{u},n}^I < 1$  then
4:     continue
5:   end if
6:   Identify all stationary points of the interval interpolant
7:   Partition the interval into monotone subintervals
8:   for monotone subintervals in increasing  $x$  do
9:     if the threshold is attained on the subinterval then
10:      Solve  $R^2 \hat{a}(x) - 1 = 0$  with a bracketed root finder
11:      return the smallest root
12:     end if
13:   end for
14: end for
15: return  $X_{\text{tr}} = \infty$ 
```

Algorithm 20 Scaled QR propagation of the stochastic factor

Require: Initial factor $\mathbf{B}_0$

Require: OWNS interval propagation operator $\mathbf{P}_{n+1,n}$

Ensure: $\mathbf{B}_n = \exp(\ell_n) \mathbf{Q}_n \bar{\mathbf{T}}_n$

```
1: Compute  $\mathbf{B}_0 = \mathbf{Q}_0 \mathbf{T}_0$  by thin QR
2:  $s_0 \leftarrow \|\mathbf{T}_0\|_{\text{scale}}$ 
3:  $\bar{\mathbf{T}}_0 \leftarrow \mathbf{T}_0 / s_0$ 
4:  $\ell_0 \leftarrow \log s_0$ 
5: for  $n = 0, \dots, N - 1$  do
6:    $\hat{\mathbf{Q}}_{n+1} \leftarrow \mathbf{P}_{n+1,n} \mathbf{Q}_n$ 
7:   Compute  $\hat{\mathbf{Q}}_{n+1} = \mathbf{Q}_{n+1} \mathbf{R}_{n+1}$  by thin QR
8:    $\hat{\mathbf{T}}_{n+1} \leftarrow \mathbf{R}_{n+1} \bar{\mathbf{T}}_n$ 
9:    $s_{n+1} \leftarrow \|\hat{\mathbf{T}}_{n+1}\|_{\text{scale}}$ 
10:   $\bar{\mathbf{T}}_{n+1} \leftarrow \hat{\mathbf{T}}_{n+1} / s_{n+1}$ 
11:   $\ell_{n+1} \leftarrow \ell_n + \log s_{n+1}$ 
12: end for
13: return  $\{(\mathbf{Q}_n, \bar{\mathbf{T}}_n, \ell_n)\}$ 
```

Algorithm 21 Stabilized OWNS propagation for first-transition analysis

Require: Initial synthesis factor $\mathbf{B}_0$

Require: OWNS marching solver

Require: Output stations $\{x_n\}_{n=0}^N$

Require: Energy metrics $\mathbf{H}(x_n)$

Require: Optional logarithmic scaling

Ensure: Propagated reduced energy matrices $\mathbf{A}(x_n)$

Ensure: Optional reconstruction factors $(\mathbf{Q}_n, \mathbf{T}_n)$

- 1: Compute a rank-revealing factorization of $\mathbf{B}_0$
 - 2: Retain the documented initial stochastic rank
 - 3: Compute $\mathbf{B}_0 = \mathbf{Q}_0 \mathbf{T}_0$ by thin QR
 - 4: Fix the QR diagonal phase convention
 - 5: **for** $n = 0, \dots, N$ **do**
 - 6: **if** $n > 0$ **then**
 - 7: March $\mathbf{Q}_{n-1}$ from x_{n-1} to x_n using OWNS
 - 8: Compute $\hat{\mathbf{Q}}_n = \mathbf{Q}_n \mathbf{R}_n$ by stable thin QR
 - 9: $\mathbf{T}_n \leftarrow \mathbf{R}_n \mathbf{T}_{n-1}$
 - 10: Apply and record scaling if needed
 - 11: **end if**
 - 12: Check $\|\mathbf{Q}_n^* \mathbf{Q}_n - \mathbf{I}\|_F$
 - 13: Form $\mathbf{H}_{Q,n} = \mathbf{Q}_n^* \mathbf{H}(x_n) \mathbf{Q}_n$
 - 14: Form $\mathbf{A}_n = \Re[\mathbf{T}_n^* \mathbf{H}_{Q,n} \mathbf{T}_n]$
 - 15: Symmetrize $\mathbf{A}_n$
 - 16: Check the positive-semidefinite defect
 - 17: Store $\mathbf{A}_n$, and store $(\mathbf{Q}_n, \mathbf{T}_n)$ if physical reconstruction is required
 - 18: **end for**
 - 19: Verify selected directional gains using direct state evaluation
 - 20: Repeat with a different QR frequency or tighter OWNS tolerance
 - 21: **return** the propagated data
-

Algorithm 22 End-to-end computation of the first-transition distribution

Require: Initial covariance or correlation matrix $\mathbf{C}_0$

Require: OWNS marching solver

Require: Streamwise domain $[0, x_{\max}]$

Require: Energy metric $\mathbf{H}(x)$

Require: Threshold $e_{\text{thres}}(x) > 0$

Require: Numerical tolerances

Ensure: Survival curve $S(x_n)$

Ensure: First-transition CDF $F_{X_{\text{tr}}}(x_n)$

Ensure: Interval probabilities and censoring probability

1: Factor the initial correlation:

$$\mathbf{C}_0 \approx \mathbf{B}_0 \mathbf{B}_0^*.$$

2: Select an initial stochastic rank large enough to retain potentially amplified low-variance directions

3: Propagate $\mathbf{B}_0$ through OWNS using stabilized QR or SVD updates from section D

4: **for** each stored station x_n **do**

5: Form the unnormalized reduced energy matrix

$$\mathbf{A}_n \leftarrow \Re[\mathbf{B}(x_n)^* \mathbf{H}(x_n) \mathbf{B}(x_n)].$$

6: Symmetrize $\mathbf{A}_n$

7: Check its positive-semidefinite defect

8: **end for**

9: Construct a fixed transition-oriented basis $\mathbf{V}_s$ using Algorithm 27

10: **for** each station x_n **do**

11:

$$\mathbf{A}_{s,n} \leftarrow \mathbf{V}_s^\top \mathbf{A}_n \mathbf{V}_s.$$

12:

$$\mathbf{G}_{s,n} \leftarrow \mathbf{A}_{s,n} / e_{\text{thres}}(x_n).$$

13: **end for**

14: Generate an initial angular rule $\{(\mathbf{u}_k, \omega_k)\}_{k=1}^K$ on $\mathbb{S}^{s-1}$

15: Resolve the continuous streamwise directional maxima using Algorithm 18

16: **for** $n = 0, \dots, N$ **do**

17:

$$S_n \leftarrow \sum_{k=1}^K \omega_k F_{\chi_s^2} \left[\frac{1}{m_k(x_n)} \right].$$

18: $F_n \leftarrow 1 - S_n$

19: **end for**

20: Refine the angular rule using Algorithm 25 or a deterministic angular convergence study

21: Repeat the reduced-rank calculation for one or more larger values of s

22: Compute

$$p_0 \leftarrow F_0, \quad p_n \leftarrow F_n - F_{n-1}, \quad p_{\text{censored}} \leftarrow S_N.$$

23: Check monotonicity, probability conservation, and $p_{\text{local}}(x_n) \leq F_n$

24: **return** $\{S_n, F_n, p_n, p_{\text{censored}}\}$

Algorithm 23 Batched explicit directional gains and CDF update

Require: Matrices $\{\mathbf{G}_{s,n}\}_{n=0}^N$

Require: Direction matrix $\mathbf{U} \in \mathbb{R}^{s \times K}$

Require: Angular weights $\boldsymbol{\omega} \in \mathbb{R}^K$

Ensure: Directional running maxima $\mathbf{m}_n$

Ensure: Transition CDF $\{F_n\}$

```

1: Normalize every column of  $\mathbf{U}$ 
2:  $\mathbf{m} \leftarrow \mathbf{0}$ 
3: for  $n = 0, \dots, N$  do
4:    $\mathbf{Z} \leftarrow \mathbf{G}_{s,n} \mathbf{U}$ 
5:    $\mathbf{a} \leftarrow \text{colsum}(\mathbf{U} \odot \mathbf{Z})$ 
6:    $\mathbf{a} \leftarrow \max(\mathbf{a}, \mathbf{0})$  ▷ Componentwise
7:    $\mathbf{m} \leftarrow \max(\mathbf{m}, \mathbf{a})$  ▷ Componentwise running maximum
8:   for  $k = 1, \dots, K$  do
9:     if  $m_k > 0$  then
10:       $\Psi_k \leftarrow \bar{F}_{\chi_s^2}(1/m_k)$ 
11:     else
12:       $\Psi_k \leftarrow 0$ 
13:     end if
14:   end for
15:    $F_n \leftarrow \sum_{k=1}^K \omega_k \Psi_k$ 
16:    $S_n \leftarrow 1 - F_n$ 
17: end for
18: return  $\{\mathbf{m}_n, S_n, F_n\}$ 

```

Algorithm 24 Blocked matrix-free first-transition calculation

Require: Propagated factor evaluator $\mathbf{B}(x_n)$

Require: Energy-weighting operator $\mathbf{W}(x_n)$

Require: Angular rule $\{(\mathbf{u}_k, \omega_k)\}_{k=1}^K$

Require: Block size K_b

Ensure: Transition CDF $\{F_n\}_{n=0}^N$

```
1: Partition the directions into  $N_b$  blocks
2: Initialize  $F_n \leftarrow 0$  for all  $n$ 
3: for  $b = 1, \dots, N_b$  do
4:   Assemble direction block  $\mathbf{U}^{(b)}$ 
5:   Initialize block running maxima  $\mathbf{m}^{(b)} \leftarrow \mathbf{0}$ 
6:   for  $n = 0, \dots, N$  do
7:      $\mathbf{V}^{(b)} \leftarrow \mathbf{B}(x_n)\mathbf{U}^{(b)}$ 
8:      $\mathbf{Y}^{(b)} \leftarrow \mathbf{W}(x_n)\mathbf{V}^{(b)}$ 
9:     for each column  $j$  in block  $b$  do
10:       $a_j \leftarrow \|\mathbf{y}_j^{(b)}\|_2^2 / e_{\text{thres}}(x_n)$ 
11:       $m_j^{(b)} \leftarrow \max(m_j^{(b)}, a_j)$ 
12:       $\Psi_j \leftarrow \bar{F}_{\chi_r^2}(1/m_j^{(b)})$ 
13:    end for
14:    Add  $\sum_j \omega_j \Psi_j$  to  $F_n$ 
15:  end for
16: end for
17: for  $n = 0, \dots, N$  do
18:    $S_n \leftarrow 1 - F_n$ 
19: end for
20: return  $\{S_n, F_n\}_{n=0}^N$ 
```

Algorithm 25 Adaptive angular convergence with scrambled RQMC

Require: Stochastic dimension r

Require: Initial point count $K_0 = 2^{p_0}$

Require: Number of scramblings J

Require: CDF tolerance $\varepsilon_F^{\text{ang}}$

Require: Standard-error tolerance $\varepsilon_{\text{SE}}^{\text{ang}}$

Require: Maximum point count K_{max}

Ensure: Angularly converged CDF estimate

1: $K \leftarrow K_0$

2: $\hat{F}_{\text{old}} \leftarrow \text{undefined}$

3: **while** $K \leq K_{\text{max}}$ **do**

4: **for** $j = 1, \dots, J$ **do**

5: Generate the first K points of scrambling j

6: Map the points to $\{\mathbf{u}_k^{(j)}\}_{k=1}^K \subset \mathbb{S}^{r-1}$

7: Compute all directional running maxima $m_k^{(j)}(x_n)$

8: **for** $n = 0, \dots, N$ **do**

9:

$$\hat{F}_K^{(j)}(x_n) \leftarrow \frac{1}{K} \sum_{k=1}^K \bar{F}_{\chi_r^2} \left[\frac{1}{m_k^{(j)}(x_n)} \right].$$

10: **end for**

11: **end for**

12: Compute the replicate mean $\hat{F}_{K,J}(x_n)$

13: Compute replicate-based standard errors at all x_n

14: $\text{SE}_{\text{max}} \leftarrow \max_n \widehat{\text{SE}}[\hat{F}_{K,J}(x_n)]$

15: **if** $\hat{F}_{\text{old}}$ is defined **then**

16: $\Delta_F \leftarrow \max_n |\hat{F}_{K,J}(x_n) - \hat{F}_{\text{old}}(x_n)|$

17: **if** $\Delta_F \leq \varepsilon_F^{\text{ang}}$ **and** $\text{SE}_{\text{max}} \leq \varepsilon_{\text{SE}}^{\text{ang}}$ **then**

18: **return** $\hat{F}_{K,J}$ and its standard errors

19: **end if**

20: **end if**

21: $\hat{F}_{\text{old}} \leftarrow \hat{F}_{K,J}$

22: $K \leftarrow 2K$

23: **end while**

24: Issue an angular-convergence warning

25: **return** the largest calculation

Algorithm 26 Adaptive projective-circle quadrature for $r = 2$

Require: Initial angular partition of $[0, \pi)$

Require: Angular tolerance $\varepsilon_{\text{cell}}$

Require: Maximum refinement level

Ensure: Adaptive angular rule and transition CDF

- 1: Evaluate $\Psi_x(\theta)$ at the initial cell midpoints
 - 2: **while** unaccepted cells remain **do**
 - 3: Select an angular cell $\Theta = [\theta_L, \theta_R]$
 - 4: Evaluate Ψ_x at its two quarter points
 - 5: Compute $I_C(x_n)$, $I_S(x_n)$, and η_Θ
 - 6: **if** $\eta_\Theta \leq \varepsilon_{\text{cell}}$ **then**
 - 7: Accept the two-half estimate
 - 8: **else**
 - 9: Replace Θ by its two half cells
 - 10: **end if**
 - 11: **end while**
 - 12: Sum all accepted cell contributions at each x_n
 - 13: **return** the CDF and the adaptive angular partition
-

Algorithm 27 Hybrid transition-oriented basis construction

Require: Matrix family $\{\mathbf{G}(x_n)\}_{n=0}^N$

Require: Integrated-basis size p_{int}

Require: Modes per record station p_{rec}

Require: Maximum rank s_{max}

Require: Residual tolerance ε_G

Ensure: Fixed basis $\mathbf{V}_s \in \mathbb{R}^{r \times s}$

1: Assemble

$$\mathbf{K}_\omega \leftarrow \sum_{n=0}^N \nu_n \omega(x_n) \mathbf{G}(x_n).$$

2: Compute the first p_{int} eigenvectors of $\mathbf{K}_\omega$

3: Initialize snapshot matrix $\mathbf{Z}$ with these eigenvectors

4: $\gamma_{\text{record}} \leftarrow -\infty$

5: **for** $n = 0, \dots, N$ **do**

6: Compute the first p_{rec} eigenpairs of $\mathbf{G}(x_n)$

7: **if** $\lambda_{\text{max}}[\mathbf{G}(x_n)] > \gamma_{\text{record}} + \varepsilon_{\text{record}}$ **then**

8: Append the leading p_{rec} eigenvectors to $\mathbf{Z}$

9: $\gamma_{\text{record}} \leftarrow \lambda_{\text{max}}[\mathbf{G}(x_n)]$

10: **end if**

11: **end for**

12: Orthonormalize $\mathbf{Z}$ to obtain $\mathbf{V}$

13: **while** number of columns of $\mathbf{V}$ is less than s_{max} **do**

14: $\mathbf{P}_\perp \leftarrow \mathbf{I} - \mathbf{V}\mathbf{V}^\top$

15: $\epsilon_* \leftarrow 0$

16: $\mathbf{v}_* \leftarrow \text{undefined}$

17: **for** $n = 0, \dots, N$ **do**

18: $\mathbf{R}_n \leftarrow \mathbf{P}_\perp \mathbf{G}(x_n) \mathbf{P}_\perp$

19: Compute the dominant eigenpair $(\epsilon_n, \mathbf{v}_n)$ of $\mathbf{R}_n$

20: **if** $\epsilon_n > \epsilon_*$ **then**

21: $\epsilon_* \leftarrow \epsilon_n$

22: $\mathbf{v}_* \leftarrow \mathbf{v}_n$

23: **end if**

24: **end for**

25: **if** $\epsilon_* \leq \varepsilon_G$ **then**

26: **break**

27: **end if**

28: Orthogonalize $\mathbf{v}_*$ against $\mathbf{V}$

29: Normalize $\mathbf{v}_*$

30: Append $\mathbf{v}_*$ to $\mathbf{V}$

31: **end while**

32: Compress the collected basis using an SVD or rank-revealing QR

33: Test nested candidate ranks using the first-transition CDF

34: Select the smallest rank satisfying both the CDF and matrix residual tolerances

35: **return** the selected fixed basis $\mathbf{V}_s$

Algorithm 28 Conditional covariance of the surviving ensemble

Require: Directions and weights $\{(\mathbf{u}_k, \omega_k)\}_{k=1}^K$
Require: Running maxima $\{m_k(x_n)\}$
Require: Stochastic dimension r
Require: Propagated factor $\mathbf{B}(x_n)$, if physical covariance is desired
Ensure: Coordinate survivor covariance $\Sigma_{w,\text{surv}}(x_n)$
Ensure: Optional physical survivor covariance

- 1: **for** $n = 0, \dots, N$ **do**
- 2: $\mathbf{M}_n \leftarrow \mathbf{0}_{r \times r}$
- 3: $S_n \leftarrow 0$
- 4: **for** $k = 1, \dots, K$ **do**
- 5: **if** $m_k(x_n) > 0$ **then**
- 6: $c_{k,n} \leftarrow 1/m_k(x_n)$
- 7: **else**
- 8: $c_{k,n} \leftarrow \infty$
- 9: **end if**
- 10: $s_{k,n} \leftarrow F_{\chi_r^2}(c_{k,n})$
- 11: $z_{k,n} \leftarrow r F_{\chi_{r+2}^2}(c_{k,n})$
- 12: $S_n \leftarrow S_n + \omega_k s_{k,n}$
- 13: $\mathbf{M}_n \leftarrow \mathbf{M}_n + \omega_k z_{k,n} \mathbf{u}_k \mathbf{u}_k^\top$
- 14: **end for**
- 15: **if** $S_n > 0$ **then**
- 16: $\Sigma_{w,\text{surv},n} \leftarrow \mathbf{M}_n / S_n$
- 17: $\mathbf{C}_{q,\text{surv},n} \leftarrow \mathbf{B}(x_n) \Sigma_{w,\text{surv},n} \mathbf{B}(x_n)^*$
- 18: **else**
- 19: Mark survivor covariance as undefined
- 20: **end if**
- 21: **end for**
- 22: **return** the conditional covariances

Algorithm 29 Sample an initial realization conditional on transition by x_n

Require: Angular rule $\{(\mathbf{u}_k, \omega_k)\}_{k=1}^K$

Require: Running maxima $m_k(x_n)$

Require: Stochastic dimension r

Ensure: Approximate conditional sample $\mathbf{w} \mid X_{\text{tr}} \leq x_n$

```

1: for  $k = 1, \dots, K$  do
2:   if  $m_k(x_n) > 0$  then
3:      $c_k \leftarrow 1/m_k(x_n)$ 
4:      $\Psi_k \leftarrow \overline{F}_{\chi_r^2}(c_k)$ 
5:   else
6:      $c_k \leftarrow \infty$ 
7:      $\Psi_k \leftarrow 0$ 
8:   end if
9:    $v_k \leftarrow \omega_k \Psi_k$ 
10: end for
11:  $F_n \leftarrow \sum_{k=1}^K v_k$ 
12: if  $F_n = 0$  then
13:   Report that the conditional event has zero represented probability
14: end if
15: Normalize  $\pi_k \leftarrow v_k/F_n$ 
16: Draw a direction index  $K_*$  from the categorical distribution  $\{\pi_k\}$ 
17: Draw  $U \sim \text{Unif}(0, 1)$ 
18: Draw

$$Z \leftarrow \overline{F}_{\chi_r^2}^{-1} [U \overline{F}_{\chi_r^2}(c_{K_*})] .$$

19: Draw a random sign  $\varsigma \in \{-1, +1\}$  with equal probabilities if the angular rule stores only one
    member of each antipodal pair
20:

$$\mathbf{w} \leftarrow \varsigma \sqrt{Z} \mathbf{u}_{K_*} .$$

21: return  $\mathbf{w}$ 

```

Algorithm 30 Sample an initial realization conditional on first transition in $(x_a, x_b]$

Require: Angular rule $\{(\mathbf{u}_k, \omega_k)\}_{k=1}^K$

Require: Running maxima $m_k(x_a)$ and $m_k(x_b)$

Require: Stochastic dimension r

Ensure: Conditional sample $\mathbf{w} \mid x_a < X_{\text{tr}} \leq x_b$

1: **for** $k = 1, \dots, K$ **do**

2: $m_{a,k} \leftarrow m_k(x_a)$

3: $m_{b,k} \leftarrow m_k(x_b)$

4: **if** $m_{b,k} > m_{a,k}$ **then**

5: $z_{L,k} \leftarrow 1/m_{b,k}$

6: **if** $m_{a,k} > 0$ **then**

7: $z_{U,k} \leftarrow 1/m_{a,k}$

8: **else**

9: $z_{U,k} \leftarrow \infty$

10: **end if**

11:

$$\Delta\Psi_k \leftarrow F_{\chi_r^2}(z_{U,k}) - F_{\chi_r^2}(z_{L,k}).$$

12: **else**

13: $\Delta\Psi_k \leftarrow 0$

14: **end if**

15: $v_k \leftarrow \omega_k \Delta\Psi_k$

16: **end for**

17: $p_{a,b} \leftarrow \sum_{k=1}^K v_k$

18: **if** $p_{a,b} = 0$ **then**

19: Report that the interval has zero represented probability

20: **end if**

21: Normalize $\pi_k \leftarrow v_k/p_{a,b}$

22: Draw direction index K_* from $\{\pi_k\}$

23: Draw $U \sim \text{Unif}(0, 1)$

24:

$$p_L \leftarrow F_{\chi_r^2}(z_{L,K_*}), \quad p_U \leftarrow F_{\chi_r^2}(z_{U,K_*}).$$

25:

$$Z \leftarrow F_{\chi_r^2}^{-1}[p_L + U(p_U - p_L)].$$

26: Draw an antipodal sign if needed

27:

$$\mathbf{w} \leftarrow \varsigma \sqrt{Z} \mathbf{u}_{K_*}.$$

28: Determine the exact first crossing within $(x_a, x_b]$ using the resolved directional gain history

29: **return** $\mathbf{w}$ and X_{tr}

Algorithm 31 Joint sample of transition interval and initial realization

Require: Interval probabilities $\{p_0, p_1, \dots, p_N, p_{\text{censored}}\}$ **Require:** Directional running maxima at every station**Require:** Angular rule**Ensure:** Transition location and compatible $\mathbf{w}$, or a censored sample

1: Draw a categorical outcome from

$$\{p_0, p_1, \dots, p_N, p_{\text{censored}}\}.$$

2: **if** the outcome is p_{censored} **then**3: Sample $\mathbf{w}$ from the survivor distribution at $x_{\max}$

4: Return a right-censored outcome

5: **else if** the outcome is p_0 **then**6: Sample $\mathbf{w}$ conditional on transition at the inlet7: Set $X_{\text{tr}} = 0$ 8: **else**9: Let the selected interval be $(x_{n-1}, x_n]$ 10: Use Algorithm 30 to sample $\mathbf{w}$ 11: Solve for the first continuous crossing location $X_{\text{tr}} \in (x_{n-1}, x_n]$ 12: **end if**13: **return** $\mathbf{w}$ and the transition outcome

Algorithm 32 Sample an initial realization conditional on survival through x_n

Require: Angular rule and running maxima at x_n **Require:** Stochastic dimension r **Ensure:** Approximate sample $\mathbf{w} \mid X_{\text{tr}} > x_n$ 1: **for** $k = 1, \dots, K$ **do**2: **if** $m_k(x_n) > 0$ **then**3: $c_k \leftarrow 1/m_k(x_n)$ 4: **else**5: $c_k \leftarrow \infty$ 6: **end if**7: $s_k \leftarrow F_{\chi_r^2}(c_k)$ 8: $v_k \leftarrow \omega_k s_k$ 9: **end for**10: $S_n \leftarrow \sum_k v_k$ 11: Normalize $\pi_k \leftarrow v_k/S_n$ 12: Draw index K_* from $\{\pi_k\}$ 13: Draw $U \sim \text{Unif}(0, 1)$

14:

$$Z \leftarrow F_{\chi_r^2}^{-1} [UF_{\chi_r^2}(c_{K_*})].$$

15: Draw an antipodal sign if needed

16:

$$\mathbf{w} \leftarrow \varsigma \sqrt{Z} \mathbf{u}_{K_*}.$$

17: **return** $\mathbf{w}$

Algorithm 33 Efficient amplitude and threshold parameter sweep

Require: Reference running maxima $m_k^{\text{ref}}(x_n)$

Require: Angular rule $\{(\mathbf{u}_k, \omega_k)\}$

Require: Amplitudes $\{\epsilon_i\}$

Require: Threshold factors $\{\alpha_j\}$

Ensure: CDF family $F(x_n; \epsilon_i, \alpha_j)$

```
1: for each amplitude  $\epsilon_i$  do
2:   for each threshold factor  $\alpha_j$  do
3:      $\rho_{ij} \leftarrow \epsilon_i^2 / \alpha_j$ 
4:     for  $n = 0, \dots, N$  do
5:        $F_n \leftarrow 0$ 
6:       for  $k = 1, \dots, K$  do
7:          $m_{k,n}^{(ij)} \leftarrow \rho_{ij} m_k^{\text{ref}}(x_n)$ 
8:
```

$$F_n \leftarrow F_n + \omega_k \overline{F}_{\chi_r^2} \left[\frac{1}{m_{k,n}^{(ij)}} \right].$$

```
9:       end for
10:    end for
11:    Store the CDF for  $(\epsilon_i, \alpha_j)$ 
12:  end for
13: end for
14: return the CDF family
```

Algorithm 34 Independent local-exceedance consistency check

Require: Matrix family $\{\mathbf{G}(x_n)\}$

Require: First-transition CDF $\{F_n\}$

Ensure: Local exceedance probabilities and consistency diagnostics

```
1: for  $n = 0, \dots, N$  do
2:   Compute eigenvalues  $\{\gamma_{j,n}\}_{j=1}^r$ 
3:   Remove only roundoff-scale negative eigenvalues
4:   Evaluate
```

$$p_{\text{local},n} \leftarrow \mathbb{P} \left[\sum_{j=1}^r \gamma_{j,n} Z_j^2 \geq 1 \right]$$

using a generalized chi-square algorithm

```
5:   if  $p_{\text{local},n} > F_n + \varepsilon_{\text{check}}$  then
6:     Issue a first-transition consistency warning
7:   end if
8: end for
9: Verify  $p_{\text{local},0} = F_0$  within tolerance
10: return  $\{p_{\text{local},n}\}$ 
```

Algorithm 35 Combined convergence and error-control workflow

Require: Target CDF error $\varepsilon_F^{\text{target}}$

Require: Initial OWNS, rank, streamwise, and angular resolutions

Ensure: Numerically converged transition CDF

Ensure: Error-budget summary

1: Allocate component tolerances:

$$\varepsilon_F^{\text{OWNS}}, \quad \varepsilon_F^{\text{red}}, \quad \varepsilon_F^{\text{stream}}, \quad \varepsilon_F^{\text{ang}}.$$

2: Perform the baseline stabilized OWNS propagation

3: Repeat the OWNS calculation with a refined wall-normal grid, marching step, or solver tolerance

4: Estimate

$$\epsilon_{\text{OWNS}} \leftarrow \max_n |F_{\text{OWNS, fine}}(x_n) - F_{\text{OWNS, base}}(x_n)|.$$

5: **if** $\epsilon_{\text{OWNS}} > \varepsilon_F^{\text{OWNS}}$ **then**

6: Update the baseline OWNS resolution and repeat

7: **end if**

8: Construct a transition-oriented reduced basis

9: Increase the rank until

$$\max_n |F_s(x_n) - F_{s-}(x_n)| \leq \varepsilon_F^{\text{red}}$$

and the matrix residual is acceptable

10: Resolve continuous streamwise maxima on a shared adaptive grid

11: Refine the streamwise grid until

$$\max_n |F_{\text{stream, fine}}(x_n) - F_{\text{stream, base}}(x_n)| \leq \varepsilon_F^{\text{stream}}.$$

12: Refine deterministic angular quadrature or RQMC until

$$\max_n |F_{K+}(x_n) - F_K(x_n)| \leq \varepsilon_F^{\text{ang}}.$$

13: Recheck stochastic-rank convergence using the final streamwise and angular settings

14: Recheck streamwise convergence using the final rank and angular settings

15: Verify probability bounds, monotonicity, complementarity, and local-exceedance inequalities

16: Compare selected cases with trajectory Monte Carlo

17: Report

$$\epsilon_{\text{OWNS}}, \quad \epsilon_{\text{red}}, \quad \epsilon_{\text{stream}}, \quad \epsilon_{\text{ang}}.$$

18: **return** the converged CDF and error summary
